



Government of the People's Republic of Bangladesh
Ministry of Health and Family Welfare

HEALTH BULLETIN 2013

Management Information System
Directorate General of Health Services
Mohakhali, Dhaka 1212
www.dghs.gov.bd



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February 2014

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Directorate General of Health Services
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We acknowledge, with gratitude, the contributions of all other personnel working in the public and private health systems of Bangladesh, who were involved directly or indirectly in the production of this Health Bulletin by sending data or providing logistical support



Honorable Minister
Ministry of Health and Family Welfare
Government of the People's Republic of Bangladesh

MESSAGE

Health Bulletin of the Directorate General of Health Services (DGHS) is one of the critical documents of health statistics in Bangladesh. It contains lots of routine health information as well as survey results, with indepth analysis. The bulletin is the witness of our many achievements towards the MDGs.

I am very happy to know that DGHS is regularly publishing this bulletin with exciting information on the health sector. Nowadays, information is easily accessible through innovative eHealth and mHealth developed by DGHS. Recently, they have started web services for many aspects of health-related information, like organization registry, HRM software, and online health bulletins. The innovations make every citizen to have free access to necessary health information. I believe this momentum will continue to realize the dream of Digital Bangladesh and to achieve the Millennium Development Goals transparently through free access to information.

Bangladesh has made service delivery available at the doorstep of people through the leadership of our Hon'ble Prime Minister Sheikh Hasina. The MIS of Health has taken steps to connect with Internet device every community clinic and almost every health worker under DGHS. This is part of the infrastructure being developed from national up to the ward level in Bangladesh. Information on every mother and her child is going to be available in near future. I am glad to acknowledge that the MIS of DGHS is the pioneer in fulfilling the targets of Digital Bangladesh through its leadership and remarkable success in recent years.

I like to extend my thanks to all MIS personnel and the development partners who are helping MIS in its endeavor to achieve these successes. I believe their good work will continue, and Health Bulletin 2013 will fulfill the information needs of all concerned.

Joy Bangla, Joy Bangabandhu.
Long live Bangladesh.

Mohammed Nasim, MP



Zahid Maleque, MP
State Minister
Ministry of Health and Family Welfare
Government of the People's Republic of Bangladesh

MESSAGE

Health Bulletin is like a mirror that reflects our commitment, policies, and efforts for improving the health scenario of the country. So, I am keen to keep an eye on this publication of the Directorate General of Health services.

Without doubt it can be emphasized that this is the most reliable source of health-related information of the country. It gives us important clues for policy-making. I think this bulletin is helping us develop a culture of evidence-based decision-making which is very important for a country like ours. I appreciate the hard work of the officials and staff involved in the production of this publication.

Although the personnel of DGHS are continuously trying for the betterment of this publication by both qualitative and quantitative improvements of data collection, I believe still there is scope for development in other areas of interest, especially about collecting data from the private sector.

I would like to give my heartiest thanks to the personnel of the Directorate General of Health Services in accomplishing this mammoth task.

Joy Bangla, Joy Bangabandhu.

Long live Bangladesh.

Zahid Maleque, MP



Secretary
Ministry of Health and Family Welfare
Government of the People's Republic of Bangladesh

MESSAGE

Health Bulletin is an important yearly publication of the Directorate General of Health Services under the Ministry of Health and Family Welfare. It provides an authentic picture that covers almost all aspects of healthcare delivery, especially by the Government. This critically helps us analyze our work and policies and also plays an important role for future planning.

I am happy to know that the efforts of the Government in improving the health scenario of the country have been reflected in the facts and figures presented in this bulletin. At the same time, it is also showing us some areas for further improvements. I am hopeful that we shall be able to take prompt actions for the betterment in those areas.

I sincerely appreciate the hard work of personnel in the Directorate General of Health Services for production of this publication. Their efforts to improve the data collection and data presentation are also praiseworthy. I hope they will continue their good work despite our limitations and constraints.

I thank those who are directly and indirectly involved in the production of this bulletin, especially its Chief Editor Professor Dr. Abul Kalam Azad.

M.M. Neazuddin



Director General of Health Services
Government of the People's Republic of Bangladesh

MESSAGE

It is, indeed, a great satisfaction for me to know that MIS-Health is going to publish the Health Bulletin 2013 soon. The bulletin, in fact, accumulates the reports of general health situation of the country as it reflects almost a complete picture of both public and most private healthcare delivery. In this 'information age', world progress is based on appropriate information, and collection of information is one of the hardest jobs in the context of Bangladesh.

It is definitely an immense pleasure for me that quality of Health Bulletin is going up year by year, particularly in terms of analysis and authenticity of data. It is worthy to note that the IT network of MIS-Health has progressed tremendously, and now it has reached up to the level of community clinic, which has made access to information easier. The rapid digitalization of MIS-Health has connected even the private health facilities for sending their data to the central database. Exciting ideas for improvement of data-collection procedure and record-keeping system by using computerized tools are utilized in service provision and monitoring of attendance in different health facilities, especially in remote areas. I applaud the MIS-Health team for their relentless valuable efforts. The availability of accurate and timely data is essential to monitor the progress of health development and to evaluate the impact. Now it is well-accepted that updated information is very important for efficient management at all levels. I hope this bulletin enriched with more updated information will be very useful to the implementers, managers, researchers, policy-makers, and to our development partners.

Finally, I would like to extend my heartiest appreciations to Professor A.K. Azad, Additional Director General (Planning & Development) and Director, MIS, the editorial board members, and to everyone who worked hard to bring out this publication.

Professor Dr. Deen Mohd. Noorul Huq



FROM THE CHIEF EDITOR

It gives me immense pleasure to finally come to a stage of presenting Health Bulletin 2013. Health Bulletin is the flagship publication of the Directorate General of Health Services (DGHS). This publication prints technical information on healthcare-related development activities of the Government of Bangladesh each year. Health Bulletin 2013 contains the remarkable achievements in our health sector during 2012 and part of 2013 in some cases.

I am very happy to put forward accomplishments of DGHS, especially in new progress in eHealth and mHealth in Bangladesh. Health Bulletin 2013 is rich in quality of contents, accurate in data presentation, has meticulous editing, and is a high-quality production which will definitely meet the needs of the readers.

I express special thanks for the support and cooperation of Mr. Mohammed Nasim, MP and Honorable Minister for Health and Family Welfare; Mr. Zahid Maleque, MP and Honorable State Minister for Health and Family Welfare; Mr. MM Neazuddin, Secretary of the Ministry of Health and Family Welfare; and guidance of my Director General Professor Dr. Deen Mohd. Noorul Huq.

I also convey my thanks to the members of the production team of Health Bulletin 2013, both in my own office and also throughout the health sector who provided data and put data into the reports. Their relentless efforts and continuous labor have made it possible for me to publish this in time. I also acknowledge the relevant personnel in all other offices and departments, both in public and private sectors, for contributing in Health Bulletin 2013.

I am hopeful that this publication will satisfy the readers, and I sincerely invite positive criticism from them, which will enable us to bring the bulletin to another height.

Professor Dr. Abul Kalam Azad
Additional Director General (Planning & Development)
and
Director, Management Information System
Directorate General of Health Services
Government of the People's Republic of Bangladesh

ACRONYMS

ADB	Asian Development Bank	CMMU	Construction, Maintenance and Management Unit
ADP	Annual Development Program	CMNS	Child and Mother Nutrition Survey
AEFI	Adverse Events Following Immunization	CMSD	Central Medical Stores Depot
AFP	Acute Flaccid Paralysis	CNP	Community Nutrition Promoter
AIDS	Acquired Immunodeficiency Syndrome	CNS	Child Nutrition Survey
ALS	Average Length of Stay	COPD	Chronic Obstructive Pulmonary Disease
AMC	Alternative Medical Care	CPR	Contraceptive Prevalence Rate
ANC	Antenatal Care	CRF	Chronic Renal Failure
APIR	Annual Program Implementation Report	CS	Civil Surgeon
APR	Annual Program Review	C-section	Cesarean Section
ARC	American Red Crescent	CSO	Community Support Organization
ARI	Acute Respiratory Infection		
BBS	Bangladesh Bureau of Statistics	DAB	Diabetic Association of Bangladesh
BCC	Behavior Change Communication	DBRH	Demand-based Reproductive Health
BCG	Bacillus Calmette Guerin	DCA	Development Credit Agreement
BCS	Bangladesh Civil Service	DCM	Dilated Cardiomyopathy
BDHS	Bangladesh Demographic and Health Survey	DDA	Directorate of Drug Administration
BEOC	Basic Emergency Obstetric Care	DDC&H	Dhaka Dental College & Hospital
BGC	Bangladesh Geographic Survey	DF	Dengue Fever
BHE	Bureau of Health Education	DFID	Department for International Development
BIDS	Bangladesh Institute for Development Studies	DG	Director General
BINP	Bangladesh Integrated Nutrition Project	DGFP	Directorate General of Family Planning
BMA	Bangladesh Medical Association	DGHS	Directorate General of Health Services
BMI	Body Mass Index	DH	District Hospital
BMMS	Bangladesh Maternal Mortality Survey	DHF	Dengue Hemorrhagic Fever
BMRC	Bangladesh Medical Research Council	DMC	Dhaka Medical College
BNHA	Bangladesh National Health Accounts	DMCH	Dhaka Medical College Hospital
BRAC	Bangladesh Rural Advancement Committee	DNS	Directorate of Nursing Services
BSA	Bangladesh Society of Anesthesiologists	DOTS	Directly Observed Treatment-Short Course
BSMMU	Bangabandhu Sheikh Mujib Medical University	DP	Development Partner
		DPA	Direct Project Aid
		DPHE	Department of Public Health Engineering
		DSF	Demand-side Financing
CBHC	Community-based Healthcare	ECNEC	Executive Committee of National Economic Council
CABG	Coronary Artery Bypass Grafting	EDPT	Early Diagnosis and Prompt Treatment
CBN	Cost of Basic Needs (method)	EmOC	Emergency Obstetric Care
CC	Community Clinic	EPI	Expanded Program on Immunization
CDC	Communicable Disease Control	EPR	Emergency Preparedness
CDD	Control of Diarrheal Diseases	ERD	Economic Relations Division
CFP	Conceptual Framework Paper	ESD	Essential Service Delivery
CGA	Comptroller General of Accounts	ESP	Essential Service Packages
CHCP	Community Healthcare Provider	ETT	Exercise Tolerance Test
CIDA	Canadian International Development Agency	EU	European Union
CIDD	Control of Iodine Deficiency Disorder		
CMCH	Chittagong Medical College Hospital	FEP	Filariasis Elimination Program
CME	Centre for Medical Education	FMAU	Financial Management and Audit Unit

FMRP	Financial Management Reforms Project	IMCI	Integrated Management of Childhood Illness
FP	Family Planning	IMED	Implementation, Monitoring and Evaluation Division
FWA	Family Welfare Assistant	IMF	International Monetary Fund
FY	Financial Year	IMHR	Institute of Mental Health and Research
GAVI	Global Alliance for Vaccine and Immunization	IMR	Infant Mortality Rate
GDP	Gross Domestic Product	IOL	Intraocular Lens
GFTAM	Global Fund to Fight AIDS, Tuberculosis and Malaria	IPGMR	Institute of Postgraduate Medicine and Research
GHDCH	Government Homeopathic Degree College Hospital	IPH	Institute of Public Health
GO	Government Organization	IPHN	Institute of Public Health Nutrition
GOB	Government of Bangladesh	IPM	Individual Performance Management
GTC	Government Tibbia College	i-PRSP	Interim Poverty Reduction Strategy Paper
GUADCH	Government Unani and Ayurvedic Degree College & Hospital	IRS	Indoor Residual Spraying
HA	Health Assistant	IST	In-service Training
HDI	Human Development Index	IT	Information Technology
HDS	Health and Demographic Survey	ITHC	Integrated Thana Health Complex
HEB	Health Education Bureau	ITMN	Insecticide-treated Mosquito-net
HEU	Health Economics Unit	IUD/IUCD	Intra-uterine (Contraceptive) Device
HFWC	Health and Family Welfare Center	IVM	Integrated Vector Management
HIES	Household Income and Expenditure Survey	IYCF	Infant and Young Child Feeding
HIU	Health Information Unit	JICA	Japan International Cooperation Agency
HIV	Human Immunodeficiency Virus	KMCH	Khulna Medical College Hospital
HIV/AIDS	Human Immunodeficiency virus/ Acquired Immunodeficiency Syndrome	LAN	Local Area Network
HKI	Helen Keller International	LBW	Low Birthweight
HLIC	High-level Inter-ministerial Committee	LD	Line Director
HSM	Hospital Services Management	LLIN	Long-lasting Insecticidal Net
HMPD	Health Manpower Development	LLP	Local-level Planning
HNP	Health, Nutrition and Population	LTSO	Long-term Strategy Options
HNPSP	Health, Nutrition and Population Sector Program	M&E	Monitoring & Evaluation
HPNSDP	Health, Population and Nutrition Sector Development Program	M/F	Male/Female Ratio
HR	Human Resource	MATS	Medical Assistant Training School
IAPB	International Association for Prevention of Blindness	MBDC	Mycobacterial Disease Control
ICOVED	Integrated Control of Vector-borne Diseases	MC	Medical College
ICT	Information and Communication Technology	MCH	Maternal and Child Health
IDA	Iron-deficiency Anemia	MCH	Medical College Hospital
IDD	Iodine-deficiency Disorder	MCWC	Maternal and Child Welfare Center
IDH	Infectious Diseases Hospital	MDA	Mass Drug Administration
IEC	Information, Education and Communication	MDG	Millennium Development Goal
IEDCR	Institute of Epidemiology, Disease Control & Research	MIS	Management Information System
IHSM	Improved Hospital Services Management	MMR	Maternal Mortality Ratio
IHT	Institute of Health Technology	MNCH	Maternal, Neonatal and Child Health
		MNH	Maternal and Neonatal Health
		MNHC	Maternal and Neonatal Healthcare
		MO	Medical Officer
		MoHFW	Ministry of Health and Family Welfare
		MoLGRDC	Ministry of Local Government, Rural Development and Cooperatives
		MoU	Memorandum of Understanding
		MP	Member of Parliament
		MSA	Management Support Agency

MSD	Medical Subdepot	PH	Public Health
MTR	Mid-term Review	PMIS	Personnel Management Information System
MWM	Medical Waste Management	PRSP	Poverty Reduction Strategy Paper
		PSM	Preventive and Social Medicine
NCD	Non-communicable Diseases		
NEMEW	National Equipment Maintenance and Engineering Workshop	RDU	Research and Development Unit
NGO	Non-governmental Organization	RADP	Revised Annual Development Program
NICRH	National Institute of Cancer Research and Hospital	RPA	Reimbursable Project Aid
NICVD	National Institute of Cardiovascular Diseases	RCHCIB	Revitalization of Community Healthcare Initiative in Bangladesh
NID	National Immunization Day	RHC	Rural Health Center
NIDCH	National Institute of Diseases of Chest and Hospital		
NIKDU	National Institute of Kidney Diseases and Urology	SBTP	Safe Blood Transfusion Program
NICRH	National Institute of Cancer Research & Hospital	SEARO	South-East Asian Regional Office
NIMHR	National Institute of Mental Health and Research	SVRS	Sample Vital Registration System
NIO	National Institute of Ophthalmology		
NIPORT	National Institute of Population Research and Training	TEMO	Transport & Equipment Maintenance Unit
NIPSOM	National Institute of Preventive and Social Medicine	TB	Tuberculosis
NITOR	National Institute of Traumatology, Orthopedics and Rehabilitation	TT	Tetanus Toxoid
NASP	National AIDS and STD Program	TTU	Technical Training Unit
NNS	National Nutrition services		
NNP	National Nutrition Program	UHC	Upazila Health Complex
NTP	National TB Program	UHFPO	Upazila Health and Family Planning Officer
OP	Operational Plan	UHFWC	Union Health and Family Welfare Center
OPD	Outpatient Department	UNICEF	United Nations Children's Fund
ORS	Oral Rehydration Salt	UNGASS	United Nations General Assembly Special Session
ORT	Oral Rehydration Therapy	USC	Union Subcenter
OT	Operation Theater	USI	Universal Salt Iodization
		VAC	Vitamin A Capsule
		VAD	Vitamin A Deficiency
		WAZ	Weight-for-age z-score
		WCBA	Women of Childbearing Age
		WHO	World Health Organization

CONTENTS

Chapter/Topic	Page
Messages	3
From the Chief Editor	7
Acronyms	8
Chapter 1 Bangladesh at a glance	13
Chapter 2 Healthcare network of Bangladesh	18
Chapter 3 Health-related millennium development goals	23
Chapter 4 Primary healthcare	28
Chapter 5 Secondary and tertiary healthcare	40
Chapter 6 Utilization of health facilities	43
Chapter 7 Morbidity profile	47
Chapter 8 Mortality profile	54
Chapter 9 Communicable disease control	61
Chapter 10 Emergency preparedness and response	71
Chapter 11 Non-communicable diseases	74
Chapter 12 Safe blood transfusion	81
Chapter 13 Nutrition situation	84
Chapter 14 Health interventions by IEDCR and IPH	89
Chapter 15 Research and development	96
Chapter 16 Health workforce situation	98
Chapter 17 Health information system, ehealth, and MBT	108
Chapter 18 Health, population, and Nutrition	119
Chapter 19 Financing healthcare	126
Annexes	131



BANGLADESH AT A GLANCE

History

Bangladesh emerged as an independent nation in 1971 before which it was a province of Pakistan and was known as East Pakistan. After the fall of Nawab Sirajuddowla in the Battle of Plassey on 23 June 1757, this part of the Indian Subcontinent was ruled by the British from 1757 to 1947. In 1947, the British colonial rule ended, and the Indian Subcontinent was divided into two independent nations—India and Pakistan. Pakistan was divided into West Pakistan (now Pakistan) and East Pakistan (now Bangladesh). Profound disparities in governance between East and West Pakistan led to the craving for independence among the East Pakistanis since the Language Movement in 1952, followed by series of mass upsurge at various times during the 1960s and early 1970s and a 9-month Liberation War under the leadership of Bangabandhu Sheikh Mujibur Rahman, and Bangladesh emerged as a sovereign nation in 1971.

Geographical location

Bangladesh has a total land area of 147,570 square kilometers (56,977 square miles). It is a low-lying country, with latitude 24 degree north and longitude 90 degree east. The country borders India on three sides with the Indian states of West Bengal, Tripura, Assam, and Meghalaya. Only a small strip in the southeast shares a border with Myanmar. The Bay of Bengal lies to the south (Figure 1.1). Bangladesh comprises primarily floodplains, with scattered hills in the eastern and northern parts. Large rivers and an intricate web of canals and rivers form this Ganges Delta, the largest delta in the world.

Climate

Bangladesh is a tropical country with a hot and rainy summer (March to June), a warm and rainy monsoon (June to October), and a mild dry winter (October to March). January is the coldest month, with an average temperature of 26°C (78.8°F). April is the hottest month, with temperatures ranging from 33°C to 36°C (91.4°F to 96.8°F). Major part of Bangladesh receives more than 1,525 mm of rain a year, with areas near the hills receiving more than 5,000 mm, primarily during the monsoon (June–September). The humidity varies from 73% to 86%



Figure 1.1. Map of Bangladesh

and is the highest during the monsoon and the lowest in the winter.

Religion and Culture

Majority (approx. 89%) of the population is Muslim. Hindus, Buddhists, and Christians comprise 9.6%, 0.6%, and 0.3% of the population respectively. Over 98% of the people speak Bangla. English is also widely spoken. Bangladesh's rich cultural traditions are displayed in archaeological sites, sculptures, terracotta, architecture, museums, archives, libraries, classical music, dance, paintings, dramas, folk arts, festivals, and ethnic cultural activities.

Population and Demography

Bangladesh Bureau of Statistics estimates population of Bangladesh as of 1 July 2011 to be 150.6 million. With annual population growth rate of 1.37% (BBS 2011), the estimated population as of 1 July 2012 should be 152.7 million and as of 1 July 2013 should be 154.8 million. Bangladesh is one of the most densely-populated countries in the world, with 1,021 people living per square km, and approximately 26% of the population lives in the urban area (BBS 2011). The average household-size is 4.5, and life-expectancy at birth for both sexes is 69 years (BBS 2011).

Governance

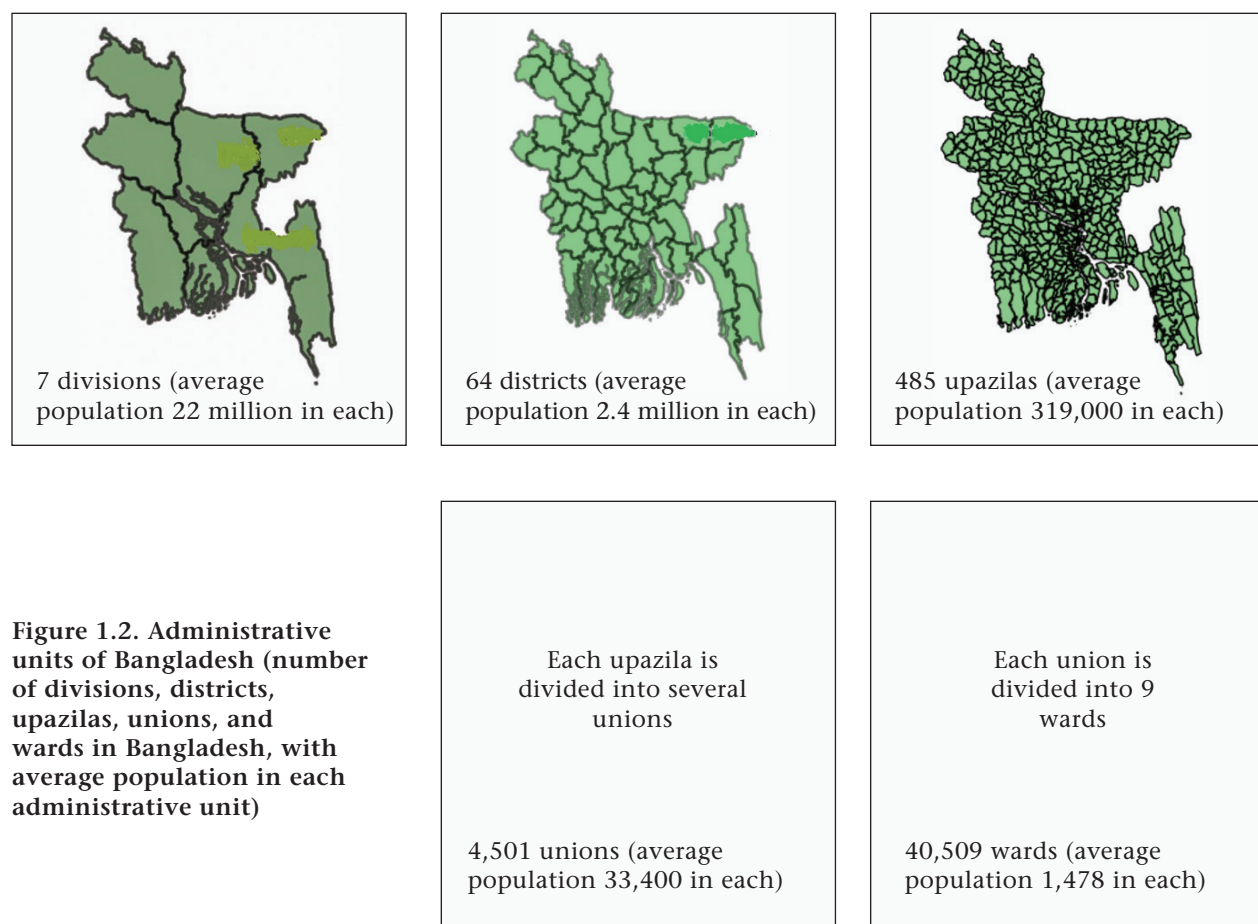
Bangladesh is divided into 7 administrative divisions. Each division is divided into several districts; each district is further subdivided into several upazilas; each upazila into several unions; and each union into nine wards (Figure 1.2).

Wards are divided into several villages. However, ward is the lowest administrative unit of the local government, having at least one representative elected for 5 years by popular vote. The city corporations and municipalities are designated as urban areas, with 11 city corporations and 310 municipalities across the country. Bangladesh is a non-federal country governed by parliamentary democracy. The National Parliament is called Bangladesh Jatiya Sangsad. There are about 40 ministries. A ministry is headed by a minister, with a secretary to head the bureaucrats. Some ministries are divided into functional divisions, with each division having a secretary to head

the bureaucrats of the respective divisions. There are 21 such functional divisions. The Ministry of Health and Family Welfare is one of the largest ministries of the Government of Bangladesh.

Economy

Despite a declining trend, the agrarian sector dominates the country's economy, accounting for majority of the rural labor force. The principal industries of the country include readymade garments, textiles, chemical fertilizers, pharmaceuticals, tea-processing, sugar, and leather goods. The principal minerals include natural gas, coal, white clay, and glass-sand. Bangladesh has been utilizing a mixed system of public and private development, which operates on free-market principles. The GDP growth rate is 6.3% (World Bank 2012). The per-capita income is estimated for the fiscal year 2012-2013 at US\$1,044 taking the fiscal 2005-2006 as the base year (BBS 2013).



Some basic information and data on Bangladesh

GEOGRAPHY

Location: Latitude 24° north and 90° east

Boundary: North and West: India; South: Bay of Bengal; East: India and Myanmar

Area (sq. km): 147,570 sq. km (56,977 sq. miles)

Territorial water: 12 Nautical miles

Standard time: GMT+ 6 hours

Rainfall: 203 mm/month

ADMINISTRATIVE UNITS (No.)

Division: 7

City corporation: 11

Metropolitan city: 4

Municipality: 310

District: 64

Upazila: 485

Union: 4,501

Ward: 40,509

Village (approx.): 87,310

DEMOGRAPHY

Population characteristics (SVRS 2011)

Population (in million): 150.6 as of 1 July 2011 (BBS 2011); Estimated as of 1 July 2013: 154.8

Population growth rate (annual): 1.37%

Sex ratio (M/F): 104.9/100.0

Urban population: 26%

Religion: Muslim: 88.8%; Other religion: 11.2%

Population by age-group

Both sexes

0-14 year(s): 33.1%

15-49 years: 53.5%

50-59 years: 7.7%

60+ years: 6.9%

Male

0-14 year(s): 32.5%

15-49 years: 52.3%

50-59 years: 8.0

60+ years: 7.2%

Female

0-14 year(s): 31.2%

15-49 years: 54.7%

50-59 years: 7.4%

60+ years: 6.7%

No. of children per 1,000 women: National: 341; Rural: 364; Urban: 303

Other population characteristics

Population density per sq. km: 1,021

Dependency ratio: National: 57%; Rural: 61%; urban: 51%

Household (approx. no.): 34.4 million

Average size of household (No. of members per household): 4.5

% of male-headed households: 86.7%

% of female-headed households: 13.3%

Crude birth rate: Per 1,000 population 19.2

Crude death rate: Per 1,000 population 5.5

Net reproduction rate (NRR) per woman (15-49 year): 1.03

Mean age at first marriage (in years): Male: 24.9; Female: 18.6

EDUCATION AND ECONOMY

Literacy rate (7+ years): Both sexes: 55.8%; Male: 58.4%; Female: 53.2% (SVRS 2011)

Adult literacy rate: (15+ years): Both sexes: 58.8%; Male: 62.5%; Female: 55.1% (SVRS 2011)

Per-capita GDP (in US\$) in FY 2012-2013: 1,044 (BBS 2013)

GDP growth rate (%): 6.3% (WB 2012)

Poverty rate: National: 26.0% (WB 2013)

Main occupations: Agriculture, forestry, fisheries: 36.1%; Non-agriculture: 63.9% (BIHES 2010)

Average monthly household income (Tk): National: 11,479; Rural: 9,648; Urban: 16,475 (HIES 2010)

Average monthly household expenditure (Tk): National: 11,200; Rural: 9,612; Urban: 15,531 (HIES 2010)

Average monthly household consumption expenditure (Tk): National: 11,003; Rural: 9,436; Urban: 15,276 (HIES 2010)

Average monthly household food expenditure (Tk): 2,491; Rural: 2,122; Urban: 3,526 (HIES 2010)

Share of food expenditure on total household expenditure (Tk): 53.85%; Rural: 57.67%; Urban: 47.40% (HIES 2010)

Per-capita daily calorie intake (kcal): National: 2,318.3; Rural: 2,344.6; Urban: 2,244.5 (HIES 2010)

Households benefiting from social safety nets: National: 24.57%; Rural: 30.12%; Urban: 9.42% (HIES 2010)

Households with access to electricity: National: 63.6% (SVRS 2011)

Households with mobile phones: National: 63.7%; Rural: 56.7%; Urban: 82.7% (HIES 2010)

HEALTH STATUS

Under-5 mortality rate (per 1,000 livebirths): 53 (BDHS 2011); 41 (UNICEF 2013)

Infant mortality rate (per 1,000 livebirths): 43 (BDHS 2013); 33 (UNICEF 2013)

Neonatal mortality rate (per 1,000 livebirths): 32 (BDHS 2011); 24 (UNICEF 2013)

Maternal mortality ratio (per 100,000 livebirths): 194 (BMMS 2010)

Life-expectancy at birth (year): Both sexes: 69.0; Male: 67.9; Female: 70.3 (SVRS 2011)

Total fertility rate (birth per women 15-49 years): 2.3 (BDHS 2011)

Contraceptive prevalence rate (%): 61.2 (BDHS 2011)

Unmet need for family planning (%): 13.5 (BDHS 2011)

Births attended by skilled health personnel (%): 26.5 (BMMS 2010); 31.7 (BDHS 2011)

Antenatal care coverage (at least one visit by skilled health professional) (%): 54.6 (BDHS 2011)

Antenatal care coverage (at least four visits) (%): 25.5 (BDHS 2011)

Postnatal care received by mothers from a trained care provider within 2 days after delivery: 27 (BDHS 2011)

Birth rate among adolescent mothers/1,000 women: 105.0 (BMMS 2010); 118.3 (BDHS 2011)

Institutional delivery rate (%): 28.8%
Public facility: 11.8%
Private facility: 15.1%
NGO facility: 1.9% (BDHS 2011)

Home delivery rate:
71.0% (BDHS 2011)

C-section rate (%): 17.1% (BDHS 2011)
Other indirect methods: 11.9%
(see Chapter 4, DGHS 2013)

Malaria prevalence/100,000 population (in endemic areas): 18.4 (DGHS 2012) estimated based on reported malaria cases

Malarial death rate/100,000 population: 0.007 (DGHS 2012) estimated based on reported malaria cases

Under-five children sleeping under insecticide-treated bednets in endemic areas (%): 94.4 (DGHS 2012)

Under-five children with fever treated with appropriate antimalarial drugs (%): 89.0 (DGHS 2011)

TB (all forms) prevalence rate/100,000 population: 411 (DGHS 2011)

TB death rate/100,000 population: 45 (DGHS 2011)

New smear+ve TB case detection rate under DOTS (%): 70 (NTP 2012)

TB cure rate (%) with DOTS: 92.0 (NTP 2012)

HIV prevalence among population aged 15-24 years (%): 0.7% (most-at-risk population group) (NASP 2011)

Population with advanced HIV infection with access to ARV drugs (%): 45.0 (UNGASS 2012)

Immunization (valid vaccination coverage) (EPI-CES 2013)

≤12 months old children: BCG 95%; OPV1 95%; OPV2 94%; OPV3 92%; Penta1 91%; Penta2 93%; Penta3 92%; Measles 86%; Full vaccination: 81%

≤23 months old children: BCG 95%; OPV1 95%; OPV2 94%; OPV3 92%; Penta1 91%; Penta2 93%; Penta3 92%; Measles 89%; Full vaccination: 84%

Tetanus toxoid coverage (%) among women of childbearing age: TT1 92%; TT2 89%; TT3 74%; TT4 52%; TT5 32% (EPI-CES 2013)

Vitamin A coverage (EPI-CES 2013)

Infant (6-11 months): 84%
Children (6-59 months): 93%
Postpartum women: 36%

Iron deficiency-related information

Anthelmintics coverage:
24-59 months old children 90% (EPI-CES 2013)

Percentage of women taking iron-folate tablets during last pregnancy: 55% (FSNSP 2010)

Iodized salt coverage

Coverage of households with iodized salt: 80% (NMSS 2011-12)

WATER AND SANITATION

Households with access to tap and tubewell water for drinking: 98.2% (SVRS 2011)

Households with access to sanitary toilet facility: 63.6% (SVRS 2011); 80.4 (MICS 2009)

HEALTH SERVICES PROVISION

No. of hospitals

Total number of government hospitals under MOHFW: 593 (DGHS 2013)

Government hospitals of secondary and tertiary levels under MOHFW: 126 (DGHS 2013)

Government hospitals at upazila and union levels: 467 (DGHS 2013)

No. of private registered hospitals and clinics under DGHS: 2,983 (DGHS 2013)

No. of private registered diagnostic centers under DGHS: 5,220 (DGHS 2013)

Hospital beds

Total functional beds (in MOHFW and registered private hospitals): 91,106

No. of hospital beds under MOHFW (functional): 45,621 (DGHS 2013)

No. of hospital beds in private-sector (in private hospitals registered by DGHS: 45,485 (DGHS 2013)

Population per hospital bed (total beds in MOHFW and registered private hospitals against estimated population as of 1 July 2013): 1,699 (DGHS 2013)

Teaching/training institutions for health (DGHS 2013)

No. of postgraduate medical teaching institutions: 33; Government: 23; Private: 10

No. of medical colleges: Total 75; Government: 22; Private: 53

No. of dental colleges: Total 27; Government: 9; Private: 18

No. of degree/diploma colleges for alternative medicine: Total 64; Government: 3; Private: 61

No. of nursing institutions offering masters degree to diploma: Total 131; Government: 57; Private: 74

No. of institutions providing midwifery training: 12

No. of training institutions for community-based skilled birth attendants: 47

No. of medical assistants training schools: Total 111; Government: 8; Private: 103

No. of institutes of health technology (IHT): Total 112; Government: 11; Private: 97; Government-Private: 4

No. of seats for medical courses

Postgraduate medical degree (MD, MS, Diploma, M.Phil. etc.): Total 2,270; Government 2,091; Private 169

FCPS, MCPS, etc.: ~400 per year

MBBS: Total 7,612; Government 2,812; Private 4,800

BDS: Total 1,597; Government 532; Private 1,065

Bachelor of Unani and Ayurvedic Medicine: 50

Bachelor of Homeopathic Medicine: 50

Nursing: Total 6,680; Government 3,915; Private 2,765

Midwifery: 320

Medical Assistant: Total 6,821; Government 716; Private 6,105

Medical technologist: Total 14,255; Government 2,684; Private 11,391; Government-Private 180

Health workforce

No. of sanctioned posts under DGHS: 115,935 (DGHS 2013)

No. of personnel and staff under DGHS (existing): 93,317 (DGHS 2013)

Physicians/dental surgeons

No. of registered physicians: 64,434 (BMDC 2013)

No. of registered dental surgeons: 6,034 (BMDC 2013)

Distribution of doctors: MOHFW 35%; Other ministries 3%; Private sector 62% (HRM unit MOHFW 2013)

No. of doctors under DGHS (existing) 15,922 (DGHS 2012)

Doctors under DGFP (existing): 511 (sanctioned posts 1,110 DGFP 2013)

Total no. of doctors in DGHS and DGFP (MOHFW): 16,433

Estimated number of doctors available in the country: 46,951 [indirect estimate from 16,433 doctors of MOHFW, which comprise 35% of country's total doctors]

Nurses

No. of registered nurses: 30,516 (BNC 2013)

No. of nurses in public sector (existing): 13,235 (DNS 2013)

No. of nurse-midwives in public sector (existing): 596 (DGHS 2013)

No. of trained skilled birth attendants: 7,265 (BNC 2013)

Family planning officers (DGFP 2013)

No. of family planning officers: 364 (sanctioned posts 485)

No. of assistant family planning officers: 280 (sanctioned posts 485)

Medical technologists under DGHS (existing) (DGHS 2012)

No. of sanitary inspectors: 502

No. of dental technologists: 495

No. of laboratory technologists: 1,610

No. of pharmacy technologists: 2,172

No. of radiographers: 634

No. of physical therapists: 147

Sub-assistant community medical officers under DGHS (existing) (DGHS 2012)

No. of sub-assistant community medical officers: 4,212

Community health workers (existing)

No. of community healthcare providers to work at community clinics 13,240 (DGHS 2013)

No. of domiciliary workers under DGHS: Total 24,062; Health inspectors (HI) 1,126; Assistant health inspectors (AHI) 3,662; Health assistants (HA) 19,274 (DGHS 2013)

No. of facility-based workers under DGFP: Total 5,358 (sanctioned posts 5,634); FWV 211 (sanctioned posts 464); Senior FWV 5,147 (sanctioned posts 5,710) (DGFP 2013)

No. of domiciliary workers under DGFP: Total 24,662 (28,000); Family planning inspectors (FPI) 3,549 (sanctioned posts 4,500);

Family welfare assistants (FWA): 21,113 (sanctioned posts 23,500) (DGFP 2013)

Total no. of community health workers under MOHFW: 67,322 (DGHS/DGFP 2013)

Population-health workforce ratio

Population per physician: 3,297 (DGHS 2013)

No. of physicians per 10,000 population: 3.0 (DGHS 2013)

Population per MOHFW's nurse: 11,696 (DNS 2013)

No. of MOHFW's nurses per 10,000 population: 0.9 (DNS 2013)

Population per MOHFW's medical technologist: 27,842 (DGHS 2013)

No. of MOHFW's medical technologists per 10,000 population: 0.4 (DGHS 2013)

Population per MOHFW's community health worker: 2,299 (DGHS/DGFP 2013)

No. of MOHFW's community health workers per 10,000 population: 4.3 (DGHS/DGFP 2013)

HEALTHCARE NETWORK UNDER MOHFW OF BANGLADESH

A well-structured model to reach out citizens

An intricate web of public health departments, NGOs, and private institutions constitutes the public healthcare network of Bangladesh. Responsibilities and functions range from policy-planning, regulation, implementation, and healthcare delivery to medical education. The Ministry of Health and Family Welfare (MOHFW) is the lead agency responsible for formulating national-level policy, planning, and decision-making in the provision of healthcare and education. The national-level policies, plans, and decisions are translated into actions by various implementing authorities and healthcare delivery systems across the country from national to the community level. The Ministry and its relevant regulatory bodies also have indirect control over the healthcare system of the NGOs and the private sector. However, this chapter gives a brief description of the organizational structure of the Directorate General of Health Services (DGHS) only. The chapter aims to describe the roles and responsibilities of the Directorate, its reporting units, and other affiliated organizations in the provision and promotion of healthcare services and health education.

Hierarchy of Personnel in the Ministry of Health and Family Welfare

The Honorable Minister for Health and Family Welfare heads the MOHFW. The Minister is assisted by the Honorable State Minister. As the principal executive of the Ministry, the Secretary works with a team of officials, including Additional Secretary, Joint Secretaries/Joint Chiefs, Deputy Secretaries/Deputy Chiefs, Senior Assistant Secretaries/Senior Assistant Chiefs, and others (Figure 2.1).

Implementing authorities

There are 9 implementing authorities under the MOHFW. These are:

1. Directorate General of Health Services (DGHS)
2. Directorate General of Family Planning (DGFP)
3. National Institute of Population Research & Training (NIPORT)

The national-level policies, plans, and decisions are translated into actions by various implementing authorities and healthcare delivery systems across the country from national to the community level ...

4. Directorate General of Drug Administration (DGDA)
5. Directorate General of Health Economics Unit (HEU)
6. Directorate General of Health Engineering Department (formerly known as Construction Management & Maintenance Unit or CMMU) (HED)
7. Directorate of Nursing Services (DNS)
8. Revitalization of Community-based Healthcare Project (Community Clinics Project) (RCHCIB)
9. Essential Drugs Company Limited (EDCL)
10. Transport & Equipment Maintenance Organization (TEMO)
11. National Electro-medical & Engineering Workshop (NEMEW)

Figure 2.2 shows the implementing authorities under the Ministry of Health and Family Welfare.

Regulatory bodies

The regulatory bodies under the MOHFW include:

1. Bangladesh Medical & Dental Council (BMDC)
2. Bangladesh Nursing Council (BNC)
3. State Medical Faculty (SMF)
4. Homeo, Unani and Ayurvedic Board
5. Bangladesh Pharmacy Council (Figure 2.3)

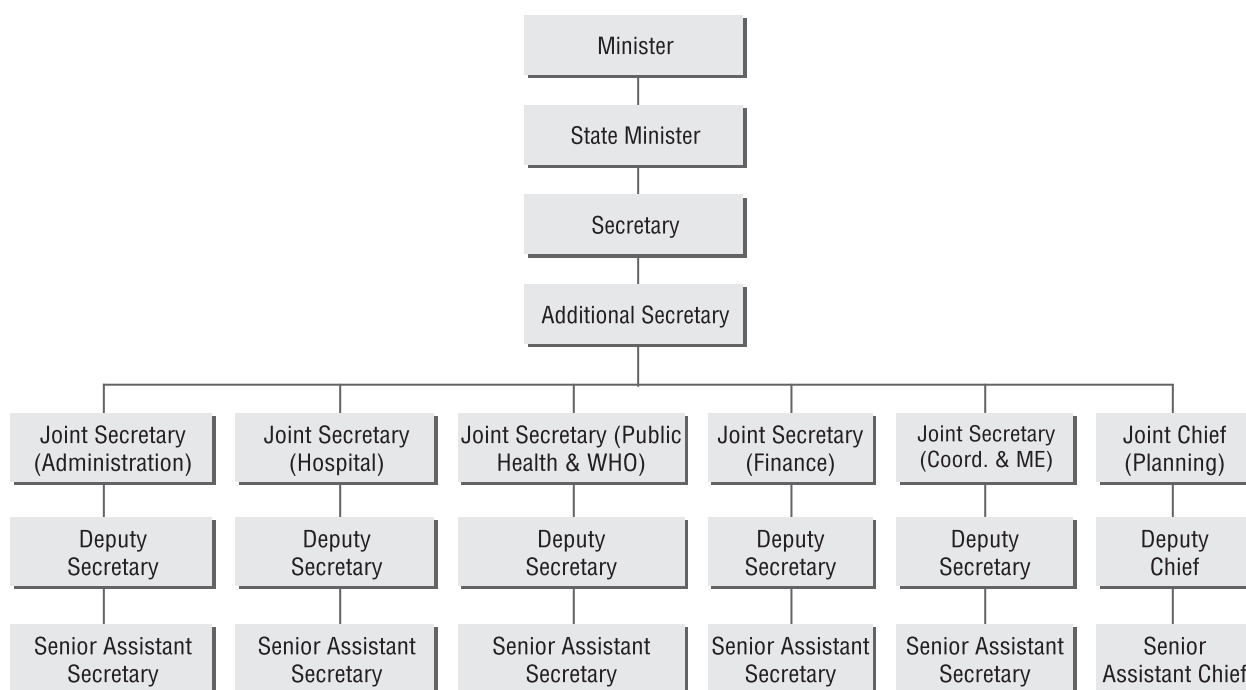


Figure 2.1. Hierarchy of personnel in the MOHFW

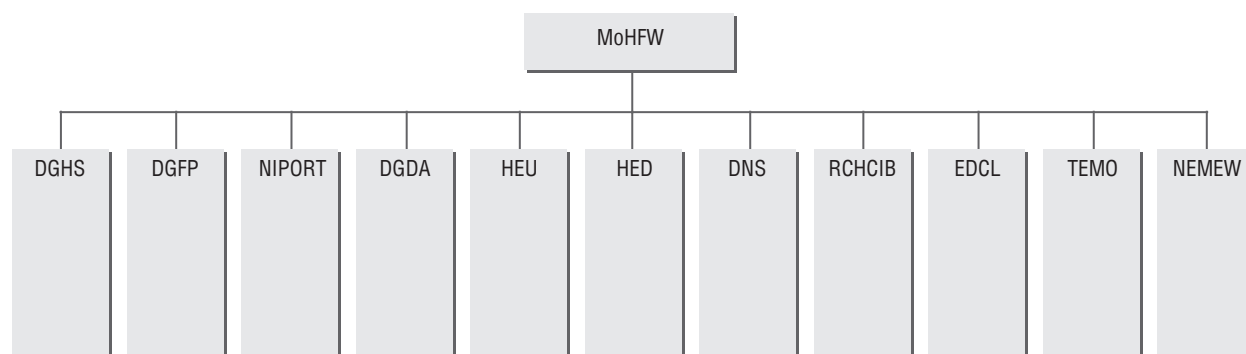


Figure 2.2. Implementing authorities under the Ministry of Health and Family Welfare (for full name of organization, see the section preceding this figure)

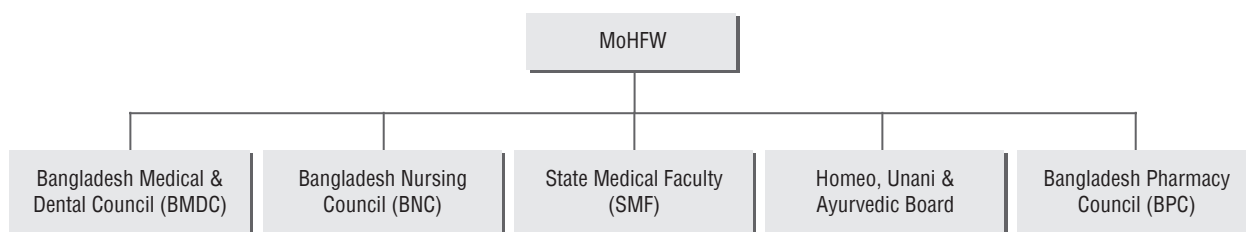


Figure 2.3. Regulatory bodies under MOHFW

Directorate General of Health Services

With more than one hundred thousand officers and staff members, the Directorate General of Health Services (DGHS) is the largest implementing authority under the MOHFW. In addition to the operation of healthcare-delivery systems in the country, the DGHS provides technical assistance to the Ministry in undertaking new programs and interventions and for improvements in the existing ones. The healthcare-delivery systems under DGHS extend from national to the community level. The activities are implemented under regular revenue setups and the development programs. The development programs are designed following a sector-wide, multi-year approach.

The administrative setup of the DGHS, as presented in Figure 2.4, indicates the diversity of activities carried out by the Directorate.

In addition to the operation of healthcare-delivery systems in the country, the DGHS provides technical assistance to the Ministry in undertaking new programs and interventions and for improvements in the existing ones ...

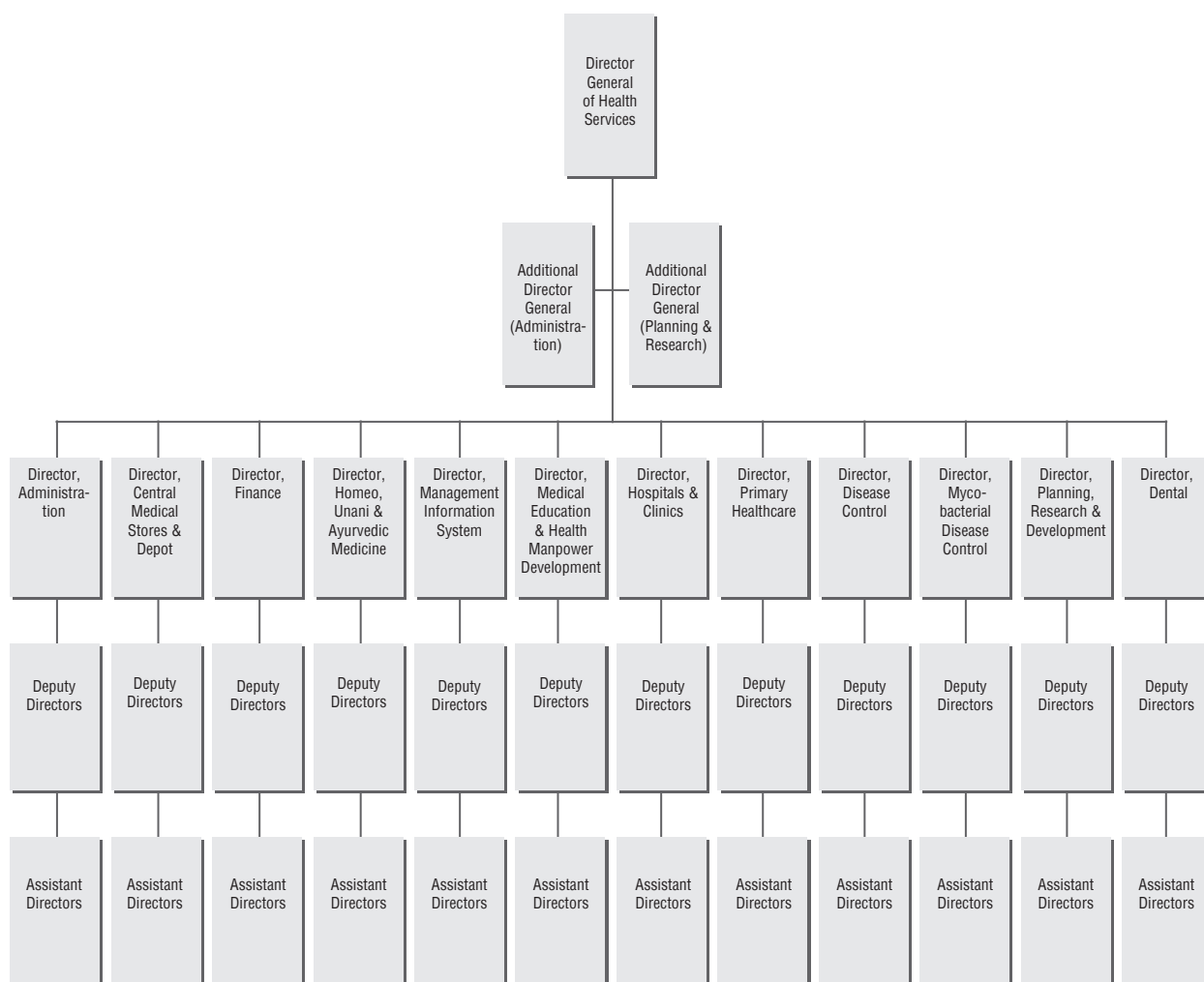


Figure 2.4. Administrative setup of the Directorate General of Health Services

Management structure and type of health facilities under DGHS

The healthcare infrastructure under the DGHS comprises six tiers: national, divisional, district, upazila (subdistrict), union, and ward. At the national level, there are institutions both for public health functions as well as for postgraduate medical education/training and specialized treatment to patients.

A divisional director for health in each division governs activities, and is assisted by deputy directors and assistant directors. There is one infectious disease hospital and one or more medical college(s) at the divisional headquarters. Each medical college has an attached hospital. Some divisional headquarters also possess general hospitals and institutes of health technologies. Divisional institutes provide basically the tertiary-level care.

The civil surgeon (CS) is the district health manager responsible for delivering secondary and primary care services. In each district, there is a district hospital. Some district hospitals have superintendents to look after the hospital management. In others, civil surgeons look after the district hospitals. Some of the district headquarters have medical colleges with attached hospitals, medical assistants training schools, and nursing training institutes.

The upazila health & family planning officer (UH&FPO) is the health manager at the upazila level. S/he manages all public-health programs, especially the primary healthcare services in the upazila and also looks after the upazila hospital (having 30 to 50 beds). The upazila where the district headquarter is located does not have an upazila hospital, and there, the upazila-hospital service is provided by the district hospital.

At the union level, three kinds of health facilities exist: rural health centers, union subcenters, and union health & family welfare centers (UHFWCs). Each union health facility employs a medical doctor among other staff. Only outdoor services are available at union level. All union facilities have

sub-assistant community medical officers to provide health services to the people.

The MOHFW established 12,527 community clinics (CCs) at the ward levels. One such community clinic is planned for every 6,000 people, with a total 18,000 CCs in the country. The existing union and upazila facilities (~4,500) also provide community clinic services. Currently, 12,527 CCs are in operation. The RCHCIB project is responsible for operationalizing the CCs. These facilities are mainly responsible for delivering primary healthcare services, like EPI, treatment for common diseases (pneumonia, fever, cough, etc), family planning services, health education, and first-aids and serve as the first contact points for patients. Some of the community clinics has also stated services for normal delivery through CSBA at community clinic. The MOHFW aims to develop the CCs as the unit of comprehensive healthcare-seeking behavior change in the respective local communities through a sense of ownership and provision of leadership by community people. At the ward or village level, there are also domiciliary health workers – one for every 5,000 to 6,000 people. There are 26,412 sanctioned posts of domiciliary workers under DGHS: 20,815 health assistants (HA), 4,198 assistant health inspectors (AHI), and 1,399 health inspectors (HI). The Directorate General of Family Planning (DGFP) also has domiciliary family planning staff working at the ward levels. Currently, the domiciliary staff members from DGHS and DGFP share the responsibility of running the independent community clinics, along with the community healthcare provider (CHCP).

Figure 2.5 shows the types of organizations and facilities under DGHS from national to the ward level, with managerial hierarchy.

The Ministry recruited 13,240 full-time community healthcare providers to run the community clinics. All of them have been trained to provide better care to the healthcare-seekers. The CHCPs have also been provided with laptop computers and Internet connection to update local health data in the online central databases.

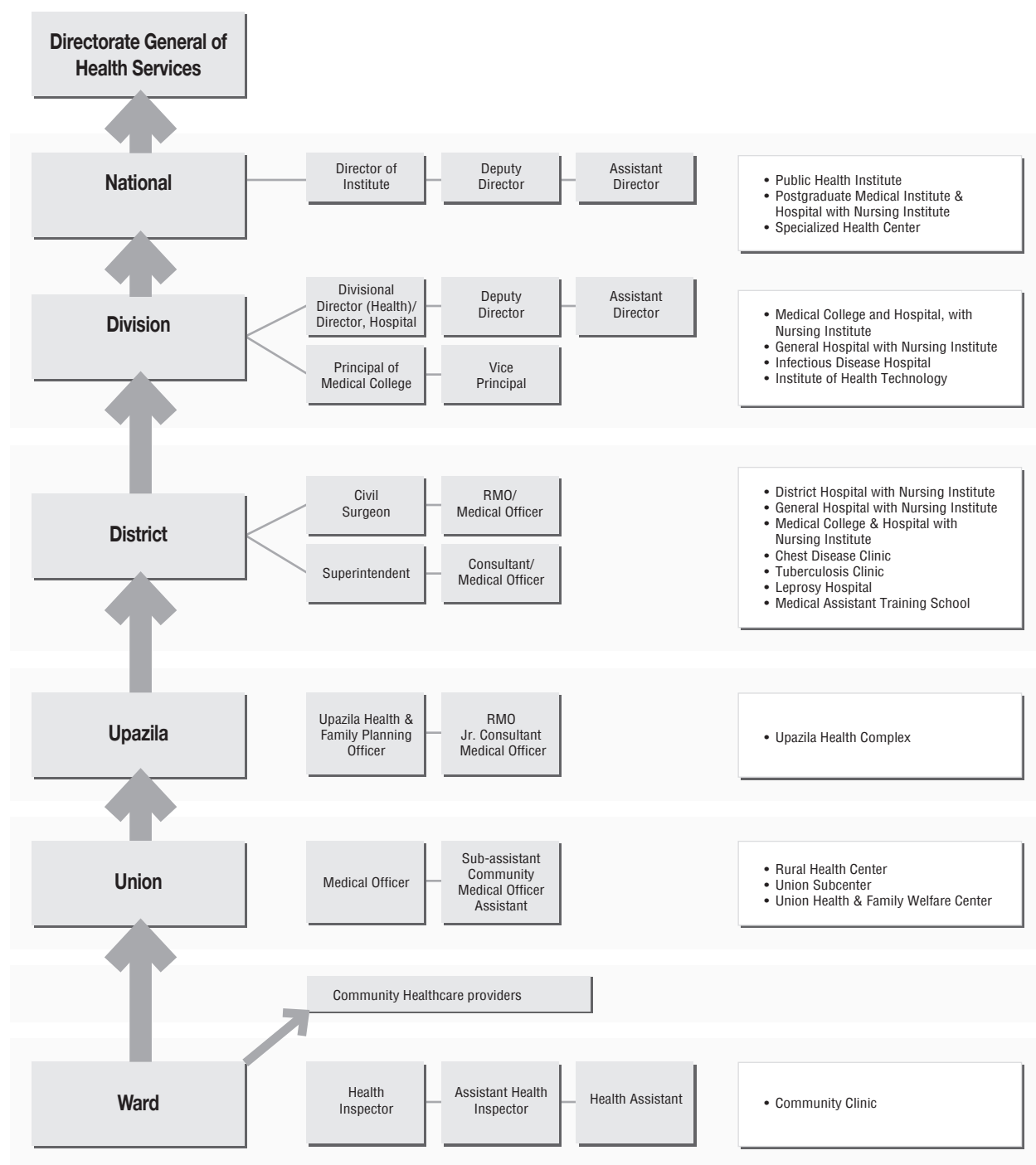


Figure 2.5. Types of facilities from national to the ward level, with managerial hierarchy

HEALTH-RELATED MILLENNIUM DEVELOPMENT GOALS

Bangladesh reduced child mortality by 71% against 2015 target of 66%

The Millennium Development Goals (MDGs) are international development goals set by the United Nations in 2000. These MDGs have targets in various human development categories, which include health improvements, poverty reduction, environmental sustainability, and developing global partnerships by 2015. In September 2000, leaders from 189 member-nations of the UN agreed to meet the following 8 development goals, five (MDG 1, 4, 5, 6, and 7) of which are health-related.

MDG 1: Eradicating extreme poverty and hunger (health-related)

MDG 2: Achieving universal primary education

MDG 3: Promote gender equality and empower women

MDG 4: Reduce child mortality (health-related)

MDG 5: Improve maternal health (health-related)

MDG 6: Combat HIV/AIDS, Malaria, and other diseases (health-related)

MDG 7: Ensure environmental sustainability (health-related)

MDG 8: Develop a global partnership for development.

Bangladesh is making good progress in almost all of the health-related MDGs. Some are already attained. In others, the country is well on track. However, a few will require attention to make substantial improvement. In this chapter, an overview of the current progress of Bangladesh on health-related MDGs is presented. The progress on each of the health-related MDGs has been shown in separate tables. The tables show the specific health-related MDGs, country benchmark, current progress, and targets. Asterisks have been used for showing status of the country. Single asterisk (*) means “on track” and double asterisks (**) mean “goal met.”

MDG 1: Eradicating extreme poverty and hunger

Table 3.1 shows the progress of Bangladesh in MDG 1. From 1990 prevalence of 66% underweight children, the figure came down to 36.4% by 2011, which is a 45% reduction against the required 50% reduction by 2015. According to the World Bank estimate, the percentage of population below minimum level of dietary energy consumption came down to 17% in 2011 from a figure of 28% in 1990. This is a 39% reduction against expected 50% by the year 2015. The progress in the latter, though appears a bit slower, it may be certainly said that Bangladesh is on track for these two indicators of Goal 1.

Table 3.1. Goal 1: Eradicate extreme poverty and hunger

Target	Indicator	Benchmark (Year)	Current progress (Reference)	Target (Year)
Reduce by half the proportion of people who suffer from hunger	Prevalence of underweight among children <5 years of age (%)	66.0 (1990)	36.4 (BDHS 2011)*	33.0 (2015)
	Population below minimum level of dietary energy consumption (%)	28.0 (1990)	17.0 (WB 2011)*	14.0 (2015)

MDG 4: Reduce child mortality

Bangladesh already attained the target of MDG 4 about reduction of child mortality rate. This is revealed in a recent publication titled “Committing to Child Survival: A Promise Renewed Progress Report 2013” by UNICEF. The report shows that Bangladesh’s under-5 mortality rate has been dropped to stunning 41 per 1,000 livebirths by 2012 from 144 per 1,000 livebirths in 1990 (Table 3.2). This is a 71% reduction against the target of 66% reduction by 2015. According to the same report, the current infant mortality rate of Bangladesh is 33 per 1,000 livebirths, and the neonatal mortality rate is 24 per 1,000 livebirths. The share of neonatal mortality rate is 59% of under-5 mortality rate and 80% of infant mortality rate. Bangladesh pledged to prevent an additional 108,000 deaths annually to reduce the national under-five mortality rate to 20 per 1,000 livebirths by 2015. One of the important indicators of MDG 4 is ensuring universal coverage of measles vaccination among the 1-year(s) old children by 2015. Table 3.2 shows that the current coverage rate is 86%. So, Bangladesh is well on track for this indicator also.

MDG 5: Improve maternal health

The latest Bangladesh Maternal Mortality Survey 2010 by NIPOORT shows Bangladesh’s current maternal mortality rate to be 194 per 100,000 livebirths which was 574 per 100,000 livebirths in 1990 (Table 3.3). Maternal mortality has been dropped by 66% against the 2015 target of 75% (143 per 100,000 livebirths). In this indicator of MDG 5, Bangladesh is well on track. However, other indicators, viz. births attended by skilled health personnel, birth rate among adolescent mothers, antenatal care coverage, and unmet need for family planning (%) will require more attention (Table 3.3). The contraceptive prevalence rate (%) is moving on track as revealed by a 61% rate against a national 2015 target of 72% (Table 3.3).

MDG 6: Combat HIV/AIDS, malaria, and other diseases

Table 3.4 shows the progress on MDG 6 relating to HIV/AIDS, malaria and tuberculosis. Bangladesh is consistently a low prevalent country with regard to HIV/AIDS and STDs. The National AIDS and STD

Table-3.2. Goal 4: Reduce child mortality

Target	Indicator	Benchmark (Year)	Current progress (Reference)	Target (Year)
Reduce by two-thirds the mortality rate among under-five children	Death rate among under-five children/1,000 livebirths	144.0 (1990)	41.0 (UNICEF 2013)** 44.0 (SVRS 2011) 53.0 (BDHS 2011)	48.0 (2015)
	Infant mortality rate/1,000 livebirths	94.0 (1990)	33.0 (UNICEF 2013)* 35.0 (SVRS 2011) 43.0 (BDHS 2011)	31.3 (2015)
	1-year old children immunized against measles (%)	52.0 (1991)	85.5% (BECES 2012)* 87.5 (BDHS 2011)	100.0 (2015)

Table-3.3. Goal 5: Improve maternal health

Target	Indicator	Benchmark (Year)	Current progress (Reference)	Target (Year)
Reduce by three-quarters the maternal mortality ratio	Maternal mortality ratio/100,000 livebirths	574.0 (1990)	194.0 (BMMS 2010)*	143.5 (2015)
	Births attended by skilled health personnel (%)	7.0 (1990)	26.5 (BMMS 2010) 31.7 (BDHS 2011)	50.0 (2015)
Ensure, by 2015, universal access to reproductive healthcare	Contraceptive prevalence rate (%)	39.9 (1991)	61.2 (BDHS 2011)* 58.4 (SVRS 2011)	72.0 (2016)
	Birth rate among adolescent mothers/1,000 women	77.0 (1990/91)	105.0 (BMMS 2010) 118.3 (BDHS 2011)	-

Table 3.3. continued

Target	Indicator	Benchmark (Year)	Current progress (Reference)	Target (Year)
	Antenatal care coverage (at least one visit by skilled health professional) (%)	27.5 (1993)	54.6 (BDHS 2011)	100.0 (2015)
	Antenatal care coverage (at least four visits) (%)	05.5 (1993)	25.5 (BDHS 2011)	100.0 (2015)
	Unmet need for family planning (%)	19.4 (1993)	13.5 (BDHS 2011)	7.6 (2016)

Program (2011) estimates a current prevalence of 0.7% HIV/AIDS and STD-affected victims among the high-risk population. There is a very slow rise in the number of victims. So, the situation is well on track. However, access to antiretroviral drugs among the population with advanced HIV infection will require improvement. In Bangladesh, malaria is endemic in only 13 districts out of 64, from where 80% of the national malaria burden is reported. The malaria prevalence and death rate have been estimated based on the reported malaria cases and deaths at the Communicable Disease Control Department of the Directorate General of Health Services of Bangladesh. Keeping in view the MDG 6 target, the national 2015 malaria target was set at 50% reduction of malaria prevalence and deaths from 2008 levels. Current estimates show that both targets have been achieved (prevalence is 18.4 per 100,000 population against the 2015 target of 29.3 per 100,000 population; malaria death rate is 0.007 per 100,000 population against the 2015 target of 0.053 per 100,000 population). Targets have also been achieved for indicators “Under-five children sleeping under insecticide-treated bednets (94.4% of under-five children sleep under insecticide-treated bednets against a target of 90%)” and for “Under-five children with fever treated with appropriate anti-malarial drugs” (89% under-five children were treated against a target of 90%). For tuberculosis, the 2015 MDG targets for case rate and cure rate have already been achieved. Current TB (all forms) prevalence is 411 per 100,000 population (against the target of 320 per 100,000 population) and TB death rate is 45 per 100,000 population (against a target of 38 per 100,000 population). For these two indicators also, the country is on track.

MDG 7: Ensure environmental sustainability

Access by all people to safe drinking-water is almost reaching the target. The Bangladesh Bureau

of Statistics in its Report 2011 of Sample Vital Registration System shows that over 98% of the Bangladesh population has now access to safe drinking-water (Table 3.5). However, two sources of estimation give controversial results about the use of sanitary latrine. The Multiple Indicators Cluster Survey 2009 showed a 80.4% coverage, whereas the SVRS 2011 showed only 63.6% coverage.

Health in the post-2015 development agenda

As the dateline for MDGs is coming to closer, there are active debates ongoing among the development organizations, countries and activists to prepare a set of development goals for the post-2015 period. The debates were triggered by the UN Secretary General in the 2010 High Level Meeting in the United Nations General Assembly on MDGs (September 2010), where he called for starting to think on the post-2015 development goals on the theme “Realizing the future we want”. Since then, events, consultations, conferences, meetings, etc. are continuing. One most important is Rio20+ held in Rio de Janeiro (June 2012). Unconventional to other years, the World Health Organization put this issue in May 2013 World Health Assembly in the plenary speeches of head of delegation of Member States. In September

There are active debates ongoing among the development organizations, countries and activists to prepare a set of development goals for the post-2015 period ...

Table-3.4. Goal 6: Combat HIV/AIDS, malaria, and other diseases

Target	Indicator	Benchmark (Year)	Current progress (Reference)	Target (Year)
Halt and begin to reverse the spread of HIV/AIDS	HIV prevalence among population aged 15-24 years (%)	0.005 (1990)	0.7% (NASP 2011)*	Halt (2015)
Ensure, by 2010, universal access to treatment for HIV/AIDS for all those who need	Population with advanced HIV infection with access to ARV drugs (%)	-	45.0 (UNGASS 2012)	100.0 (2015)
Halt and begin to reverse the incidence of malaria and other major diseases	Malaria prevalence/100,000 population	58.5 (2008)	18.4 (DGHS 2012)** Estimated based on reported malaria cases	29.3 (2015)
	Malarial death rate/100,000 population	0.106 (2008)	0.007 (DGHS 2012)** Estimated based on reported malaria cases	0.053 (2015)
	Under-five children sleeping under insecticide-treated bednets in endemic areas (%)	81.0 (2008)	94.4 (DGHS 2012)**	90.0 (2015)
	Under-five children with fever treated with appropriate antimalarial drugs (%)	60.0 (2008)	89.0 (DGHS 2011)*	90.0 (2015)
	TB (all forms) prevalence rate/100,000 population	639.0 (1990)	411.0 (DGHS 2011)*	320.0 (2015)
	TB death rate/100,000 population	76.0 (1990)	45.0 (DGHS 2011)*	38.0 (2015)
	New smear+ve TB case detection rate under DOTS (%)	21.0 (1994)	70.0 (NTP 2012)**	>70.0 (2015)
	TB cure rate (%) with DOTS	73.0 (1994)	92.0 (NTP 2012)**	>85.0 (2015)

Table 3.5. Goal 7: Ensure environmental sustainability

Target	Indicator	Benchmark (Year)	Current progress (Reference)	Target (Year)
Reduce by half the percentage of people without sustainable access to safe drinking-water and basic sanitation (%)	Population using improved drinking-water sources (%)	78.0 (1990)	98.2 (SVRS 2011)*	100.0 (2015)
	Population using improved sanitation facility (%)	39.2 (2006)	80.4 (MICS 2009)*	100.0 (2015)
			63.6 (SVRS 2011)	

2012, a high level dialogue was held in Botswana in a global thematic consultation on this issues. The informal consultation was co-led by WHO and UNICEF in collaboration with Governments of Botswana and Sweden, and a UN inter-agency

group, including UNAIDS, UNFPA, UNDP, UNDESA and OHCHR. The dialogue discussed four key thematic issues, viz. lessons learnt from the health-related MDGs; health priorities post-2015; defining future health goals; and measuring

progress towards achievement. In March 2013, the South-East Asia Regional Office of the WHO organized a regional consultation in Bangkok to discuss and compile recommendations of different national consultations held on this issue and in Bangladesh, India, Indonesia, and Timor-Leste. The UN Secretary General Ban Ki Moon constituted a High Level Panel of Eminent Persons (HLP-EP) to give him advice on the same. An overview of all the discussions shows a common focus to issues, like poverty, hunger, water, sanitation, education, and healthcare to go beyond goals set in 2015-MDGs for promoting sustainable development goals. The high level panel in its report (May 2013) recommended 2030 as the dateline for the next round of development goals. It recommended 12 goals and 54 targets. The health goal is recommended as “ensure healthy lives”, with the proposed outcomes of ending preventable infant and under-5 mortality, increasing coverage of immunization, decreasing maternal mortality, ensuring universal sexual and reproductive health and rights, and reducing burden of diseases from HIV/AIDS, TB, malaria, neglected tropical diseases and priority NCDs. The panel also put MDG 8, i.e. a stronger global partnership at the heart of all recommendations. Another inter-governmental committee of experts mandated by Rio20+ proposed for holding the global temperature rise to less than 2°C. There is strong recommendation from health communities to include universal health coverage. Bangladesh is also sounding its voice to consider as guiding principle the definition of health to set health agenda in the post-2015 development goals. Health is defined as “a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity”, inclusive of every citizen of the world. Bangladesh is also putting emphasis on target setting pointing to fallacies of MDG targets. The

argument which Bangladesh is pushing forward is inherent inequality principle of MDG targets. For example, the 1990 maternal mortality ratio (MMR) baseline figures of Uzbekistan, Viet Nam, Sierra Leone, and Bangladesh were 59; 240, 1,300 and 574 respectively. By 2015, if all of these countries attain MDG 5, the MMR of these countries will be 15; 60; 325 and 144 respectively. That means even after 2015, more mothers per 100,000 livebirths will be allowed to die in Sierra Leone, and Bangladesh than in Uzbekistan and Viet Nam. Bangladesh is proposing for setting target in a way that will say: “All countries of the world will drop MMR to X per 100,000 by 2030.” If this kind of target is adopted, countries far lagging behind, would need more effort, more resource and more attention to upgrade their situation, both from national leaderships as well as from global bodies. However, citizens of every country of the world will enjoy equal opportunity and human rights. A common agreement is assumed to take shape to develop an overarching top post-2015 goal as “Well-being for all.” Bangladesh recommended to include health as one of the priority areas, and to include within health “unfinished health-related MDGs, NCDs, Universal Health Coverage, Poverty & Hunger Elimination, and Primary Healthcare, inclusive of community participation and empowerment as well as holistic multisectoral engagement.” Bangladesh further emphasized to include “information system—both health and non-health, to capture all the basic information under a broadened CRVS system to register and track each and all citizens fully, timely, and accurately to generate evidence as systematic continued process, locally, subnationally, nationally, and globally as an integral component.” It is hoped that a globally-agreed effective and comprehensive set of UN post-2015 development goals will be available before we reach 2015.

PRIMARY HEALTHCARE

Comprehensiveness of essential healthcare further improved at grassroots level

Bangladesh is one among the few countries of the world that provides free medical services to the people at the community level through various public health facilities. The primary healthcare is provided through an extensive network of health facilities extended down to the community level with upward referral linkage and a set of government-funded permanent community healthcare workers.

The public-sector healthcare network under the Ministry of Health and Family Welfare is shown in Table 4.1.

The community clinics are the lowest-level static health facilities located at the ward level. These have upward referral linkages with health facilities located at the union and upazila levels. There are 467 government hospitals at the upazila level and below, which altogether have

Table 4.1. Primary healthcare centers run by DGHS at the upazila level and below (2012)

Level	Type of facility	Type of service	Total No. of facilities	Total beds
Upazila	Upazila health complex (50-bed)	Hospital	268	13400
	Upazila health complex (31-bed)	Hospital	145	4495
	Upazila health complex (20-bed)	Hospital	1	20
	Upazila health complex (10-bed)	Hospital	11	110
		Subtotal of upazila health complexes	425	18025
	31-bed hospital	Hospital	5	155
	30-bed hospital	Hospital	1	30
		Subtotal of hospitals outside health complexes	6	165
	Trauma center (20-bed)	Hospital	5	100
		Total of upazila-level hospitals	436	18290
Union	20-bed hospital	Hospital	18	360
	10-bed hospital	Hospital	13	130
		Subtotal of union-level hospitals	31	490
	Union subcenter	Outpatient only	1275	-
	Union health and family welfare center	Outpatient only	87	-
		Subtotal of union outpatient centers	1362	-
		Total of union-level facilities	1393	490
Ward	Community clinic	Outpatient only	12527	-
		Grand total of hospitals (upazila and below)	467	-
		Grand total of health facilities (upazila and below)	14356	18780

18,780 hospital beds. At the upazila level, there are 436 hospitals with 18,290 beds. At the union level, there are 31 hospitals with 490 beds and 1,362 health facilities for outpatient services only. So, at the union level, there are 1,393 health facilities. At the ward level, there are 12,527 community clinics in operation till date.

Community clinics deserve special mention due to flagship nature of the program. In addition to the community clinics, important components of primary healthcare, among others, include domiciliary healthcare, essential service delivery, along with urban primary healthcare, maternal healthcare (inclusive of some screening programs for women's health), child healthcare, school health program, and adolescent health program. This chapter makes an overview of these components.

Community clinics

The Government of Bangladesh, in 1996-2001, planned to establish 18,000 community clinics (CCs) for provision of primary healthcare services to rural people—13,500 as independent new clinics and 4,500 in the existing union and upazila-level health facilities. From 1998 to 2003, 10,723 community clinics were constructed, of which 8,000 were made functional. The Government resumed the CC project with the name “Revitalization of Community-based Healthcare in Bangladesh.” As of now, 12,527 independent community clinics have been started. Keeping in view the target of setting up 13,500 independent CCs, initiative was undertaken to recruit 13,500 community healthcare providers (CHCP), one for each CC. However, due to administrative reasons, 13,240 could be recruited. In addition to the CHCP, the existing domiciliary staff members of DGHS and DGFP also provide service to the community clinics three working days a week alternately. The community clinics provide the basic healthcare package to the community people, viz. maternal and child healthcare, reproductive health and family-planning services, immunization, nutrition education, micronutrient supplementation, health education and counseling, communicable disease control, treatment for minor ailments and first-aid, and referral to higher-level health centers. The community clinics are managed by a 9 to 13-member management committee selected from the respective communities. At least 4 members must be female. There are also three community support groups each comprising 15 members to work as community health

The community clinics provide the basic healthcare package to the community people, viz. maternal and child healthcare, reproductive health and family-planning services, immunization, nutrition education, micronutrient supplementation, health education and counseling, communicable disease control, treatment for minor ailments and first-aid ...

volunteers (non-paid) to assist the management committee and community clinic. The local government representatives are included in the management committee. By October 2013, all the community clinics will have Internet connection through a laptop and wireless modem to help collection of local health-related data, provide telemedicine service, community health education, and certain other ICT-based health solutions.

Figure 4.1 shows the government expenditure for supply of medicines to the community clinics in different fiscal years. The amount of allocation per community clinic for medicine supply was BDT 0.07 million in 2009-2010, BDT 0.09 million in 2010-2011, and BDT 0.11 million in each of 2011-2012 and 2012-2013. Items of medicines supplied in 2009-2010 were 25. In 2010-2011,

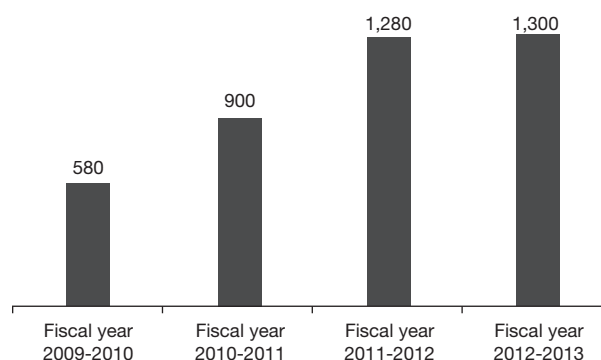


Figure 4.1. Government expenditure (BDT in million) for supply of medicines to community clinics

the list included 28 items but from 2011-2012, items were increased to 30.

Figure 4.2 shows the number of patients given treatment in and referred from community clinics in different time periods. Number of service-recipients is sharply increasing and almost doubled from 2011 to 2012. Since re-inception in 2009, about 15 million patients received service from community clinics. It is estimated that, in 2012, 35 patients, on average, received service from each community clinic daily, which was 12 patients per day per community clinic in 2009.

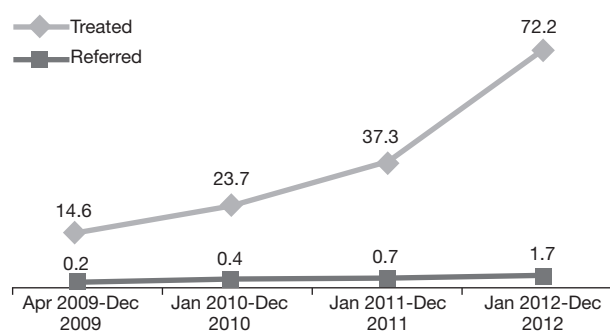


Figure 4.2. No. of clients (in million) treated (total 147.9 million) and referred (total 3.0 million) from community clinics in different time period

Community clinic is an unprecedented instance of community participation and public-private partnership. Being inspired by community participation, some UN agencies and NGOs have started working with the community clinic project. Many other organizations are also coming forward to working as the days are passing. A summary of the key examples of such partnership is provided in the annex.

Community clinic is certainly a pro-people health initiative led by the Government. If quality health services can be ensured near doorsteps even at the remotest corner of the country, people will spontaneously seek necessary service from the well-trained care providers at the health facilities, instead of the untrained traditional healers. It is expected that community clinics will ensure provision of quality healthcare for the mass people of rural Bangladesh, particularly the poor, vulnerable, and the underprivileged and will contribute to the achievement of the health development goals under MDGs and in the post-2015 period.

Domiciliary service in rural Bangladesh

There are domiciliary workers—one for every 5 to 6 thousand people at the ward or village level. Under DGHS, there are 26,412 sanctioned posts of domiciliary workers, of whom 20,815 are for health assistants (HA), 4,198 for assistant health inspectors (AHI), and 1,399 for health inspectors (HI). As of now, 91% posts were filled up. Like DGHS, the DGFP also has domiciliary workers to work at the ward or village level. These staff members are called family planning inspectors (FPI) and family welfare assistants (FWA).

Essential service delivery and urban primary healthcare

Under the Health, Nutrition and Population Sector Development Program (HPNSDP) 2011-2016, there is an operational plan, namely “Essential Service Delivery” mainstreamed under DGHS to help improve service, particularly at the upazila level and below and complement urban primary healthcare. The areas of services include limited curative care, support services and coordination, medical waste management, urban health, mental health, and tribal health. The urban primary healthcare in Bangladesh is principally the responsibility of the Ministry of Local Government, Rural Development and Cooperatives (MOLGRD) carried out through the city corporations and municipalities. These local bodies run a number of small to medium-sized hospitals and outdoor facilities. Besides, two large-scale primary healthcare projects, viz. Urban Primary Healthcare Project (UPHCP) and Smiling Sun Franchise Program run by NGOs in collaboration with the city corporations and with the financial assistance from donors. The clients in these latter projects also share a part of the cost through service-charge. There is a concern among the public health communities that there is a need

The urban primary healthcare in Bangladesh is principally the responsibility of the Ministry of Local Government, Rural Development and Cooperatives (MOLGRD) carried out through the city corporations and municipalities ...

for better coordination between the two ministries, viz. MOHFW and MOLGRD, with regard to urban primary healthcare, although MOHFW contributes to urban primary healthcare through outpatient services distributed through its secondary, tertiary and specialized hospitals located in the urban settings. Besides, there are 35 urban dispensaries and 23 school health clinics in some of the bigger cities and municipalities. To respond to the concerns for need of better coordination between MOHFW and MOLGRD with regard to urban primary healthcare, the MOHFW included in its HPNSDP 2011-2016 a component named “urban health” under the operational plan “Essential Service Delivery.” This urban health component aims at designing programs through maintaining better coordination and collaboration with the city corporations, municipalities, UPHCP, Smiling Sun Franchise Program, other NGOs and stakeholders.

Maternal healthcare

The Ministry of Health and Family Welfare of Bangladesh, in collaboration with UNICEF, is undertaking facility-based Emergency Obstetric Care (EOC) Program in all the districts of Bangladesh to improve the maternal health situation, targeting to achieve the Millennium Development Goal 5. All the government medical college hospitals, district hospitals, upazila hospitals, and maternal and child welfare centers (MCWCs) provide obstetric care service, inclusive of emergency obstetric care. A number of private clinics or hospitals and health-related NGOs are also partners of this program. Obstetric care is classified into two categories in this program, viz. Comprehensive Emergency Obstetric Care (CEmOC) and Basic Emergency Obstetric Care (BEOC). Currently, all medical college hospitals, 59 district hospitals, 3 general hospitals, 132 upazila health complexes, and 63 MCWCs provide CEmOC, and rest of the upazila health complexes provide BEOC. The list also includes NGOs and private care providers from a number of districts. Under a program, jointly operated by the Management Information Systems (MIS) of DGHS and UNICEF, data are collected from the EOC facilities. For this publication, data from 627 sources, including 14 medical college hospitals, 62 district/general hospitals, 404 upazila health complexes, 55 maternal and child welfare centers, private hospitals from 53 districts, NGOs from 36 districts, and 3 other types of hospitals have been used for analysis to translate into a format called United Nations Process Indicators. Table 4.2 summarizes the sources of data.

Table 4.2. Number of data sources used for this publication on emergency obstetric care (2012)

Type of hospital	No.	Percentage
Medical college hospital	14	2.2
District and general hospital	62	9.9
Upazila health complex	404	64.4
MCWC	55	8.8
Districts from where private care providers sent data	53	8.5
No. of districts from where NGO care providers sent data	36	5.7
Other health facilities	3	0.5
Total	627	100.0

Figure 4.3 shows the number of different obstetric care encounters and clients served by the emergency obstetric facilities in Bangladesh. Detailed information with division-wise disaggregation is provided in the annex.

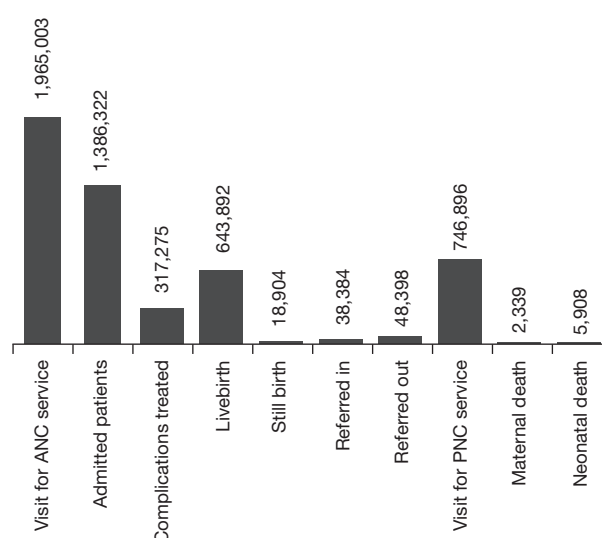


Figure 4.3. Number of different obstetric care encounters or clients served by the emergency obstetric care facilities in Bangladesh (2012)

Figure 4.4 reveals that 653,642 institutional deliveries were reported in 2012, of which normal delivery accounted for 56%, cesarean section accounted for 42%, and vaginal breech and forceps delivery collectively accounted for 2%.

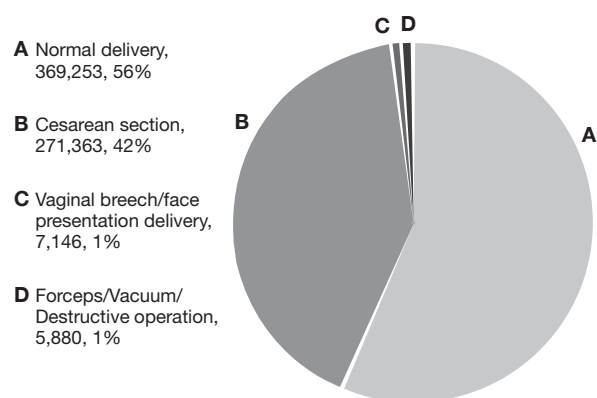


Figure 4.4. Distribution of reported institutional deliveries by type (n=653,642)(2012)

Figure 4.5 shows the outcome of obstetric care of the emergency obstetric care facilities for the year 2012. Maternal death as % of total deliveries was 0.4%, neonatal death as % of all livebirths was 0.9%, birth at EmOC facilities as % of all births was 21.9%, percentage of all births at EmOC facilities was 21.9%, percentage of met need for EmOC was 70.8%, and case-fatality rate (CFR) as % of all maternal complications treated was 0.7%.

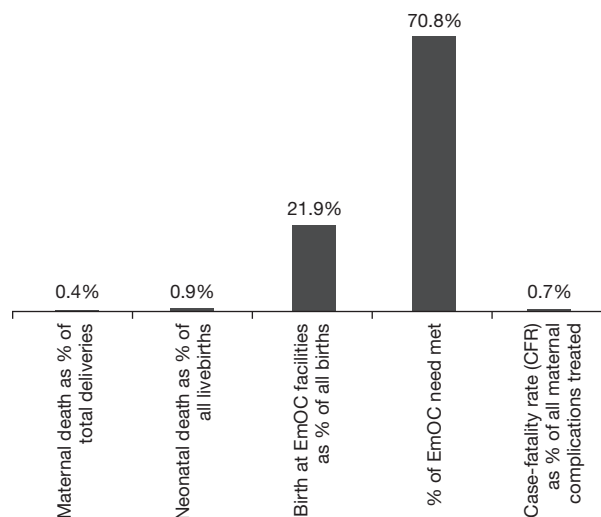


Figure 4.5. Outcome of obstetric care in EmOC facilities (2012)

Table 4.3 shows the distribution of obstetric care services provided by the government and non-government emergency obstetric care facilities. Detailed profile has been provided in the annex.

Table 4.3. Obstetric care services provided by the government and non-government emergency obstetric care facilities (2012)

UN process indicator	Government facility		Non-government facility		Total	
	No.	%	No.	%	No.	%
Visit for ANC service (N)	1307769	66.6	657234	33.4	1965003	100.0
Admitted patients (N)	1118048	80.6	268324	19.4	1386372	100.0
Complications (N)	216198	68.1	101077	31.9	317275	100.0
Livebirth (N)	395969	61.5	247923	38.5	643892	100.0
Stillbirth (N)	15716	83.1	3188	16.9	18904	100.0
Other surgeries (N)	23199	85.9	3809	14.1	27008	100.0
Referred in (N)	25277	65.9	13107	34.1	38384	100.0
Referred out (N)	40730	84.2	7668	15.8	48398	100.0
PNC service (N)	480498	64.3	266388	35.7	746886	100.0
Maternal death (N)	2106	90.0	233	10.0	2339	100.0
Neonatal death (N)	5504	93.2	404	6.8	5908	100.0

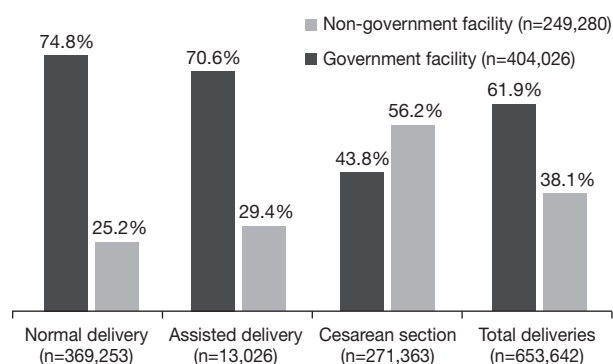


Figure 4.6. Distribution of delivery by type between government and non-government (private, NGO, other) health facilities (2012)

Figure 4.6 shows the distribution of normal, assisted, cesarean and total deliveries between government and non-government health facilities in 2012. While most of the total (61.9%) and normal deliveries (74.8%) took place in the government facilities, the percentage of cesarean section was more in private health facilities (56.2%).

Table 4.4 shows the distribution of normal, assisted, cesarean and total deliveries within government and non-government emergency obstetric care facilities (2012). Of the total 808,724 deliveries in government health facilities, 22.1% took place in medical college hospitals, 26.7% in district hospitals, and the largest proportion (42.0%) took place in upazila health complexes;

Table 4.4. Distribution of normal, assisted, cesarean and total deliveries within government and non-government emergency obstetric care facilities (2012)

Type of delivery	Government facility										Non-government facility							
	MCH		DH		UHC		MCWC		Total		NGO facility		Private facility		Other		Total	
	No.	%	No.	%	No.	%	No.	%	No.	%	No.	%	No.	%	No.	%	No.	%
Normal delivery	39635	14.3	62740	22.7	144235	52.2	29763	10.8	276373	100.0	38145	41.1	52555	56.6	2180	2.3	92880	100.0
Assisted delivery	2934	31.9	1899	20.6	3766	40.9	601	6.5	9200	100.0	1008	26.3	2813	73.5	5	0.1	3826	100.0
Cesarean section	46618	39.2	43295	36.4	21958	18.5	6918	5.8	118789	100.0	18475	12.1	134007	87.8	92	0.1	152574	100.0
Total deliveries	89187	22.1	107934	26.7	169959	42.0	37282	9.2	404362	100.0	57628	23.1	189375	76.0	2277	0.9	249280	100.0
Total	178374	-	215868	-	339918	-	74564	-	808724	-	115256	-	378750	-	4554	-	498560	-

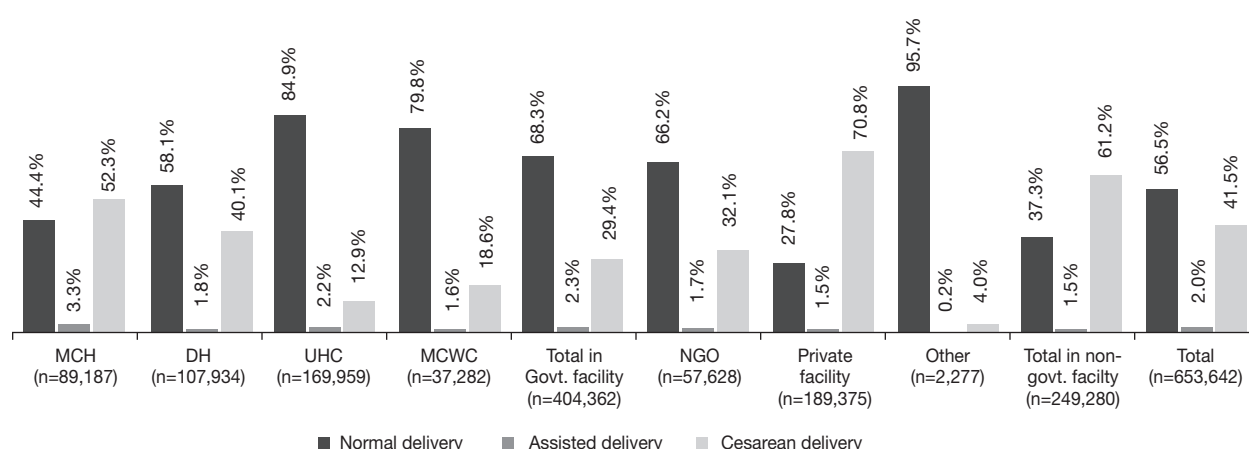


Figure 4.7. Distribution of normal, assisted and cesarean delivery by type of health facility (2013)

9.2% of deliveries took place in maternal and child health centers. Of the total 498,560 deliveries in non-government facilities (NGO, private, others), 23.1% were done at NGO facilities and 76.0% at private clinics/hospitals. Table 4.4 also reveals that there were 118,789 cesarean sections in public health facilities and 152,574 in non-government health facilities. Of the total cesarean sections at public facilities, 39.2% were done in medical college hospitals (n=46,618), 36.4% in district hospitals (n=43,295), 18.5% in upazila health complexes (n=21,958), and 5.8% in maternal and child welfare centers (n=6,918). Of the total cesarean sections done in non-government health facilities 12.1% were done at NGO facilities (n=18,475), 87.8% at private clinics/hospitals (n=134,007), and 0.1% at other private health facilities (n=92).

Figure 4.7 shows the distribution of normal, assisted and cesarean deliveries by type of health facility.

An indirect national estimate of cesarean section rate can be made from the available institutional data. Data from the Bangladesh Demographic and Health Survey 2011 show an institutional delivery rate of 28.8%. So, against a total 653,642 (Figure 4.7) reported institutional deliveries as mentioned above, we extrapolated the total number of deliveries both at home and health facility. This number should be $[(653,642 \div 28.8) \times 100] = 2,269,590$ total deliveries. Cesarean section always takes place in health facility. This report shows the total cesarean sections of 272,363 (Figure 4.6). Therefore, the estimated national cesarean section rate should be $[(272,363 \div 2,269,590) \times 100] = 12.0\%$.

Voucher scheme for maternal health

The Ministry of Health and Family Welfare, in collaboration with WHO, introduced in 2007 an innovative maternal health voucher scheme, a demand-side financing (DSF) initiative, to improve access to and use of quality maternal health services. Currently, the program is being implemented in 46 upazilas of 41 districts and 7 upazilas of 4 MNHI (maternal and newborn health initiative) districts. Poor women defined by specific criteria (roughly 50% of the pregnant women) and validated by local government representatives are eligible for the voucher. Half of the target population qualifies as poor. The total number of cumulative beneficiaries reached 870,423 (Figure 4.8). In 2012, a total of 152,401 pregnant women received the benefit. A voucher entitles its holder for specific health services free of charge, viz. antenatal and postnatal

Poor women defined by specific criteria (roughly 50% of the pregnant women) and validated by local government representatives are eligible for the voucher ...

care, safe delivery, treatment for complications, including cesarean section, transportation cost, and laboratory tests. If delivery is attended by skilled staff, voucher-holders get unconditional cash benefits for nutritious food and gift-box. Safe delivery rate is now at impressive 85% amongst the voucher recipients. Both public and non-public healthcare providers (NGO and private facilities) participate in the DSF scheme. There is a target to scale the program up to 100 upazilas, with a 20% increase each year. Strikingly, the maternal mortality rate among the voucher-holder women is 12 per 100,000 livebirths, in sharp contrast to the national rate of 194 per 100,000 livebirths.

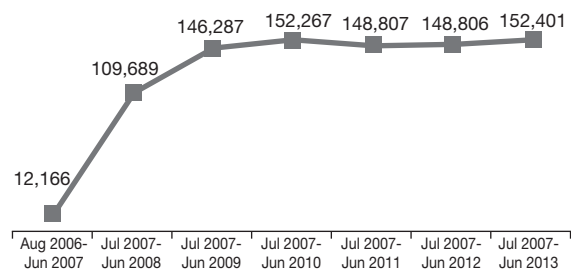


Figure 4.8. Number of DSF (Demand-side Financing) beneficiary pregnant women by year (total 870,423)

Maternal and Newborn Health Initiative (MNHI)

The Maternal and Newborn Health Initiative is being implemented by the Director of Primary Healthcare of DGHS in four districts of Bangladesh, with the assistance of UNFPA, UNICEF, and WHO and funded by EC and DFID. The districts are Thakurgaon, Jamalpur, Narail, and Maulvibazar. All the upazilas under these four districts are included. The program focuses on saving maternal and newborn lives through creating need-based demand and priority-based actions. The broad principle of this program is Local-level Planning

(LLP) and decentralization. The civil surgeon and deputy directors of family planning of the respective districts serve as the local focal points for the program. The project has a number of “novel and innovative” approaches, based on global best practices, to accelerate progress toward achievement of MDG 4 and 5, having elements, such as: (i) a district-focused approach with direct resource allocation to identified cost-centers and the application of WHO’s problem-solving techniques to develop, monitor, and implement the plans; (ii) continuum of care that links the mothers and newborns and addresses the three delays model; (iii) rights-based equitable approach in planning, monitoring, implementation, and supervision through involvement of consumer groups and public-health watch groups to ensure accountability to women, families, and communities; (iv) piloting initiatives, such as contracting private practitioners to provide specialized services in an attempt to improve human resources for MNH at the district and upazila levels; (v) pilot-testing of demand-side financing schemes (vouchers and other means), targeting the vulnerable and marginalized households to address equity; and (vi) pilot-testing of ARH community-based and clinic-based ‘youth-friendly’ services and Voluntary Confidential Counselling and Testing (VCCT) centres in selected districts with high risks of HIV and STIs.

Tetanus Toxoid (TT) for women of childbearing age

Table 4.5 shows the tetanus toxoid coverage among the women of childbearing age in Bangladesh in 2012. The country is maintaining the maternal and neonatal tetanus-free status since 2008. The immunization program of Bangladesh aims to immunize the women of childbearing age by administering tetanus toxoid vaccine (TT) before the age of 18 years. A period of 2 years and 7 months is required to complete all the 5 doses of TT vaccines. If a woman starts TT vaccination at the age of 15 years and maintains the exact interval, she would be able to complete all the

Table 4.5. Tetanus toxoid coverage (%) in Bangladesh among women of childbearing age (2012)

Area	TT1	TT2	TT3	TT4	TT5
National	92%	89%	74%	52%	32%
Rural	92%	89%	74%	52%	33%
Urban	92%	89%	74%	52%	32%

doses before she reaches the age of marriage, ensuring protection for her entire reproductive life. The data shown in Table 4.5 have been excerpted from EPI Coverage Evaluation Survey 2013 (in press at the time of preparing this report), which shows a consistent rate of TT coverage among the target women both in rural and urban areas. However, the coverage gradually falls for the subsequent TT doses and is remarkably lower for the fourth and final doses. This aspect needs attention to ensure effective coverage.

Community-based skilled birth attendants and midwives

Shortage of skilled manpower in the remote areas to extend obstetric care is one of the major barriers to improving maternal health. The Ministry of Health and Family Welfare undertook a short-term measure to tackle the problem by producing trained manpower for fulfilling the gap in the interim period. Young medical doctors were given 6 months’ training on obstetrics and anesthesiology. The Directorate General of Health Services is also implementing community-based skilled birth attendant (CSBA) training program since 2003, with the goal to train and educate the family welfare assistants/female health assistants and similar health workers in NGOs and private sector, with midwifery skills. The CSBAs are trained to conduct the normal safe deliveries at home and to identify the risks and complicated cases so that they can motivate the women and their family members to refer to the nearby health facilities where comprehensive EOC services are available. The CSBA training program is now organized in 342 upazilas of 60 districts.

Obstetric fistula program

In Bangladesh, obstetric fistula and other maternal morbidities affect thousands of women. It is estimated that approximately 71,000 women are currently living with fistula in the country (1.69 per 1,000 ever-married women). The UNFPA has been assisting the Government of Bangladesh in strengthening quality service delivery and capacity development of service providers at 10 medical college hospitals and 4 private hospitals. Since 2003, 24 doctors and 253 nurses have been trained; 3,050 complicated obstetric fistula surgeries were performed. National Fistula Center has been established in Dhaka Medical College Hospital.

Cervical and breast cancer screening program

The cervical and breast cancers are significant disease burdens in Bangladesh. The United Nations

Population Fund has been assisting the Ministry of Health and Family Welfare to run cervical and breast cancer screening program. The program is being coordinated from the Department of Obstetrics and Gynecology at the Bangabandhu Sheikh Mujib Medical University (BSMMU), Dhaka. Beginning from 2005, about 300 centers have been established throughout the country to conduct the cervical and breast cancer screening program. A list of the centers has been provided in the annex. Ninety-seven service providers from 64 districts were trained in 2012 on cervical and breast cancer screening, based on visual inspection with acetic acid (VIA) and clinical breast examination (CBE). Figure 4.9 shows that a total 539,338 VIA screening tests were done in 8 years (from 2005 to 2012) throughout the country, using the screening centers and, on average, 5.0% of them were found positive. The screening tests coverage is increasing every year. In 2012, 112,091 VIA screening tests were done, with 5% showing VIA+ve. All VIA+ve cases were referred to colposcopy clinic at BSMMU and different medical college hospitals. It is reported that the colposcopy clinics at BSMMU and various medical college hospitals were attended by 6,785 referred VIA+ve women in 2012. Analysis of data on 10,075 VIA+ve women for whom colposcopy was done between 2005 and 2012 at BSMMU shows that 7% (n=737) women had cervical cancer, 9% (n=944) had CIN II and III, 34% (n=3,381) had CIN I, and 47% (n=4,740) had no problem (appeared healthy) in their cervix. A list of the colposcopy clinics with number of colposcopies done in 2012 is provided in the annex.

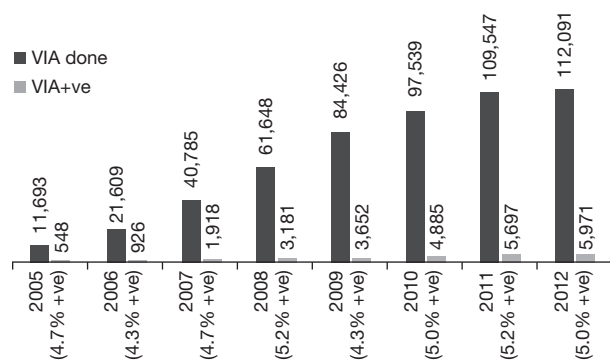


Figure 4.9. Number of VIA tests done and their results (2005 to 2012) (total 539,338 tests done in 8 years; 5.0% found positive)

Figure 4.10 shows that a total of 460,533 screenings for CBE were done in 6 years (from 2007 to 2012) throughout the country, using the screening centers and, on average, 1.5% of them were found positive. The number of screening tests

coverage is increasing every year. In 2012, 110,733 screening tests were done, with 1.1% showing positive.

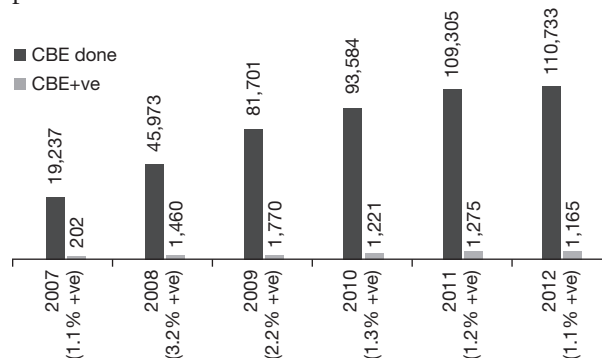


Figure 4.10. Number of CBE tests done and their results (2007 to 2012) (total 460,533 tests done; 1.5% found positive)

Child healthcare

This report on child healthcare in Bangladesh covers information on universal routine child immunization, integrated management of childhood illness (IMCI), scaling up of newborn health interventions, and special-care newborn unit (SCANU), school health and adolescent health programs.

Universal routine child immunization

This chapter excerpts data from the manuscript of EPI Coverage Evaluation Survey 2013 (EPI-CES 2013) Report as the book was awaiting publication. The percentage of children aged ≤12 months covered with all vaccinations was 81.0% in 2013 (Figure 4.11). The EPI CES 2013 validated the immunization coverage both by EPI cards and history. Figure 4.11 also shows the trend of immunization coverage from 2011 to 2013 among the same age-group of children.

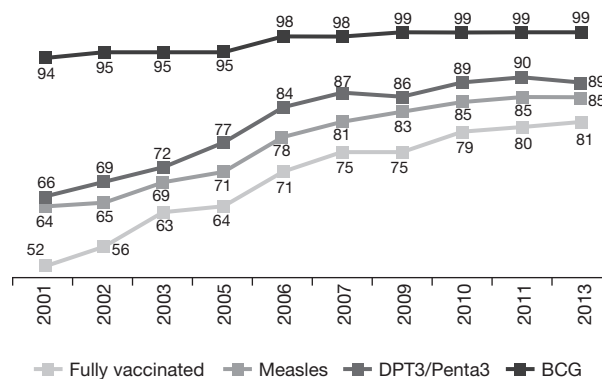


Figure 4.11. Trend of child (≤12 months) immunization coverage (%) (card plus history)

Figure 4.12 shows the trend of immunization coverage from 2005 to 2013 among ≤ 12 and ≤ 23 months old children.

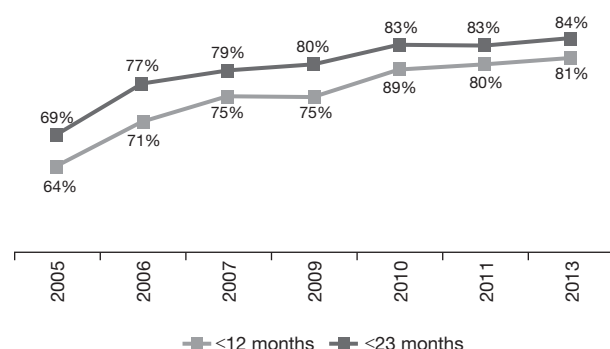


Figure 4.12. Trend of valid vaccination coverage among children aged ≤ 12 months and ≤ 23 months (Ref. EPI-CES 2013)

Table 4.6 shows the valid vaccination coverage of ≤ 12 and ≤ 23 months old children as found in EPI-CES 2013. Measles vaccine coverage is 86% and 89% among ≤ 12 and ≤ 23 months old children respectively. Full vaccination coverage among these two groups of children is 81% and 84% respectively.

Bangladesh showcases a success story on polio eradication. The country is polio-free from 2001, with 18 exceptional cases of wild polio imported from neighboring India in 2006. National Immunization Day is observed every year. The current valid national OPV3 coverage rate is 92%, with each district having coverage of more than 80%. The polio eradication program in Bangladesh illustrates Government's commitment through providing 100% cost of routine polio immunization and 95% cost of supplementary polio immunization activities. Bangladesh, despite being free from polio for a long time, could not achieve polio-free certification as two countries

of the WHO South-East Asia Region, viz. India and Nepal, could not eliminate the last traces of polio. However, for the 36 months, no new polio case was reported from these two countries. Based on this progress, the WHO completed necessary scientific observations and the South-Asia Region, including Bangladesh, is expected to obtain the polio-free certification in February 2014. As per the WHO recommendation, Bangladesh is trying to introduce inactivated poliovirus vaccine (IPV) by withdrawing the type 2 component of oral poliovirus vaccine. Bangladesh is also satisfactorily progressing toward achieving the measles elimination goal of the WHO's South-East Asia Region by 2020. National measles control activities have been accelerated since 2004 and already implemented all recommended strategies for measles elimination and rubella/congenital rubella syndrome control. Valid measles vaccination coverage among ≤ 12 months old children is 86% according to EPI-CES 2013. Coverage of the first dose (MCV1) of measles vaccine was estimated to increase from 81% in 2004 to 96% in 2012. In 2012; the second dose of measles vaccine and sentinel surveillance for congenital rubella syndrome, covering all districts, were introduced in the national immunization program.

Bangladesh, despite being free from polio for a long time, could not achieve polio-free certification as two countries of the WHO's South-East Asia Region, viz. India and Nepal, could not eliminate the last traces of polio ...

Table 4.6. Valid vaccination coverage of ≤ 23 and ≤ 12 months old children as found in EPI-CES 2013

Age-group	BCG	OPV1	OPV2	OPV3	Penta1	Penta2	Penta3	Measles	Full vaccination
≤ 12 months	95%	95%	94%	92%	91%	93%	92%	86%	81%
≤ 23 months	95%	95%	94%	92%	91%	93%	92%	86%	81%

Table 4.7. Valid full vaccination coverage differentials by sex, area of residence, and division as found in EPI-CES 2013

Age-group	Sex		Residence		Division						
	Boy	Girl	Rural	Urban	Barisal	Chittagong	Dhaka	Khulna	Rajshahi	Rangpur	Sylhet
≤12 months	81%	80%	81%	80%	78%	79%	79%	85%	85%	81%	81%
≤23 months	84%	83%	84%	84%	80%	82%	83%	87%	87%	85%	83%

Table 4.7 shows the valid full vaccination coverage differentials by sex, area of residence, and division as found in EPI-CES 2013 among ≤12 and ≤23 months old children.

In the measles campaign (follow-up), high-potency vitamin A and antihelminthes distribution are also included. Table 4.8 shows the Penta vaccine, vitamin capsule and antihelminthic coverage among the under-five children.

Integrated management of childhood illness (IMCI)

The relevant section of DGHS, with assistance from UNICEF, WHO, and other partners, is implementing IMCI program since 1998. Both facility and community IMCI have been scaled up—facility IMCI to 425 upazilas initially in all districts with high child mortality, and community IMCI to 150 upazilas mainly in the low-performing districts. Given the neonatal mortality showing slow declining rate, neonatal health has been incorporated in both facility and community IMCI programs. The care-seeking from trained providers for pneumonia and diarrhea has increased remarkably over the last few years. Antibiotic treatment for pneumonia, ORT-use for diarrhea, and exclusive breastfeeding have been increased to 71%, 80%, and 64% respectively (BDHS 2011). More than 4,000 doctors, 17,000 paramedics, 8,500 basic health workers, and 15,600 skilled birth attendant have been given training on different aspects of IMCI (3,500 doctors, 9,500 paramedics on IMCI clinical management; 500 doctors and 7,500 paramedics on sick newborn

care; 8,500 basic health workers on community case management; and 15,600 skilled birth attendants on helping babies breathe). Save The Children and UNICEF are supporting MOHFW for the national scale-up of Helping Baby Breathe program for the prevention and management of newborn deaths due to birth asphyxia. As of April 2013, 40 districts and 2 city corporations have been covered. Logistics, like penguin sucker, bag and mask, have been distributed. Large-scale maternal, neonatal and child health (MNCH) program, along with newborn health interventions, are being carried out in 31 out of 64 districts. This program is supported by UNICEF, UNFPA, JICA, USAID, Save The Children, BRAC, UNDP, PLAN, and other partners. Moreover, the MOHFW has established special-care newborn unit (SCANU) in 14 hospitals (medical college and district hospitals) with support from UNICEF; additional 11 will be established in 2013-2014 with support from UNICEF; all district and upazila hospitals of 10 more districts will be added by 2016 through assistance from SAARC Development Fund.

Before inception of IMCI program in Bangladesh, there were separate vertical child health programs, viz. Control of Diarrheal Diseases (CDD) and Acute Respiratory Infections (ARI). IMCI addresses morbidities which are responsible for almost 75% of under-five deaths. To simplify case management in the primary healthcare settings, diseases and problems covered by IMCI program in Bangladesh have been classified into 10 broad categories, viz. (i) very severe disease, (ii) pneumonia, (iii) cough and cold-not pneumonia, (iv) diarrhea, (v) fever-malaria, (vi) fever-not malaria, (vii) measles, (viii)

Table 4.8. Coverage of Penta vaccine, vitamin A capsule, and anti-helminthes among under-five children (EPI CES 2013)

Residence	Penta vaccine (coverage) (0-59 months)	Vitamin A capsule			Antihelminthes (24-59 months)		
		Infant (6-11 months)	Children (12-59 months)	Postpartum woman	Boy	Girl	Total
National	96%	84%	93%	36%	88%	90%	90%
Rural	97%	83%	92%	35%	90%	90%	90%
Urban	96%	87%	94%	37%	95%	89%	90%

ear problem, (ix) PEM (protein-energy malnutrition), and (x) others. The Management Information System (MIS) of DGHS, with support from UNICEF, collects data on IMCI services being provided. For 2012, data on 4,601,431 children aged 0 to 5 year(s) who received treatment from the IMCI facilities of the 64 districts have been received. Figure 4.13 shows distribution of the children by division.

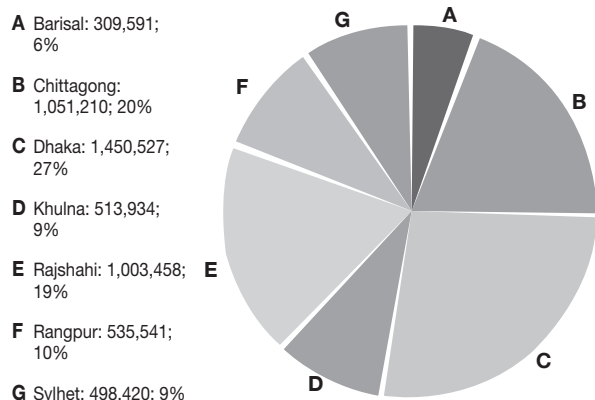


Figure 4.13. Division-wise distribution of children aged 0 to 5 year(s), who received treatment from IMCI facilities in 2012 (total 5,362,681)

Figure 4.14 shows the age distribution of the children. It is seen that children aged 1 to 5 year(s) constituted the largest IMCI service recipients (58%), followed by 2 to 12 months age-group (34%). Of the total under-five children, 3% were at the neonatal age. Age-group 29 to 59 days comprised 5%.

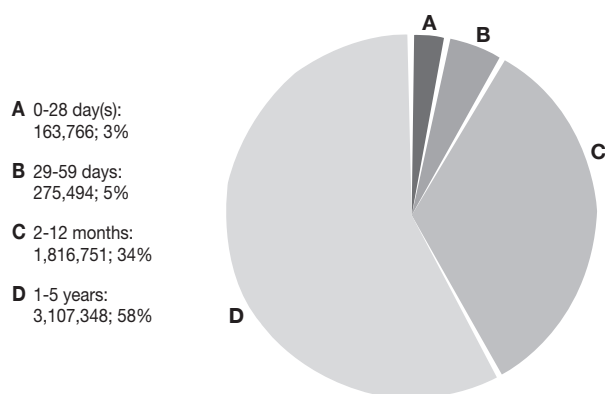


Figure 4.14. Distribution by age group of the children (0-5 years) who received treatment from the IMCI facilities in 2012 (total 5,362,681)

Further disaggregation of the children who received treatment from the IMCI facilities in 2012 is shown in the annex.

School health program

In Bangladesh, school health program began in 1951 in Dhaka and Chittagong and gradually expanded by 1972 to a network of 23 school health clinics located mainly in one school campus of the then district headquarters. Currently, two medical officers are assigned to each of the clinics. The school health clinics provide clinical services to pupils of the schools. In addition, the clinics help in improvement of the school environment, improvement of school health and nutrition services, and health education to the pupils. The HPNSDP 2011-2016 has broadened the scope of school health program to expand all over the country, to provide preventive and promotive health services through health education; screening for eye, ENT, nutrition and dental health; first-aid, and referral. School health program includes training of school teachers for teaching on first-aid to the school students, personal hygiene, handwashing, nutrition, safe water and sanitation, and provision of first-aid box.

Adolescent health program

The adolescents (10-19 years) constitute about 23% of the population in Bangladesh. The annual growth rate of the adolescent population is 4.3% compared to 1.37% general population growth rate. Early marriage and motherhood are common in Bangladesh. About 50% of all 15-19 years old females are married, of whom about 33% are already mothers, and another 6% are pregnant having risks to their health. Their knowledge on unprotected sex is also limited that may expose them to STDs, unwanted pregnancies, and abortions. In consideration of the above facts, the adolescent health program has been incorporated into school health program under HPNSDP running 2011-2016. The objectives of the program include; (i) improvement of knowledge of adolescents on adolescent reproductive health issues; (ii) creation of positive changes in the behavior and attitude of the gate-keepers of the adolescents toward reproductive health; (iii) providing easy access of all adolescents to adolescent-friendly and related health and other services. The adolescent health program provided training of trainers in 2011-2012 to 121 officers of both health and education departments to train the field-level health workers, teachers, and students, provided training to 1,118 health personnel and 1,889 secondary and higher secondary teachers of Kushtia, Rangpur, and Mymensingh district to develop their skills for fostering the objectives of adolescent health program. Secondary school students (n=1,560) were also provided training for creating peer-group learning.

SECONDARY AND TERTIARY HEALTHCARE

Population-bed ratios higher in medical college hospitals in Khulna, Chittagong and Rajshahi divisions

Secondary and tertiary-care health facilities are those that provide more advanced or specialty care than the primary healthcare facilities at the ward, union and upazila levels. However, many of the upazila health complexes (UHCs) have clinical specialists who provide specialty care to the patients. The district hospitals are usually termed secondary hospitals as these have fewer facilities for specialty care compared to many in the medical college hospitals. There are also different types of specialty-care centers, such as infectious disease hospitals, tuberculosis hospitals, leprosy hospitals, which fall under the health facilities of secondary care. The medical college hospitals are located at the regional level, one for few districts and are affiliated with medical colleges and provide specialty care in many disciplines. These hospitals are called tertiary hospitals. Tertiary hospitals also include the super-specialty hospitals at the national level or centers that provide high-end medical services in a specific field.

A recap of primary healthcare

An overview of primary-care hospitals and health centers located at the upazila level and below is given in Chapter 4. To summarize, there are 467 hospitals (425 upazila health complexes and 6 hospitals outside UHC at the upazila level; and 31 hospitals at the union level), with a total of 18,780 beds (18,290 at the upazila-level hospitals and 490 at the union-level hospitals), and 13,889 health facilities for outpatients only (1,362 at the union level and 12,527 at the ward level), yielding a total of 14,356 health facilities in rural setting

Tertiary hospitals also include the super-specialty hospitals at the national level or centers that provide high-end medical services in a specific field ...

for primary healthcare. Chapter 4 also mentions that primary healthcare in the rural setting is principally the responsibility of the Ministry of Local Government, Rural Development and Cooperatives (MOLGRD) and is delivered through the city corporations and municipalities which run a number of small to medium-sized hospitals and outdoor facilities. Additionally two large-scale donor-financed primary healthcare projects, viz. Urban Primary Health Care Project (UPHCP) and Smiling Sun Franchise Program run by NGOs under stewardship of the city corporations provide urban primary healthcare. The MOHFW owns 35 urban dispensaries and 23 school health clinics in some of the bigger cities and municipalities.

Secondary and tertiary hospitals

Table 5.1 summarizes the number and bed-capacity in different types of secondary and tertiary hospitals/health centers under DGHS. Further details, including the list and bed-capacity of each type of hospitals, are provided in the annexure.

Table 5.1. Secondary and tertiary hospitals/health centers under DGHS (as of December 2012)

Type of hospital/health center	No. of facilities	No. of beds				
		Functional	Revenue	Development	Proposed	Will increase by
Specialized hospital affiliated with postgraduate teaching institute	7	2,300	1,550	750	250	100

Table 5.1. continued

Type of hospital/health center	No. of facilities	No. of beds				
		Functional	Revenue	Development	Proposed	Will increase by
Medical college hospital/Dental college hospital/Hospital affiliated with college for alternative medicines (some colleges have postgraduate teaching facilities)	22*	11960	9430	2530	3500	2100
Specialized center	3	200	0	200	200	100
Specialized hospital	2	750	400	350	0	0
Infectious disease hospital	5	180	180	0	0	0
District hospital	53	7850	5670	2180	2150	1050
General hospital	11	1350	1150	1200	250	150
Chest disease/TB hospital	13	816	566	250	-	-
Leprosy hospital	3	130	130	0	0	0
Other hospital	6	305	250	55	100	50
Subtotal (hospital currently functional)	124	26841	19326	7515	6450	3550
Proposed new hospital	2	0	0	0	600	600
Total (currently functional + proposed)	126	26841	19326	7515	7050	4150

*Hospitals have not been established yet in five new medical colleges (Cox's Bazar, Jessore, Noakhali, Kishoreganj, and Pabna). Hence, the actual number of medical and dental college hospitals is 22)

Bangabandhu Sheikh Mujib Medical University (BSMMU)

The Ministry of Health and Family Welfare provides financial assistance to BSMMU as well as to its affiliated hospital. The BSMMU is the only Medical University in Bangladesh. Both university and its affiliated hospital are autonomous. The hospital has 1,212 beds which includes 452 free beds. The hospital has 48 clinical departments, 105 cabins, and 18 operation theaters.

Distribution of public hospitals and hospital beds by division

Figure 5.1 shows the distribution of the secondary and tertiary hospitals by administrative division. Dhaka division has the highest number (46) of secondary and tertiary hospitals, followed by Chittagong and Khulna divisions with 17 in each. Barisal division has the lowest number (8) of such hospitals.

Table 5.2 shows that, of the total 22 medical college hospitals, 9 are in Dhaka division (40.9%), followed by 4 in Chittagong division (18.2%), 3 in Rajshahi division (13.6%), 2 each in Khulna and Rangpur division (9.1%), and one in each of Barisal and Sylhet division (4.5%). There is no infectious disease hospital in Barisal and Rangpur divisions but each of the other 5 divisions has one infectious disease hospital. Rajshahi and Chittagong divisions have

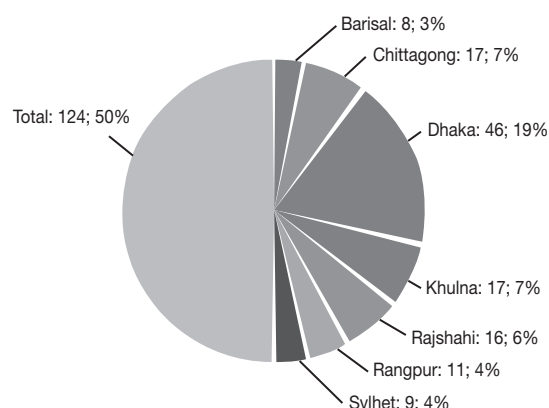


Figure 5.1. Distribution (number and percentage) of government-owned secondary and tertiary hospitals by administrative division of Bangladesh (2012) (total 124)

the highest number of chest hospitals (3 in each out of 13 in the country). Each of Khulna and Dhaka divisions has 2 chest hospitals. Each of Barisal, Rangpur and Sylhet divisions has one chest hospital.

Available number of beds for the population in the catchment areas is one of the good proxies for measuring the strength of healthcare infrastructure.

Figure 5.2 shows the distribution of government-owned secondary and tertiary-care hospital beds by administrative division of Bangladesh. It

Table 5.2. Distribution of secondary and tertiary public hospitals under DGHS by division (2012)

Type of hospital	No. and %	Barisal	Chittagong	Dhaka	Khulna	Rajshahi	Rangpur	Sylhet	Total
Medical college hospital	No.	1	4	9	2	3	2	1	1
	%	4.5	18.2	40.9	9.1	13.6	9.1	4.5	4.5
Infectious disease hospital	No.	0	1	1	1	1	0	1	1
	%	-	20.0	20.0	20.0	20.0	-	20.0	20.0
Chest hospital	No.	1	3	2	2	3	1	1	1
	%	7.7	23.1	15.4	15.4	23.1	7.69	7.7	7.7

is not surprising to see that 45% of beds in all government-owned secondary and tertiary hospitals and 46% of beds in the government medical college hospital are concentrated in Dhaka division.

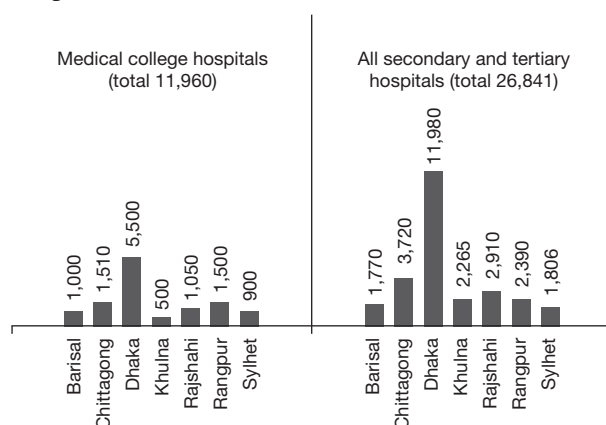


Figure 5.2. Distribution of the number of secondary and tertiary hospital beds by administrative division of Bangladesh (2012)

Figure 5.3 shows the population-bed ratio for the secondary and tertiary-care hospitals by administrative division of Bangladesh. The Khulna division has the paucity of beds in its medical college hospital (only one bed per 33,069 people) compared to other divisions. In Chittagong and Rajshahi divisions also, there is paucity beds in medical college hospital compared to the national average of one bed per 12,696 people. With the addition of beds from the new medical college hospitals, the population-bed ratios in Khulna and Chittagong divisions are expected to be reduced. However, Rajshahi and these two divisions would still need attention to increase beds in medical college hospitals. Increasing beds in medical college hospitals will contribute to less requirement of people of these regions to travel to other regions or in reducing waiting time for getting a bed in medical college hospitals for admission. However, further analysis of bed-occupancy and turn-over rates will provide better understanding of the need.

Available number of beds for the population in the catchment areas is one of the good proxies for measuring the strength of healthcare infrastructure ...

Private hospitals, clinics, and diagnostic centers

As of 2012, DGHS provided registration to 8,203 private hospitals, clinics, and diagnostic centers in Bangladesh. The number of registered private hospitals and clinics is 2,983 and that of registered private diagnostic centers is 5,220. The total number of beds in these registered private hospitals and clinics is 45,485. The MIS-DGHS received information on the number of sanctioned beds, free beds, departments, wards, cabins, and operation theaters in some of the private and non-profit hospitals, which is summarized in the annex.

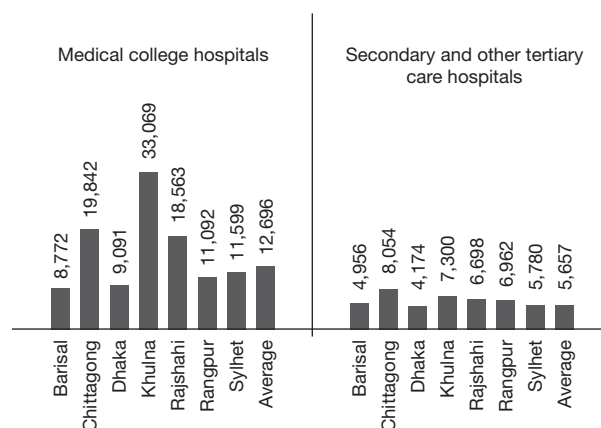


Figure 5.3. Population per government secondary and tertiary hospital bed by administrative division of Bangladesh (2012)

UTILIZATION OF HEALTH FACILITIES

More use of public facilities in 2012 than previous years

Along with the public hospitals and health centers, huge numbers of private and NGO-run health facilities are increasingly participating in sending patient-related data to the MIS-DGHS.

Public hospitals

For 2012 (January to December), we received data from quite a good number of public hospitals and health centers (Table 6.1). From these health facilities, reportedly 138,281,115 patients received healthcare from their outpatient departments. The number of children (both sexes) was 26,385,230. The number of male adults was 42,564,781, and the number of female adults was 69,331,104. The number of reported admissions was 4,564,318, of which 1,694,966 were for males and 2,018,786 for females. The number of hospital deaths was 87,238, of which 39,180 were among males, and 27,730 among females; the number of deaths of children (both sexes) was 20,328. The average hospital

death rate was 1.9%. Detailed information on each hospital/health facility is given in the annex.

Figure 6.1 compares the outdoor visits in all health facilities under DGHS in 2011 and 2012. The number of reported total outdoor visits increased by 22.8% in 2012 compared to that in 2011. Visits by male, female and child patients to the outdoor increased by 21.9%, 26.3%, and 16.0% respectively.

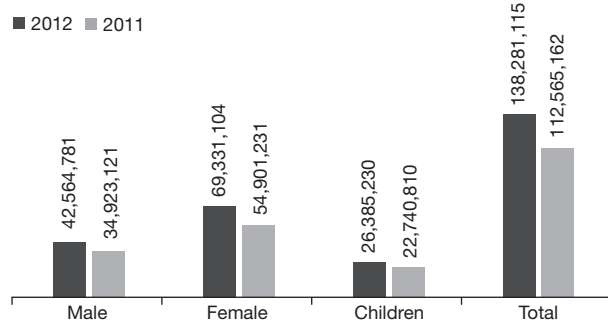


Figure 6.1. Number of reported outdoor visits in all DGHS health facilities (2011 and 2012)

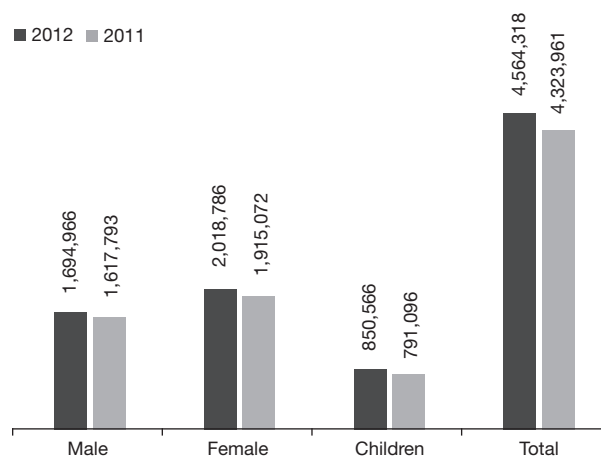


Figure 6.2. Number of reported admissions in all DGHS hospitals in 2011 and 2012

Figure 6.2 compares the number of admissions in all hospitals under DGHS in 2011 and 2012. The number of reported total admissions increased by 5.3% in 2012 compared to that in 2011. Admission of male, female and child patients increased by 4.6%, 5.1%, and 7.0% respectively.

Table 6.1. Number of reported admissions, deaths, and outpatient visits in public health facilities (2012)

Type of service-use and death	Male	Female	Children	Total
Outdoor visit	42564781	69331104	26385230	138281115
Admission	1694966	2018786	850566	4564318
Hospital death	39180	27730	20328	87238

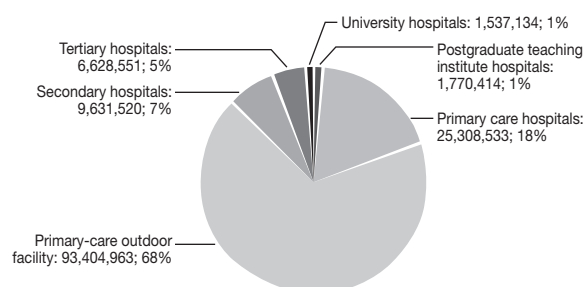


Figure 6.3. Distribution of outdoor patients by type of health facility in 2012 (total patients 138,281,115)

Figure 6.3 shows distribution of the OPD patients by type of health facilities. Of the total reported 138,281,115 outdoor patients, 68% were seen in the primary-care outdoor-only health facilities (trauma center, union subcenter, health and family welfare center, community clinic); 18% patients were seen in the primary-care hospitals (upazila health complex and rural health center). In the secondary-care hospitals (district or general hospital), 7% patients were seen. The tertiary-care hospitals (medical college hospital, dental hospital, and mental hospital), and the postgraduate

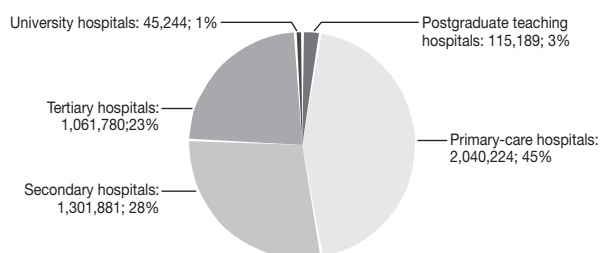


Figure 6.4. Distribution of admissions by type of hospital in 2012 (total admissions 4,564,318)

teaching hospitals served 5% and 1% of the outdoor patients respectively. The Bangabandhu Sheikh Mujib Medical University Hospital served 1% of the total reported outdoor patients.

Figure 6.4 shows distribution of admissions in 2012 by type of hospital. Of the total 4,564,318 reported admissions, 45% were in the primary-care hospitals (upazila hospitals) and 28% in the secondary-care hospitals (district and general hospitals); the tertiary-care hospitals (medical and dental college hospitals, mental hospital) and the postgraduate teaching hospitals had 23% and 3% of admissions respectively. Bangabandhu Sheikh Mujib Medical University Hospital had 1% of the total admissions.

Table 6.2 shows the average length of stay, bed-occupancy rate, hospital death rate, average daily admission, and average daily outdoor patient visit in different types of public hospitals. Detailed information is given in the annex.

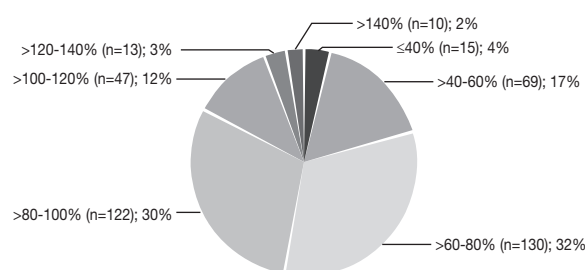


Figure 6.5. Distribution of 406 upazila hospitals by bed-occupancy rate (%) in 2012

Figure 6.5 shows the distribution of 406 upazila health complexes by bed-occupancy rate in 2012. It revealed that bed-occupancy rate of 47% at

Table 6.2. Average length of stay, bed-occupancy rate, hospital death rate, average daily admission, and average daily outdoor patient visits in different types of public hospitals (2012)

Type of health facility	Average length of stay (day)	Bed-occupancy rate (%)	Hospital death rate (%)	Average daily admissions (N)	Average daily outdoor visits (N)
University hospital (BSMMU)	-	98.0	2.4	124	4199
Specialized postgraduate teaching hospital	6.9	87.5	4.2	35	537
Medical college hospital	5.0	129.4	5.0	205	1276
Dental college hospital, Dhaka	23.0	70.0	0.0	29	208
Mental hospital, Pabna	114.0	83.9	0.2	4	83
District hospital	3.0	114.4	1.6	58	372

Table 6.3. Distribution (%) of the upazila hospitals by bed-occupancy rate in the last 5 years (2008 to 2012)

Bed-occupancy rate (%)	2008		2009		2010		2011		2012	
	No.	%	No.	%	No.	%	No.	%	No.	%
≤40	10	2.5	8	2.0	40	9.7	24	5.9	15	3.7
>40–50	42	10.4	23	5.6	23	5.6	19	4.7	23	5.7
>50–60	59	14.6	36	8.8	52	12.6	42	10.4	46	11.3
>60–70	74	18.3	75	18.4	48	11.6	50	12.4	53	13.1
>70–80	76	18.8	113	27.7	73	17.7	63	15.6	77	19.0
>80–90	61	15.1	60	14.7	56	13.6	64	15.8	61	15.0
>90–100	48	11.9	48	11.8	35	8.5	47	11.6	61	15.0
>100–110	16	4.0	20	4.9	30	7.3	36	8.9	36	8.9
>110–120	16	4.0	12	2.9	25	6.1	27	6.7	11	2.7
>120–130	0	-	9	2.2	14	3.4	17	4.2	9	2.2
>130–140	0	-	4	1.0	6	1.5	7	1.7	4	1.0
>140	2	0.5	0	-	11	2.7	8	2.0	10	2.5
Total	404	100.0	408	100.0	413	100.0	404	100.0	406	100.0

upazila health complexes was more than 80%; at 32% of upazila health complexes, it was between >60 to 80%; and at 21% of upazila health complexes, it was ≤60%.

Table 6.3 shows the distribution (%) of the upazila hospitals by bed-occupancy rate in the last 5 years (2008 to 2012).

Private hospitals

The private hospitals are increasingly providing data on hospital utilization. The annex provides a list of private hospitals, with the number of admissions, hospital deaths, outdoor visits, bed-occupancy rates, hospital death rates, average daily admissions, and average daily outdoor visits in 2012.

USAID-supported NGO Health Service Delivery Project

The NGO Health Service Delivery Project (NHSDP) is a follow-up project to the Smiling Sun Franchise Program (SSFP), started in January 2013 as one of the USAID's largest investments in the health sector of Bangladesh. NHSDP supports the delivery of an essential service package (ESP) of primary healthcare through a nationwide network of 26 local NGOs, 327 static clinics, 8,838 satellite clinics, and 6,320 community service providers. USAID's NHSDP serves approximately 20 million people of Bangladesh (13% of the total population) who

have had 26.5 million service encounters from the *Surjer Hashi* (SH) clinics. The project complements GOB efforts to maximize the outreach to the poor and underserved population in the country, with quality services at an affordable fee or no cost. There are four dimensions of performance by NGOs under NHSDP, viz. coverage and uptake, quality, equity, and institution strengthening through three elements, such as improved reproductive and MNCH outcomes, sustainability, and delivery of quality ESP to the poor.

Table 6.4 shows the number of patients served by the USAID's NGO Health Service Delivery Project partners in 2012. A total of 108,242 newborn contacts for essential newborn care, 3,478,010 immunization service, treatment to 2,504,304 ARI cases, ANC and PNC service to 1,202,433, and 399,160 maternal encounters were reported. The total maternal health and family planning service contact was 3,622,819 and 14,485,472 respectively. Out of the total service encounters, 8,522,758 were for the poor.

Table 6.5 shows that 327 SH clinics provided treatment to 25,657,339 patients. The number of admissions was 20,270 in 52 ultra-clinics (38 CEmOC and 14 BEmOC) for childbirth. There was no death as in the previous year. In 52 ultra-clinics, bed-occupancy rate was 63%, and daily admission rate was 62. A total of 23,270 child deliveries were conducted by the SBAs both in facilities and at home.

Table 6.4. Number of patients served under USAID's NHSDP (former SSFP) partners in 2012 shown by type of service

Month	Newborn contact	EPI	ARI	ANC	PNC	Maternal contact	FP	Poor customer
January	6272	316126	196316	93020	31510	294134	1253953	702121
February	6958	303209	195316	92870	31712	292503	1219197	691948
March	8805	300618	198011	104028	34371	307964	1241610	719989
April	8546	306084	200363	103516	34180	310294	1228164	722926
May	9159	295020	207968	107853	34081	316714	1210552	737026
June	9007	284258	209347	106352	33594	314067	1190102	721320
July	9383	290907	213109	108340	34564	321867	1196552	728032
August	9184	252381	211117	90697	30797	277545	1145656	674842
September	11267	297979	215947	109197	35483	321592	1233996	728756
October	9913	265087	216873	94613	32364	286200	1173673	689759
November	9964	283534	216336	96185	32791	290016	1180655	699514
December	9784	282807	223601	95762	33713	289923	1211362	706525
Total	108242	3478010	2504304	1202433	399160	3622819	14485472	8522758

Table 6.5. Some parameters of hospitalized patients served by USAID's NHSDP (former SSFP) partners in 2012

Month	Total patients (N)	Admission (N)	Death (N)	Average length of stay (day)	Bed-occupancy rate (%)	Hospital death rate (%)	Average daily admission (N)	Total skilled delivery (N)
January	2146744	1560	0	3	63	0	61	1724
February	2115708	1482	0	3	62	0	61	1661
March	2160334	1435	0	3	62	0	61	1688
April	2159565	1405	0	3	62	0	62	1578
May	2174888	1445	0	3	62	0	61	1650
June	2169393	1395	0	3	62	0	61	1571
July	2166389	1628	0	3	63	0	62	1781
August	2040321	1872	0	3	64	0	63	2033
September	2196867	2238	0	3	65	0	64	2716
October	2092006	2096	0	3	64	0	63	2345
November	2097096	1868	0	3	64	0	63	2267
December	2138028	1846	0	3	64	0	63	2256
Total	25657339	20270	0	3	63	0	62	23270

MORBIDITY PROFILE

Road traffic accident emerging as one of the leading causes of hospitalization

As in the previous year, the morbidity profile for 2012 has been estimated based on admitted patients in the upazila health complexes, district hospitals, and medical college hospitals of Bangladesh. Admitted patients were categorized using a uniform online form of DHIS 2.12 software. The form contains a list of 156 diseases/conditions plus another category called 'other'. Forms were filled directly in computers in each of the health facilities of the country on a monthly basis. The data were accumulated automatically for analysis. For 2012, beginning from January and ending in December, the MIS-Health received data on morbidity profile of indoor patients from all 421 upazila health complexes, 58 district and general hospitals, 5 of 21 medical college hospitals, and 6 postgraduate institute hospitals. The number of institutions from which data were collected is shown in Table 7.1.

Limitation

The precision of diagnosis may not be equally sound among various types of hospitals. In the upazila and district-level hospitals, diagnoses mainly depend upon clinical judgment with little help from laboratory investigations. On the other hand, hospitals of academic institutions, like medical college hospitals and postgraduate institute hospitals, have more qualified consultants, availability of wider range of laboratory investigations, and patients remain relatively longer in this type of hospitals.

Therefore, precision of diagnosis is better in these hospitals.

Use of the same data-collection format, irrespective of the type of hospital, appeared as another limitation. For some types of hospitals, particularly specialized hospitals, many of the diagnoses constituting large percentage of the disease burden, fell under 'other' category. As there was no specific mention about what the 'other' stands for, it was not possible to find out from the report which diagnoses belonged to the 'other' category. However, we introduced ICD 10 based reporting system and, hopefully, subsequent reporting will be of improved quality.

In our country, patients usually make their own choices in selecting hospital for services. Therefore, proportion of different types of morbidities varies between types of hospitals. Moreover, as explained earlier, the precision of diagnosis also varies between levels of hospitals, lower-level hospitals having lower precision. Considering these facts, summaries have been made separately for upazila hospitals, district-level hospitals, and medical college hospitals. As the postgraduate institute hospitals are specialized in nature and each hospital treats specific disease of only one specialized discipline, summary for each specialized hospital has been presented individually.

This year, we are including year-wise comparison of the percentages of top-seated diseases in

Table 7.1. Type of government hospitals from which indoor morbidity data for 2012 were received

Type of hospital	Total number of hospitals	No. and % of hospitals which provided indoor morbidity data	
		Number	Percentage
Upazila health complex	424	421	99.29
District and general hospital	64	58	93.55
Medical college hospital	21	6	28.57
Postgraduate teaching institute hospital	7	6	85.71

different types of institutions. This will show the trends of disease patterns over the last three years.

Morbidity profile of patients in upazila hospitals

Figure 7.1 shows the morbidity profile in terms of the top 15 diseases/causes of hospitalization among patients admitted in the upazila hospitals. We received data from 421 upazila hospitals out of 424. Total number of patients included in the analysis

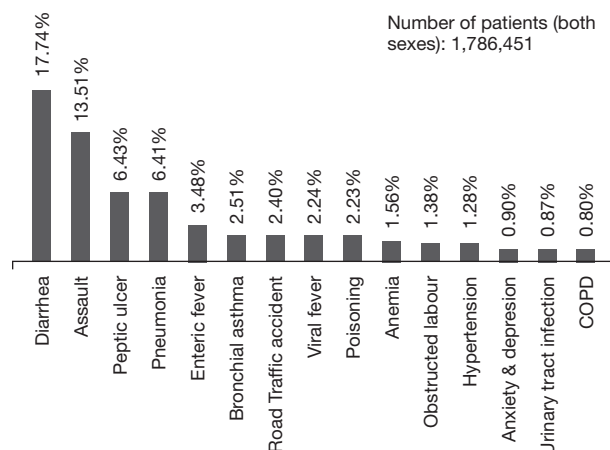


Figure 7.1. Top 15 diseases/causes of admission among the indoor patients of upazila hospitals (n=421) in 2012

was 1,786,451. The 'top 15' have been selected based on the percentage of prevalence in both sexes.

Diarrhea (17.74%) was the commonest cause for which the patients were admitted, followed by assault (13.51%). The other three most common diseases were peptic ulcer diseases (6.43%), pneumonia (6.41%), and enteric fever (3.48%).

The comparison of percentages and ranking of the top 10 diseases with those of the previous year are given in Table 7.2. In 2012, percentages of diarrhea, assault, bronchial asthma, and road traffic accident as causes of admissions in the upazila health complexes increased from those in 2011. On the other hand, percentages of peptic ulcer, pneumonia, poisoning, and anemia decreased in 2012. There was no major shift in the ranking within the top 10 causes of admissions, except for bronchial asthma which moved to 6th position in 2012 from its 9th place in 2011.

Morbidity profile of patients in district and general hospitals

Figure 7.2 shows the morbidity profile of patients in terms of the top 15 diseases/causes of admission among the indoor patients in the district and general hospitals. This year, we received data from 58 district and general hospitals out of 62. The total number of patients included in the

Table 7.2. Top 10 diseases/causes of admission in upazila health complexes in 2011 and 2012

Percentage trends	Disease/Cause of admission	2012*		2011**		Ranking changes
		Ranking	Percentage	Ranking	Percentage	
↑	Diarrhea	1	17.74	1	15.54	-
↑	Assault	2	13.51	2	12.63	-
↓	Peptic ulcer	3	6.43	4	7.43	+1
↓	Pneumonia	4	6.41	3	7.46	-1
↔	Enteric fever	5	3.48	5	3.48	-
↑	Bronchial asthma	6	2.51	9	2.16	+3
↑	Road traffic accident	7	2.40	6	2.34	-1
↑	Viral fever	8	2.24	10	1.80	+2
↓	Poisoning	9	2.23	7	2.27	-2
↓	Anemia	10	1.56	10	2.19	-

*2012: Number of patients: 1,786,451; Number of UHCs: 421; **2011: Number of patients in 2011:1,857,267; Number of UHCs: 418

analysis was 955,127. Similar to upazila hospitals, diarrhea (12.24%) was the commonest cause for which the patients were admitted, followed by assault (7.91%). The next three causes in order

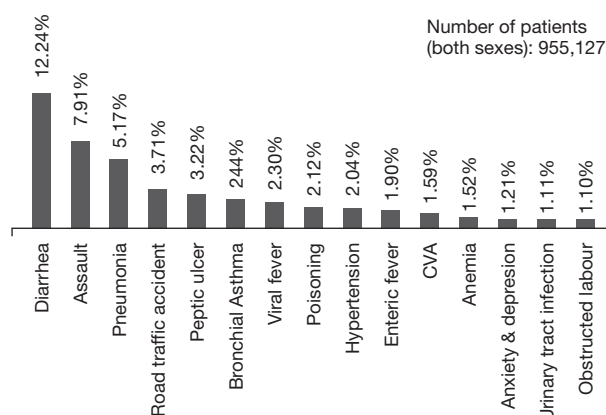


Figure 7.2. Top 15 diseases/causes of admission among the indoor patients in district and general hospitals (n=58) in 2012

of percentages in both sexes were pneumonia (5.17%), road traffic accidents (3.71%), and enteric fever (3.22%).

The percentage and ranking of top causes of admissions in district level hospitals in 2012 were comparable with those in 2011 (Table 7.3). Increased percentages were noticed for diarrhea,

assault, road traffic accident, bronchial asthma, and viral fever. Interestingly, percentages of exactly these five causes were increased also in the upazila health complexes. Percentages decreased for the cases of pneumonia, poisoning, hypertension, and enteric fever. The rankings of the causes of admission were almost identical with those in the previous year.

Morbidity profile of patients in medical college hospitals

Figure 7.3 shows the morbidity profiles of admitted patients in terms of the top 15 diseases/causes of admission reported from the medical college hospitals. At the time of analysis, we received

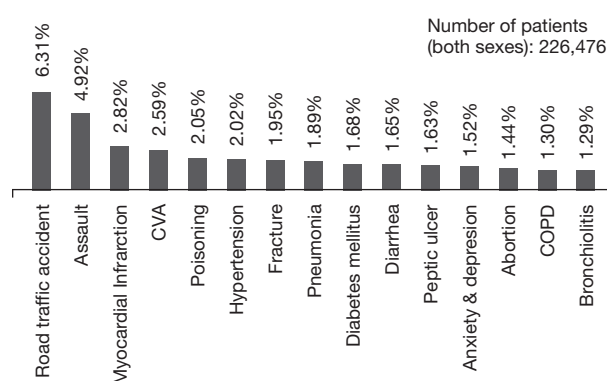


Figure 7.3. Top 10 diseases/causes of admission of patients in medical college hospitals (n=5) in 2012

Table 7.3. Top diseases/causes of admission in district-level hospitals in 2011 and 2012

Percentage trends	Disease/Cause of admission	2012*		2011**		Ranking changes
		Ranking	Percentage	Ranking	Percentage	
↑	Diarrhea	1	12.24	1	11.10	-
↑	Assault	2	7.91	2	7.42	-
↓	Pneumonia	3	5.17	3	5.73	-
↑	Road traffic accident	4	3.71	4	3.43	-
↑	Bronchial asthma	6	2.44	6	2.43	-
↑	Viral fever	7	2.30	7	2.36	-
↓	Poisoning	8	2.12	9	2.00	+1
↓	Hypertension	9	2.04	8	2.24	-1
↓	Enteric fever	10	1.90	11	1.78	+1

*2012: Number of patients: 955,127; Number of hospitals: 58; **2011: Number of patients: 876,691; Number of hospitals: 57

data from 5 medical college hospitals out of 21. Total number of patients included in the analysis was 226,475. The medical college hospitals from which data were available at the time of analysis were: Comilla Medical College Hospital; Faridpur Medical College Hospital; Dinajpur Medical College Hospital; Rangpur Medical College Hospital; Shaheed Suhrawardy Medical College Hospital, Dhaka; Shahid Ziaur Rahman Medical College Hospital, Bogra; and M.A.G. Osmani Medical College Hospital, Sylhet.

The most frequent cause for admission (in both sexes) was road traffic accident (6.31%). This was followed by assault (4.92%), myocardial infarction (2.82%), CVA (2.59%), poisoning (2.05%), and hypertension (2.02%).

Compared to the previous year, percentages of road traffic accident, myocardial infarction, cerebrovascular accident, poisoning, and hypertension increased this year while percentages of assault, fracture, pneumonia, diabetes, and diarrhea decreased as causes of admission in the medical college hospitals (Table 7.4).

As in other organizations, percentage of road traffic accident increased also in medical college hospitals and, this year, it grabbed the top position replacing assault.

Among other changes, the most notable is the ranking increment of myocardial infarction

Although diarrhea remained in the top position as a cause of admission in upazila and district-level hospitals, its ranking went down to the 10th position in medical college hospitals this year from its 3rd position in 2011. This may indicate that now fewer patients need to go to a tertiary facility for the treatment of diarrheal diseases ...

(commonly known as 'heart attack'); it was not in the top 10 positions in 2011 but, this year, it took the 3rd place.

Another interesting change is related to diarrheal diseases. Although diarrhea remained in the top position as a cause of admission in upazila and district-level hospitals, its ranking went down to the 10th position in medical college hospitals this year from its 3rd position in 2011. This may

Table 7.4. Top 10 diseases/causes of admission in medical college hospitals in 2011 and 2012

Percentage Trends	Disease/Cause of admission	2012*		2011**		Ranking changes
		Ranking	Percentage	Ranking	Percentage	
↑	Road traffic accident	1	6.31	2	5.58	+1
↓	Assault	2	4.92	1	5.90	-1
↑	Myocardial infarction	3	2.82	Was not in "top 10"		N/A
↑	Cerebrovascular accident (CVA)	4	2.59	5	2.31	+1
↑	Poisoning	5	2.05	7	1.98	+2
↑	Hypertension	6	2.02	9	1.88	+3
↓	Fracture	7	1.95	4	2.64	-3
↓	Pneumonia	8	1.89	6	2.14	-2
↓	Diabetes mellitus	9	1.68	10	1.70	+1
↓	Diarrhea	10	1.65	3	2.82	-7

*2012: Number of patients: 226,476; Number of hospitals: 6; **2011: Number of patients: 407,437; Number of hospitals: 10

indicate that now fewer patients need to go to a tertiary facility for the treatment of diarrheal diseases.

Morbidity profile of patients in postgraduate institute hospitals: From local health bulletins

From 2012, all public hospitals and health organizations of Bangladesh publish their local health bulletins online. An innovative software automatically creates the health bulletins as soon as the local organizations enter their data, and the bulletins are instantly published on the web site for global access. The morbidity profiles of indoor patients in the postgraduate teaching hospitals, namely National Institute of Traumatology, Orthopedics and Rehabilitation (NITOR), National Institute of Kidney Diseases and Urology (NIKDU), National Institute of Diseases of Chest and Hospital (NIDCH), National Institute of Cardiovascular Diseases (NICVD), National Institute of Cancer Research and Hospital (NIRCH), and National Institute of Mental Health & Research (NIMHR) have been taken from their respective local health bulletins published in 2013. Local health bulletins of 2013 used ICD 10 codes for morbidity reporting. As this was the first attempt of using the ICD 10 codes for the institutions, some mistakes were anticipated. So, we have included in the analysis only those data that appeared relatively error-free.

Table 7.5 summarizes the morbidity profile of the indoor patients in National Institute of Traumatology, Orthopedics and Rehabilitation

Table 7.5. Morbidity profile of the indoor patients (all ages) in National Institute of Traumatology, Orthopedics and Rehabilitation (NITOR) in 2012

Disease/ Cause of admission	Number of patients	Percentage
Road traffic accident	9664	46.00
Fracture	1470	7.00
Assault	840	4.00
Spinal cord injury	210	1.00
Burn	109	0.52
Rickets	63	0.30
Gangrene	63	0.30
Osteomyelitis	42	0.20
Osteosarcoma	126	0.60
Others	8420	40.08

Total patients: 21,009; Male: 12,923; Female: 8,086

An innovative software automatically creates the health bulletins as soon as the local organizations enter their data, and the bulletins are instantly published on the web site for global access.

(NITOR). The table shows that among the 21,009 patients, 12,923 were male, and 8,086 (61.51%) were female (38.49%). Road traffic accident was the leading cause of admission (46%). This condition was also the leading cause of admission in 2011, although the percentage was slightly less (44.7%) in that year. Fractures, assault, and spinal cord injury were responsible for 7%, 4%, and 1% of admissions respectively. The percentages and rankings of the top causes of admission were almost the same as those in the previous year.

Table 7.6 summarizes the morbidity profile of the indoor patients in the National Institute of Kidney Diseases and Urology (NIKDU). The table

Table 7.6. Morbidity profile of the indoor patients (all ages and both sexes) in National Institute of Kidney Diseases and Urology (NIKDU) in 2012

ICD 10 code	Disease/ Cause of admission	Number of patients	Percentage
N185	Chronic kidney disease, Stage 5	1565	35.26
N04	Nephrotic syndrome	576	12.98
N18	Chronic renal failure	524	11.80
N17	Acute renal failure	428	9.64
N00	Acute nephritic syndrome	212	4.78
N13	Obstructive and reflux uropathy	348	7.84
N20	Calculus of kidney and ureter	395	8.90
N40	Hyperplasia of prostate	255	5.74
C64	Malignant neoplasm of kidney, except renal pelvis	236	5.32

Total patients: 4,439; Male: 2,891; Female: 1,548

shows that among the 4,439 patients, 2,891 were male (65.12%), and 1,548 were female (34.88%). Chronic kidney disease of Stage 5 was the leading cause of admission (35.26%), followed by nephrotic syndrome (12.98%). Chronic renal failure appeared again in the third position but this was without specific staging. Acute renal failure and acute nephritic syndrome appeared in the fourth and fifth position and were responsible for 9.64% and 4.78% of admissions respectively.

Table 7.7 summarizes the morbidity profile of the indoor patients in the National Institute of Diseases of Chest and Hospital (NIDCH). The table shows that among 9,983 patients, 7,618 were male (76.31%), and 2,365 were female (23.69%). Pulmonary tuberculosis was the leading cause of admission (26.85%), followed by pleural effusion (17.41%) and bronchial asthma (14.52%).

Table 7.7. Morbidity profile (%) of the indoor patients (all ages and both sexes) in National Institute of Diseases of Chest and Hospital (NIDCH) in 2012

Disease/Cause of Admission	Number of patients	Percentage
Tuberculosis (pulmonary with or without extra-pulmonary)	2680	26.85
Pleural effusion (tubercular/malignant/empyema)	1738	17.41
Bronchial asthma	1450	14.52
COPD	1150	11.52
Pneumothorax	818	8.19
Bronchiectasis	636	6.37
Chest trauma (due to RTA/assault/gun-shot injury/fall from height)	581	5.82
Ca-Lung	424	4.25
Ca-Esophagus	294	2.95
Foreign body airway/Esophagus	212	2.12

Total patients: 9,983; Male: 7,618; Female: 2,365

Table 7.8 summarizes the morbidity profile of the indoor patients in the National Institute of Mental Health & Research (NIMHR). Total number of patients were 2,030, among whom 1,315 were male (64.78%), and 715 were female (35.22%). Bipolar affective disorder was the leading cause of admission (39.61%), followed by schizophrenia (37.54%). Drug or other substance-abuse appeared next (5.22%) in the ranking.

Table 7.8. Morbidity profile of the indoor patients (all ages and both sexes) in National Institute of Mental Health & Research (NIMHR) in 2012

ICD 10 code	Disease/Cause of admission	Number of patients	Percentage
F31	Bipolar affective disorder	804	39.61
F20	Schizophrenia	762	37.54
F19	Mental and behavioral disorders due to multiple drug-use and use of other psychoactive substances	106	5.22
F41	Other anxiety disorders	93	4.58
F79	Unspecified mental retardation	89	4.38
G40	Epilepsy	50	2.46
F00	Dementia in Alzheimer's disease	30	1.48
R401	Stupor	26	1.28
F42	Obsessive-compulsive disorder	23	1.13
F07	Personality and behavioral disorders due to brain disease, damage, and dysfunction	18	0.89

Total patients: 2,030; Male: 1,315; Female: 715

Table 7.9 summarizes the causes of admissions in the National Institute of Cancer Research and Hospital. In total, 3,360 patients were admitted in this hospital and, among them, 1,434 were female (42.68%). The table shows that carcinoma of lungs or bronchus ranked the first position in terms of the number of admitted patients (12.83%). Breast and cervical cancers appeared in the 2nd and 3rd position respectively.

Table 7.10 shows the causes of admissions in the National Institute of Cardiovascular Diseases in 2011. Total number of admitted patients was 44,559 and, among them, 13,588 were female (30.49%). Acute myocardial infarction was the leading cause of admissions (27.75%) among both sexes, followed by hypertension (14.24%), multiple valve disease (9.01%), and heart failure (8.53%).

Table 7.9. Morbidity profile of the indoor patients (all ages and both sexes) in National Institute of Cancer Research (NICR) and Hospital in 2012

ICD 10 code	Disease/ Cause of admission	Number of patients	Percentage
C349	Carcinoma of bronchus or lung, unspecified	431	12.83
C509	Carcinoma of breast, unspecified	297	8.84
C539	Carcinoma of cervix uteri, unspecified	217	6.46
C169	Carcinoma of stomach, unspecified	178	5.3
C859	Non-Hodgkin's lymphoma, unspecified type	155	4.61
C699	Carcinoma of eye, unspecified	149	4.43
C56	Malignant neoplasm of ovary	127	3.78
C189	Carcinoma of colon, unspecified	118	3.51
C069	Carcinoma of mouth, unspecified	111	3.3
C20	Malignant neoplasm of rectum	93	2.77

Total patients: 3,360; Male: 1,926; Female: 1,434

Table 7.10. Morbidity profile of the indoor patients (all ages and both sexes) in National Institute of Cardiovascular Diseases (NICVD) in 2012

ICD 10 code	Disease	Percentage
I21	Acute myocardial infarction	27.75
I11	Hypertensive heart disease	14.24
I089	Multiple valve disease, unspecified	9.01
I50	Heart failure	8.53
Q20	Congenital malformations of cardiac chambers and connections	8.51
I70	Atherosclerosis	8.05
I1330	Acute and subacute infective endocarditis	6.21
I501	Left ventricular failure	3.65
J440	COPD with acute lower respiratory infection	3.53
C759	Diseases of endocrine gland, unspecified	0.84

Total patients: 44,559; Male: 30,971; Female: 13,588

MORTALITY PROFILE

More accuracy achieved through using ICD 10 codes

An initiative has been undertaken by the Management Information System of DGHS to use ICD 10-based reporting of mortality and morbidity from hospitals. The Local Health Bulletins 2013 (online health bulletins uploaded by all health organizations of Bangladesh) present all data on mortality and morbidity among their patients according to ICD 10 codes. Data on death for 2012 reports from government health facilities were extracted from 504 Local Health Bulletins 2013. Each hospital published one Local Health Bulletin (Table 8.1). In these 504 hospitals, 95,256 deaths occurred in 2012. The largest proportion of those deaths was reported from medical college hospitals (39.3%), followed by upazila health complexes (38.3%), and district/general hospitals (18.5%).

Data on deaths collected from four postgraduate institute hospitals showed that 3.6% died in those hospitals.

Table 8.1 shows the type of hospitals where from mortality data for 2012 were collected.

Distribution of causes of deaths according to ICD 10 major groups (all types of hospitals)

The Local Health Bulletins 2013 mentioned causes of deaths, using ICD 10 on top 10 diseases, which took place in the respective hospitals. Out of total 99,256 deaths, 39,593 (40.0%) were reported in the top 10 disease categories. Table 8.2 shows the distribution of these 39,593 deaths in ICD 10

Table 8.1. Reported deaths from different types of hospitals

Type of hospital	Existing	Data from hospitals		Total reported deaths	
	No.	No.	%	No.	%
Upazila Health Complex	421	345	81.9	38058	38.3
District and General Hospital	62	49	79.0	18359	18.5
Medical College Hospital	17	12	70.6	39044	39.3
National institute of Cardiovascular Diseases (NICVD)	1	1	100.0	3085	3.1
National Institute of Kidney Diseases & Urology (NIKDU)	1	1	100.0	185	0.2
National Institute of Cancer Research & Hospital (NICRH)	1	1	100.0	65	0.1
Institute of Child & Mother Hospital (ICMH)	1	2	200.0	460	0.5
Total	504	411	81.5	99256	100.0

Table 8.2. Distribution of top 10 causes of deaths according to ICD 10 major disease groups across 504 public hospitals (total deaths 99,256; top 10 causes included 39,593 deaths) (2012)

ICD 10 disease group	No. of deaths	%
Diseases of the circulatory system	12149	12.2
Certain conditions originating in the perinatal period	6858	6.9
Diseases of the respiratory system	5956	6.0
Injury, poisoning and certain other consequences of external causes	4557	4.6
External causes of morbidity and mortality	3025	3.0
Certain infectious and parasitic diseases	2590	2.6
Diseases of the nervous system	1569	1.6

Table 8.2. continued

ICD 10 disease group	No. of deaths	%
Diseases of the genitourinary system	1133	1.1
Symptoms, signs, and abnormal clinical and laboratory findings, not elsewhere classified	497	0.5
Diseases of the digestive system	464	0.5
Endocrine, nutritional and metabolic diseases	178	0.2
Neoplasms	155	0.2
Diseases of the blood and blood-forming organs and certain disorders involving the immune mechanism	155	0.2
Pregnancy, childbirth, and the puerperium	142	0.1
Congenital malformations, deformations, and chromosomal abnormalities	113	0.1
Diseases of the skin and subcutaneous tissue	15	0.0
Mental disorders	13	0.0
Factors influencing health status and contact with health services	9	0.0
Diseases of the eye and adnexa	8	0.0
Diseases of the musculoskeletal system and connective tissue	7	0.0
Total	39593	39.9

major disease groups. The top 5 groups included diseases of the circulatory system (12,149 deaths, 12.2%); certain conditions originating in the perinatal period (6,858, deaths; 6.9%); diseases of the respiratory system (5,956 deaths, 6.0%); injury, poisoning, and other consequences of external causes (4,557 deaths, 4.6%); and external causes of morbidity and mortality (4,557 death, 4.6%).

Top 10 causes of deaths according to ICD 10 three-digit codes (all types of hospitals)

Figure 8.1 shows the top 10 causes of deaths across 504 public hospitals according to ICD 10 three-digit codes. Acute myocardial infarction and birth asphyxia appeared as the leading causes of deaths.

Causes of deaths by type of hospitals

Upazila health complexes

Table 8.3 shows the distribution of top 10 causes of deaths according ICD 10 major disease groups in 345 upazila health complexes (total deaths 38,058; top 10 causes included 5,770 deaths). The top five disease groups included diseases of the respiratory system (2,099 deaths, 68.6%); diseases of the circulatory system (1,385 deaths, 45.3%); injury, poisoning, and certain other consequences of external causes (411 deaths, 13.4%); certain infectious and parasitic diseases (367 deaths, 12.0%); and certain conditions originating in the perinatal period (357 deaths, 11.7%).

Figure 8.2 shows the top 10 causes of deaths in

345 upazila health complexes according to ICD 10 three-digit codes. Asthma and pneumonia appeared as the leading causes of deaths.

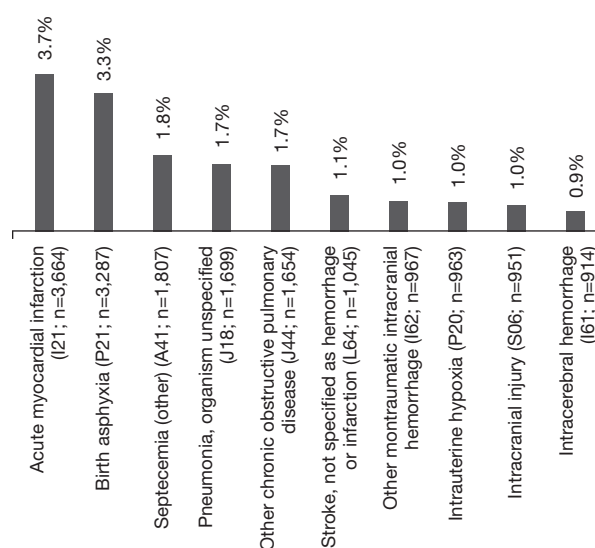


Figure 8.1. Top 10 causes of deaths according to ICD 10 three-digit codes across 504 public hospital (n=99,256 deaths)(2012)

District and general hospitals

Table 8.4 shows the distribution of top 10 causes of deaths according ICD 10 major disease groups in 49 district and general hospitals (total deaths 18,359; top 10 causes included 10,318 deaths). The

Table 8.3. Distribution of top causes of deaths according ICD 10 major disease groups in 345 upazila hospitals (total deaths 38,058; top 10 causes included 5,770 deaths) (2012)

ICD 10 major disease group	No. of deaths	%
Diseases of the respiratory system	2099	68.6
Diseases of the circulatory system	1385	45.3
Injury, poisoning, and certain other consequences of external causes	411	13.4
Certain infectious and parasitic diseases	367	12.0
Certain conditions originating in the perinatal period	357	11.7
External causes of morbidity and mortality	310	10.1
Symptoms, signs, and abnormal clinical and laboratory findings, not elsewhere classified	250	8.2
Diseases of the digestive system	130	4.3
Diseases of the nervous system	113	3.7
Diseases of the blood and blood-forming organs and certain disorders involving the immune mechanism	72	2.4
Diseases of the genitourinary system	66	2.2
Neoplasms	53	1.7
Endocrine, nutritional and metabolic diseases	51	1.7
Pregnancy, childbirth, and the puerperium	48	1.6
Diseases of the skin and subcutaneous tissue	15	0.5
Congenital malformations, deformations, and chromosomal abnormalities	14	0.5
Mental disorders	11	0.4
Factors influencing health status and contact with health services	9	0.3
Diseases of the musculoskeletal system and connective tissue	7	0.2
Diseases of the eye and adnexa	2	0.1
Total	5770	188.9

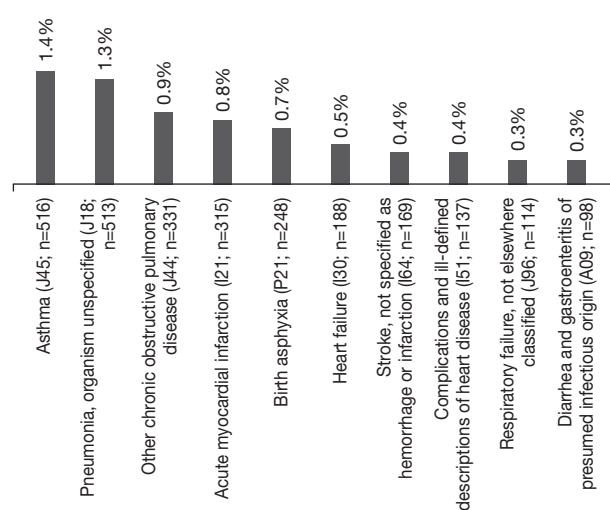


Figure 8.2. Top 10 causes of deaths according to ICD 10 three-digit codes in 345 upazila health complexes (n=38,085 deaths)(2012)

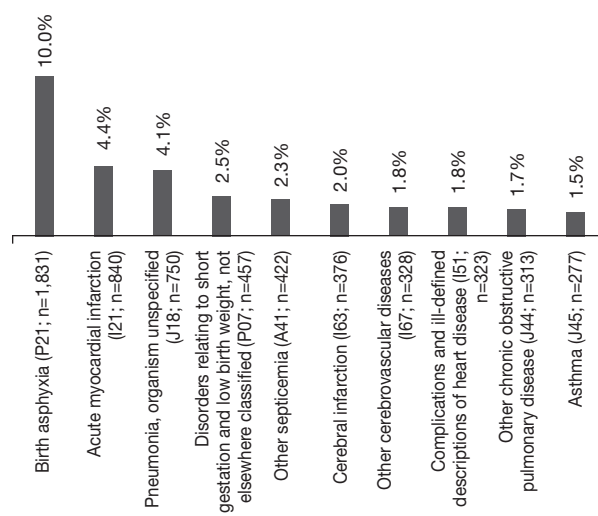


Figure 8.3. Top 10 causes of deaths according to ICD 10 three-digit codes in 49 district and general hospitals (n=18,359 deaths) (2012)

Table 8.4. Distribution of top causes of deaths according ICD 10 major disease groups in 49 district and general hospitals (total deaths 18,359; top 10 causes included 10,318 deaths) (2012)

ICD 10 major disease group	No. of deaths	%
Diseases of the circulatory system	3293	17.9
Certain conditions originating in the perinatal period	2528	13.8
Diseases of the respiratory system	1908	10.4
Injury, poisoning, and certain other consequences of external causes	689	3.8
Certain infectious and parasitic diseases	592	3.2
External causes of morbidity and mortality	413	2.2
Diseases of the nervous system	211	1.1
Symptoms, signs, and abnormal clinical and laboratory findings, not elsewhere classified	171	0.9
Diseases of the genitourinary system	151	0.8
Diseases of the digestive system	85	0.5
Congenital malformations, deformations, and chromosomal abnormalities	83	0.5
Diseases of the blood and blood-forming organs and certain disorders involving the immune mechanism	83	0.5
Endocrine, nutritional and metabolic diseases	43	0.2
Neoplasms	37	0.2
Pregnancy, childbirth, and the puerperium	25	0.1
Diseases of the eye and adnexa	6	0.0
Total	10318	56.1

top five disease groups included diseases of the circulatory system (3,293 deaths, 17.9%); certain conditions originating in the perinatal period (2,528 deaths, 13.8%); diseases of the respiratory system (1,908 deaths, 10.4%); injury, poisoning, and certain other consequences of external causes (689 deaths, 3.8%); and certain infectious and parasitic diseases (592 deaths, 3.2%).

Figure 8.3 shows the top 10 causes of deaths in 49 district and general hospitals according to ICD 10 three-digit codes. Birth asphyxia appeared as the leading cause of death.

Medical college hospitals

Table 8.5 shows the distribution of top causes of death according ICD 10 major disease groups in 12 medical college hospitals (total deaths 39,044; top 10 causes included 20,409 deaths). The top five disease groups included diseases of the circulatory system (5,043 deaths, 12.9%); certain conditions originating in the perinatal period (3,457 deaths, 8.9%), external causes of morbidity and mortality (2,302 deaths, 5.9%); and diseases of the respiratory system (1,911 deaths,

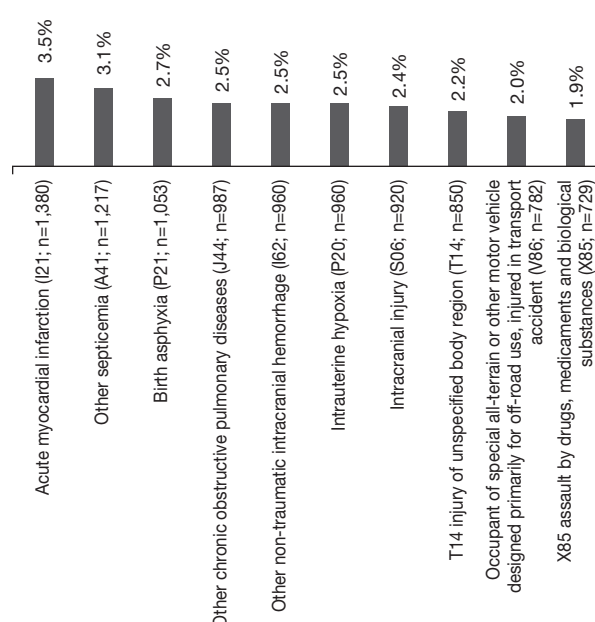
**Figure 8.4. Top 10 causes of deaths according to ICD 10 three-digit codes in 12 medical college hospitals (n=39,044 deaths) (2012)**

Table 8.5. Distribution of top causes of deaths according ICD 10 major disease groups in 12 medical college hospitals (total deaths 39,044; top 10 causes included 20,409 deaths) (2012)

ICD 10 major disease group	No. of deaths	%
Diseases of the circulatory system	5043	12.9
Certain conditions originating in the perinatal period	3770	9.7
Injury, poisoning, and certain other consequences of external causes	3457	8.9
External causes of morbidity and mortality	2302	5.9
Diseases of the respiratory system	1911	4.9
Certain infectious and parasitic diseases	1517	3.9
Diseases of the nervous system	1229	3.1
Diseases of the genitourinary system	764	2.0
Diseases of the digestive system	249	0.6
Pregnancy, childbirth, and the puerperium	69	0.2
Symptoms, signs, and abnormal clinical and laboratory findings, not elsewhere classified	56	0.1
Endocrine, nutritional and metabolic diseases	27	0.1
Neoplasms	15	0.0
Total	20409	52.3

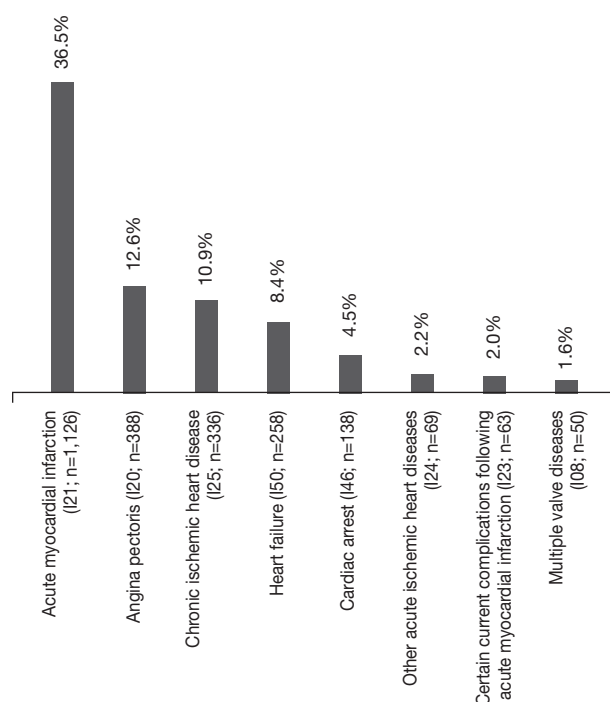


Figure 8.5. Distribution of causes of deaths according to ICD 10 three-digit codes in the National Institute of Cardiovascular Diseases (NICVD)(3,085 deaths)(2012)

4.9%). Detailed mortality profiles from medical college hospitals are provided in the annex to this chapter.

Figure 8.4 shows the top 10 causes of deaths in 12 medical college hospitals according to ICD 10 three-digit codes. Acute myocardial infarction and other septicemia appeared as the leading causes of deaths.

Postgraduate Institute Hospitals

National Institute of Cardiovascular Diseases (NICVD)

Figure 8.5 shows the distribution of causes of deaths in the National Institute of Cardiovascular Diseases (NICVD) according to ICD 10 three-digit codes. In more than one-third of the cases (36.5%), acute myocardial infarction was the cause of death.

National Institute of Kidney Diseases & Urology (NIKDU)

Figure 8.6 shows the distribution of causes of deaths in the National Institute of Kidney Diseases and Urology (NIKDU) according to ICD 10 three-digit codes. Chronic and acute renal failures appeared as the leading causes of deaths (collectively 72% deaths).

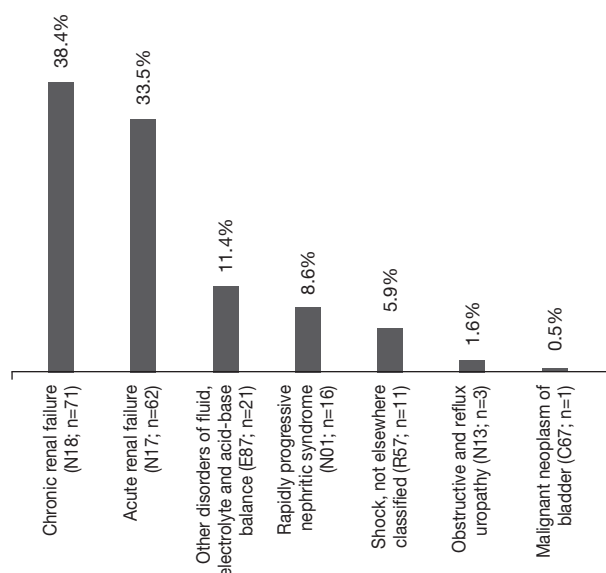


Figure 8.6. Distribution of causes of deaths according to ICD 10 three-digit codes in the National Institute of Kidney Diseases & Urology (NIKDU)(185 deaths)(2012)

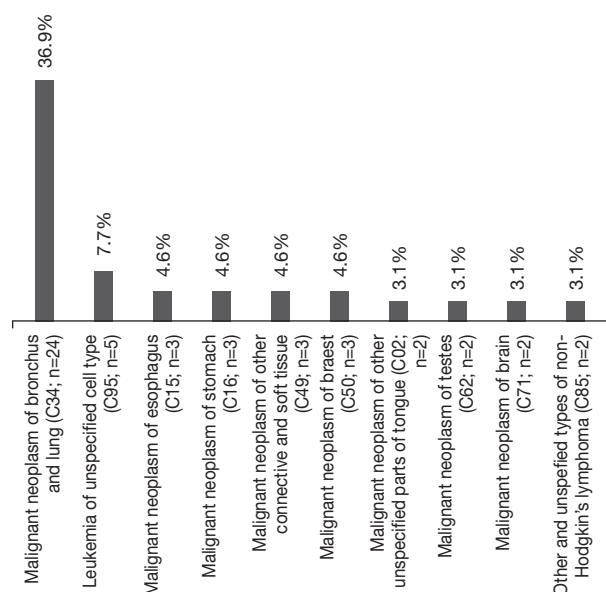


Figure 8.7. Distribution of causes of deaths according to ICD 10 three-digit codes in the National Institute of Cancer Research & Hospital (NICRH)(65 deaths)(2012)

National Institute of Cancer Research & Hospital (NICRH)

Figure 8.7 shows the distribution of causes of deaths in the National Institute of Cancer Research & Hospital (NICRH) according to ICD 10 three-digit codes. Malignant neoplasm of

bronchus and lung was responsible for 37% of deaths while all of the rest caused between 3.1 and 7.7% of deaths.

National Institute of Mother & Child Hospital

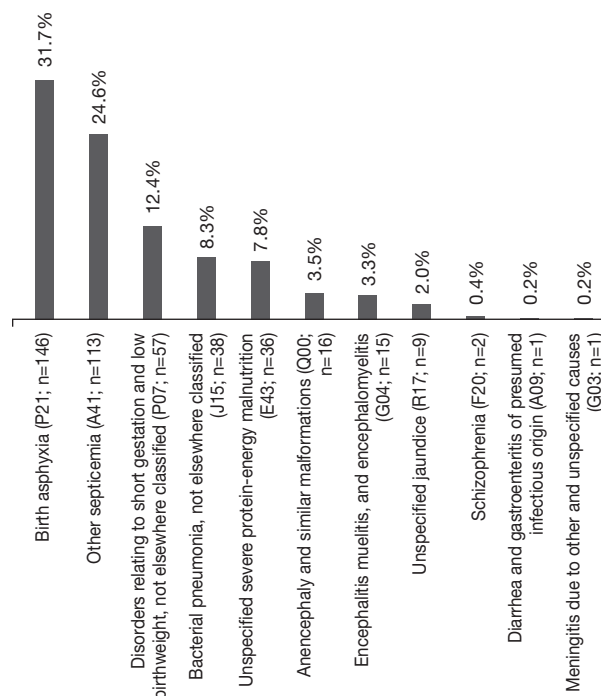


Figure 8.8. Distribution of causes of deaths according to ICD 10 three-digit codes in the National Institute of Mother & Child Health (ICMH)(460 deaths)(2012)

(ICMH)

Figure 8.8 shows the distribution of causes of deaths in the National Institute of Mother & Child Hospital (ICMH) according to ICD 10 three-digit codes. Birth asphyxia (31.7%) and other septicemia (24.6%) were leading causes of deaths.

National Institute of Diseases of Chest & Hospital (NIDCH)

Figure 8.9 shows the distribution of causes of deaths in the National Institute of Diseases of Chest and Hospital (NIDCH) according to ICD 10 three-digit codes. Chronic obstructive pulmonary diseases (27.3%) and tuberculosis (22.6%) were the leading causes of deaths.

Causes of deaths among under-five children

A total 21,567 deaths among under-five children were reported–10,212 from upazila and district hospitals and 11,355 from tertiary-level hospitals.

Table 8.6. Causes of death among the under-five children

ICD 10 major disease group	No. of deaths	%
Certain conditions originating in the perinatal period	7029	73.6
Diseases of the respiratory system	1310	13.7
Certain infectious and parasitic diseases	592	6.2
Congenital malformations, deformations, and chromosomal abnormalities	254	2.7
Endocrine, nutritional and metabolic diseases	113	1.2
Symptoms, signs, and abnormal clinical and laboratory findings, not elsewhere classified	95	1.0
Injury, poisoning, and certain other consequences of external causes	91	1.0
Diseases of the digestive system	62	0.6
Total	9546	100.0

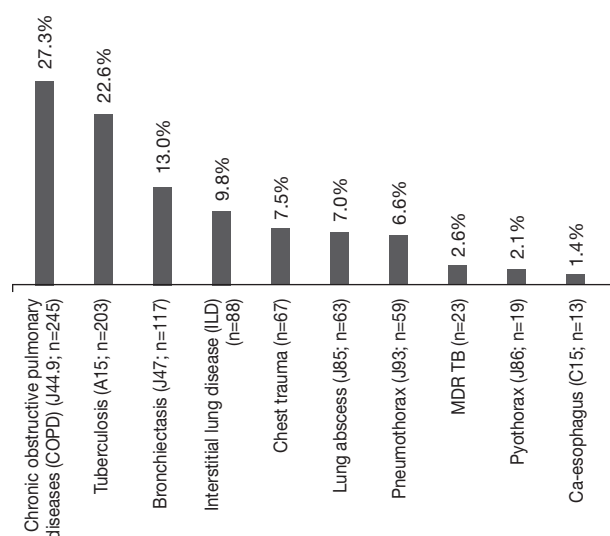


Figure 8.9. Distribution of causes of deaths according to ICD 10 three-digit codes in the National Institute of Diseases of Chest & UHospital (NIDCH)(897 deaths)(2012)

Causes of 9,546 deaths were analyzed. Perinatal conditions were the cause of most deaths (74%) (Table 8.6). These conditions included birth asphyxia, septicemia, prematurity of the baby, etc. The second cause was pneumonia (14%). Infectious diseases include mainly encephalitis and diarrhea.

The figure 8.10 shows the distribution of causes of deaths among under-five children. Birth asphyxia was the leading cause of death (31.7%), followed by low birthweight (17.7%) and septicemia (17.6%).

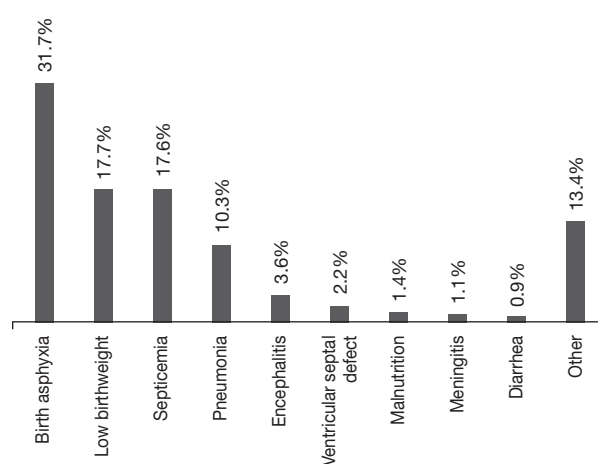


Figure 8.10. Distribution of causes of deaths among under-five children (2012)

Limitations of mortality analysis

In this chapter, data for mortality analysis were extracted from DHIS software as routine health information and from online Local Health Bulletins 2013. In 2012, mortality reporting was started using ICD 10 coding system. In the initial phase, the respective organizations were asked to provide only ICD 10 codes for top 10 diseases in aggregated form. Therefore, age and sex disaggregation of the mortality data could not be shown in this report. Effort to collect more comprehensive mortality data, using ICD 10 coding system, is ongoing, and in future, reporting on age, sex, and other disaggregation would be possible.

COMMUNICABLE DISEASE CONTROL IN BANGLADESH

Innovations and leadership covered diverse areas of communicable diseases control

Communicable disease control continues to remain a public-health priority both nationally and internationally. The communicable diseases of public-health importance include malaria, kala-azar, infestation with filarial and other worms, avian influenza, and influenza by novel virus, tuberculosis, HIV/AIDS and STDs, emerging zoonotic diseases, like Nipah, anthrax, brucellosis, food and waterborne diseases, like hepatitis due to viruses, diarrheal disorders, enteric fever, and leptospirosis, arthropodborne diseases, like dengue and chikungunya. Bangladesh is a signatory of International Health Regulation (IHR) 2005 and has to build its capacity in terms of detection and response to case detection and outbreak of emerging diseases. IHR 2005 also requires strengthening of national capacity to deal with the public-health emergency with national and international concern. Bangladesh has a number of target-oriented programs on malaria, kala-azar, filarial and intestinal worms, avian and pandemic influenza and achieved visible progress. This chapter describes the country situation with some of the major communicable diseases as mentioned above.

Malaria

Malaria is now a localized disease of Bangladesh and is endemic in 13 eastern and northern districts of the country. However, the three hilly districts account for 80% of the total burden of malaria. During the last decade, 82% of the total annual reported cases were due to *Plasmodium falciparum*. The remaining cases are due to *Plasmodium vivax*, and few are due to mixed infection. The overall malaria prevalence in the 13 endemic districts was 3.1% (Ref: Malaria Baseline Socioeconomic and Prevalence Survey 2007).

Over 10.9 million people of Bangladesh are at high risk of malaria, with the utmost vulnerability of the under-five children and pregnant women. Malaria claims about 0.10% lives annually. The national malaria control has achieved remarkable success in terms of reduction in the number of cases and deaths. Early diagnosis and prompt treatment through doorstep facilities provided

Bangladesh has a number of target-oriented programs on malaria, kala-azar, filarial and intestinal worms, avian and pandemic influenza and achieved visible progress ...

by GO-NGO partnership proved effective. This measure is being supported by global fund. The use of insecticide-treated bednets has supplemented the effort. The malaria-prone areas in the country



Map. Malaria-endemic 13 districts of Bangladesh

experience a perennial transmission with two peaks in pre-monsoon (March-May) and post-monsoon (September-November) periods. Outbreaks are also reported from districts bordering India in the north and northeast parts of the country. Bangladesh replaced the firstline drug Chloroquine by Artemisinin-based Combination Therapy (ACT) for the treatment of falciparum malaria cases since 2004. *Anopheles dirus*, *An. Minimus*, and *An. philipinensis* are the principal vectors, and all are susceptible to malathion and synthetic pyrethroid. Promotion and use of insecticide-treated nets/ long-lasting insecticide-treated nets (ITNs/ LLINs), selective indoor residual spraying (IRS) for containment of outbreaks and intensive IEC (information, education, and communication) for increasing awareness of the people are the main components for vector control.

Figure 9.1 shows that the numbers of malaria cases (*P. falciparum* and *P. vivax*) are declining every year. In 2012, only 29,518 malaria cases were reported. The number was 84,690 in 2008. The 5-year average of malaria cases shows that 90% of the cases are due to *Plasmodium falciparum*. In 2012, only 11 deaths were reported, giving a malaria death rate of 0.04%. Between 2011 and 2012, remarkably 42% and 70% reductions in malaria cases and deaths were witnessed. Data on the incidence and death due to malaria from 2000 to 2012 have been provided in the annex.

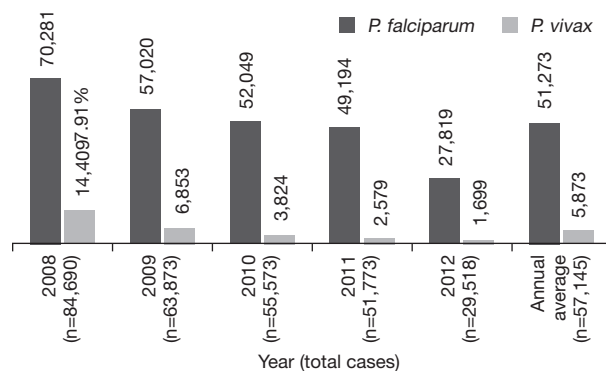


Figure 9.1. No. of *P. falciparum* and *P. vivax* malaria cases reported in Bangladesh by year

Figure 9.2 shows the distribution of reported malaria cases (number and percentage) in 2012 by endemic district. It is revealed that, out of the endemic districts, malaria cases are particularly concentrated in four districts (89% of total cases), namely Bandarban (29%), Rangamati (27%), Khagrachhari (20%), and Cox's Bazar (13%). Detailed information is provided in the annex.

The Department of Control of Communicable

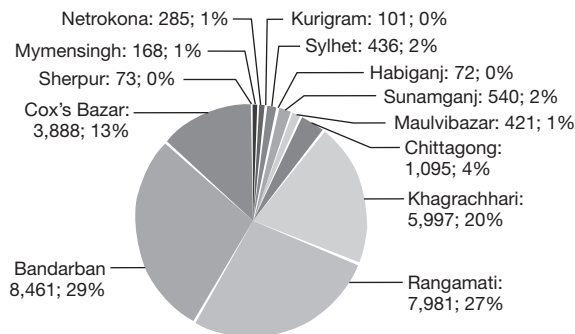


Figure 9.2. Distribution of malaria cases by endemic district (total cases 29,518) (2012)

Disease of DGHS informed that, in fiscal 2011-2012, 3.224 million LLINs, 2.785 million ITNs, 1.57 million anti-malaria tablets (K-O Tab 123), 0.662 million rapid diagnostic test-kits, 0.304 million Coartem (Artemether-Lumefantrin), and 0.20 million PAN-RDTs were distributed.

Dengue

In Bangladesh, dengue re-emerged in 2000 after an earlier outbreak as Dhaka Fever in the 1960s. The re-emergence resulted in huge number of cases, along with morbidity and mortality of public-health concern. Bangladesh demonstrated appreciable competence in managing the cases, with continued decline in mortality. Training of healthcare providers on dengue management and increased awareness might have contributed to the achievement. The prevalence shows rise in cases twice a year—once in May-June and the other in September-October. In other months of the year, low prevalence exists. Dengue is concentrated in urban settings due to unplanned urbanization with difficult-to-maintain sanitation, offering abundance of breeding place for *Aedes* mosquito. Table 9.1 shows the distribution of dengue cases, deaths, and case-fatality rates in Bangladesh by year beginning from 2000.

Filariasis

Filariasis is a global problem and a major social and economic burden in the tropics and subtropics of Asia, Africa, Western Pacific, and parts of Americas, affecting over 120 million people in 83 countries. It is estimated that about 700 million people are living in areas endemic for lymphatic filariasis in South-East Asia Region of the WHO. About 60 million people are infected in the region, and about 31 million people have clinical manifestation of the disease. Bangladesh is surrounded by endemic areas for filariasis, particularly by the northeast areas bordering

Table 9.1. Distribution of dengue cases, deaths, and case-fatality rates by year

Year	No. of cases	Death	Case-fatality rate (%)
2000	5551	93	1.7
2001	2430	44	1.8
2002	6132	58	0.9
2003	486	10	2.1
2004	3934	13	0.3
2005	1048	4	0.4
2006	2200	11	0.5
2007	466	0	0.0
2008	1153	0	0.0
2009	474	0	0.0
2010	409	0	0.0
2011	1362	6	0.4
2012	671	1	0.1

India, adjacent to the states of Assam, Bihar, and West Bengal. Filariasis is seen all over Bangladesh, with the highest endemicity in northern part of the country. The exact prevalence of filariasis in Bangladesh is not known; studies (using ICT card tests by lot quality assurance surveys- LAQAS) done in 2002 and 2004 reveal that it is endemic in 34 out of 64 districts (list shown in the annex). Ten districts, viz. Nilphamari, Thakurgaon, Dinajpur, Rangpur, Panchagarh, Kurigram, Gaibandha, Chapainowabganj, Rajshahi, and Lalmonirhat show particularly high endemicity. It is estimated that about 70 million are at risk of infection while 10 million people are with various forms of clinical deformity, and another 10 million

people are microfilaremics. To control filariasis, mass drug administration (MDA) was introduced in 2001 in Panchagarh district and gradually scaled to 18 more districts. Five districts (Rajshahi, Dinajpur, Meherpur, Patuakhali, and Barguna), after several rounds of MDA, have shown in 2011 a microfilaria prevalence coming down to less than 1%. Given this result, MDA has been stopped officially from these 5 districts. In 2012, five other districts (Sirajganj, Pabna, Kushtia, Chuadanga, and Pirojpur) fulfilled the criteria of stopping MDA. Remaining nine districts are receiving MDA, and the level of microfilaria rate has been declining to achieve elimination within 2015. Detailed information on year-wise MDA rounds is provided in the annex. After several rounds of MDA, microfilaria prevalence has been declining significantly in the endemic districts from 16% to less than 3%. The number of new clinical cases, including hydrocele and elephantiasis, has also been declining steadily in the area.

Figure 9.3 shows the effects of MDA in the endemic districts.

Soil-transmitted helminthes (STH) control and nationwide deworming program

Both filariasis and intestinal worm infestation are caused by soil-transmitted helminthes (STH). So, albendazole is distributed as additional filariasis elimination strategy. The same drug also serves the purpose of intestinal deworming. Survey in 2005 showed the prevalence of intestinal worm infestation among school children to be 79.8%. As a response, nationwide deworming program was launched in 2008 through primary schools of all kinds, viz. government, non-government, registered, formal, non-formal, kindergarten, madrasa, moktab, etc. The deworming tablets are distributed among all 5 to 12 years old

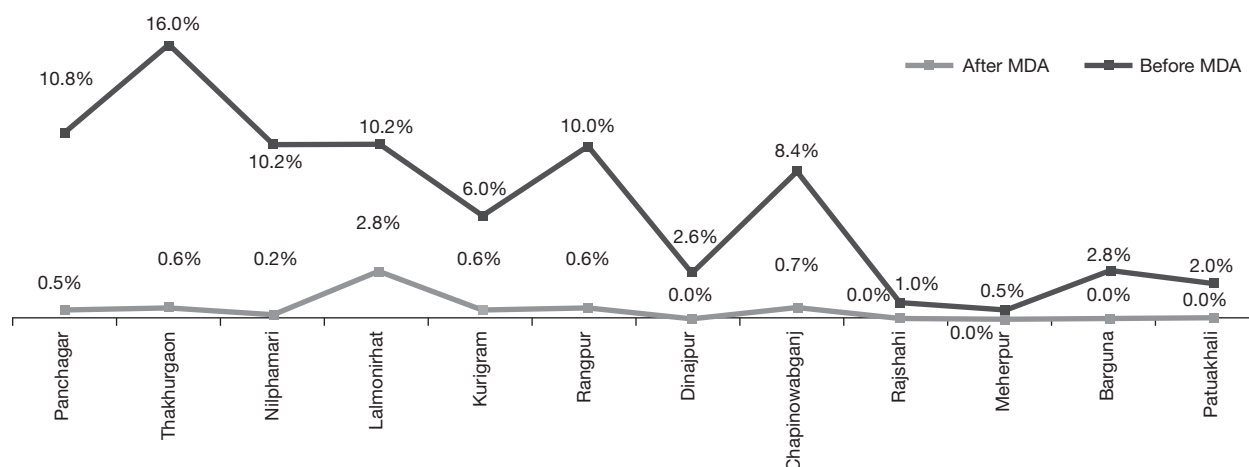


Figure 9.3. Impact of mass drug administration on filariasis prevalence in the endemic districts (2008-2009)

children twice a year by school teachers in each national round of deworming. So far, nine rounds of deworming have been completed with over 98% reported coverage. From May 2011, a new 'Little Doctors' initiative was launched to engage the senior pupils of primary level schools for peer education for STH control and developing healthy personal hygiene practice. There are about 82,000 primary-level schools in Bangladesh, with about 20 million pupils. Eighteen senior pupils (viz. from Class IV and V) of each school are selected as 'Little Doctors', divided in six groups—one group for one class and are engaged in self-motivated practice of healthy habits and educating other pupils to practise the same, including intake of albendazole. The 'Little Doctors' initiative aims at creating 1.23 million little doctors each year with a view to making the program sustainable. It is assumed that adoption by even a small percentage of the 20 million children or the little doctors for healthy lifestyle throughout their

lives will have huge positive impact on future health of the nation. There is a guideline for selection, orientation, and functioning of little doctors. A summary of year-wise deworming coverage with albendazole among school children in 2012 is provided in the annex.

Kala-azar

Kala-azar is prevalent in about 100 countries, threatening 350 million people living in these countries. However, only five countries, viz. India, Bangladesh, Nepal, Sudan, and Brazil, bear 90% of the global kala-azar burden. It is estimated that collectively around 147 million people of Bangladesh, India, and Nepal are at kala-azar risk, with an annual incidence of about 100,000 cases. In Bangladesh, kala-azar is endemic in 100 upazilas; the country is making progress in kala-azar elimination and has set target of kala-azar elimination by 2015, with a goal to reduce the annual incidence to less than 1 case per 10,000 population through implementing objectives of early diagnosis and complete management of the cases, integrated vector management, patient and vector surveillance, and operational research. Kala-azar patients are detected and treated in Bangladesh at upazila health complexes. ICT cards, based on 'rK39', are used for diagnosis and oral miltefosine for treatment. Injection Sodium Stibogluconate (SSG), which was earlier used, is being phased out. Very recently, single-dose AmBisome has been piloted in Fulbaria and Trishal upazila, with a plan to scale up soon in all the remaining endemic upazilas. In 2011, vector management with indoor residual spray (IRS) was piloted in five unions of Fulbaria upazila. Village-wise Blanket IRS was scaled up in all villages of 8 hyperendemic upazilas in 2012. One round of pre-monsoon IRS has also been completed in 26 low-endemic upazilas. There is a plan to scale up both focal and blanket IRS in all the remaining moderate and low-endemic upazilas. Kala-azar cases are detected through both active and passive surveillances. In addition, pre- and post-IRS vector bionomics and susceptibility tests are done on regular basis. These activities are further backed by operational research, like that on pharmacovigilance, vector bio-assay test, etc. The kala-azar control program has succeeded to reduce the number of both post-kala-azar dermal leishmaniasis (PKDL) and kala-azar cases. In 2006, the number of reported kala-azar cases was 10,000, which has been dropped to 10,000 in 2012. Figure 9.4 shows the year-wise kala-azar cases and related deaths in Bangladesh.

Figure 9.5 shows the number of kala-azar cases reported in 2012 by district.



'Little Doctors' in one of the schools under the initiative

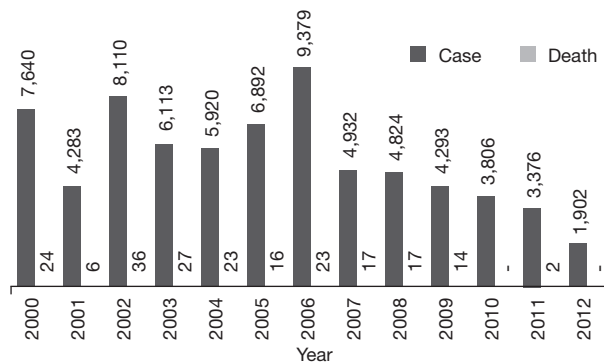


Figure 9.4. Year-wise number of kala-azar cases and death

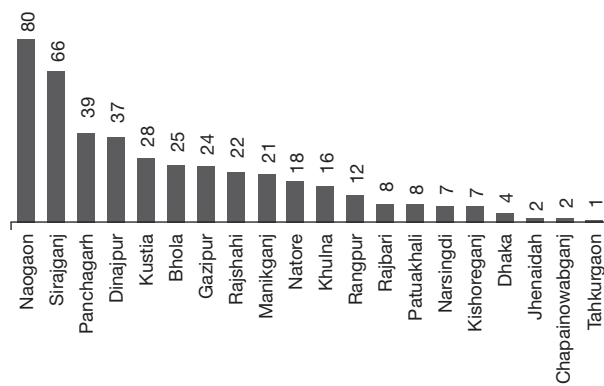


Figure 9.5. Number of kala-azar cases by district (2002)

Emerging and re-emerging disease control program

The emerging and re-emerging disease control program of DGHS is currently prioritizing the control of rabies, anthrax, Nipah, chikunguniya, and antimicrobial resistance. A goal has been set to eliminate rabies from the country by 2020. Collaborative action plan is also underway to combat other zoonotic diseases.

Rabies

Rabies was a neglected tropical zoonotic disease. It claims more than 2,000 lives annually in the country. This is the highest for any single infectious disease. Rabies, if manifested, is nearly 100% fatal but 100% preventable as well if appropriate preventive measures are taken. Annual number of dog-bites in Bangladesh varies between 200,000 and 300,000, and 95% of rabies occur due to dog-bites. Only suspected rabid dogs are thought to be responsible for rabies as dogs rarely can remain healthy. Other animals that occasionally transmit rabies in Bangladesh are cat, fox, monkey, jackal, and mongoose. Other than humans, an

estimated 25,000 or more cattle succumb to rabies every year. In the national rabies elimination goal by 2020 a number of activities are being conducted, such as, national rabies survey, setting up of national and district rabies prevention and control centers, mass dog vaccination, and dog population management. To estimate the total dog population and annual events of dog-bites and rabies, a national rabies survey was conducted in 600 clusters all over the country, each cluster comprising 300 households and altogether covering a total of 180,000 households. A national rabies prevention and control center has been established at Infectious Diseases Hospital, Mohakhali, Dhaka, where about 350 to 450 dog-bite victims receive the service daily. Anti-rabies vaccines and rabies immunoglobulin are distributed free of charge from this center. In addition to the national center, 65 rabies prevention and control centers have also been established at the district level where also dog-bite victims are receiving modern management. These district centers also distribute anti-rabies vaccines and rabies immunoglobulin free of charge. Mass dog vaccination, launched in 2011 in Cox's Bazar Sadar municipality as a pilot, is a unique idea to make the dog population immune from rabies so that any dog-bite does not transmit rabies to humans. If 70% of the existing dog population can be vaccinated, there will be herd immunity among the dog population, keeping them protected from rabies as well. Large-scale mass dog vaccination activities have been scaled to 37 districts of four divisions (Dhaka, Rajshahi, Sylhet, and Rangpur). The list of these districts is given in the annex. Intra-dermal tissue culture-based rabies vaccines are now being used replacing the locally-produced nerve tissue vaccines for dog-bites. For all these measures, the case-fatality rates are declining. The annex provides data on the year-wise rabies-related deaths at the Infectious Disease Hospital, Mohakhali, Dhaka.

Anthrax

Anthrax is caused by *Bacillus anthracis*, a bacterium that can form spores. Spores allow it to survive in the soil for long periods. Anthrax is primarily a disease of herbivorous mammals, such as cattle, sheep, goats, and buffaloes, which may have chance to ingest anthrax spores while grazing. Humans generally acquire the disease directly or indirectly from infected animals or from occupational exposure to infected or contaminated animal products. Control in livestock is, therefore, the key to reducing incidence. There is no documented case of person-to-person transmission. The impact of the disease on animal and human

health can be devastating. The disease exists in animals and humans in most countries of Africa and Asia, in several southern European countries, in the Americas, and certain areas of Australia. Anthrax outbreaks in animals also occur sporadically in other countries. Prevalence of anthrax in Bangladesh was not well-documented earlier. However, since August 2009, the Institute of Epidemiology, Disease Control and Research (IEDCR) investigated 14 outbreaks of cutaneous anthrax in three districts of Bangladesh (Pabna, Sirajganj, and Tangail). Recently, more and more outbreaks are being reported. In 2012, a total of 176 cases of anthrax were reported from 5 districts—74 from Sirajganj, 67 from Meherpur, 16 from Bogra, 14 from Tangail, and 5 from Kushtia.

Nipah virus infection

Nipah virus infection in humans is an emerging zoonotic disease in Bangladesh. First recognized in a large outbreak with 276 reported cases in Malaysia and Singapore (between September 1998 and May 1999), Bangladesh identified the first cases in 2001. Encephalitis and respiratory distress are common presenting symptoms and signs of Nipah infections. In India, two outbreaks in humans during 2001 and 2007 were reported from West Bengal. Bangladesh encountered 11 Nipah outbreaks between 2001 and 2012 in 20 districts, all occurring between December and May. The districts where the outbreaks occurred are Meherpur (2001), Naogaon (2003), Rajbari (2004), Faridpur (2004), Tangail (2005), Thakurgaon (2007), Kushtia (2007), Manikganj, Rajbari (2008), Faridpur (2010), Lalmonirhat (2011), and Joypurhat (2012). In all these outbreaks, a total of 215 human cases of Nipah infections in Bangladesh were recognized, with 164 deaths, indicating a very high case-fatality rate (76.3%). Among the Bangladeshi patients, frequency of respiratory problems, including pneumonia, was more than among patients in Malaysia. This may be attributed to genetic diversity of the virus strains. Human-to-human transmission, rare earlier, was also noticed, probably due to more involvement of respiratory tract infections.

Chikungunya

Chikungunya fever, a dengue-like disease, is emerging alarmingly in the country in recent years. It is caused by an insectborne virus. In 2011 (August to October), suspected Chikungunya fever outbreaks were detected in Dohar upazila of Dhaka district and Shibganj upazila of Chapainowabganj district. Recently, two other outbreaks in Rajshahi and Pabna districts were identified. In no outbreak,

case-fatality rate was reported other than persistent arthralgia causing patients' sufferings. Diagnosis of Chikungunya is important to distinguish from dengue. Chikungunya is caused by mosquitoes; so, reducing mosquito breeding is an important public-health measure to control the disease.

Diarrhea

In 2012, report on 2,630,470 diarrhea cases and 43 deaths (0.002% death rate) was received. Of the total cases, 144,244 (5.5%) were from Dhaka Hospital of icddr at Mohakhali. Figure 9.6 shows the distribution of reported diarrhea cases in 2012 by division. The Dhaka division had the highest reported cases (n=1,118,399, 43%), followed by Khulna (n=465,336, 18%) and Chittagong (n=449,481, 17%). Of the 24 reported diarrheal deaths, 24 were from Chittagong, 8 from Khulna, 5 from Dhaka, and 3 from each of Rangpur and Sylhet division.

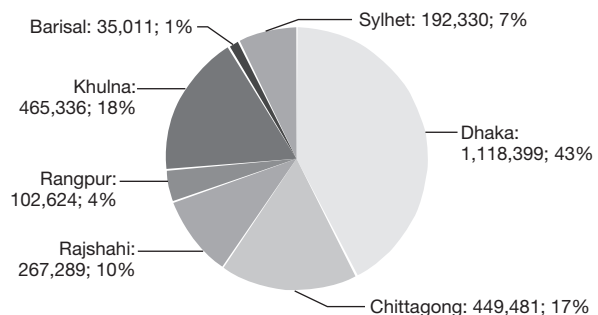


Figure 9.6. Distribution of reported diarrhea cases by division in 2012 (total cases 2,630,470)

Figure 9.7 shows the number of diarrhea cases and deaths in 2012 by month.

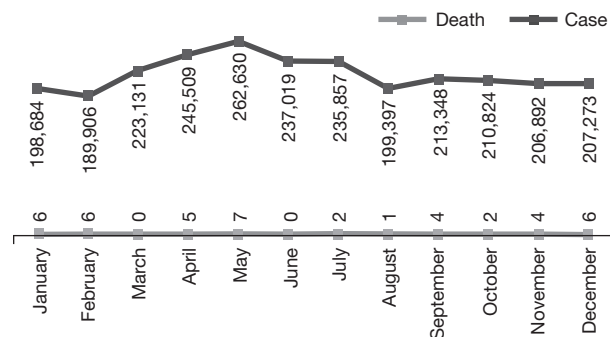


Figure 9.7. Number of diarrhea cases and deaths in 2012 by month (total cases 2,630,470; total deaths 43)

Avian and pandemic influenza

The population-size and density in Bangladesh encouraged proliferation of poultry industry, producing 220 million chickens and 37 million ducks annually. Poultry industry in the country is one of the largest in the world. Approximately 50% of the local-breed hens and ducks are raised in backyard farms, and another 50% comprise broilers and layers reared in commercial farms. The risk of avian influenza and other zoonotic diseases increases due to (i) high density of humans and animals; (ii) high number of backyard farms and live poultry markets; (iii) mixed farming practice with low biosecurity; (iv) limited control over poultry movements; (v) inadequate regulation of slaughtering and processing of animal products; and (vi) suboptimal veterinary public health infrastructure, surveillance systems, and laboratory facilities. External risk factors include: (i) long porous border (with significant cross-border movements of people and a lack of animal quarantine stations); (ii) importation of avian species, particularly breeder chickens; and (iii) regular movement of companion animals from overseas by expatriates. To contain the disease in poultry, awareness development, capacity-building in management, multi-sectoral coordination and cooperation, hygienic measures at commercial farms, backyard, and wet markets, surveillance and regulation of culling activities in livestock sector. So far, seven cases of human influenza infection (H5N1) have been identified in Bangladesh—the first in 2008, second and third in 2011, fourth, fifth, and sixth in 2012, and the seventh in 2013. Out of the seven cases, only one (the seventh) experienced death. Bangladesh developed modern diagnostic facility for pandemic influenza, with real-time polymerized chain reaction (RT-PCR) and biosafety level 3 (BSL 3) laboratory. The

country reports to WHO on any event of avian influenza, following the guidelines of International Health Regulation 2005 (IHR 2005). An Avian and Pandemic Influenza Ward at Asthma Center of National Institute of Diseases of Chest and Hospital (NIDCH), Dhaka, with support from WHO and isolation units in 64 district hospitals have been established. There is a national guideline to operate the isolation units both when human pandemic avian influenza or other pandemic influenza case is detected. When there will be no such cases, the units are specified to use for other infectious diseases, like Nipah virus infection, meningitis, diphtheria, encephalitis, etc. The Infectious Disease Hospital (IDH) at Mohakhali, Dhaka, has been also made ready for the management of pandemic influenza patients if needed. In such cases, the provision has been made to open separate outdoor and indoor facilities in medical college hospitals and also in district and upazila hospitals. In all hospitals having more than 200-bed capacity, there is an instruction to create pandemic influenza management committee.

Tuberculosis

According to the revised estimates by WHO, the prevalence and death rates of all forms of tuberculosis (TB) in 2011 were 411 and 45 per 100,000 people respectively. As of 2012, the detection rate of new smear-positive TB cases and cure rate with DOTS (directly observed treatment short course) was estimated at 70.0% and 92.0% respectively. The estimate for incidence rate of HIV-positive TB cases was 0.40 per 100,000 people. The proportion of multidrug-resistant tuberculosis (MDR-TB) among the new TB cases and retreatment cases was 1.4% and 29.0% respectively. Figure 9.8 shows the percentage of smear-positive (total, new, and relapse) and new extrapulmonary cases out

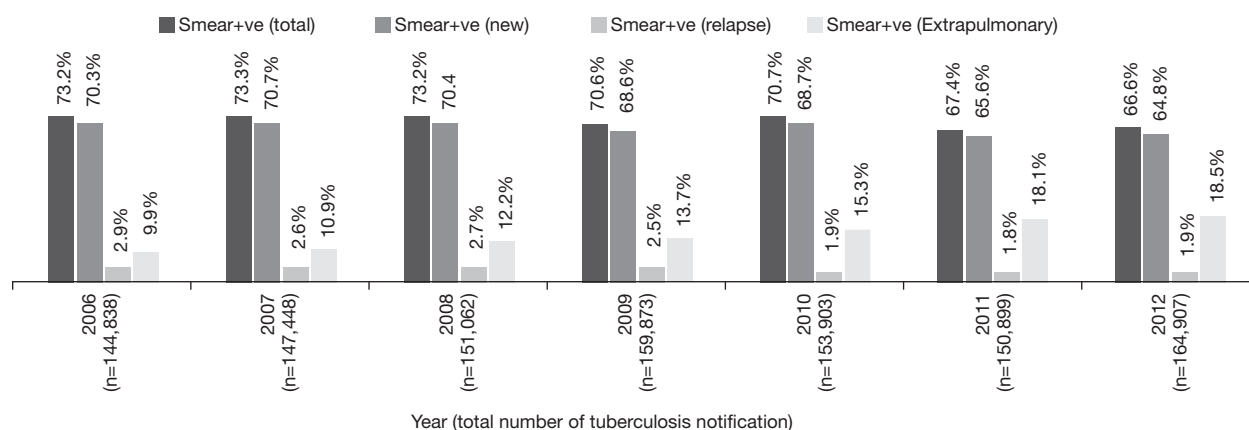


Figure 9.8. Percentage of smear+ve (total, new, and relapse) and new extrapulmonary tuberculosis cases detected out of the tuberculosis notification (2006-2012)

of the tuberculosis notification in different years (2006-2012). Detailed data are provided in the annex.

The emergence of resistance in antiTB drugs, particularly the multidrug-resistant tuberculosis (MDR-TB), has become a significant public-health threat globally against effective TB control. According to the WHO estimates, there were an estimated 630,000 cases of MDR-TB in 2011 among 12 million cases of TB prevalent in the world. Globally-estimated number of incident MDR-TB cases was 600,000 in 2011, and the proportions of the new and retreatment cases that have MDR-TB are 3.7% and 29% respectively. In Bangladesh, a national TB reference laboratory (NTRL) has been established in the National Institute of Diseases of Chest and Hospital (NIDCH), Dhaka, for diagnosis and management of MDR-TB. The NTRL started functioning since 27 June 2007 for culture and drug-sensitivity test (DST). It is linked with supranational reference laboratory (SRL) in Antwerp, Belgium. By the end of December 2012, a total of 971 MDR-TB patients have been enrolled. Treatment success rate in MDR-TB patients (2008 and 2009 cohort) enrolled in NIDCH was 64%. The MDR-TB patients are also managed in the chest disease hospital (CDH) of Rajshahi division, supported by Damien Foundation, Bangladesh but with a shorter regimen of 9 months. Since May 2005, this center has been managing MDR-TB patients, and by the end of December 2012, a total of 980 MDR-TB patients have been enrolled. In this center, treatment success rate of MDR-TB patients under 9 months' regimen is over 80%. Two more regional TB reference laboratories have been established in the chest disease hospitals of Rajshahi and Chittagong respectively. From October 2010 to December 2012, a total of 127 MDR-TB patients have been enrolled in these two facilities.

A drug resistance survey has been completed by the national TB control program. The national tuberculosis program follows a public-private mix strategy to link the resources of public and private healthcare providers to achieve the targets of national TB control program. The private care providers for TB have important and strategic roles in reaching groups of the population, particularly those who bypass the public healthcare delivery system.

Leprosy

In 1991, the prevalence of leprosy in Bangladesh was 13.6 cases per 10,000 people. The country adopted national leprosy elimination goal as per WHO criteria (<1 case per 10,000 people within 2000). Bangladesh has achieved this goal by

December 1998 (0.87 case per 10,000 people), 2 years ahead of WHO deadline. In that time, leprosy was limited only in 15 districts. Through further consolidation of the national elimination program, the prevalence came down to 0.57 cases per 10,000 people by the end of December 2004, with endemic districts becoming 10 in number. However, in the past decade, decline rate was slow, and both in 2011 and 2012 (by the end of December), the registered prevalence was recorded at 0.22 cases per 10,000 people (Figure 9.9).

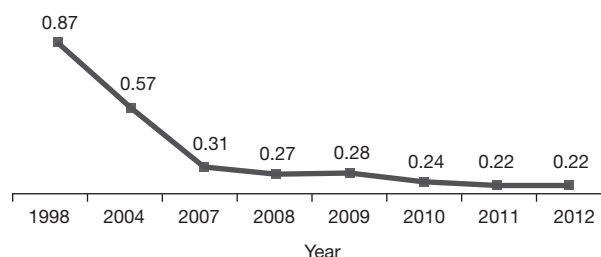


Figure 9.9. National leprosy prevalence (number of cases per 10,000 population (1998-2012))

However, in December 2012, the endemic district was only one (Gaibandha), with a registered prevalence of 1.39 cases per 10,000 people. Another important indicator for evaluating leprosy program is the disability Grade 2 rate among newly-detected cases per year, which was 21.4% in 1993, the year of launching the revised national leprosy elimination program (NLEP). The Grade 2 deformity rate is still 11.5% (December 2012) against the target of 2% to less than 5%. A distribution of the newly-detected leprosy cases (2009-2012) and patients receiving multidrug treatment (2011-2012) by division of Bangladesh is provided in the annex.

HIV/AIDS

Bangladesh is still considered a low-prevalence country for HIV/AIDS but the country remains extremely vulnerable to HIV epidemic because of the high prevalence in neighboring countries and the high mobility of people inside the country and beyond. Inadequacy in correct knowledge about HIV and AIDS due to illiteracy and ignorance and gender inequity aggravate this vulnerability. High rate of needle-sharing among injecting drug-users (IDUs) and low rate of condom-use and high prevalence of STIs among the most-at-risk population are the most important factors that may contribute to a potential HIV epidemic. The Government of Bangladesh has been undertaking precautionary measures to limit the spread of HIV infection since the detection of the first HIV-positive case in 1989. National AIDS Committee (NAC) was formed in 1985 and reconstituted in

2010. The MOHFW is playing the leading role in the prevention of HIV and AIDS. The National AIDS/STD Program (NASP) is implementing HIV and AIDS prevention activities in Bangladesh through a coalition of three functionaries, namely National AIDS Committee (NAC), MOHFW, and DGHS. The NASP is one of the wings of DGHS responsible for coordinating activities of all stakeholders and development partners involved in the HIV and AIDS program. As of December 2012, a total of 2,871 HIV+ve cases have been identified. Among them, 1,204 developed AIDS; among the latter group, 390 had died. Although HIV prevalence among the general population as well as among the most-at-risk population remains at a very low level (<0.01% and <1.0% respectively), the UNAIDS and WHO (2012) estimate that there may be about 8,000 persons living with HIV as of 2012. Figure 9.10 shows the year-wise cumulative number of HIV+ve cases, AIDS patients, and AIDS-related deaths in Bangladesh.

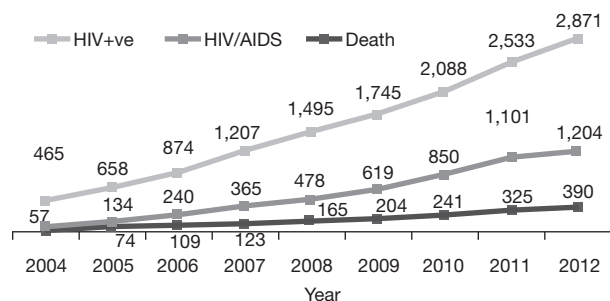


Figure 9.10. Cumulative number of HIV+ve cases, AIDS patients, and AIDS deaths by year in Bangladesh (2004-2012)

Figure 9.11 shows the number of new AIDS cases from 1990 to 2012, revealing drastic reduction in the number of new AIDS cases in 2012.

The National AIDS/STD Program introduced a surveillance system since 1998 using facility-based data from HIV/AIDS and STD service providers.

The most-at-risk populations, including female and male sex workers (FSW and MSW), men who have sex with men (MSM), transgendered people (Hijra), and people who inject drugs (PWID) are covered in the surveillance. The 9th round of serological surveillance (2011) was conducted in populations most-at-risk of HIV, i.e. sex workers, PWID, heroin-smokers, combined PWID and heroin-smokers, MSM, and Hijra. This round was conducted during December 2010 to June 2011; 12,894 individuals were sampled from 36 geographical areas of Bangladesh. The overall prevalence of HIV and active syphilis was found to be 0.7% and 3.0% respectively. The population group with the highest rate of HIV continues to be PWID in Dhaka but the prevalence declined to 5.3% from 7% (in the 8th round). However, the decline is not statistically significant. Fortunately, the localization of the PWID epidemic to one neighborhood of Dhaka observed in previous years has also remained static. HIV was also detected in another four groups of people who use drugs (PWUD), e.g. male PWID from Narayanganj (1.5%) and Satkhira (0.4); combined female PWID and heroin-smokers from Dhaka, Narayanganj, Tongi (1.2%), and Benapole (1%). Active syphilis rate at >5% was detected among six groups of PWUD, and the highest proportion was found in male PWID in Narsingdi (7.9%), followed by PWID in Chandpur (6.1%) and female PWUD in Dhaka, Tongi, and Narayanganj (5.9%). High prevalence of active syphilis indicates practice of unsafe sex. Antibodies to hepatitis C virus (HCV) were measured in all PWID and groups of combined PWID and heroin-smokers but not in the groups consisting of only heroin-smokers. The rates varied in different cities, and in six cities, HCV-positive cases were found to be >50% (Rangpur was included in Rajshahi). The higher prevalence of HCV was found among PWID from several towns of Rajshahi division, with Kanshat having the highest prevalence (95.7%). In Dhaka, HCV rates have declined significantly ($p < 0.05$) over the rounds of the surveillance. A

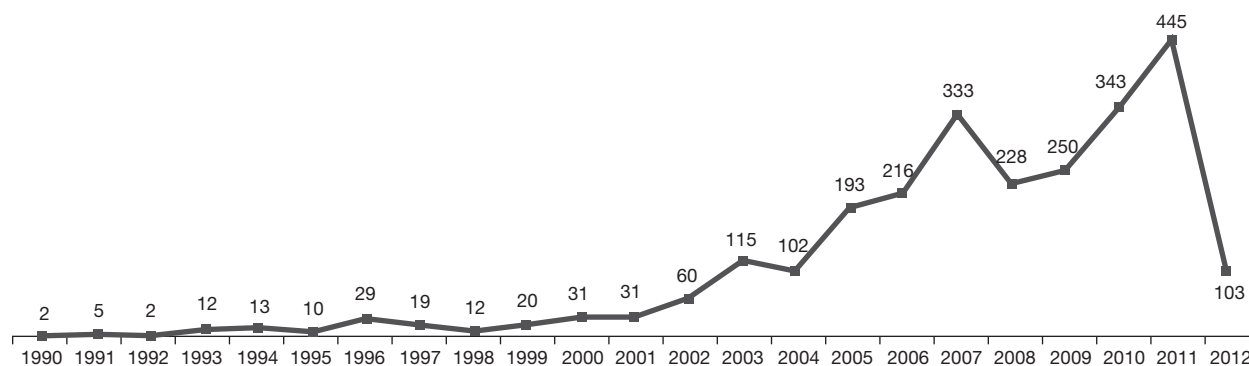


Figure 9.11. Number of new AIDS cases from 1990 to 2012 in Bangladesh

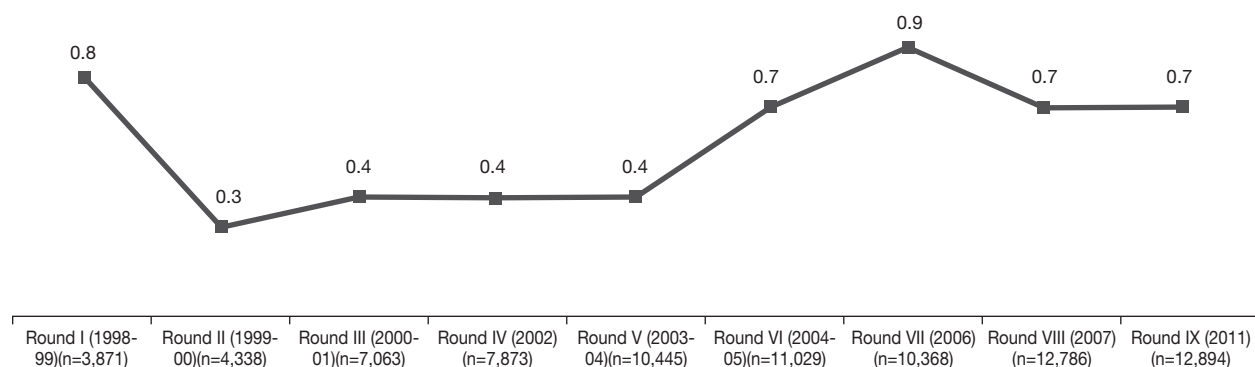


Figure 9.12. Prevalence (%) of HIV infections among most-at-risk population in different rounds of HIV sero-surveillance of Bangladesh

total of 3,568 female sex workers were sampled from 13 areas of Bangladesh. Overall, HIV prevalence was low (<1%) in all groups of female sex workers, except in casual sex workers from Hili where two in 125 samples were positive (1.6%). Active syphilis rates at >5% was detected in three groups and sites—street-based sex workers of Hili (12.5%) and Chittagong (10.3%) and hotel-based sex workers of Sylhet (9.3%). Male sex workers (MSW) and MSM continue to have low levels of HIV and active syphilis. Approximately 1% of Hijra had HIV. Among the group of MSM, MSW, and Hijra, active syphilis rates were the highest in Hijra of Dhaka and Manikganj (6.1%). High rates of active syphilis highlight the need for intensification of ongoing HIV/STI prevention programs in these population groups and sites. In this round of surveillance, more population groups were sampled from border towns, and Benapole was included for the first time. Male PWID, street-based female sex workers, combined MSM/MSW and Hijra of Hili, combined group of male and female PWID and heroin-smokers of Benapole, casual female sex workers of Burimari, combined group of hotel- and residence-based female sex workers of Teknaf were sampled. However, for all these groups, the sample-size was small. HIV was

detected in two groups in Hili and Benapole each but the numbers were small. Cross-border mobility was more commonly reported by respondents in Hili. Figure 9.12 shows the trend in HIV prevalence among the most-at-risk population.

The National AIDS/STD Program carries out annual case reporting. The 2012 report of the Program has made an occupational distribution of the new cases of HIV infections. The summary is provided in the annex. It is seen that, among the new cases of HIV infections, 63.0% are male, 34.9% are female, and 2.1% are transgendered population. Good progress has been made to increase the coverage of antiretroviral drugs (ARVs) for people living with HIV. Supported by the Global Fund and USAID, the current coverage is 100% for brothel-based sex workers, >60% for street-based sex workers, >45% for hotel- and residence-based sex workers, >25% for MSM, >80% for Hijra, and >75% for people who inject drugs.

It may be concluded that the communicable disease control programs, particularly that by the relevant department of DGHS, have provided strong leadership and also introduced innovative and diverse approaches to containing the burden of communicable diseases in the country.

PREPAREDNESS AND RESPONSE FOR HEALTH EMERGENCIES

Experience guides preparedness for better response

Mass health emergencies, principally due to natural calamities, are common in Bangladesh. Due to geographical location and topographical feature, natural calamities, like cyclone, floods, and tornado, etc. occur in Bangladesh almost as an annual phenomenon. The country is also vulnerable to earthquake. Bangladesh is the most densely-populated country in the world. This creates vulnerability of more victims to road, rail and river traffic accidents. The overall disaster situation of the country causes high burden of disaster-related diseases, disabilities, and deaths. Skilled manpower, uninterrupted supply of logistics, and availability of guidelines for the best public health practices at adequate level are required to reduce the adverse health impact of these disasters to a great extent.

The health emergency preparedness and response program is active in DGHS, which focuses on adequate disaster preparedness and quick response. Two programs work in collaboration. One is the National Health Crisis Management Center and Control Room (NHCMC&CR) of DGHS supported under the HPNSDP 2011-2016, and the other one is the WHO-supported Emergency Preparedness and Response (EPR).

The NHCMC&CR operates round-the-clock all seven days a week to receive reports of any health emergency relating to disaster or accident. Some of the recent experiences will clarify how it works. A devastating tornado ripped through more than a dozen of rural villages in the Brahmanbaria district of Bangladesh, affecting seven unions of three upazilas (subdistricts) in the evening of Friday, 22 March 2013. The tornado claimed 37 lives and injured 971 persons. The NHCMC&CR immediately started coordination and mobilization of local medical teams to start rescue operations and medical relief program. A team of NHCMC&CR rushed to the area and took notes on deaths and injured persons and visited admitted patients in Brahmanbaria Sadar Hospital and in Akhaura and Cosba Upazila Health Complexes. Local health authorities and medical teams were given guidance on taking adequate care of patients. Subsequently, NHCMC&CR maintained contacts

Skilled manpower, uninterrupted supply of logistics, and availability of guidelines for the best public health practices at adequate level are required to reduce the adverse health impact of the disasters to a great extent ...

with the district health manager's office regularly and communicated the information to the higher authority and policy-makers. The NHCMC&CR coordinated the activities relating to management of buffer stocks of emergency drugs and logistics and ensured replacement at the quickest possible time.

On 24 April 2013, one of the greatest tragedies in the history of Bangladesh occurred. A nine-storied building called 'Rana Plaza' located in peri-urban Dhaka (Savar area) collapsed. The building housed five garment factories, a shopping complex, and branch of a bank. Many garment workers died immediately, others were injured, and many more were trapped inside the collapsed building. The number of dead bodies found was 1,132, and 2,755 injured persons were registered. The NHCMC&CR received information immediately from the local authority and categorized it as a high-priority national health emergency. So, the Director General of Health Services, the Minister for Health, the Secretary to the MOHFW, and all other allied directors and departments, including local hospitals and clinics, were immediately informed. A rapid response team rushed to the site of incidence and started coordinating and organizing medical rescue operations, resuscitation, first-aid, and transfer of injured persons to appropriate hospitals. Ten ambulances were engaged for transportation of injured persons. Communication and coordination with armed forces, fire brigade,

disaster management bureau, WHO, UN agencies, development partners, and relevant NGOs were established. Government provided all the medicines and logistics support to all hospitals, public or private, for treatment of the admitted victims. Many victims suffered serious limb, bone, kidney and other internal injuries. Some of them are still in the hospitals. The NHCM&CR maintained tracking activities several months after the incidence for follow-up of these patients, coordinating their treatment, and communicating the reports to the higher authority and policy-makers.

in the morning of 16 May 2013, a warning came from the meteorological department that a cyclone called Mohasen was about to hit the coastal belt of Bangladesh. This was also a high-priority national emergency. Considering the possible health implications, the NHCMC&CR communicated the information to the Director General of Health Services, Honorable Health Minister, and Health Secretary as well as to the local health facilities in the coastal area. Medical teams, buffer stocks of medicines and logistics, and community awareness measures to minimize morbidity and mortality were mobilized. However, cyclone Mohasen was weakened during its long travel and hit Patuakhali, Pirojpur, and Barguna coastal districts of Barisal division. There were 16 deaths and 81 injuries. Although it was not required to make fully functional, the NHCMC&CR activated 1,327 government medical teams to provide post-cyclone medical operations.

The EPR program of DGHS is supported by the WHO. The regular activities include capacity-building of the health managers and raising awareness of community people. The program activities operate round-the-clock all seven days a week during emergencies. The program's primary goal is to reduce avoidable and preventable morbidities, disabilities, and deaths during emergencies through strengthening overall capacity of the health sector to prevent and mitigate the adverse health consequences of emergencies and disasters. During the normal period, it develops plans, policies, guidelines, IEC materials (viz., training modules, leaflets, posters, etc.), collects disaster-related information, and conducts other coordination function with the NHCMC&CR and other government and NGO stakeholders.

The EPR has a number of institutional capacity-building activities, such as formation and training of "Disaster Health Management Committees" at all levels; conducting training of trainers (TOT)/workshop/mock drills/simulation exercises on

search, rescue, evacuation, first-aid, psychosocial support, risk communication, and mass casualty management for health professionals and workers; provision of emergency supplies (first-aid kits, rain-coats, umbrellas, solar lamps, safety rubber boots, jackets, caps, whistles, etc.) for the first-level health responders; and provision of emergency drugs (maintaining buffer stock) and medical equipment/supplies. During the crises, strategic priority functions are undertaken. These are: (i) Assessment and monitoring; (ii) Critical gap-filling; (iii) Coordination through cluster approach; and (iv) Capacity-building during emergency.

For assessment and monitoring, a team is formed and sent immediately to the affected areas as and when an emergency situation arises after cyclones, floods, earthquakes, etc. The team measures the health status of the victims and promptly makes an assessment of their needs, identifying priority actions to address the health problems and avert deaths.

For critical gap-filling, the EPR program ensures that critical gaps in health responses are rapidly identified and filled based on the needs assessment report and available resources.

For coordination through cluster approach, the emergency situations are assessed by conducting cluster meetings participated by invited humanitarian actors for joint planning and joint response and actions. The participants with multi-disciplinary knowledge and experience share the observations and identify the under-served or over-served areas. All participants are made aware about "who does what and where" to ensure a holistic collaborative effort.

For capacity-building during emergency, DGHS, on an urgent basis, conducts some sessions of relevant training for the health managers on "how to manage and overcome ongoing devastating situation." Providing training to staff and the local people in the community is helpful in identifying a sustainable strategy to be adopted in future emergencies both for public health interventions and related areas, e.g. water quality surveillance, mental health counseling, and so on.

For facing the disaster efficiently and effectively, the concerned department of DGHS has formed the Disaster Health Management Committee at all levels of health facilities and health management tiers, trained 1,485 doctors, 44,550 paramedics and other health and family planning workers, and 5,940 volunteers at the union level. A well-developed buffer stock system is working for making logistics available.

The EPR is part of a strong coordinated response to emergencies, along with DGHS, armed forces, UN agencies, fire brigade, Red Cross, development partners, and NGOs ...

There are 5 divisional health management committees, with 50 trained staff, 44 district committees, 880 trained staff, 9 medical college-level committees, and 297 upazila health complex-level committees having 5,940 trained staff.

All these activities of the EPR program are supported by the Emergency and Humanitarian Action (EHA) program of the WHO and other health cluster members. In addition to the Government and other organizations, CDMP, and European Union Narre consortium also provide technical and logistic support to the program for strengthening disease surveillance and emergency supplies, like drugs for replenishing buffer stocks, medicines, laboratory reagents, and related goods for proper investigation and case management. The EPR is part of a strong coordinated response to emergencies, along with DGHS, armed forces, UN agencies, fire brigade, Red Cross, development

partners, and NGOs. Table 10 shows some recent capacity-building activities carried out by the EPR program from January 2012 to June 2013.

Some identified challenges include (i) replenishment of buffer stocks at regular intervals; (ii) full functioning of Emergency Medical Services (EMS); (iii) achievement of 12 SEARO-EHA benchmarks; and (iv) adequate coordination among concerned agencies.

The EPR program plans, in future, to: (i) prepare comprehensive national disaster management plan for the health sector; (ii) operationalize the national disaster management institute in the health sector; (iii) strengthen National Crisis Management Center and Control Room; and (iv) procure make-shift hospitals and river ambulances.

Bangladesh is the world's worst victim of climate change and the most densely-populated country. There are obvious resource constraints. However, the use of experience to guide emergency preparedness for health helped the health authority to ensure better response.

The use of experience to guide emergency preparedness for health helped the health authority to ensure better response ...

Table 10. Number of training program held with support from WHO from January 2012 to June 2013

Activity	No. of training sessions held	No. of participants
Workshop on comprehensive health-sector emergency preparedness and response for health and disaster management professionals	7	245
Training on public-health risks and interventions in emergencies for health professionals	6	210
Training on prevention and control of post-disaster communicable diseases for health professionals	4	140
Health cluster meeting	6	180
Training on risk communication for health professionals	2	74
TOT on emergency medical services for master trainers	3	110
Training on search, rescue, and evacuation for community-level health workers	3	105
Advanced training on psychosocial health for health professionals	2	68

NON-COMMUNICABLE DISEASES

Burden of non-communicable diseases is on the rise

Non-communicable diseases (NCDs) in Bangladesh occupy a major share of the disease burden and mortality. The middle-aged and the elderly populations are the most affected group. Factors responsible for non-communicable diseases include changing dietary habits and lifestyle, rapid urbanization, growth of commuting, tobacco-use, uncontrolled growth, and consumption of processed foods and beverages, indoor air pollution, road-traffic injuries, lack of awareness about healthful behavioral patterns, and psychological pressure. In the current Health, Population and Nutrition Sector Development Program (HPNSDP) 2011-2016, control of non-communicable diseases was given one of the topmost priorities. The NCD operational plan categorized NCDs into two major groups, viz. conventional and non-conventional NCDs. The conventional group includes major NCDs, like cardiovascular diseases (CVDs), peripheral vascular diseases (PVDs), cerebrovascular disease (stroke), cancer, diabetes, chronic obstructive pulmonary diseases (COPDs), arsenicosis, renal diseases, deafness, osteoporosis, congenital anomalies, oral health, and thalassemia. The non-conventional group includes issues, like road safety and traffic injury prevention; child injury (including drowning); sports injury; snake-bite; suicide and related injury; violence against women; acid burn; occupational health and safety; industrial and agricultural health hazards; strengthening Institute of Public Health (IPH); climate change; air pollution; water, sanitation and other environmental health issues; emergency preparedness and response; post-disaster health management and emergency medical services; mental health; autism; tobacco, alcohol and substance-abuse, etc.

The national NCD risk factor survey (2010) revealed some critical facts, such as (i) NCDs may account for 61% of the total disease burden; (ii) among the sampled adult population (15+ years), 97% had at least one risk factor, half of whom had two risk factors; (iii) the country has 40 million adult smokers and smokeless tobacco-users; (iv) 64.5 million people are not taking adequate fruits

and vegetables; (v) 17 million people are not doing adequate physical activity; (vi) 18% adults have hypertension; and (vii) 4% have self-reported documented diabetes.

Given the above NCD situation in the country, summary of data gathered from different specialized hospitals is presented below to understand the volume of patient-loads these hospitals are dealing with.

National Institute of Cardiovascular Diseases

Figure 11.1 shows the number of outdoor visits and admissions in the National Institute of Cardiovascular Diseases (NICDV) in the last five years (2008-2012). More detailed information from 2002 to 2012, with male, female and child disaggregation, including average daily outdoor visits and admissions, average length of stay, and bed-occupancy rate, is provided in the annex.

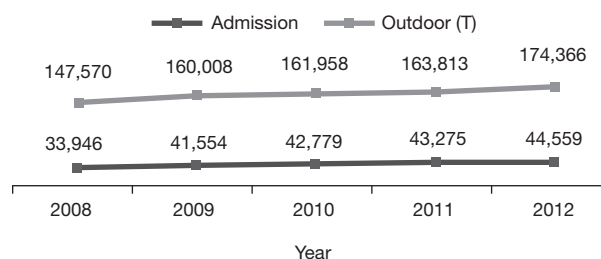


Figure 11.1 Number of outdoor visits and admissions in NICVD (2008-2012)

In 2012, a total of 2,442 exercise tolerance tests (ETTs) were done in the Institute; the recipients of services included 79.6% males (n=1,945) and 20.4% females (n=497). Detailed data on ETTs from 2001 to 2012 are provided in the annex. In 2012, 147 myocardial perfusion imaging was done. More data on myocardial perfusion imaging from 2003 to 2012 are provided in the annex.

Table 11.1 shows the number of cath lab procedures done in NICVD in 2012. In total, 4,881 coronary angiography, 256 cardiac cath, 121 angiography, 28 angioplasty, and 3,522 other

Table 11.1. Number of cath lab procedures performed in NICVD in 2012

Coronary angiography	Cardiac cath	Angiography	Angioplasty	Other interventions						Total
				PCI	PTMC	TPM	PPM	EPS&RFA	Other	
4881	256	121	28	1681	137	1090	461	56	97	3522

Table 11.2. Heart and vascular surgeries performed in NICVD in 2012

Open-heart surgery					Closed-heart surgery	Vascular surgery		
CABG	Valve	Congenital	Other	Total		Routine	Emergency	Total
175	249	468	57	949	82	254	1274	1528

procedures were done. Detailed data on the various cath lab procedures done in the Institute from 2003 to 2012 are given in the annex.

Table 11.2 shows the number of heart and vascular surgeries done in the NICVD in 2012. These included a total of 949 open-heart surgeries, 82 closed-heart surgeries, and 1,528 vascular surgeries. Detailed data on heart and vascular surgeries from 2002 to 2012 are provided in the annex.

National Center for Control of Rheumatic Fever and Heart Diseases

The National Center for Control of Rheumatic Fever and Heart Diseases (NCCRFHD) takes care of the patients suffering from rheumatic heart diseases and related conditions. There were 27,479 outdoor visits in 2012, with 10,603 (38.6%) males and 16,876 (61.4%) females. Out of all these 27,479 patients, 11,626 (42.3%) were new and 15,853 (57.7%) were old patients. A detailed age- and sex-disaggregation by monthly outdoor attendance of patients is shown in the annex. Figure 11.2 shows the monthly distribution of the number of prophylactic antibiotic injections (n=8,057) given, ECGs (n=1,733), and echocardiograms (1,230) done on patients at the NCCRFHD in 2012.

National Institute of Kidney Diseases & Urology

National Institute of Kidney Diseases & Urology (NIKDU) is a specialized postgraduate institute and training center. It offers postgraduate courses, like MD (Nephrology), MD (Pediatric Nephrology), and MS (Urology) and provides postgraduate training on nephrology, urology, pediatric nephrology,

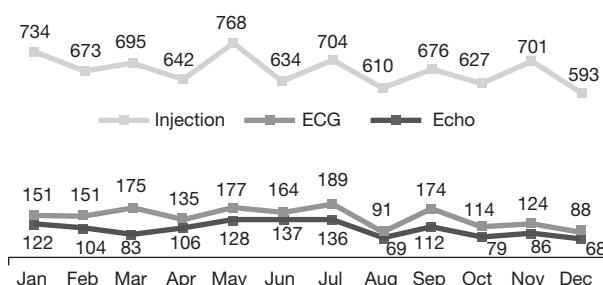


Figure 11.2. Monthly distribution of number of prophylactic antibiotic injections (n=8,057) given, ECGs (n=1,733) and echocardiograms (1,230) done on patients at NCCRFHD in 2012

radiology and imaging, biochemistry, histopathology, microbiology, immunology, hematology, and anesthesiology. Figure 11.3 shows the number of outdoor and indoor patients treated in this institute from 2009 to 2012. Detailed disaggregated data on male, female, and children, including old and new patients, is provided in the annex.

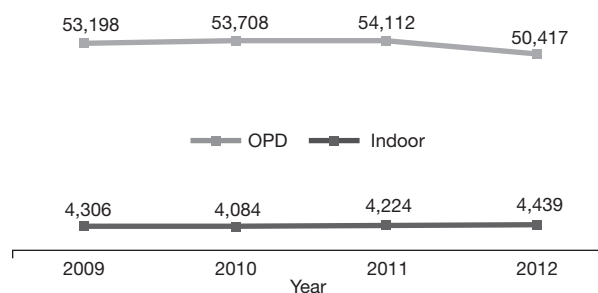


Figure 11.3. Number of outdoor and indoor patients at the National Institute of Kidney Diseases & Urology (NIKDU) from 2009 to 2012

National Institute of Cancer Research & Hospital

National Institute of Cancer Research and Hospital (NICRH) is the only tertiary-level cancer hospital in the country, which acts as the focal point for cancer treatment. In 2012, NICRH provided services to 61,024 outdoor, 3,784 emergency and 3,501 indoor patients. Figure 11.4 shows the data for 2011 and 2012, with sex disaggregation.

Table 11.3 shows the age distribution of cancer patients drawn from the cancer registry of the Institute for 2010. The age-group 45-54 years (24.3%), followed by 55-64 years (20.6%) comprised the majority of cancer patients.

Table 11.4 shows the distribution of admitted cancer patients at NICRH in 2012 by department.

Majority (40.5%) of the patients were admitted to the Medical Oncology Department, followed by Pediatric Oncology (14.0%), Surgical Oncology (14.0%), Radiation Oncology (11.5%), Hematological Oncology (4.8%), Urological Oncology (4.4%) and Gynecological Oncology (4.3%).

Table 11.5 shows the top five types cancer among males and females. Among the males, lung cancer was predominant (27%). Lymphoma occupied the second-highest position (9.1%), followed by esophageal cancer (6.1%), stomach cancer (5.1%), and connective tissue cancers (4.5%). Among the females, breast cancer was the leading type (25.8%), followed by cervical cancer (20.5%), lymphoma (5.7%), ovarian cancer (5.3%), and gall bladder cancer (4.7%).

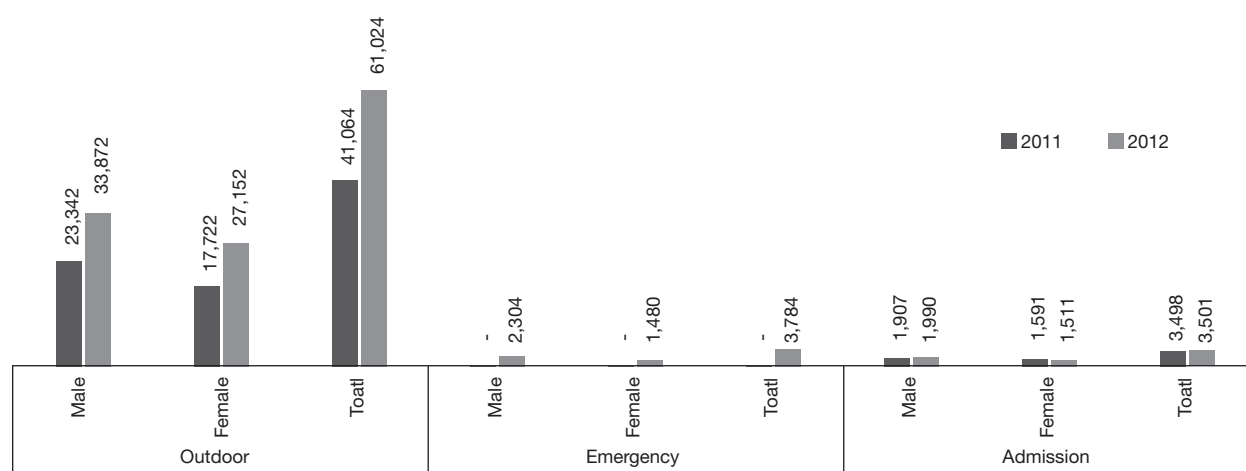


Figure 11.4. Number of outdoor and emergency visits and admissions at the National Institute of Cancer Research & Hospital (2011 to 2012)

Table 11.3. Distribution of cancer patients by age-group drawn from NICRH cancer registry (2010)

Age-group (years)	≤14	15-24	25-34	35-44	45-54	55-64	65-74	75-84	85-94	95+	Total
No.	503	517	1,372	1,992	2,967	2,511	1,598	629	87	22	12,198
%	4.1	4.2	11.2	16.3	24.3	20.6	13.1	5.2	0.7	0.2	100.0

Table 11.4. Distribution of patients in various departments of NICRH (2012)

Department	Medical	Pediatric	Surgical	Radiation	Hematological	Urological	Gynecological	ENT	Maxillo-facial surgery	Plastic and reconstructive surgery	Total
No.	1,362	471	469	385	161	147	143	126	53	43	3360
%	40.5	14.0	14.0	11.5	4.8	4.4	4.3	3.8	1.6	1.3	100.0

Table 11.5. Distribution of cancer patients by top five types of cancer

Male			Female		
Type of cancer	No.	%	Type of cancer	No.	%
Lung	1843	27.0	Breast	1389	25.8
Lymphoma	619	9.1	Cervix	1102	20.5
Esophagus	418	6.1	Lymphoma	305	5.7
Stomach	347	5.1	Ovary	287	5.3
Connective tissue	309	4.5	Gall bladder	253	4.7

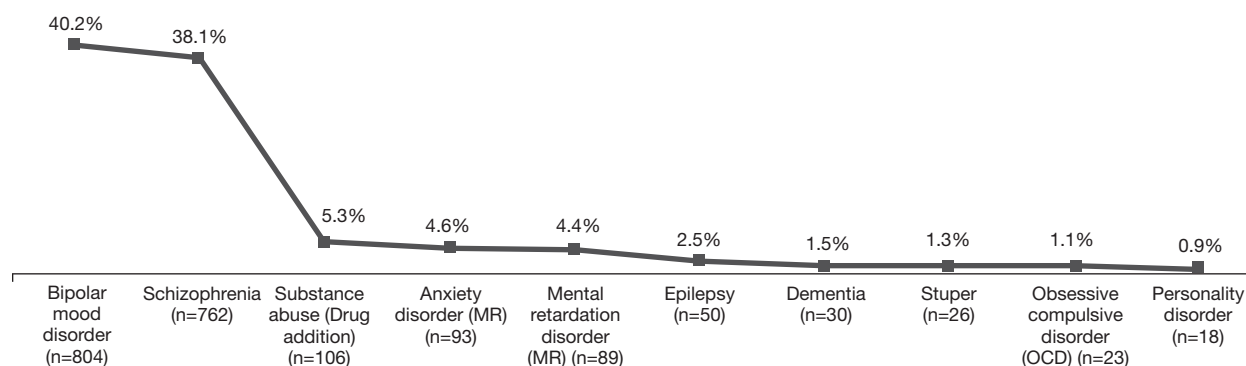


Figure 11.5. Distribution of the admitted patients in NIMHR (2012) by descending order (total patients 2,001)

National Institute of Mental Health & Research

In 2012, the National Institute of Mental Health & Research (NIMHR) provided treatment to 25,508 new outdoor patients, 2,030 emergency patients, and 1,953 indoor patients. Among the outdoor patients, 14,959 (58.7%) were males, 8,939 (35.0%) were females, and 1,610 (6.3%) were children. Among the emergency patients, 1,249 (61.6%) were males, 679 (33.4%) were females, and 102 (5.0%) were children. Among the indoor patients, 1,159 (59.3%) were males, 667 (34.2%) were females, and 127 (6.5%) were children. A detailed profile of patients from 2007 to 2012 is given in the annex.

Figure 11.5 shows the percentage distribution of the admitted patients in NIMHR in 2012. Bipolar mood disorder (40.2%) and schizophrenia (38.1%) accounted for 78.3% of the total admitted patients.

Arsenic in groundwater: Mitigation program of DGHS

The presence of arsenic in harmful level (unsuitable for human consumption) was first detected in 1993 by the Department of Public Health Engineering (DPHE) in 4 tubewells of

Chamagram village of Chapainowababganj Sadar upazilla, According to a report published by the National Arsenic Mitigation Center in 2003, water-samples from 4.95 million tubewells were tested for the presence of arsenic. Of these, 1.44 million (29.1%) showed the evidence of arsenic contamination. The DPHE reported in 2006 that arsenic contamination in tubewell water was present in 62 of the 64 districts of Bangladesh. The first detection of health problems in 8 persons due to drinking of arsenic-contaminated water was recorded in 1994 by the Department of Occupational and Environment Health (OHE) of the National Institute of Preventive and Social Medicine (NIPSOM). Since then, the National Arsenic Program of the Directorate General of Health Services has been conducting several key activities, viz. (i) consultations and workshops for the development of methodology, tools, database, and data sources; (ii) orientation training of government and non-government health service providers, like nurses, medical assistants, technologists, and field-level health and family planning workers; (iii) mass awareness programs on consumption of arsenic-free safe drinking-water; (iv) testing tubewells water at

health facilities for prevention of arsenicosis; (vi) screening of patients through house-to-house searching programs; (vii) identification, diagnosis, and management of arsenicosis patients; (viii) capacity-building of human resources and improving facilities for effective case management and referral; (ix) establishment of rehabilitation centers for disabled arsenicosis patients; (x) conducting surveys and research on arsenicosis; (xi) updating national arsenic mitigation policy and strategy; (xii) strategic partnership with local bodies and community-based organization regarding the mitigation of arsenicosis; (xiii) further collaboration between DGHS and DPHE at the field level to strengthen water screening at the community level; (xiv) strengthening of the existing BAN-net and InfoBase and further inclusion of electronic database at the DGHS (logistics, human resource, and IT network); and (xv) strengthening routine MIS for hospital statistics on arsenicosis and interlinking with MIS-DGHS.

Figure 11.6 shows the cumulative number of arsenic patients from 1996 to 2012. In 2012, the cumulative number stood at 65,910.

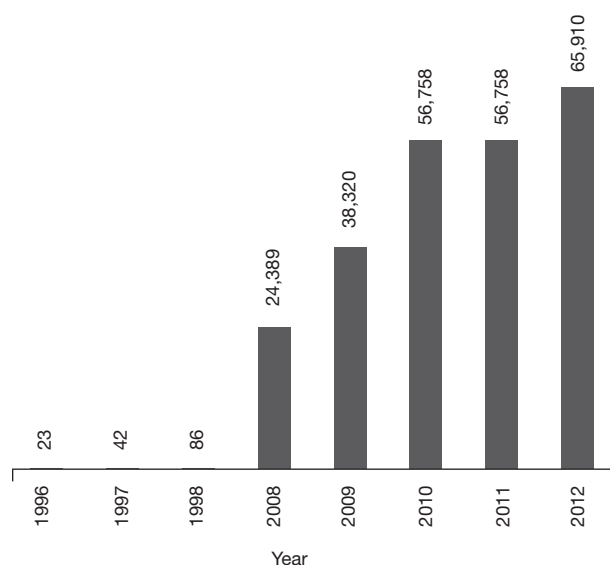


Figure 11.6. Cumulative number of arsenic patients in Bangladesh year by year detected by National Arsenic Program of DGHS

Figure 11.7 shows the distribution of arsenicosis patients by division of Bangladesh. Chittagong division shows nearly half of the identified patients (48%; n=31,230), followed by Khulna (21%; n=13,598), Dhaka (17%; n=11,090), Rajshahi (9%; n=6,120), Rangpur (3%; n=1,674), Barisal (2%; n=1,512), and Sylhet division (1%; n=686).

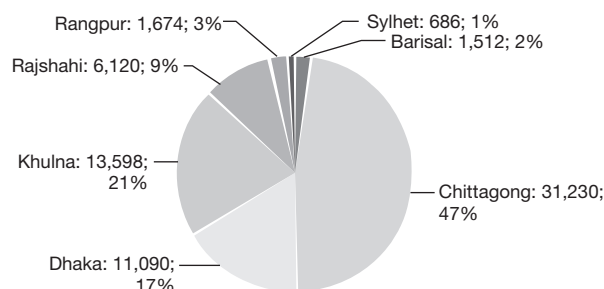


Figure 11.7. Distribution of arsenicosis patients by division of Bangladesh (2012)

Climate change and response

To address climate change-related health issues, the non-communicable disease control unit of DGHS developed and published training modules—one for doctors and another for nurses, paramedics, health assistants, and family welfare assistants. Several training and orientation sessions have been organized at the central, district and upazila levels. A total of 18,450 field-level personnel have been trained on climate change, health impact, and management. A total of 3,690 teachers have been oriented. After receiving training, the field-level health workers are now organizing courtyard sessions at the communities. They are also registering climate-sensitive diseases in their respective areas. Necessary collaborations are being made to strengthen the activities of Climate Change Health Promotion Unit (CCHPU) of the MOHFW.

Injury situation in Bangladesh

Injury has drawn attention of policy-makers and development activists of Bangladesh since the Bangladesh Health and Injury Survey (BHIS 2005) which revealed an annual estimate of 70,000 injury-related deaths in Bangladesh, the children being the worst victims (43% of the total injury-related deaths). The Center for Injury Prevention and Research, Bangladesh (CIPRB) has been involved in the injury prevention initiatives since 2005 and has a sentinel surveillance system to monitor the injury situation in its intervention areas—Raiganj and Sirajganj. Through the surveillance system, data are collected at a six-month interval from over 31,500 households representing more than 145,000 people. The injury rates in this surveillance area are expected to be lower than the national estimates due to the ongoing interventions in the area. However, this is the only community-based active injury surveillance that exists in the country to measure the injury burden. The mortality rates due to

injury in all ages were found to be 37.4 and 43.6 per 100,000 people in 2011 and 2012 respectively. The rate of injury-related mortality in children below 18 years of age was 39.9 and 37.8 per 100,000

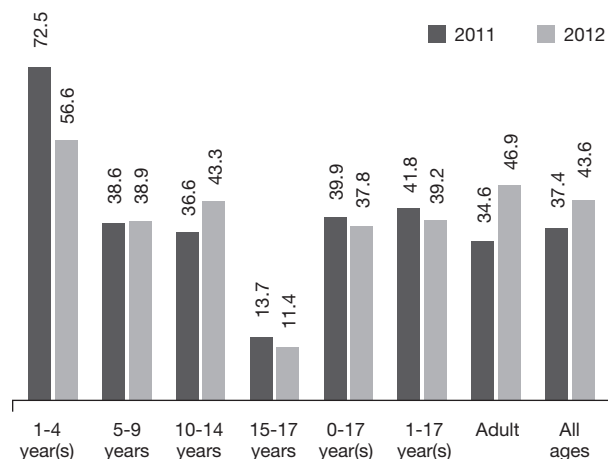


Figure 11.8. Injury mortality per 100,000 population by age-group in the surveillance area of Bangladesh (2011 and 2012)

people in 2011 and 2012 respectively. Among all children, those aged 1-4 year(s) were the worst victims (72.5 and 56.6 per 100,000 people in 2011 and 2012 respectively). Among adults, the rates were 34.6 and 46.9 per 100,000 people in 2011 and 2012 respectively (Figure 11.8).

Figure 11.9 summarizes the injury-related mortality rate by cause per 100,000 people as found in the surveillance area of Bangladesh in 2011 and 2012. Drowning was the leading killer of children below 18 years of age, claiming about 23 lives per 100,000 people.

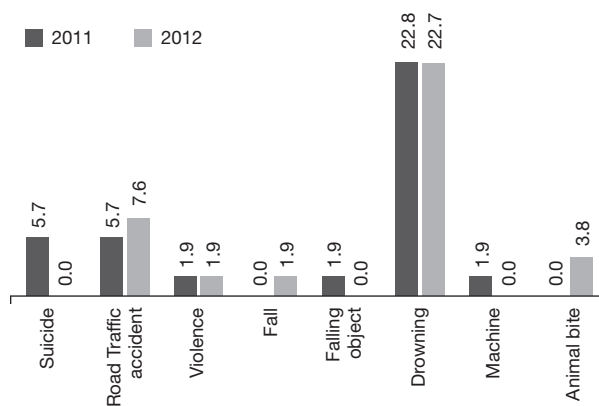


Figure 11.9. Injury-related mortality rate per 100,000 population by cause among children below 18 years in the surveillance area of Bangladesh (2011 and 2012)

Figure 11.10 shows the injury-related mortality rate by cause per 100,000 people among adults as found in the surveillance area of Bangladesh in 2011 and 2012. Road-traffic injury was the most common cause of injury-related deaths.

Autism

It is estimated that every day 360,000 babies are born in the world. Sadly, one in every 110 of them is born with an autism spectrum disorder, i.e. more 3,000 new autistic children are added to the existing number every day. The challenges faced by families living in poverty are already immeasurable; it is even more challenging when that child has a complex disability, such as autism. One report from the Department of Social Welfare of Bangladesh estimates that around 280,000 children in Bangladesh are facing autism-related health problems, yielding

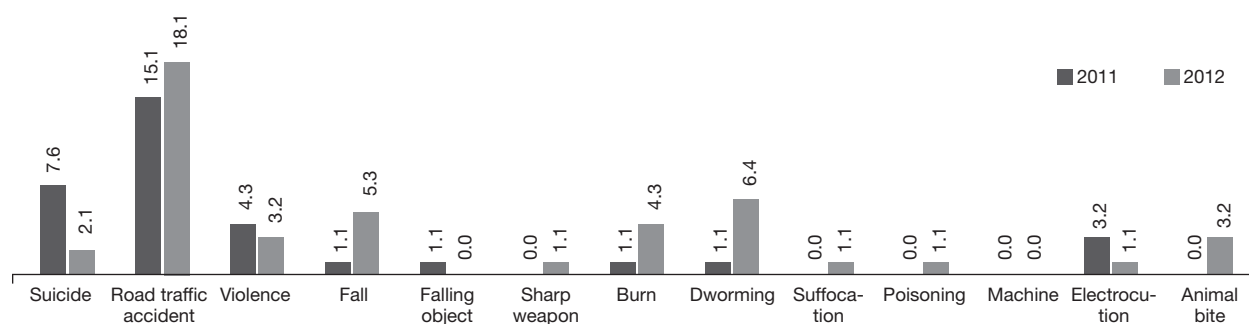


Figure 11.10. Injury-related mortality rate by cause per 100,000 population among adults as found in the surveillance area in 2011 and 2012

one in every 500 children. The national health program has identified this problem as a priority and has undertaken the following activities:

(i) national advisory committee on autism and neurodevelopment has been constituted; (ii) a national steering committee on autism by the involvement of 7 relevant lead ministries has been created; (iii) a national strategic plan on autism has been formulated, along with a short-term and a long-term action plan; (iv) autism has been incorporated in undergraduate medical curriculum; (v) child development centers (Sishu Bikash Kendro) have been established in 15 medical college hospitals; (vi) piloting of home-based screening of autism and neurodevelopmental disorders in children aged 0-9 year(s) at selected 7 upazila, one in division, has been conducted; (vii) nurses, paramedics, and doctors have

been trained on autism; (viii) IEC materials on autism have been developed; and (ix) Center for Neurodevelopment and Autism in children has been established at Bangabandhu Sheikh Mujib Medical University. Bangladesh remains in the leadership and forefront position in global awareness creation on autism. Under the initiative of Bangladesh, resolutions on autism have been approved by the United Nations General Assembly (2012), Regional Committee Meeting of the WHO South-East Asia (2012), and the Executive Board of the WHO (May 2013).

It is concluded, the wide variety of non-communicable diseases in Bangladesh continues to show increasing disease burden on the country. More population-based actions are needed for primary prevention and control of these diseases.

SAFE BLOOD TRANSFUSION

No paid donor and blood screening reduced risk of transfusion-transmitted infections

Establishment of the first blood transfusion service in Bangladesh dates back to 1950 in Dhaka Medical College Hospital. Under Health and Population Sector Program (HPSP) 1998-2003, the Safe Blood Transfusion Program (SBTP) was launched in 2000 with the assistance of UNDP to ensure supply of safe blood for humans through screening. Under this program, blood-screening facilities were developed in 99 blood transfusion centers. In 2004, the activities of the Safe Blood Transfusion Program received financial support from the World Bank and DFID through IDA credit. A Memorandum of Understanding (MoU) was signed between MOHFW and WHO under HIV/AIDS Prevention Project (HAPP) with technical assistance from the latter. This continued till 2007. Since then, the activities were being implemented under the Health, Nutrition and Population Sector Program (HNPSPP) 2003-2011. The activity is being continued also under the current Health, Population and Nutrition Sector Development Program (HPNSDP) 2011-2016.

The Safe Blood Transfusion Program (SBTP) made a good progress over the past years through reduction in the number of paid donors from 70% to 0%, capacity-building for blood screening in all blood transfusion centers for HIV, hepatitis B and C, syphilis, and malaria and expansion of activities up to the upazila health complex level. Currently, 203 blood transfusion centers, with 89 in the upazila level, are functional under the program. Blood component separation facilities have been developed in 18 blood transfusion centers. Six centers have been equipped with modern mobile vans for outdoor blood collection. Following is a profile of the SBTP as of 2012:

No. of blood transfusion centers supported currently by SBTP: 203

No. of blood transfusion centers at the upazila level: 89

No. of centers where blood-component separation facilities exist: 18

No. of centers with mobile vans for blood collection: 6

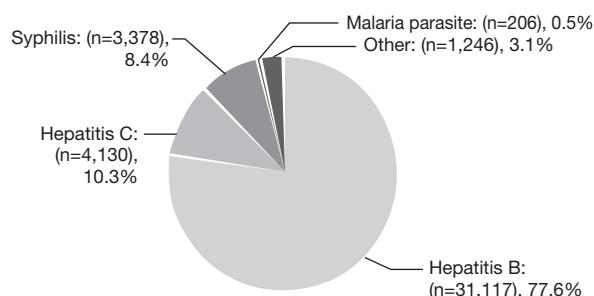


Figure 12.1. Percentage of units of blood rejected due to various reasons from 2001 to 2012 (Total units rejected=40,077)

During 2001 to 2012, a total of 33,97,150 units of blood were tested in 203 centers, out of which 40,077 units were rejected (1.2%) due to the evidence of transfusion-transmitted infections (TTIs). Of the rejected units, 31,117 (77.6%) were for hepatitis B, 4,130 (10.3%) for hepatitis C, 3,378 (8.4%) for syphilis, 1,246 (3.1%) for malarial parasites, and 206 (0.5%) for HIV (Figure 12.1).

Table 12 shows the year-wise distribution of the number of units of rejected blood due to various reasons based on the screening tests. Similar to other years, hepatitis B continues to remain the leading cause of transfusion-transmitted infections in 2012 (0.9%), followed by hepatitis C (0.12%) and syphilis (0.07%) among the screened blood samples.

In 2012, a total of 91,557 units of blood components were produced by that blood centers. These included 40,889 units of red blood cell concentrate, 29,753 units of fresh frozen plasma, 20,265 units of platelet concentrate, 382 units Platelet Rich Plasma (PRP), 43 units of Fresh Plasma (FP) and 205 units of cryoprecipitate 3,78,622 units of blood components upto 2012.

The Safe Blood Transfusion Act 2002 of Bangladesh is in place that circulated the rules and regulations in 2008. There is a reference laboratory for blood transfusion at the new Annex Building of Dhaka Medical College Hospital. The functions of

Table 12. Cumulative screening report for blood with TTIs, 2001–2012

Year	No. of units tested	HIV+ve*		Hepatitis B+ve		Hepatitis C+ve		Syphilis+ve		Malarial parasite+ve	
		No.	%	No.	%	No.	%	No.	%	No.	%
2001	99653	2	0.002	1381	1.4	82	0.08	290	0.29	7	0.01
2002	170948	4	0.002	2433	1.4	246	0.14	655	0.38	53	0.03
2003	180015	1	0.001	1900	1.1	1024	0.57	428	0.24	13	0.01
2004	121993	36	0.030	1284	1.1	251	0.21	257	0.21	8	0.01
2005	203575	8	0.004	1689	0.8	201	0.10	305	0.15	6	0.00
2006	228127	20	0.009	1814	0.8	242	0.11	209	0.09	1	0.00
2007	324005	27	0.008	2764	0.9	251	0.08	215	0.07	1013	0.31
2008	369026	13	0.004	2996	0.8	309	0.08	143	0.04	4	0.00
2009	358067	9	0.003	2135	0.6	181	0.05	115	0.03	7	0.00
2010	384447	6	0.002	3313	0.9	374	0.10	182	0.05	37	0.01
2011	415372	21	0.005	4356	1.0	272	0.07	179	0.04	39	0.01
2012	541682	56	0.010	5052	0.9	676	0.12	399	0.07	58	0.01
Total	3396910	203	0.006	31117	0.9	4109	0.12	3377	0.10	1246	0.04

*HIV-reactivity done by rapid test

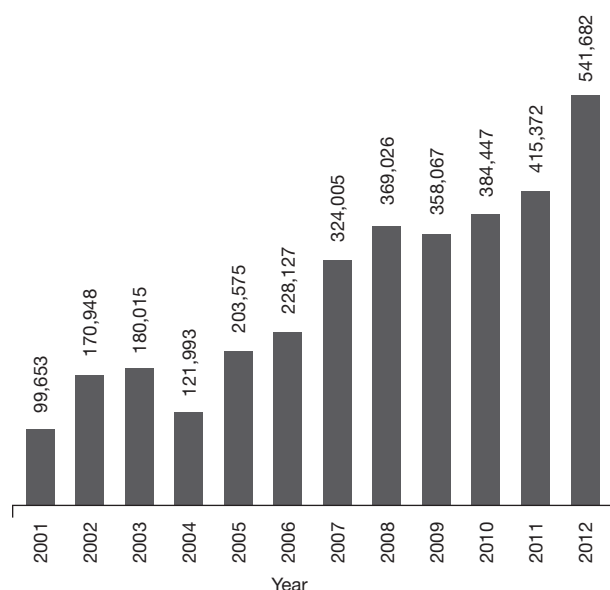


Figure 12.2. Number of blood units collected year-wise by the blood centers under SBTB (n=3,396,910)

the reference laboratory are to support various organizations for training and monitoring. The reference laboratory is also testing the referred samples and validation of kits. The professionals engaged in the safe blood transfusion program deeply feel that a National Blood Center should

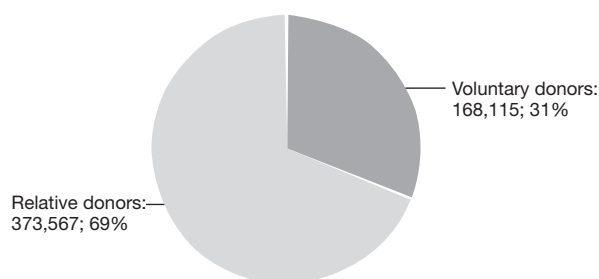


Figure 12.3. Distribution of voluntary and relative blood donors in 2012 (n=541,682)

be established as soon as possible to further streamline the stewardship role and coordination functions for the current fragmented blood transfusion services operating throughout the country.

The blood centers under the Safe Blood Transfusion Program collectively gathered a total of 33,96,910 units of blood from 2001 to 2012. The year-wise distribution of collection is shown in Figure 12.2. In 2012, the program personnel collected 5,41,682 units of blood. The distribution of voluntary (1,68,115) and relative (3,73,567) blood donors is shown in Figure 12.3.

A number of voluntary or non-profit organizations also contribute to encouraging healthy donors for donating blood voluntarily. Some of these organizations have their own set-up for collecting,

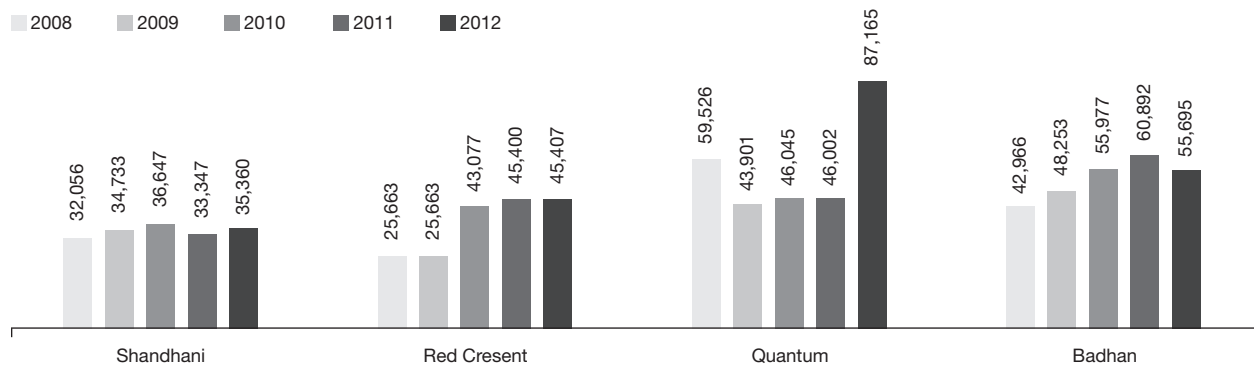


Figure 12.4. Number of units of blood collected by different voluntary blood-donation organizations during the last 5 years

testing, storing, and distributing blood or blood products. Figure 12.4 shows the year-wise collection by the major voluntary blood-donation organizations.

It is pleasing to note that the Safe Blood Transfusion Program and the voluntary blood-donation organizations could consistently keep the percentage of paid donors to zero. The paid donors

are sources of transfusion-transmitted infections (TTIs), viz. hepatitis B and C, syphilis, malarial parasites, HIV, etc., and they dominated the blood donors prior to inception of Safe Blood Transfusion Program and emergence of the voluntary blood-donation organizations. Currently, the absence of paid donors and screening of collected blood prior to transfusion have reduced the risk of transfusion-transmitted infections.

NUTRITION SITUATION IN BANGLADESH

NNS catches up mainstreaming the national nutrition service

Although Bangladesh has made considerable progress during the past two decades in many health indicators, the prevalence of malnutrition is still one of the highest among the developing countries. The country is on track to attain most of the Millennium Development Goals relating to human development, such as reducing child and maternal mortality. However, these improvements are yet to be translated into positive changes in terms of maternal and child nutrition.

Public health nutrition program under MOHFW

Under the HNPS (2003-2011), there were two operational plans (OPs), namely National Nutrition Program (NNP) and Micronutrient Supplementation (MNS). Limited facility-based nutrition services were provided through MNS, and community-based nutrition services were delivered by NNP. However, the services provided by NNP were independent of mainstream health services, with duplication of field-level activities under these two operational plans. Scaling up of services throughout the country appeared very costly. Consequently, under the current sector program HPNSDP (July 2011-June 2016), the MOHFW started to accelerate reduction of the persistently high rates of maternal and child undernutrition by mainstreaming implementation of high-impact, evidence-based nutrition services into the health and family planning services, along with scaling up of the provision of community-based nutrition services throughout the country. Under the National Nutrition Services (NNS), which is housed in the Institute of Public Health Nutrition (IPHN), both DGHS and DGFP are streamlining and strengthening the nutrition services by using regular manpower.

Four hundred and eighty-two medical officers of the upazila health complexes are already designated as Medical Officer (Public Health Nutrition) who will be responsible for coordinating all activities of NNS at the upazila level. The NNS caters all the services of NNP and MNS. The components of NNS include: (i) facility-based services; (ii) community/area-based nutrition

Under the current sector program HPNSDP (July 2011-June 2016), the MOHFW started to accelerate reduction of the persistently high rates of maternal and child undernutrition by mainstreaming implementation of high-impact, evidence-based nutrition services into the health and family planning services, along with scaling up of the provision of community-based nutrition services throughout the country ...

activities; (iii) human resource development; (iv) providing micronutrients to people; (v) operational research and surveys; (vi) monitoring and evaluation; and (vii) nutrition information system. Capacities of the upazila health complexes, district hospitals, and community clinics as well as of the facilities under DGFP, e.g. of MCWCs, are now in the process of strengthening. The NNS aims to cater nutrition services through establishing IMCI-nutrition corners in all the health facilities where IMCI corners are already established. Mass awareness is also being created through behavior change communication (BCC).

Current nutrition situation in Bangladesh

The Bangladesh Demographic and Health Survey (BDHS) 2011 shows that 41% of the under-five children are stunted, with 15.3% severely stunted (Figure 13.1). The age pattern of stunting shows that it increases with age, being 18% among <6 months children to 52% among children aged 18-23 months. Among children of 48-59 months, the prevalence is 42%. Rural children are more likely to be stunted than the urban children (43% compared

to 36%). Stunting is the lowest in Khulna and Rajshahi divisions (34%) of Bangladesh. The prevalence of wasting among the under-five children is 16%, with 4% being severely wasted. Wasting peaks at 18-23 months (17%) of age. The prevalence of underweight children is 36%, with 10% being severely underweight. The proportion of underweight in children peaks at age 36-47 months (43%). There have been some improvements in nutritional status of children over the past four years. The level of stunting has declined from 51% in 2004 and 43% in 2007 to 41% in 2011. Wasting has declined from 17% in 2007 to 16% in 2011. The level of underweight has declined to 36% in 2011 from 41% in 2007. Therefore, it is expected that the HPNSDP 2011-2016 goal for reduction of underweight (33%) and stunting (38%) will be possible to achieve.

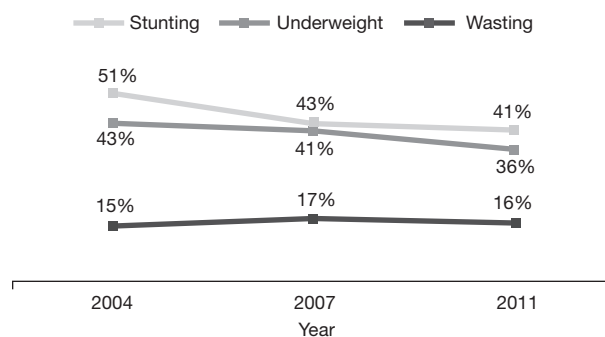


Figure 13.1. Trend of protein energy malnutrition among under-5 children in Bangladesh

Figure 13.2 shows the findings of Survey 2012 of Food Security and Nutrition Surveillance Project (FSNSP), jointly conducted by BBS, HKI, and BRAC University. The survey reveals that prevalence of wasting is high in eastern hills (13%) compared to national average (10%). Rate of severe wasting is also found comparatively higher (5%) in the same area than in other areas/zones. Interestingly, wasting rate in *haor* area is comparatively lower (9%) but the underweight rate is the highest (40%) there. In contrast, the national rate is around 33%. The BDHS 2011 has also shown 36% national rate.

The recent National Micronutrients Status Survey (MNSS-2011-2012), jointly conducted by the Institute of Public Health and Nutrition (IPHN), UNICEF, icddr, and GAIN (January 2013), shows that underweight, stunting, and wasting rates are comparatively higher in slum area than urban and rural areas (Figure 13.3).

The FSNS surveillance (mentioned above) in its round 9 (October-December 2012) shows that 11% of adolescent girls and 18% of women were severely and moderately underweight (Figure 13.4).

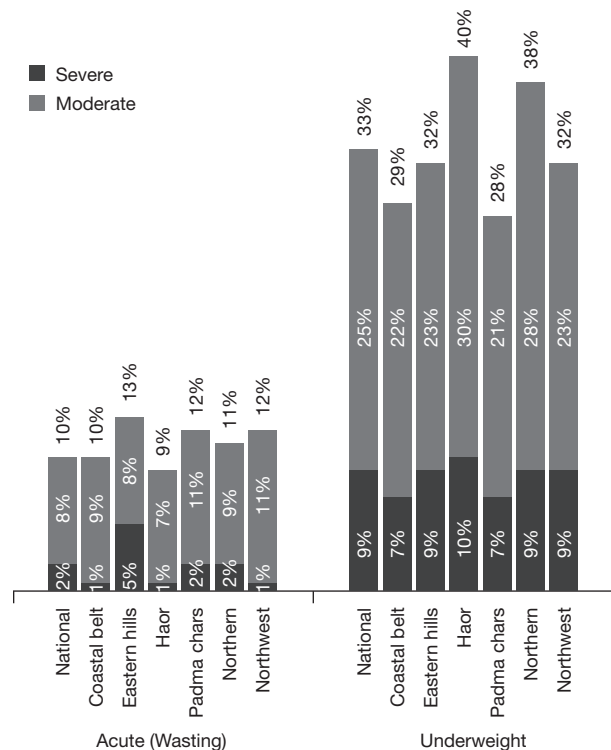


Figure 13.2. Trends in nutritional status by region (October-December 2012)

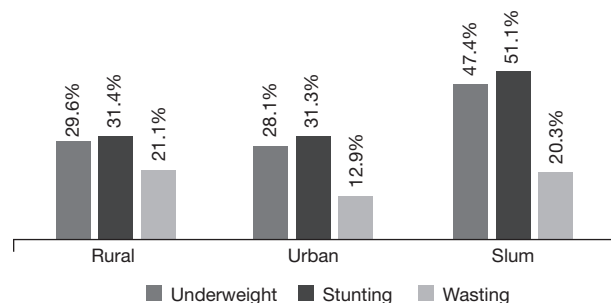


Figure 13.3. Prevalence of underweight, stunting, and wasting among under-5 children between rural, urban and slum areas of Bangladesh (January 2013)

The figure also shows the prevalence in different areas of Bangladesh.

Figure 13.5 shows that there is substantial variation in rates of underweight (severe and moderate) among adolescent girls and women among divisions (FSNSP 2011), with the highest rate among adolescent girls in Khulna division and among women in Sylhet division.

Figure 13.6 shows the maternal nutrition status by place of residence (urban-rural) and division.

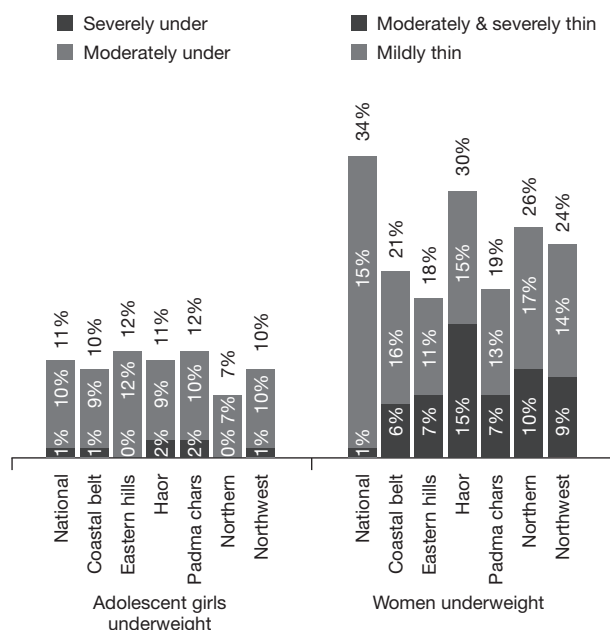


Figure 13.4. Prevalence of undernutrition among adolescent girls and women in different areas of Bangladesh (FSNS 2012)

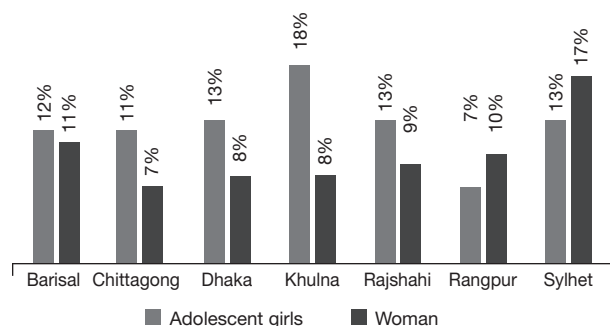


Figure 13.5. Regional variation in prevalence of undernutrition (severe and moderate) among adolescent girls and women

Undernutrition rate varies between 17% and 36% by place of residence. Prevalence of overweight among urban mothers is 38% compared to rural counterparts (18%). However, in three divisions, viz. Chittagong, Dhaka, and Khulna, the prevalence of overweight exceeded undernutrition.

Breastfeeding practices

The BDHS 2011 shows that the exclusive breastfeeding rate among children below 6 months increased from 43% in 2007 to 64% in 2011. Intensive government programs undertaken with the focus on maternal, newborn and childcare, working in synergy with the health programs undertaken by other stakeholders,

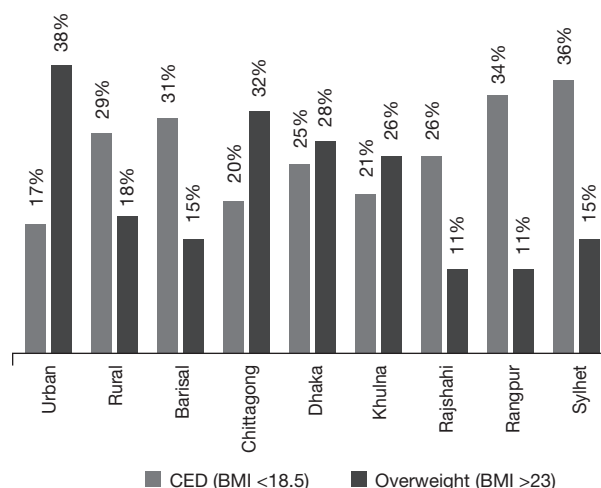


Figure 13.6. Maternal nutrition status by place of residence

have contributed largely to this improvement. It is expected that, if the current initiatives, like promotion of Exclusive Breastfeeding (EBF), Baby-friendly Hospital Initiative (BFHI), and other MNCH programs be scaled up successfully, the HPNSDP 2011-2016 target of exclusive breastfeeding to 50% infants up to 6 months of age would be achieved. The FSNS (October-December 2012) findings show early initiation of breastfeeding for 47% of the 0-5 month(s) old children, and 46% were continuing exclusive breastfeeding.

Infant and young child-feeding practices

Infant and Young Child-feeding (IYCF) practices include timely initiation of solid, semi-solid or soft foods from six months of age. Overall, 21% of children aged 6-23 months were fed appropriately

if the current initiatives, like promotion of Exclusive Breastfeeding (EBF), Baby-friendly Hospital Initiative (BFHI), and other MNCH programs be scaled up successfully, the HPNSDP 2011-2016 target of exclusive breastfeeding to 50% infants up to 6 months of age would be achieved. ...

Table 13.1. Prevalence of vitamin A and zinc deficiency among three subsets of population by place of residence (2012)

Micronutrient type	Children preschool age			Children of school age			Non-pregnant and nulliparous women		
	Rural	Urban	Slum	Rural	Urban	Slum	Rural	Urban	Slum
Subclinical vitamin A deficiency	19.4%	21.2%	38.1%	20.2%	22.1%	27.1%	5.4%	4.9%	6.9%
Zinc deficiency	48.6%	29.5%	51.7%	-	-	-	57.5%	54.5%	66.4%

according to standard IYCF practices. It is a drop from 42% observed in 2007. However, this is due to change in the definition of IYCF in the demographic health survey tool. The NNS is scaling up IYCF services in the facilities and community through community clinics, with the aim of achieving HPNSDP 2011-2016 target of IYCF (52%). The FSNS surveillance in its round 9 (October-December 2012) shows that solid and semi-solid foods have been given to 6-8 months old children in 90% cases, and 31% of 6-23 months old children have been given minimum acceptable diet. However, 13% of 0-23 month(s) old children were bottle-fed.

Micronutrient status

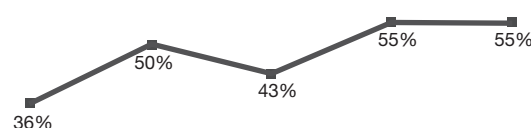
Vitamin A supplementation

The Government of Bangladesh prioritizes vitamin A supplementation as an important public-health program and is distributing vitamin A capsules to children of 6-59 months through National Vitamin A Campaign (NVAC). In the past, the success of NVAC was highly appreciated. The coverage was 84% according to BDHS 2011 and 88% according to Utilization of Essential Service Delivery Survey 2008. Every year, two rounds of vitamin A capsule supplementation to children aged 6-59 months have been conducted. Health workers and volunteers administer vitamin A capsules to around 20 million children at 140,000 sites located in health facilities, health centers, schools as well as in mobile sites (bus, boat, and railway stations) throughout the country. The last campaign was held on 12 March 2013. Next campaign is due on 5 October 2013. The national estimate of the coverage was 77.0%, with 77.9%, 73.1%, and 72.4% rural, urban and the slum coverage respectively. According to asset index, the coverage was 76.4% in the 'poorest' section and 87.5% in the 'richest' section of the population.

Table 13.1 shows the subclinical vitamin A and zinc status among three subsets of population.

Control and prevention of iron-deficiency anemia

In a study by Helen Keller International (HKI) done in 2006, nearly half of pregnant women in Bangladesh were found anemic. In Food Security and Nutrition Surveillance Project, jointly done by BBS, BRAC University, and HKI, it was seen that the proportion of women taking iron folic acid (IFA) supplementation during the last pregnancy was slowly rising. Figure 13.7 shows the trend. A survey, jointly done by IPHN and HKI in 1999, showed that 49.2% of pregnant women and 52.7% of preschool children in rural Bangladesh were suffering from iron-deficiency anemia. It seems that, iron-deficiency anemia is remaining in the same level. Control and prevention of iron-deficiency and other nutritional anemia was broadly made through the country's routine service-delivery network and the NNS. In 2011, the NNS distributed 100 million iron-folate tablets to the community clinics. Control of nutritional anemia is done by treating intestinal parasites through distribution of albendazole tablets during vitamin A campaign and separate Deworming Week.



BDHS 2000 BDHS 2004 CMNS 2005 BDHS 2007 FSNSP 2007

Figure 13.7. Percentage of mothers who took iron-folate during last pregnancy

A recent survey (NMSS 2011-2012) shows that the prevalence of anemia among the preschool children is 33.1%–37.0% in rural and 22.8% in urban areas. In 2011, the prevalence was 47.0% (NSP 2001).

Control of iodine-deficiency disorders and other micronutrient problems

The NNS provides training to doctors and other health staff on iodine-deficiency disorders. The NNS also provides training on capacity-building of managers, chemists, and relevant persons in different zones in collaboration with Bangladesh Small and Cottage Industries Corporation (BSCIC). The IPHN laboratory for testing the iodine level in salt has also been strengthened. A recent survey (NMSS 2011-12) on urinary iodine concentration shows 40% of the school-age children to have a iodine deficiency, implying an improved situation from 2004-2005 (33.8%). Among the non-pregnant and nulliparous women, the prevalence of iodine deficiency was 42.1%, showing an improvement from 2004-2005 figures (38.0%). About 80% of the households used iodized salt (iodine level ≥ 5 ppm) while 57.6% of the households used adequately-iodized salt (iodine level ≥ 15 ppm). In the rural area, the use of adequately-iodized salt was 51.8%. The national rate of the use of 'brand' salt is 75.8%; however, a substantial proportion (30%) of households in rural area still uses 'open' salt. The use of 'open' salt is 37.0% and 17.0% in the 'poorest' and the 'richest' households respectively. The proportion of retailer salt samples with adequately-iodized salt (≥ 20 ppm) was 66.4%.

Micronutrients consumption from foods

Survey shows that consumption of foods of animal source is increasing in the country (Household Income & Expenditure Survey of Bangladesh 2010). However, data of the National Micronutrients Survey suggest that the population of Bangladesh is still well short of the recommended daily allowance (RDA) for the key micronutrients. In the case of vitamin A, the median daily consumption, as expressed by Retinol Equivalents (RE) is 270.0, 318.0, and 372.0 respectively among the preschool and school-age children, and the non-pregnant and nulliparous women, which are certainly short of the RDA they need. Daily median consumption of iron from foods is 4.17, 5.21, and 6.64 mg

Data of the National Micronutrients Survey suggest that the population of Bangladesh is still well short of the recommended daily allowance (RDA) for the key micronutrients ...

among the preschool and school-age children; non-pregnant and nulliparous women, which are also short from the RDAs they require. The consumption of iron from animal source, the form of dietary iron that is readily absorbed in the body, is low in terms of the total iron consumption. The share of iron from animal source is 23.0%, 24.0%, and 18.0% of the total iron consumption respectively among the population groups mentioned above. The median daily consumption of zinc from foods among the preschool children is 3.2 mg and 2.6 mg respectively in the urban and slum area against the RDA of 3 to 5 mg.

NNS catching up mainstreaming nutrition service

The current sector-wide approach of the MOHFW under HPNSDP 2011-2016 has made a great stride towards mainstreaming the nutrition activities through the Ministry's two major agencies, viz. DGHS and DGFP, through one operational plan, namely National Nutrition Service (NNS). This is a major shift in nutrition program from the one in HNPS 2003-2011, under which nutrition program was implemented through two operational plans, viz. National Nutrition Program (NNP) and Micronutrient Supplementation (MNS). From a more than one year's experience, it is now safe to comment that the NNS is gradually catching up. It is hoped that nutrition situation in the subsequent months and years will benefit from coordinated activities and services of NNS.

HEALTH INTERVENTIONS BY IEDCR AND IPH

Both institutions fulfilled their public-health commitments to the nation

The Institute of Epidemiology, Disease Control and Research (IEDCR) and the Institute of Public Health (IPH) are the two national institutions that undertake significant public-health interventions on behalf of the Ministry of Health and Family Welfare of Bangladesh. This chapter highlights some of the interventions undertaken by these two institutions.

Institute of Epidemiology, Disease Control and Research (IEDCR)

Mandate of IEDCR

The IEDCR, since its inception in 1976, continues to be a national institution responsible for conducting disease surveillance and outbreak investigations. The Institute is the WHO-designated National Influenza Centre (NIC) in Bangladesh. With a staff strength of 115, the IEDCR has eight departments, viz., Biostatistics, Epidemiology, Medical Entomology, Medical Social Science, Microbiology, Parasitology, Virology, and Zoonosis. The specific objectives of IEDCR are to (i) conduct disease surveillance; (ii) conduct outbreak investigation and response; (iii) conduct epidemiological research; and (iv) provide training to concerned staff.

Laboratories of IEDCR

The Institute established a biosafety level 3 (BSL3) laboratory. The other laboratories of the Institute are: Medical Entomology Laboratory, Microbiology Laboratory, Parasitology Laboratory, Virology Laboratory, Zoonosis Laboratory, RT-PCR Laboratory, and BSL2 Laboratory. The laboratories have wide-ranging diagnostic facilities for testing parasitic and fungal pathogens that cause visceral leishmaniasis (kala-azar, PKDL), malaria, and intestinal diseases, along with dermatophytes, and candida; those that cause viral diseases, e.g. Nipah encephalitis; influenza; hepatitis A, B, C, and E virus (HAV, HBV, HCV, and HEV); HIV; dengue; and chikungunya; bacterial diseases, e.g. enteric fever; brucellosis; rickettsial diseases; and other aerobic and anaerobic bacterial infections. These laboratories also conduct biochemical tests. The IEDCR laboratories have facilities for cell-culture

also. The Department of Entomology regularly performs biological efficacy tests for insecticides.

Outbreak investigations

The IEDCR conducted 107 outbreak investigations since 2007 (11 in 2007, 29 in 2008, 10 in 2009, 17 in 2010, 22 in 2011, and 18 in 2012; <www.iedcr.gov.bd>). The outbreak investigations in 2007 were notable for identification of Nipah virus outbreak, mass psychogenic illness in 18 districts, and outbreak of illness due to toxic *Ghagra shak* in Sylhet. The 2008 outbreak investigations were notable for identification of Nipah virus, puffer-fish poisoning, mass psychogenic illness, first human case of avian influenza (AI), and chikungunya. The investigations in 2009 were notable for identification of pandemic influenza A H1N1 (swine flu), chikungunya, cutaneous anthrax, pesticide poisoning, and mass psychogenic illness. Notable outbreaks investigated in 2010 were for Nipah virus, pneumonia, bronchiolitis, chicken pox (in Lama), suspected water contamination (in Boropukuria Power Plant), anthrax (in Tangail, Sirajganj, and Pabna), suspected insecticide poisoning (in Naogaon), mass psychogenic illness and rabies (in Narsingdi). Investigations for Nipah virus, cholera and shigellosis outbreaks, detection of the second and third human cases of avian influenza (H5N1), the first case of H9N2, investigation for suspected pesticide poisoning, cutaneous anthrax, suspected rubella encephalitis, hepatitis E, H5 outbreak among ducks and waterfowl (for human contact investigation), influenza B outbreak, respiratory virus cluster, and unknown diseases were the major activities of IEDCR in 2011. The prominent outbreaks investigated in 2012 were for Nipah virus, detection of avian Influenza H5N1, mass psychogenic illness, cutaneous anthrax, Japanese encephalitis, chikungunya, dengue, and suspected pesticide victims.

Routine disease surveillance by IEDCR

Disease surveillance is one of the main activities of IEDCR. Routine and disease-specific surveillances are conducted round the year. The routine surveillances include: (i) priority

communicable diseases; (ii) sentinel surveillance; and (iii) institutional disease surveillance. The disease-specific surveillances are: (i) event-based surveillance–outbreak investigation and response; (ii) Nipah surveillance; (iii) acute meningo-encephalitis syndrome surveillance; (iv) hospital-based influenza surveillance in 12 (tertiary-care hospitals) sites; (v) national influenza surveillance–Bangladesh (NISB) in 14 district hospitals; (vi) sero-prevalence of antibodies to avian influenza A viruses among Bangladeshi workers of poultry markets; (vii) surveillance for human infections with avian influenza A viruses among workers of live bird markets and their household members in Dhaka city area; (viii) community-based avian/human influenza surveillance among poultry workers in H5-infected poultry farms; (ix) high-risk group surveillance in wet markets under Dhaka City Corporation; (x) surveillance for hospital-acquired respiratory infections in patients and healthcare workers in three tertiary-care facilities; (xi) web-based integrated disease surveillance up to upazila level; (xii) behavioral risk factor surveillance (BRFSS), Bangladesh, through telephone interviews; and (xiii) hospital-based dengue surveillance.

Web-based disease surveillance

The IEDCR uses e-connection with offices of civil surgeons and conducts web-based disease surveillance covering whole of Bangladesh. This surveillance has been extended to the upazila level. Data-entry is done directly at the sources in coordination with district- and tertiary-level hospitals. Private and NGO health facilities will also participate in this endeavor.

Training/workshop conducted by IEDCR

The training programs and workshops conducted by the institute in 2012 include: (i) Workshop on development of SOPs and updating of formats for integrated disease surveillance and early warning and alert system for epidemic-prone diseases; (ii) Workshop on review and updating of strategies and guidelines of integrated disease surveillance, with development of plan of action: Collaborative workshop with International Association of National Public Health Institutes (IANPHI) on mentorship and scientific writing; (iii) Refreshers' training for physicians on SOPs on avian influenza in humans; (iv) Training of medical technologists on detecting emerging and re-emerging diseases; (v) Orientation of physicians, nurses, and auxiliary staff on the strategy and guideline for prevention, control, and management of Nipah and other encephalitis; (vi) Advocacy of multisectoral partners for International Health Regulations (IHR) 2005 at all levels, including Points of Entries

(POEs); (vii) Workshop on development of strategy and guideline for networking among public health laboratories; (viii) Orientation of journalists on emerging diseases; (ix) Training of Upazila Rapid Response Team (URRT) members on emerging infectious diseases and outbreak investigation; (x) Training of physicians of urban health centers on emerging infectious diseases; (xi) Consultative meeting for inclusion of IHR in medical curriculum; (xii) Advocacy on IHR 2005 for policy-makers, health service providers, and relevant agencies at all levels; (xiii) Advocacy meetings on IHR 2005 for academicians of all medical colleges and Institutes; (xiv) Capacity-building of technical personnel under health, customs and immigration of PoEs for implementation of IHR 2005 at designated PoEs; (xv) Capacity-building of NGO/DCC health personnel for strengthening wet market surveillance in city corporation areas; (xvi) Workshop on Envisioning One Health for emerging infectious diseases and beyond: Developing the country-level strategic framework and multi-year roadmap for Bangladesh; and (xvii) Workshop on One Health for infectious diseases in Bangladesh: Validating the country-level strategic framework and action plan.

Other activities done by IEDCR in 2012

The IEDCR developed a number of policies, strategies, and guidelines in 2012, which include (i) Strategy and guideline for prevention, control and management of Nipah and other encephalitis; (ii) Strategy and guideline for networking among public health laboratories; (iii) SOP for web-based integrated disease surveillance and early warning and alert system for epidemic-prone diseases; (iv) Guidelines on emerging infectious diseases and outbreak investigation; (v) Updating health rules for designated POEs and national legislation, regulations, and other instruments for IHR 2005 implementation; and (vi) Development of new law for implementations of IHR 2005.

Research conducted by IEDCR in 2012

The research conducted by IEDCR include: (i) Assessment of the impact of hepatitis B vaccination in Bangladesh, a sero-prevalence study; (ii) Safety and efficacy of Liposomal Amphotericin B (Ambisome) in Bangladeshi patients with visceral leishmaniasis—a Phase III clinical trial; (iii) HIV, syphilis, and hepatitis among pregnant women in selected health facilities of Greater Sylhet area in Bangladesh; (iv) Assessment of the vulnerability of population and the health system in Bangladesh to the impact of climate change; (v) Effectiveness of *Ghagra shak* poisoning prevention campaign in Sylhet: a post-test only intervention control study; (vi) Mitigating the impact of climate change to

reduce the burden of climate-sensitive illnesses; (vii) Assessing prevalence and risk factors of mild/asymptomatic H5N1 infections among persons exposed to H5N1-infected poultry; and (viii) Estimate the risk of mild human infection among persons exposed to H5N1-infected poultry.

International affiliation

The IEDCR is a member of the International Association of National Public Health Institutes (IANPHI), Global Outbreak Alert Response Network (GOARN). Director of IEDCR is a member of the Executive Board of IANPHI. He is also a member of the 11-member Emergency Committee on Middle East Respiratory Syndrome Coronavirus (MERS-CoV) formed by Director General of WHO. The IEDCR has collaborative activities with the Centers for Disease Control and Prevention (CDC) of USA, Rockefeller Foundation, icddr, and IANPHI. The

Institute is supported by WHO, UNICEF, FAO, USAID, and other UN and international agencies.

Institute of Public Health

Mandate of IPH

The Institute of Public Health (IPH), established in 1953, is responsible for ensuring the (i) quality of food and water; (ii) production of vaccines, intravenous fluids, antisera and diagnostic reagents; and (iii) diagnosis of infectious diseases. The IPH also conducts research on the above disciplines. The IPH has five major sections and several units to perform the different activities.

Production of intravenous fluids

Table 14.1 shows the quantity of different types of intravenous fluids produced by IPH over the last 5 years (2008 to 2012).

Table 14.1. Production of intravenous fluids by IPH over the last 5 years (2008 to 2012)

Item	Pack- size (ml)	2008	2009	2010	2011	2012
Glucose saline	1000	130799	107724	87040	70700	81590
	500	110179	180489	139630	143225	127255
Glucose aqua	1000	134416	86243	85894	66225	82810
	500	110006	154894	125044	120235	112785
Normal saline	1000	-	50978	64471	51078	61471
	500	54379	91854	93291	101394	12179
Cholera saline	1000	108521	80665	67440	56367	81227
	500	69401	135443	121350	107320	124120
P.D. fluid	1000	52481	46085	30110	21192	25530
	500	3640	-	-	-	-
3% Normal saline	1000	-	-	-	-	-
	500	7700	10674	7740	11130	10479
Baby saline	1000	-	-	-	-	-
	500	-	26120	21560	30475	36355
Hemodialysis fluid	1000	10500	12600	8150	7830	10700
	-	-	-	-	-	-
Hartman's Solution	1000	-	-	-	-	-
	500	97752	144943	124040	110305	86465

Production of blood bags and related accessories

Table 14.2 shows the quantity of blood-bags and related accessories produced by IPH over the last 5 years (2008 to 2012).

Production of antirabies vaccines

Table 14.3 shows the quantity of antirabies vaccines produced by IPH over the last 5 years (2008 to 2012).

Production of diagnostic reagents

Table 14.4 shows the quantity of different types of diagnostic reagents produced by IPH from 2008 to 2012.

Production of oral rehydration salt

Figure 14.1 shows the quantity of oral rehydration salt (ORS) produced and distributed by IPH from 2008 to 2012.

Table 14.2. Production of blood-bags and accessories by IPH over the last 5 years (2008 to 2012)

Item	Pack type	2008	2009	2010	2011	2012
CPD blood-bag	Single	55060	85800	83890	62272	105523
Baby bag	150 mL	-	-	-	1400	-
Transfusion set	-	7925	-	-	3800	58000
Infusion set	-	30400	-	-	10200	75600

Table 14.3. Production of antirabies vaccine by IPH over the last 5 years (2008 to 2012)

Year	For humans (5 mL)			For animals (10 mL)		
	mL	Ampoule	Course	mL	Ampoule	Course
2008	3017125	2848440	83793	619620	51990	15570
2009	2895500	579100	41365	543800	54380	1295
2010	2296100	459220	32802	324500	32450	773
2011	1296370	259274	18519	368300	36830	877
2012	476800	95360	6811	124300	12430	296

Table 14.4. Production of diagnostic reagents by IPH (2008 to 2012)

Item	2008	2009	2010	2011	2012
Benedict's Solution (L)	560	457	460	360	400
ESR fluid (L)	380	155	160	89	160
20% Sulfuric acid solution (L)	40	-	30	NIL	-
N/10 Hydrochloric acid solution (L)	190	60	50	41	70
Acetone alcohol (L)	-	25	NIL	NIL	60
5% Acetic acid solution (L)	170	80	40	10	60
WBC fluid (L)	100	70	100	NIL	40
RBC fluid (L)	80	-	30	NIL	30
30% Suplhosalicic acid (L)	07	-	04	NIL	10
20% Sodium hydroxide solution (L)	-	-	NIL	NIL	-

Table 14.4. continued

Item	2008	2009	2010	2011	2012
20% Potassium hydroxide solution (L)	-	-	NIL	NIL	-
Semen analysis fluid (L)	-	-	NIL	10	-
Normal saline (L)	60	90	20	NIL	20
Methylene Blue (L)	25	20	05	NIL	-
Crystal violet (L)	05	27	05	NIL	-
Basic fuchsin (L)	22	5	NIL	NIL	10
Carbol fuchsin (L)	-	22	NIL	NIL	-
Gram iodine (L)	-	20	NIL	5	5
Lugol's iodine (L)	40	50	05	15	10
Leishman stain (L)	104	61.8	43	49.6	72
Giemsa stain (L)	60	71.9	34	43	40
Glucose kits	100	100	NIL	48	99
Bilirubin kits	151	96	97	NIL	149
Creatinine kits	69	47	49	250	298
Uric Acid kits	-	-	NIL	NIL	-
EDTA vials	-	511	NIL	NIL	-
Urea kits	-	-	-	48	50

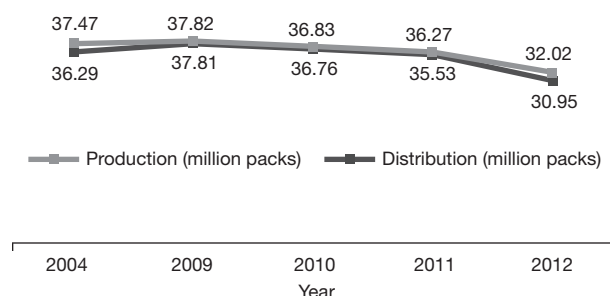


Figure 14.1. Production and distribution of oral rehydration salt (ORS) by IPH over the last 5 years (2008 to 2012)

Testing of food, water, drug and stool samples

Table 14.5 shows the number of food samples tested by IPH from 2008 to 2012. The table also shows the distribution of the genuine and adulterated samples out of the total samples tested each year.

Figure 14.2 shows the results of water samples tested by chemical means by IPH during 2008-2012.

Figure 14.3 shows the results of water-samples for which bacteriological test was done by IPH during 2008 to 2012.

Table 14.5. Food samples tested IPH over the last 5 years (2008-2012)

Year	Total samples	Genuine		Adulterated	
		No.	%	No.	%
2008	8734	5066	58.0	3668	42.0
2009	6338	3356	52.9	2982	47.1
2010	5749	2759	48	2990	52
2011	5812	2671	45.96	3147	54.04
2012	5322	2734	51.37	2558	48.63

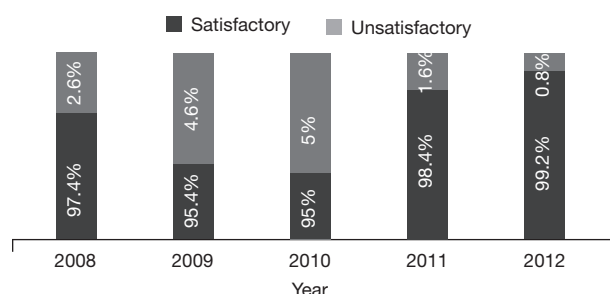


Figure 14.2. Result of water samples tested chemically by IPH in the last 5 years (2008 to 2012)

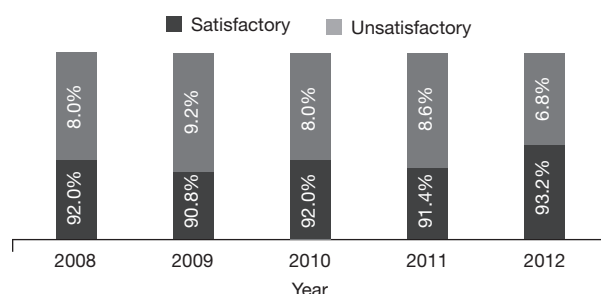


Figure 14.3. Result of water samples tested bacteriologically by IPH over the last 5 years (2008 to 2012)

Table 14.6 shows the number of drug samples received from 2008 to 2012 by IPH and their test results.

The National Polio Laboratory of IPH is a WHO-accredited laboratory established to assist the eradication of wild polio virus from the country. It is a partner of SEARO-WHO Polio Network. Table

14.7 shows the number of stool samples tested by IPH for polio virus from 2002 to 2012 and the results of the tests.

Serological tests for measles and rubella

The Measles Laboratory of IPH is involved with the serological study of measles and rubella to support measles control program in the country. Table 14.8

Table 14.6. Number of drug samples received and tested by IPH with their results (2008 to 2012)

Year	Samples received (N)	Satisfactory (N)	Unsatisfactory (N)	Not analyzed	Feedback given to senders
2008	4589	3639	100	-	-
2009	3145	3078	67	-	-
2010	5006	3833	82	1091	-
2011	3720	2583	104	1033	-
2012	4239	2276	95	1868	264

Table 14.7. Number of stool samples tested by IPH for polio virus from 2008 to 2012 and their results

Item	2008	2009	2010	2011	2012
AFP cases (N)	1809	1522	1541	1600	1570
Samples (N)	4356	3483	3464	3619	3450
Polio virus isolates (N)	80	56	72	75	84
Wild polio viruses (N)	-	-	-	-	-
Vaccine (Sabin) viruses (N)	76	56	72	75	84
NPEV (Non-polio enteroviruses) (N)	1012	684	645	638	489
Negative samples (N)	3264	2743	2746	2906	2877
Total (N)	10597	8544	8540	8913	8554

Table 14.8. Number of measles-positive, rubella-positive (IgM antibody) and total negative blood samples (both measles and rubella) tested by the Measles Laboratory of IPH from 2008 to 2012

Test	2008	2009	2010	2011	2012
Measles-positive	16	35	51	1788	714
Rubella-positive	243	1,133	1,425	672	481
Total negative	529	769	817	1633	1359
Total samples	788	1937	2293	4093	2590

Table 14.9. Number of routine tests done by IPH from 2008 to 2012

Test	2008	2009	2010	2011	2012
Biochemical (blood)	-	-	-	-	09
Serological	3,293	-	-	11043	6449
Routine examination (stool, blood-CP, urine, sputum)	123	82	162	157	79
Culture and sensitivity (stool, blood, urine, sputum, throat-swab, ear-swab)	98	78	30	-	-

shows the numbers of measles-positive, rubella-positive and negative blood samples tested by the Measles Laboratory of the Institute from 2008 to 2012.

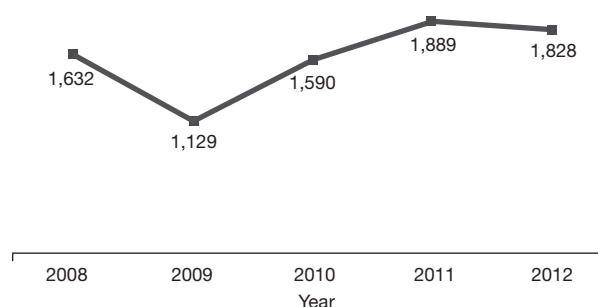
Routine blood, serum, stool, urine, sputum, throat-swab, and ear-swab tests

The IPH performs routine tests on blood, serum, stool, urine, sputum, throat-swab, ear-swab, etc. Table 14.9 shows a summary of the tests done on such samples by the Institute from 2008 to 2012.

Visits by medical and dental students

The IPH is a designated site for field visit by medical students in the country. Graduate medical and dental students from almost all medical colleges come to see the activities of IPH

for learning. Figure 14.4 shows the number of medical/dental students who visited IPH during 2008 to 2012).

**Figure 14.4. Number of medical/dental students who visited IPH during 2008 to 2012**

RESEARCH AND DEVELOPMENT

Important health topics researched by premier institutions

Bangladesh has huge scope for health research and development. There are several ongoing research projects on health, nutrition, and environment in collaboration with renowned public health universities in the world. Several research organizations, both in the private and public sectors, are conducting research on various health topics. In 2012, MIS-DGHS received information on research from several public, private and autonomous institutions, which include Bangladesh Medical Research Council (BMRC), Institute of Epidemiology, Disease Control and Research (IEDCR), National Institute of Preventive and Social Medicine (NIPSOM), Institute of Mother and Child Health (ICMH), International Centre for Diarrheal Disease Research, Bangladesh (icddr,b), and the James P. Grant School of Public Health under BRAC University.

Bangladesh Medical Research Council

Bangladesh Medical Research Council (BMRC) was established in 1972 by order of the President as an autonomous body under the Ministry of Health and Family Welfare (MOHFW). The objectives, rules, and regulations of BMRC were formulated by the Ministry's resolution in 1974 and 1976. The resolution states BMRC as the focal point for health research in Bangladesh. The main activities of BMRC include providing research fund, publication of journals and research bulletins, provision of training, and issuing ethical clearance. The list of BMRC seminars, research, and publications is provided in the annex.

International Centre for Diarrheal Disease Research, Bangladesh

The International Centre of Diarrheal Disease Research, Bangladesh (icddr,b) is an international health research institution located in Dhaka. Being dedicated to saving lives through research and treatment, the organization addresses some of the most critical health concerns ranging from improving neonatal survival to HIV/AIDS. In collaboration with academic and research institutions throughout the world, icddr,b conducts research, training, and extension activities as well as program-based activities to develop and share knowledge for global lifesaving

solutions. The organization translates knowledge from research into policy, using strategic health programs. This allows basic research to influence policy applications and actions rapidly if the evidence supports meaningful public-health benefit. Research priorities at icddr,b are cross-cutting, covering child health, infectious diseases and vaccine sciences, reproductive health, nutrition, population, HIV/AIDS, and safe water. The organization published several internal publications, journal, reports, and abstracts in 2012, which are listed in the annex.

James P. Grant School of Public Health, BRAC University

The James P. Grant School of Public Health (JPGSPH) under BRAC University was established in 2004 as an international educational and research institution focusing on the integral areas of teaching, research, and services. The goal of the School is not only to impart knowledge but also to act as a center of excellence in knowledge creation through research and training that connect with practice. Aside from the flagship educational programs, i.e. Masters of Public Health (MPH), JPGSPH also provides short courses on public health for health professionals through the Continuing Education Program (CEP). Additionally, JPGSPH possesses a burgeoning research portfolio conducting innovative and pioneering studies on public health issues funded by multiple international donors. JPGSPH also provides services in the form of training, advocacy workshops, and seminars with a special focus on the rights of the marginalized and vulnerable population of Bangladesh. The School serves as the secretariat for an initiative called Bangladesh Health Watch (BHW), a multi-stakeholder civil society advocacy and monitoring network dedicated to improving the health system in Bangladesh through critical review of policies and programs and recommendations of appropriate actions for change. The main activity of the Bangladesh Health Watch is to publish an annual report on the state of health in Bangladesh every year by commissioning researches. A working group consisting of researchers and activists from different organizations carry out various activities.

The list of studies and publications of JPGSPH is provided in the annex.

Institute of Epidemiology, Disease Control and Research

The Institute of Epidemiology, Disease Control and Research (IEDCR) conducted quite a good number of studies in collaboration with national and international public health institutions and universities. It has also PhD researchers. All research activities and publications by IEDCR are listed in the annex.

National Institute of Cardiovascular Diseases

The National Institute of Cardiovascular Diseases (NICVD) has been conducting three postgraduate courses under the University of Dhaka/BSMMU. The courses are Masters of Surgery (MS-Cardiovascular & Thoracic Surgery), Doctor of Medicine (MD-Cardiology), and Diploma in Cardiology (D-Card). Each student of the final part courses for MS and MD has to submit a thesis for final examination to the relevant university. This work is done under the guidance of a professor of the relevant department. The list of research activities by NICVD is provided in the annex.

Institute of Child & Mother Health

The Institute of Child & Mother Health (ICMH) is a national-level institute in Bangladesh committed to being a center of excellence in the South-East

Asia. The Institute is working for the improvement of health and nutrition of children and mothers in the country through its three objectives: human resource development, conducting research, and patient-care. This Institute was made autonomous through an Act in the Parliament in 2002 and is now administered through a Board of Governors. All research activities by ICMH are listed in the annex.

National Institute of Preventive & Social Medicine

The National Institute of Preventive & Social Medicine (NIPSOM), the only national-level public health institute under the University of Dhaka, Bangladesh, was established in 1978 with the aim to produce postgraduates in public health, capable of satisfying the needs of the community in promoting and restoring health. The Institute is also supporting various health policy formulations of the government and community health programs through research, training, and services. It conducts eight Masters of Public Health (MPH) courses, each of one-year duration and one M. Phil course of two-year duration. The NIPSOM's dissertation and research list was published in Health Bulletin 2012, which can be found in our website www.dghs.gov.bd. Students' dissertations for ongoing courses have not been completed yet. So, to avoid repetition, the earlier list is not published in this year's Health Bulletin.

HEALTH WORKFORCE SITUATION

Between 2011 and 2012, over 11,000 newly-sanctioned posts added under DGHS

This chapter covers the health workforce situation, particularly of DGHS. Summary of information of allied departments, viz. Directorate General of Family Planning, Directorate General of Drug Administration, and Directorate of Nursing Services is also provided. An overview of the number of sanctioned and filled-up posts, along with vacancies is presented; medical teaching/training institutions and programs, along with courses and training, have also been shown. Health workforce deployment and redeployment are ongoing processes; attrition due to death, retirement, resignation, termination, and migration; transfer, replacement, and filling-in are constantly occurring. Therefore, the status of health workforce as shown in this report may not remain the same by the time this health bulletin is published.

Overall health workforce situation in DGHS

Table 16.1 shows a summary of health workforce situation in DGHS. A division-wise distribution is shown in the annex to this chapter.

It is revealed from Table 16.1 that, out of 115,935 sanctioned posts under DGHS, about half (44.7%) are of class III category, with physicians (class I) comprising 19.1%, class II comprising 13.3%, and class IV comprising 22.4%. Of the available 93,317 health workforce, 47.1% are of class III, 17.1% are doctors (class I), 13.3% are of class II, and 22.3% are of class IV. The class I non-doctors comprise 0.4% of the sanctioned posts and 0.2% of the available staff. Table 16.1 also shows that 22,618 sanctioned posts remained vacant as of December 2012, which constituted 20% of the total sanctioned posts. Vacancy rate was 28% (6,198 posts) for doctors, 15% (7,918 posts) for class III staff, 20% (5,218 posts) for class IV staff, 20% (3,010 posts) for class II staff, and 56% (274 posts) for class I non-doctors.

Figure 16.1 shows the percentage distribution of male and female staff under DGHS as of December 2012. Two-thirds (66.4%) of the total available staff members (n=93,317) were male and one-third (33.6%) were female. Among the doctors (class I),

Table 16.1. Number of sanctioned, filled-up and vacant posts under DGHS (December 2012)

Class	Sanctioned No. (%)	Filled-up				Vacant	
		Male No. (%)	Female No. (%)	Total No. (%)	Filled-up as % of sanctioned posts	No.	Vacant as % of sanctioned posts
Class I Doctors	22120 (19.1%)	12356 (19.9%)	3566 (11.4%)	15922 (17.1%)	72%	6198	28%
	Non-doctors 490 (0.4%)	163 (0.3%)	53 (0.2%)	216 (0.2%)	44%	274	56%
Class II	15421 (13.3%)	968 (1.6%)	11443 (36.5%)	12411 (13.3%)	80%	3010	20%
Class III	51878 (44.7%)	33152 (53.5%)	10808 (34.5%)	43960 (47.1%)	85%	7918	15%
Class IV	26026 (22.4%)	15358 (24.8%)	5450 (17.4%)	20808 (22.3%)	80%	5218	20%
Total	115935 (100.0%)	61997 (100.0%)	31320 (100.0%)	93317 (100.0%)	80%	22618	20%

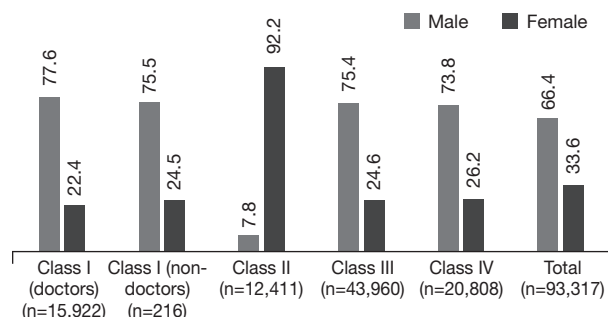


Figure 16.1. Percent distribution of male and female staff in DGHS (December 2012)

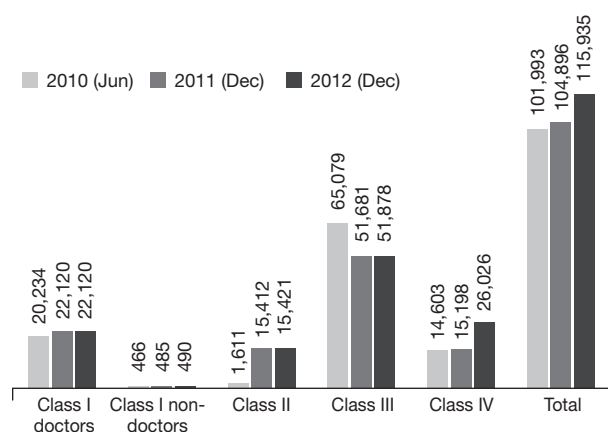


Figure 16.2. No. of sanctioned pots of doctors by year

three-quarters (77.6%) were male, and one-quarter (22.4%) were female. Class I non-doctors also had similar distribution (75.5% male vs 24.5% female). Of the class II staff, completely reverse picture was seen (7.8% male vs 92.2% female). Among the class III staff, 75.4% were male, and 24.6% were female. Among the class IV staff, 73.8% were male, and 26.2% were female.

Figure 16.2 shows the total number of sanctioned posts of DGHS in 2010, 2011, and 2012. Between 2011 and 2012, there was an increase of 11,039 posts comprising 10,828 class IV posts, 197 class III posts, 9 class II posts, and 5 class I posts for non-doctors.

Administrative, managerial, academic and clinical positions

Table 16.2 shows the number of sanctioned, filled-up and vacant posts at administrative, managerial, academic and clinical positions under DGHS as of December 2012.

Medical technologists

Table 16.3 shows the number of sanctioned, filled-up and vacant posts of medical technologists as in three years (2010, 2011, and 2012).

Figure 16.3 shows the percentage of vacancies among different disciplines of medical technologists under DGHS as of December 2012. Overall, 21% posts were vacant.

Table 16.2. Number of sanctioned, filled-up and vacant posts at administrative, managerial, academic and clinical positions under DGHS (December 2012)

Post	Sanctioned	Filled-up		Vacant	
		No.	Sanctioned (%)	No.	Sanctioned (%)
DG/ADG/equivalent	6	5	83	1	17
Director/Principal/Vice Principal/equivalent	89	85	96	4	4
Deputy Director/equivalent	105	84	80	21	20
Assistant Director/Civil Surgeon/equivalent	203	156	77	47	23
Deputy Civil Surgeon/UHFPO	505	485	96	20	4
Professor	470	275	59	195	41
Associate Professor	686	395	58	291	42
Assistant Professor	946	771	82	175	18
Senior Consultant	439	226	51	213	49
Senior Lecturer	8	8	100	0	0
Junior Lecturer	32	27	84	5	16
Junior Consultant/equivalent	3829	1972	52	1857	48
Assistant Surgeon/equivalent	14802	11433	77	3369	23
Total	22120	15922	72	6198	28

Table 16.3. Number of sanctioned, filled up and vacant posts of medical technologists by discipline in three years

Month/ Year	Post	Pharmacy	Lab	Radiography	Radiotherapy	Physiotherapy	Dental	Total
2010 (Jun)	Sanctioned	2881	1776	713	56	199	525	6150
	Filled-up	2115	1349	634	31	33	496	4658
	Vacant	766	427	79	25	166	29	1492
2011 (Dec)	Sanctioned	2934	1990	715	57	201	531	6428
	Filled up	2174	1613	635	38	147	498	5105
	Vacant	760	377	80	19	54	33	1323
2012 (Dec)	Sanctioned	2934	1990	715	57	201	531	6428
	Filled up	2172	1610	634	38	147	495	5096
	Vacant	762	380	81	19	54	36	1332

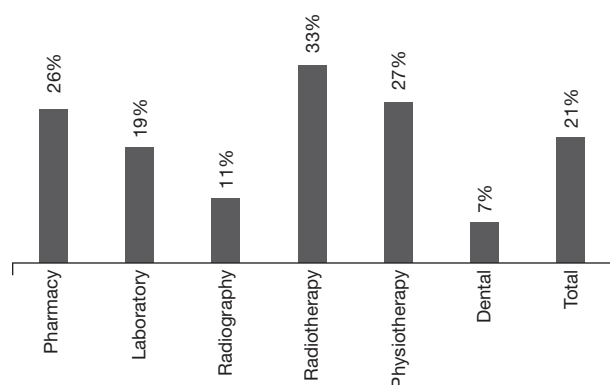


Figure 16.3. Percentage of vacancies among different disciplines of medical technologists under DGHS (December 2012)

Sub-assistant Community Medical Officers (SACMOs)

Table 16.4 shows the number of sanctioned, filled-up and vacant posts of Sub-assistant Community Medical Officer (SACMO) in the three-year period (2010, 2011, and 2012) under DGHS. Percentage of vacancy has been reduced in 2011 and 2012 to a figure of 22% from 32% in 2010.

Domiciliary staff (Health Inspectors, Assistant Health Inspectors, and Health Assistants)

Table 16.5 shows the number of sanctioned, filled-up and vacant posts of domiciliary staff (Health Inspectors, Assistant Health Inspectors, and Health Assistants) under DGHS in the three-year period (2001, 2011, and 2012). The rate of vacancy has been dropped to 9% in 2011 and 2012 from 12% in 2010.

Table 16.4. Number of sanctioned, filled-up and vacant posts of Sub-Assistant Community Medical Officer (SACMO) in the three-year period (2010, 2011, and 2012) under DGHS

Year	No. of posts			Vacancy (%)
	Sanctioned	Filled-up	Vacant	
2010 (Jun)	5411	3694	1717	31.7
2011 (Dec)	5411	4216	1195	22.1
2012 (Dec)	5411	4212	1199	22.2

Officers and staff in alternative medicines

Table 16.6 shows the number of sanctioned, filled-up and vacant posts of various officers and staff in alternative medicines under DGHS as of December 2012. For medical officers, vacancy turned to 58%.

Directorate General of Family Planning

Table 16.7 shows the number of sanctioned, filled-up and vacant posts under the Directorate General of Family Planning (DGFP) in the three-year period (2010, 2011, and 2013). Percentage of vacancy has been increased compared to 2010.

Directorate of Nursing

Table 16.8 shows the number of sanctioned, filled-up and vacant posts under the Directorate of Nursing Services (DNS) in the three-year period (2010, 2011, and 2013).

Directorate General of Drug Administration

Table 16.9 shows the number of sanctioned, filled-up and vacant posts under the Directorate General of Drug Administration (DGDA) in the three-year

Table 16.5. Number of sanctioned, filled-up and vacant posts of domiciliary staff (Health Inspectors, Assistant Health Inspectors, and Health Assistants) under DGHS in the three-year period (201, 2011, and 2012)

Month/ Year	Post	Health Inspector	Assistant Health Inspector	Health Assistant	Total field staff	Vacancy (%)
2010 (Jun)	Sanctioned	1399	4198	20819	26416	11.9
	Filled-up	1124	3444	18717	23285	
	Vacant	275	754	2102	3131	
2011 (Dec)	Sanctioned	1399	4198	20815	26412	8.8
	Filled-up	1132	3669	19279	24080	
	Vacant	267	529	1536	2332	
2012 (Dec)	Sanctioned	1399	4198	20815	26412	8.9
	Filled-up	1126	3662	19274	24062	
	Vacant	273	536	1541	2350	

Table 16.6. Number of sanctioned, filled-up and vacant posts (revenue and development) of various officers and staff in alternative medicines under DGHS (December 2012)

Name of post	Sanctioned post		Fill-up post		Vacant post	
	Revenue	Development	Revenue	Development	Revenue	Development
Medical officers for unani medicine	51	122	-	15	51	90
Medical officers for ayurvedic medicine	51	122	-	15	51	90
Medical officers for homeopathic medicine	51	113	-	15	51	90
Compounders for alternative medicine	153	334	-	64	153	270
Herbal assistants for herbal gardens	-	467	-	467	0	0
Total	306	1158	-	576	306	540

Table 16.7. Number of sanctioned, filled-up and vacant posts under DGFP in the three-year period (2010, 2011, and 2013)

Year	Class	Sanctioned	Filled-up	Vacant	Vacancy (%)
2010 (Jun)	Class I	1925	1242	683	35.5
2011 (Dec)		1954	1049	905	46.3
2013 (Apr)		1954	1021	933	47.7
2010 (Jun)	Class II	1023	446	577	56.4
2011 (Dec)		1022	401	621	60.8
2013 (Apr)		1074	401	673	62.7
2010 (Jun)	Class III	40430	38906	1524	3.8
2011 (Dec)		16937	14646	2291	13.5
2013 (Apr)		16886	14760	2126	12.6
2010 (Jun)	Class IV	8959	8044	915	10.2
2011 (Dec)		32507	29845	2662	8.2
2013 (Apr)		32516	29103	3413	10.5

Table 16.8. Number of sanctioned, filled-up and vacant posts under the DNS in the three-year period (2010, 2011, and 2013)

Year	Category	Sanctioned	Filled-up	Vacant	Vacancy (%)
Class I					
2010 (Jun)	Nursing	161	6	155	96.3
	Non-nursing	1	-	1	100.0
2011 (Dec)	Nursing	171	2	169	98.8
	Non-nursing	1	-	1	100.0
2013 (Jun)	Nursing	174	1	173	99.4
	Non-nursing	1	-	1	100.0
Class II					
2010 (Jun)	Nursing	463	150	313	67.6
	Non-nursing	20	8	12	60.0
2011 (Dec)	Nursing	16207	14041	2166	13.4
	Non-nursing	20	7	13	65.0
2013 (Jun)	Nursing	21052	12609	8443	40.1
	Non-nursing	20	7	13	65.0
Class III					
2010 (Jun)	Nursing	16559	13327	3232	19.5
	Non-nursing	353	214	139	39.4
2011 (Dec)	Nursing	1375	837	538	39.1
	Non-nursing	353	202	151	42.8
2013 (Jun)	Nursing	1375	625	750	54.5
	Non-nursing	358	204	154	43.0
Class IV					
2010 (Jun)	Non-nursing	704	633	71	10.1
2011 (Dec)	Non-nursing	863	620	243	28.2
2013 (Jun)	Non-nursing	863	614	249	28.9

period (2010, 2011, and 2013). The vacancy for class I officers has been significantly dropped in 2013.

Institutions offering postgraduate medical degrees

Table 16.10 shows the number of institutions, both in the government and private sector, providing postgraduate medical degrees. Thirty-three institutions—23 in public sector and 10 in private sector—offer such degrees. The table also shows the titles of the courses offered by each institution, along with the number of seats in each course.

This report is based on information provided by Director, Medical Education, as of December

2011. One institution—Bangladesh College of Physicians and Surgeons (BCPS)—offers FCPS (Fellow of the College of Physicians and Surgeons) and MCPS (Member of the College of Physicians and Surgeons) degrees. Any eligible candidate can sit for the examinations, and results depend on the candidate's competence shown in the examinations. The number of seats is, therefore, variable. Other institutions offer courses, like MS, MD, M.Phil, Diploma, MPH, MTM, and MMED. The number of seats is shown in Table 16.10. The detailed list of the organizations with courses and number of seats is shown in the annex.

Figure 16.4 shows the number of doctors who obtained FCPS and MCPS degrees from the

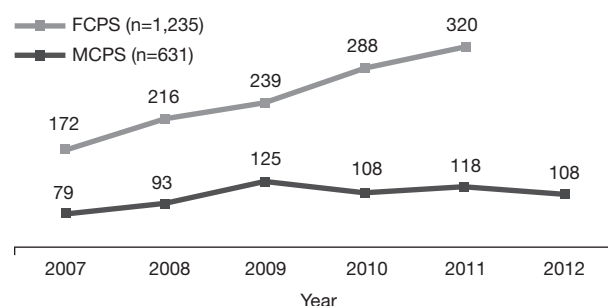
Table 16.9. Number of sanctioned, filled-up and vacant posts under the DGDA in the three-year period (2010, 2011, and 2013)

Class	Year	Class	Sanctioned	Filled-up	Vacant	Vacancy (%)
Class I	2010 (Jun)	Class I	118	28	90	76.3
	2011 (Dec)	Class I	118	27	91	77.1
	2013 (Jun)	Class I	118	78	40	33.9
Class II	2010 (Jun)	Class II	25	7	18	72.0
	2011 (Dec)	Class II	25	7	18	72.0
	2013 (Jun)	Class II	25	4	21	84.0
Class III	2010 (Jun)	Class III	115	58	57	49.6
	2011 (Dec)	Class III	115	55	60	52.2
	2013 (Jun)	Class III	115	53	62	53.9
Class IV	2010 (Jun)	Class IV	112	39	73	65.2
	2011 (Dec)	Class IV	112	40	72	64.3
	2013 (Jun)	Class IV	112	39	73	65.2

Table 16.10. Type of institutions offering postgraduate medical courses, with number of seats (December 2012)

Type of organization	No. of organizations	No. of seats							Total
		MS	MD	M. Phil	Diploma	MPH	MTM	MMED	
Government (BSMMU)	1	140	150	70	106	0	10	0	477
Government	22	312	360	242	478	185	0	15	1614
Private	10	21	38	15	95	0	0	0	169
Total	33	473	548	327	679	185	210	15	2270

Bangladesh College of Physicians and Surgeons (BCPS) from 2007 to 2012. Detailed data are given in annex.

**Figure 16.4. Number of FCPS and MCPS postgraduates by year**

Institutions offering graduate medical degrees

Table 16.11 shows the number of institutions, along with total number of seats both in the

government and private sectors, which offer MBBS degrees. Detailed list of institutions, with number of seats in each, is provided in the annex.

Table 16.11. Government and private institutions offering MBBS degree, with number of seats (December 2012)

Type of organization	No. of institutions	No. of seats
Government	23	2862
Private	64	4850
Total	77	7712

Institutions offering undergraduate dental degrees

Table 16.12 shows the number of institutions, along with the total number of seats, both in the

Table 16.12. Government and private institutions offering BDS degrees, with number of seats (December 2012)

Type of organization	No. of institutions	No. of seats
Government	9	532
Private	18	1065
Total	27	1597

government and private sectors, which offer BDS degrees. Detailed list of institutions, with number of seats in each, is provided in the annex.

Institutions offering degrees and diplomas in alternative medicines

Table 16.13 shows the list of academic institutions, along with the number of seats, both in the government and the private sectors, offering degrees and diplomas in alternative medicines.

Institutions offering nursing degrees

Table 16.14 shows the number of institutions, along with the number of seats, in both government and private sectors, offering different types of nursing degrees. Detailed list of institutions and the number of seats in each institution are shown in the annex.

Table 16.13. Institutions for teaching and training of alternative medicines in Bangladesh

Name of institution	Total	Govt.	Private	Duration of course	Duration of internship	Degree offered	No. of seats
Govt. Unani and Ayurvedic Medical College	1	1	0	5 years	1 Year	BUMS (Bachelor of Unani Medicine and Surgery); BAMS (Bachelor of Ayurvedic Medicine and Surgery)	50
Homeopathic Medical College	2	1	1	5 years	1 Year	BHMS (Bachelor of Homeopathic Medicine and Surgery)	50
Tibbia College/Unani Diploma College	12	1	11	4 years	6 Months	DUMS (Diploma in Unani Medicine and Surgery)	25*
Ayurvedic Diploma College	8	0	8	4 years	6 Months	DAMS (Diploma in Ayurvedic Medicine and Surgery)	-
Homeopathic Diploma College	41	0	41	4 years	6 Months	DHMS (Diploma in Homeopathic Medicine and Surgery)	-
Total	64	3	61				

*Only in government institutions

Table 16.14. Number of nursing institutions, along with the number of seats, offering different types of nursing degrees (December 2012)

Course	Ownership	Affiliation	No.	Seats
BSc	Government	MOHFW	7	700
		Armed Forces Medical Institute, Dhaka Cantonment, Dhaka	1	60
		Faculty of Nursing BSMMU, Dhaka	1	25
		Sub-total (Government)	9	785
	Private	Private (one institution has been closed)	13	430
		Sub-total (BSc in Nursing)	22	1215
Post-basic BSc	Government	MOHFW	4	500
	Private	Private	10	345
		Sub-total (Post-basic BSc in Nursing)	14	845

Table 16.14. continued

Course	Ownership	Affiliation	No.	Seats
Specialized	Private	Private	4	80
		Sub-total (Specialized education in Nursing)	4	80
Diploma	Government	MOHFW	43	2580
		Armed Forces Medical Institute	1	50
		Sub-total (Government)	44	2630
	Private	Private	47	1910
		Sub-total (Diploma in Nursing)	91	4540
		Total	131	6680

Table 16.15. Junior midwifery institutions, with number of seats in each (December 2012)

Division	Name of junior midwifery institution	No. of seats
Chittagong	1. Junior Midwifery Institute, Red Crescent Matrisadan Hospital, Chandpur	20
	2. Jemison Red Crescent Midwifery Institute, Agrabad, Chittagong	50
	3. Christian Hospital, Chandroghona, Rangamati	20
	4. Junior Midwifery Institute, Memon Hospital, City Corporation, Chittagong	30
Dhaka	5. Junior Midwifery Institute, Holy Family Red Crescent Hospital, Dhaka	60
	6. Junior Midwifery Institute, Shaheed Moyez Uddin Memorial Red Crescent Matrisadan Hospital, Bangla Bazar, Dhaka	20
	7. Junior Midwifery Institute, Kumudini Hospital, Mirzapur, Tangail	20
	8. Central Hospital Nursing Institute, Green Road, Dhanmondi, Dhaka	20
Khulna	9. Junior Midwifery Institute Ad-Din Matrisadan Hospital, Jessore	20
	10. Junior Midwifery Institute, Fatema Hospital, Jessore	20
Rajshahi	11. Junior Midwifery Institute, Christian Hospital, Bogra	20
Rangpur	12. Prime Nursing College, Rangpur	20
	Total seats	320

Table 16.16. Training institutions for production of community-based skilled birth attendants (December 2012)

Ownership	Type of facility	Location	No. of facilities
Government	CSBA institutions run by civil surgeon and attached with general hospital/district hospital	Narayanganj (WHO), Manikganj, Kishoreganj, Jamalpur, Habiganj, Gopalganj, Narsingdi, Nilphamari, Natore, Naogaon, Kurigram, Panchagarh, Gaibandha, Jhenaidah, Bagerhat, Rajbari, Madaripur, Munshiganj, and Chandpur	19
	Family Welfare Visitor Training Institute	Tangail (WHO), Barisal, Faridpur, Comilla (WHO), Kushtia, Sylhet, Rangamati, Dhaka, Rajshahi, Bogra, Dinajpur, and Khulna (WHO)	12
	CSBA institutions attached with nursing institutions	Noakhali, Jessore, Satkhira, Thakurgaon, Feni, Joypurhat, Pabna, Brahmanbaria, Netrakona, Chuadanga, Cox's Bazar, Patuakhali, Chapainowabganj, and Sirajganj	14
Private	CSBA Institution	Kumudini Hospital, Mirzapur, Tangail and Lamb Hospital, Parbotipur, Dinajpur	2
		Total	47

Institutions to produce midwives

There are 12 junior midwifery institutions in the private sector, with total seats of 320, to produce midwifery professionals (18-month course). Table 16.15 shows the list.

Training facilities for production of community-based skilled birth attendants

To facilitate attendance at childbirths by skilled health personnel, the Ministry of Health and Family Welfare has a program to produce community-based skilled birth attendants. There are 47 facilities—45 in the government sector and 2 in the private sector to provide such training. Table 16.16 shows the location of the training facilities.

Training schools for production of medical assistants

Medical assistants (now to be designated as Sub-assistant Community Medical Officer) assist the medical doctors posted at health facilities at the upazila health complex level and below. Medical assistants are produced by Medical Assistant Training School (MATS) through a three-year academic course comprising theoretical and

practical classes. Currently, there are 8 MATS in the government sector and 103 MATS in the private sector (total 111). Total annual production-capacity is 6,821, of which 716 are produced by the government MATS and 6,103 by the private MATS (Table 16.17). Detailed list of institutions, with the number of seats in each institution, is shown in the annex.

Institutes of Health Technology (IHT) for production of medical technologists

Medical technologists are laboratory technologists or staff responsible for technical jobs under the supervision of medical experts. There was an acute shortage of medical technologists in the country. However, a steady growth of private institutions has been noticed and, by now, there are 112 institutions (Table 16.18). Eleven government institutions and 97 private institutions offer diploma and/or BSc/MSc courses. Four institutions (government plus private) offer certificate course in medical technology. The total number of seats in diploma, BSc and MSc courses is 14,075, and that for certificate course is 180. The two have a grand total of 14,255 seats. The detailed list of IHTs, along with the number of seats in each, is given in the annex.

On-the-job training

Under the Operation Plan of In-service Training, a large number of health personnel and support staff receive on-the-job training each year. A summary of the types of training programs and the number of participants in these training programs is given in Table 16.19. Details of the on-the-job training are provided in the annex.

Table 16.17. Government Medical Assistant Training Schools (MATS), with the number of seats (December 2012)

Ownership	No. of MATS	No. of seats
Government	8	716
Private	103	6105
Total	111	6821

Table 16.18. Number of institutions of health technology, along with the number of seats (December 2012)

Ownership	Type of course	No. of institutions	Discipline	No. of seats
Government	Diploma	8	Lab (365); Radiology (305); Physiotherapy (300); Sanitary inspection (300); Dentistry (305); and Radiotherapy (120) <i>Note: Offspring of freedom fighters and tribal students have 41 reserved seats</i>	2419
Private	Diploma	82	Lab; Radiology; Physiotherapy; Sanitary Inspection; Dentistry; Pharmacy; and Radiotherapy <i>Note: Offspring of freedom fighters and tribal students have reserved seats</i>	10231
Government	BSc	3	Lab (60); Physiotherapy (85); and Dentistry (120)	265
Private	BSc + MSc	15	Lab (395); Physiotherapy (320); Dentistry (320); and others (115)	1160
Total (diploma + BSc + MSc)				14075
Government + Private	Certificate	4	Optometrist, refraction, ophthalmic assistant, ophthalmic nursing assistant, cath lab technician	180
Total		112	Grand-total	14255

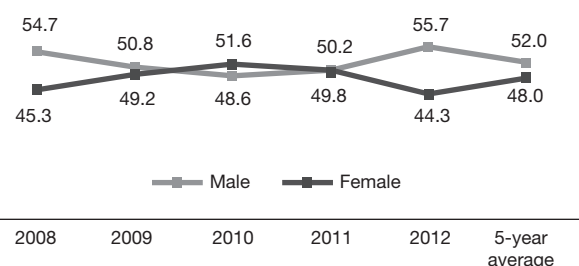
Table 16.19. Number of participants in on-the-job training given under Operational Plan of In-service Training in FY 2011-2012

Area/subject of the training/ workshop/seminar	Duration	No. of batches	No. of participants
Local training (short-term)			
Essential service delivery		652	15212
Management training		428	9585
Sub-total		1219	27665
Overseas training			
Different clinical specialties	1 week to >6 months	26	216
Different management and public health specialty	1 week to >6 months	12	120
Specialized overseas training/exchange visit	1-4 week(s)	0	225
Sub-total		38	561
Total		1257	28226

Yearly output from medical and dental colleges of Bangladesh

Table 16.20 shows the year-wise number of new doctors produced from various medical and dental colleges of Bangladesh.

Figure 16.5 shows the percentage distribution of new male and female medical and dental doctors produced from various medical and dental colleges during the 5-year period from 2008 to 2012. The number of male and female doctors is almost equal each year. Detailed data are provided in the annex.

**Figure 16.5. Percent distribution of new male and female doctors produced in Bangladesh during a 5-year period (2008 to 2012)****Table 16.20. Year-wise number of new doctors produced from various medical and dental colleges of Bangladesh**

Name of medical or dental college	No. of students graduated					Total
	2008	2009	2010	2011	2012	
Dhaka Medical College	207	147	132	178	202	866
Sir Salimullah Medical College	153	155	146	170	167	791
Rajshahi Medical College	147	139	170	164	205	825
Rangpur Medical College	83	-	185	23	214	505
Mymensingh Medical College	124	162	184	155	207	832
Chittagong Medical College	125	142	181	147	225	820
M.A.G. Osmani Medical College, Sylhet	127	161	160	155	203	806
Sher-e-Bangla Medical College	136	143	166	164	190	799
Faridpur Medical College	42	59	59	60	117	337
S.Z.R. Medical College, Bogra	46	-	56	93	130	325
Dinajpur Medical College	41	50	52	75	119	337
Khulna Medical College	47	61	46	80	116	350
Comilla Medical College	40	52	60	49	147	348
Dhaka Dental College	69	79	97	129	84	458
Chittagong Dental College	-	18	36	59	43	156
Rajshahi Dental College	-	37	33	13	47	130
Total	1387	1405	1763	1714	2416	8685

HEALTH INFORMATION SYSTEM

eHEALTH AND MBT

Ambitious initiative to prepare longitudinal health-records of citizens

The MIS-DGHS, in 2012, further scaled up eHealth, reaching up to the level of community clinics and community health workers, thus integrating more solutions and robustness to the IT system. This created an avenue to the standardization and enhancement of inter-operability of the system, with a vision to maintain electronic records of all citizens for their lifetime sharing. On 16 July 2013, Honorable Prime Minister Sheikh Hasina inaugurated the distribution of laptop computers to community clinics and handheld android tablet devices to community health workers. All community clinics and union-level health facilities located in every small rural community of Bangladesh have been provided laptop computers and broadband wireless Internet modems. This is going to create a new era of the population-based data collection and analysis through routine health information system for correctly understanding, planning, intervening, and reviewing the actual health problems prevailing at the community level. Telemedicine service and other innovative ICT-based citizen-centric health services, such as pregnancy and child registration and follow-up, citizens' grievance against management, health awareness creation, etc. will be massively scaled up. This chapter discusses current situation and

progress of various HIS and eHealth programs of MIS-DGHS. The HIS-EH operational plan has three components, viz, Health Information System, eHealth, and Medical Biotechnology (MBT). This chapter makes an overview of all three components.

Health Information System: New COIA Initiative

The health information system of MIS-DGHS collects data from various sources and cleans, analyzes, and summarizes the data to generate and distribute information through routine administrative reports, website, health bulletins, newsletters, etc. Data are collected electronically, covering all health facilities and health administrative points from the national to the community level. The community clinics and community health workers have been provided with laptop computers and android tablet devices. An initiative has been undertaken to register and track every pregnant mother and under-five child, using 11 core indicators (see annex to this chapter) suggested by the Commission on Information and Accountability (COIA) of the United Nations, of which the WHO is one of the key partners. The WHO has undertaken an initiative to measure the



Honorable Prime Minister Sheikh Hasina is seen distributing laptop computers to community clinics (August 2013)

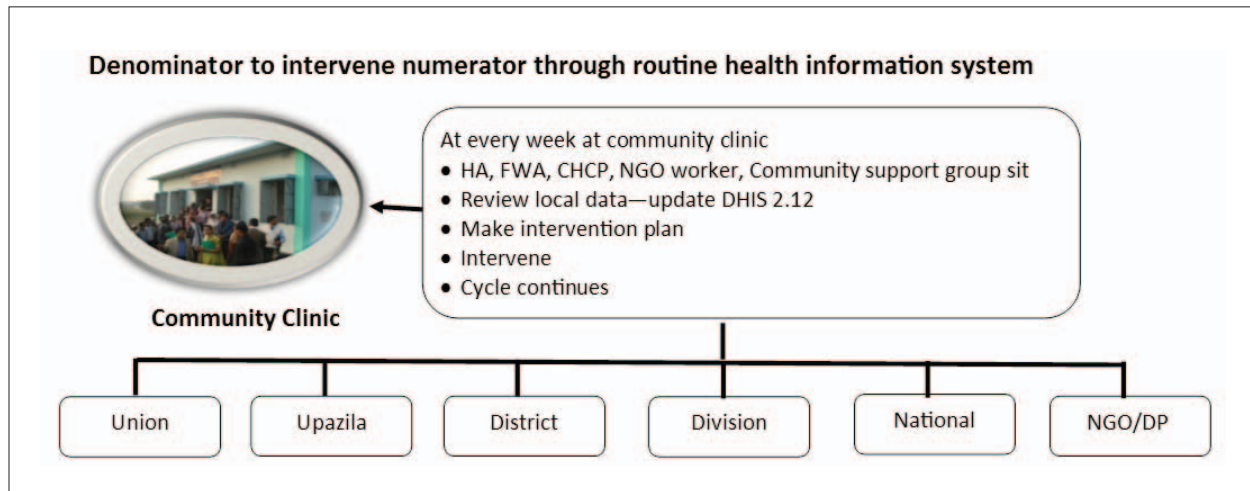


Figure 17.1. The COIA framework for registering and tracking every pregnant woman and every under-five child

progress, real time, of MDGs 4 and 5 through using 11 COIA indicators in 75 countries, including Bangladesh. This would need to reach the MDG 4 or 5 targets by 2015. Bangladesh is receiving technical assistance to pilot this initiative.

Although two upazilas have been planned for COIA pilot study, MIS-DGHS has made preparation to roll out this initiative to all community clinics. A commonly-shared online data-collection form has been designed and tested in District Health Information System (DHIS 2.12) (version 2.12), with assistance from icddr, b-Measure Evaluation and in consultation with line directors, community clinic project, MIS-DGFP, development partners (particularly UNICEF and WHO), and NGOs (Figure 17.1).

Community health workers and community clinics will register every pregnant woman and every under-five child living in the respective community catchments, using this form online. There will be a routine weekly meeting, in the community clinic, of the government community health workers (CHCP, HA, and FWA), NGO healthcare workers, and members of the community clinic management committee and community support group, where the local data will be reviewed, cleaned and further updated if required; maternal and child health situation will be reviewed, intervention plan for the next week will be chalked out and implemented. This routine cycle (Figure 17.2) will continue to track, follow-up, and improve maternal and child health. This local health review, planning, and intervention cycle will be scaled to other health parameters gradually.

Population register for lifetime shared health records of citizens—basic soft data nearly ready for use

The data collected on 120 million rural citizens, using machine-readable paper forms are now towards completion of digitization in searchable database format. These data, which comprise basic health-records of the citizens, will make the foundation of future lifetime shared health-records. With technical assistance from DFID, a software consortium comprising both local and non-local IT houses is now engaged in developing an integrated national eHealth enterprise architecture (eHEA) (Figure 17.2) that will combine individual records of all citizens, lifetime health records, organizations registries, information on public health programs (DHIS 2.12), hospital information system (Open MRS), health workforce registry (HRIS), and inventory system for major equipment to virtually function as one system through a data exchange system, popularly known as Electronic Health Information Exchange (eHIE). All hospitals, no matter public or private, will have the ability to post records of each patient to the national data warehouse through eHEA. The first usable version will be ready by one year. The eHEA will work in unison with the electronic Civil Registration and Vital Statistics System and National Population Register System to be jointly developed by the MOHFW, Ministry of Local Government, Rural Development and Cooperatives (MOLGRDC), and National Statistics Office (BBS). For this reason, MIS-DGHS is working with support from the UN-ESCAP with the Birth and Death Registration Project of MOLGRDC, and BBS. Geo-location and organization registries have already been completed and are downloadable from DGHS

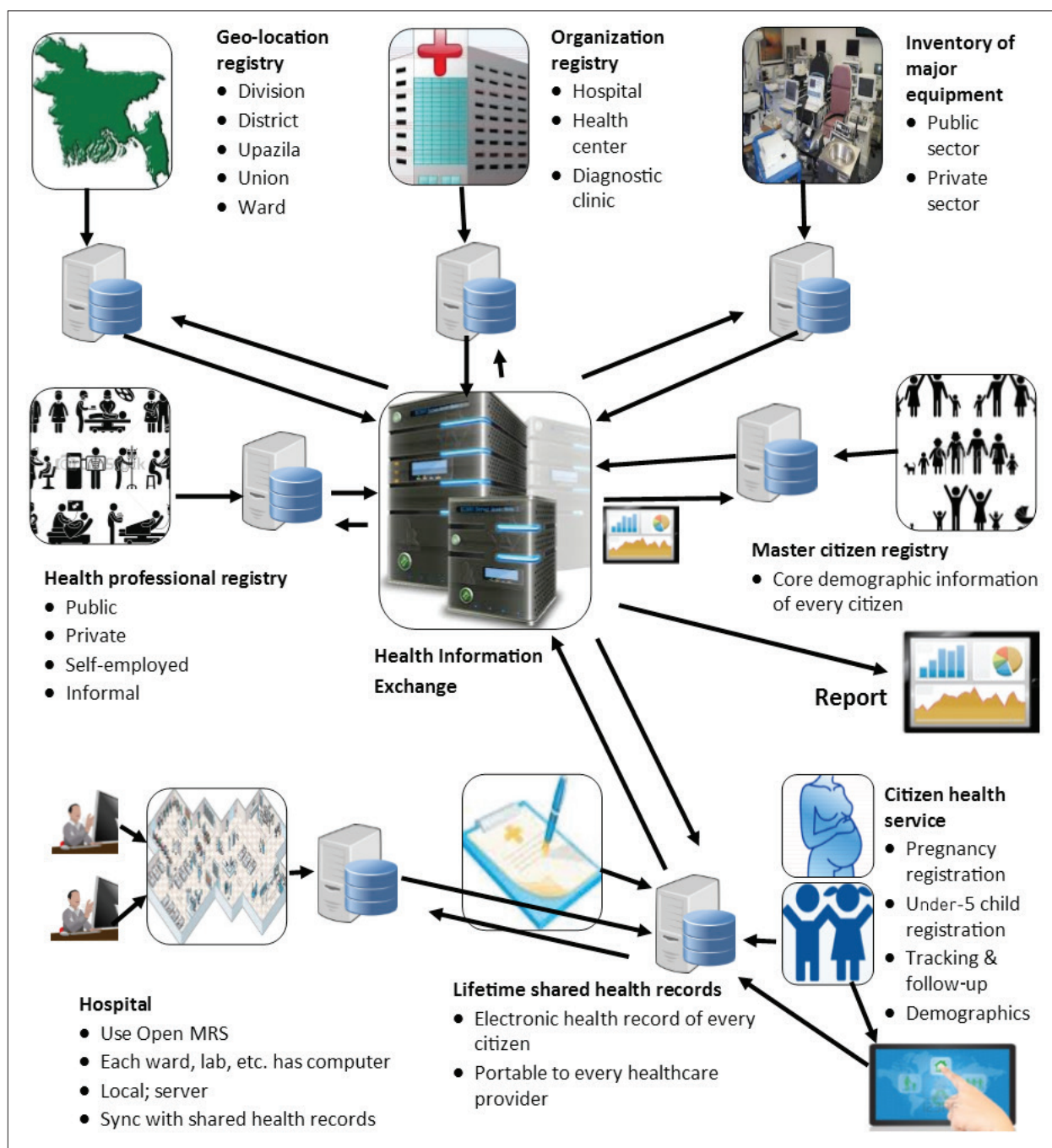


Figure 17.2. Proposed design of the eHealth Enterprise Architecture which is being developed with technical assistance from DFID

website (www.dghs.gov.bd). Health professional registry for DGHS is almost completed. Other organizations, ministries, and agencies in private sector will be added incrementally.

National eHealth Policy and Strategy

There is a growing international effort to develop long-term country-specific vision, goal, policy,

and strategy for eHealth. To keep pace with global trend, MIS-DGHS hired an international expert, with technical assistance from DFID to draft the National eHealth Policy and Strategy. WHO-ITU has developed a standard toolkit, with guidance on how to proceed step-by-step to build up national consensus and develop a well-represented national eHealth strategy.

Remote data disaster recovery center as back-up for the national health data center

For the planned future integrated and large-scale national eHealth Enterprise Architecture, the MIS-DGHS requires a robust, highly-secured, and never-sleep data center with plenty of storage space. Such a world-class state-of-the-art data center already exists at the MIS-DGHS. Equipped with RAID servers, firewalls, VMware, underground cable system, automatic fire protection, and humidity control, four tiers of power supply system, antispay and antihacking system to prevent unauthorized entry, remote monitoring system, text alerts by



Figure 17.3. The data center of MIS-DGHS at headquarter (left); Disaster Recovery center in Khulna (right)

mobile phone, etc., the data center has further been improved with Internet bandwidth load management systems, new servers, additional storage, sever intelligence systems, etc. (Figure 17.3). In addition to multiple methods of existing data back-up system, a data disaster recovery system has been established in Khulna, 400 kilometers away from the main data center in Dhaka.

Strengthening local information culture

Local Health Bulletins have been well-appreciated. Annual MIS Conference is helping improve data quality of Local Health Bulletins and DHIS 2.12 (Figure 17.4).

Publication of online Local Health Bulletins by each health organization under DGHS was a remarkable milestone of fiscal 2011-2012. The initiative was further vitalized in fiscal 2012-2013. About 550 health organizations at all levels (national and subnational down to upazila) published their respective health bulletins online, using the web-based template provided by MIS-DGHS. These Local Health Bulletins are accessible through MIS-DGHS website (www.dghs.gov.bd). Thanks to our talented developer who created a reporting tool of data summary that enables display of aggregate reports interactively by geographic location or by type of organization. At the organization level, the respective copies of

Local Health Bulletin are printed, multiplied for wider circulation, and a local publication ceremony is organized. This creates an indirect pressure on the local authority to remain careful about data quality because, in the ceremony, they have to answer about the conflicting data. For further improvement in data quality, the MIS-DGHS organizes Annual MIS Conference where all local health managers, local MIS focal points, and

local statistical staff as well as technical experts from MIS-DGHS, development partners, and major NGOs remain present. In the Annual MIS Conference, open discussion and critical expert and peer-review are held on Local Health Bulletins on year-wise case-by-case basis. Experts provide feedback and technical guidance to the local authorities to improve data and reporting quality of the respective Local Health Bulletins. In respective divisional headquarters, the Annual MIS Conferences for health organizations and hospitals under the divisions (divisional, district and upazila levels) were held. So, there were seven divisional Annual MIS Conferences outside Dhaka. In Dhaka, the Annual MIS Conference for all the tertiary hospitals in the country was held in 2013. In total, eight Annual MIS Conferences were held. Local Health Bulletins 2013 have new elements, viz. exhaustive report on equipment, morbidity, and mortality, using ICD 10 (International Classification of Diseases, version 10) codes. Local Health Bulletins are increasingly drawing attention of the senior policy-makers and development partners as a source of subnational health data for making location-specific plans and decisions.

Human resource information system (HRIS)—a complete overhauling done

The need of detailed human resource information is increasingly being felt for making long-term human resource plan, production, placement, and management. It is felt that it is important to know information on the health workforce of MOHFW and of other ministries and the private sector, inclusive of on-the-job and self-employed ones. The HRIS should include even informal healers as a reasonable number of people in Bangladesh consult informal healers for health healthcare. Making and maintaining

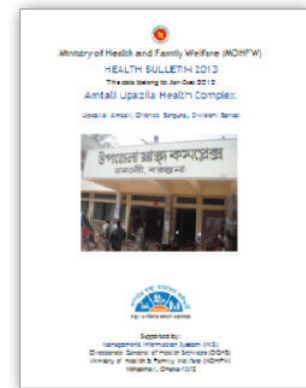


Figure 17.4. Each health organization publishes an online Local Health Bulletin

an integrated HRIS in Bangladesh is a challenging task as the existing human resource management processes among different ministries, agencies, and organizations are diverse and are managed by thousands of national and subnational authorities. Standardization and harmonization of data will be a fundamental requirement for managing the HRIS through a single inter-operable electronic system. Currently, several vertical HRISs are being operated by different agencies of MOHFW. The systems are not inter-operable for using different definitions of data and coding for the same data elements; none of these can track relationship between sanctioned and filled-up posts; none of these have health workforce information of other ministries or of the private sector. All HRISs are updated at periodic interval by the respective health organizations and suffer inadequate compliance. Therefore, the existing systems are incapable of satisfying full and timely need of human resource information. For a solution, MIS-DGHS has developed an iHRIS (integrated Human Resource Information System) as building block of a future large-scale integrated national eHealth Enterprise Architecture (eHEA). MIS-DGHS reviewed several globally-renowned human resource software, viz. iHRIS, OrangeHRM, etc. However, these software did not appear suitable to manage the complexities and diversities of health workforce management processes at MOHFW and other organizations. In this context, our developers built own customized system as building block of the national eHEA. Data are now being imported from the existing legacy databases–HRISs. Sanctioned posts of DGHS country-wide are already entered in the new system. This system meets international standards and inter-operability framework and is ready for capturing health workforce data under any ministry, agency, sector, or organization, inclusive of self-employed care providers and informal healers incrementally. The software is capable of producing disaggregated report of any type. The software has been built on the framework of process automation so that all human resource information can be updated at real time as the process will happen. However, making a good software or system is not all–the implementation by strong policy support will be the key to yielding results of the new iHRIS.

Procurement portal of MOHFW is in full use

The procurement portal of MOHFW, built with technical assistance from Management Sciences for Health (MSH) supported by USAID, is in full use by now. All procurements under HPNSDP (2011-2016), beginning from fiscal 2013-2014 are being done by the online procurement portal through shared use by line directors, MOHFW, CMSD, and World Bank. The initial experience shows significant speeding up of different steps of procurement processes.

Monthly and annual reporting for Cabinet Division

The MIS-DGHS provided routine monthly and annual report to Cabinet Division of the Government of Bangladesh, using the standard proforma. The report contains exhaustive information items on overall health sector.

Data on different health programs and organizations

In addition to the routine data from various organizations and health facilities, the MIS-DGHS collected specific program and performance-related data from Community Clinics (CCs); Essential Service Delivery (ESD); Maternal, Neonatal, Child and Adolescent Health (MNCAH); Communicable Disease Control (CDC), Non-communicable Disease Control (NCDC); National Nutrition Service (NNS); National Tuberculosis Control Program (NTP); and from organizations and agencies, such as Institute of Epidemiology, Disease Control and Research (IEDCR); Institute of Public Health (IPH); National Institute of Preventive and Social Medicine (NIPSOM); postgraduate teaching institutes and hospitals; Directorate General of Family Planning (DGFP); Directorate General of Drug Administration (DGDA); Directorate of Nursing Services (DNS); and from a number of non-governmental organizations. Health Bulletin 2013 summarizes those data.

Data collection on shared software resource and use

One of the successes of our HIS initiatives is: increasing convergence of all vertical programs or line directors to use our shared resource (data center and DHIS 2.12) for collection of program data. The shared data include those from community clinics collected by Community Clinics Project; data on obstetric care, IMCI, and EPI by MNCAH; data on tuberculosis by Tuberculosis Control Program; data on malaria, kala-azar, rabies by CDC; data on HIV/STD by NASP and partners; data on nutrition by NNS; data on pregnancy and under-5 child tracking by Community Clinics Project, DGFP, MNCAH, NNS, NGOs, and DPs; data on cervical cancer and breast cancer screening by an MOHFW-supported program based in BSMMU; data on obstetric fistula screening and care by National Fistula Program; data on neonatal health surveillance program by BSMMU; urban health data managed by DMIS, etc. MIS-DGHS is helping IEDCR in customizing DHIS 2.12 for disease surveillance. The list is growing further. Moreover, MIS-DGHS has developed a Progress Monitoring software under Annual Development Program (ADP) for MOHFW, which is now being used routinely by all line directors and MOHFW, and also by MOF and WB through i-bus.

Improving GIS-based data reporting

In the growing list of geo-spatial data, which were earlier up to upazila level, the MIS-DGHS started collection of geo-spatial data of community clinics with cooperation from Community Clinics Project in the last fiscal year. Over 90% of the community clinics have been covered by now. The MIS-DGHS has given more attention to improving GIS (geographical information system)-based reporting through its website. The readers are invited to visit www.dghs.gov.bd to have GIS information service.

DGHS website redesigned

The website of MIS-Health remains a vibrant platform for information dissemination as a focal point for the DGHS. The website has been completely redesigned to adjust with the new trend and taste. Social media interfaces, viz. Facebook, Twitter, Google+, YouTube Channel, Blog, etc. have been added to attract wide range of audiences.

Dissemination of information and publications

Several seminars and discussions were held to disseminate information and progress. Annual MIS Conference 2013 held in eight locations (seven in divisions and one in MIS-DGHS office) were attended by more than 2,000 health managers, MIS focal points, statisticians, and experts from across the country. Print and electronic media featured our success stories a number of times. The website has been completely redesigned and multiple social media channels added. A separate menu on "Citizens' Right to Information" to promote the cause of "Right to Information Act 2009" has been added to the website. Routine publications, like Health Bulletin, IMCI Newsletter, EOC Newsletter (Voice of MIS), etc. have been continued. A separate publication called "Annual Report 2012" to comply with guidance of Information Commission was printed and distributed. Online Local Health Bulletins have met further subnational information need as before.

eHealth

The MIS-DGHS uses the term "eHealth" to describe health services to citizens delivered through the use of ICT. Like the HIS, the eHealth service being provided by MIS-DGHS has also seen new dimensions as described in the following sections.

Mobile phone health service being rolled out

The mobile phone health service, being provided by each upazila health complex and each district hospital (grand total 482), has been continued as a sustainable service. Each of the hospitals has a mobile phone which an on-duty doctor carries

and is accessible for incoming calls round-the-clock. People living in the catchment areas call the doctor, if need arises, and the doctor answers to give appropriate medical advice. The service is free of charge. The service could be more helpful if rolled out further down to health workers who live close to the people's houses. Given this context, it has been planned to engage the community healthcare providers (CHCPs) also to provide mobile phone health service. As of now, 12,527 community clinics were running, and each by a CHCP. The mobile phone health service received recognition through ICT4 Development Award (2010) and special mention in Manthan India Award (2011).

Telemedicine-28 new centers added

In fiscal 2009-2010, equipment for 8 telemedicine centers and one coordination center at MIS-DGHS were procured, resulting in setting up of telemedicine centers in 8 hospitals of Bangladesh. Honorable Prime Minister of Bangladesh Sheikh Hasina formally inaugurated the telemedicine service on 6 July 2011 from the National Digital Innovation Fair held in Bangabandhu Novo-theater. In 2012, ten more telemedicine centers in 10 different hospitals were opened. In 2013, another 10 new telemedicine centers were established in 10 different hospitals. Therefore, current number of telemedicine centers is 28. In fiscal 2013-2014, additional 15 telemedicine centers are planned. These telemedicine centers are equipped with high Internet bandwidth, large screen display, good-quality telemedicine camera, and telemedicine peripherals. The Bangladesh Computer Council is expected to provide equipment support to MIS-DGHS for establishing more telemedicine centers in Bangladesh.



Hon'ble Prime Minister Sheikh Hasina inaugurated the telemedicine service of MIS-Health on 6 July 2011 at the Digital Innovation Fair 2011 organized by the Access to Information Program of Prime Minister's Office. The Hon'ble Minister for Health and Family Welfare Professor A.F.M. Ruhul Haque and Professor Dr. Abul Kalam Azad, Additional Director General (Planning & Development) and Director, MIS-DGHS, were present

All rural citizens covered by telemedicine

All functioning community clinics and union health centers have been brought under coverage of Internet connectivity through provision of one laptop and one broadband wireless Internet modem in each. These ICT gadgets are going to explore a new horizon of expanding medical consultation by qualified physicians to patients visiting community clinics and union health centers. A total of 12,527 community clinics were functioning, while writing this report all over the rural areas of Bangladesh—one for about 6,000 to 10,000 people. There are about 4,500 unions, each having an outpatient health center. In community clinics or most of the union health centers, no qualified doctor is posted. However, there may be occasions when some patients need to consult a qualified medical practitioner. In such cases, a Skype video-conferencing will be setup to hook the community clinic or union health center to a doctor sitting in the nearby upazila hospital to have a direct conversation between the patient and doctor. The laptop computers in the community clinics and union health centers will be used for multiple purposes, viz. telemedicine, updating community health data, health education to people, training of health staff, monitoring of clinic operation time, email communication, and Internet-browsing. The telemedicine project of MIS-Health received the National ICT4 Development Award in 2011.

Telemedicine in Union Information and Service Centers (UISC)

With support from A2I (Access to Information) project of the Prime Minister's Office, the Ministry of Local Government, Rural Development and Cooperatives started UISCs in all 4,501 unions. These centers are run by local young entrepreneurs as a self-earning project for promoting ICT-based delivery of services to citizens. They sell computer- and Internet-based services to citizens at nominal fees. The services include composing and printing, email checking, faxing, and video-conferencing. In 30 UISCs, the entrepreneurs started telemedicine service. They take some nominal fees from the clients, and the MIS doctors provide medical advice free of charge, using Skype platform. The service appears very popular and most wanted among the people of those localities.

Citizens' feedback system: Complaint-suggestion box

The recently-concluded mission of the joint expatriate national independent review team for the second year of HPNSDP 2011-2016 recommended promotion of the SMS-based complaint-suggestion box run by MIS-DGHS to improve accountability



Figure 17.5. A display board mounted on hospital wall. The top section says how to get free medical consultation round-the-clock from a doctor of this hospital by calling to a mobile phone number. The bottom section describes how to make complaints or suggestions

and transparency of public hospitals. In each of about 800 public hospitals/organizations, a display board is mounted, which describes how to send complaints about quality of services or suggestions for service improvement (Figure 17.5). Clients of the hospitals or health organizations make complaints or suggestions to a particular mobile number. A webserver located at MIS-DGHS receives the complaints-suggestions. The responsible staff members at MIS-DGHS check the complaints and suggestions instantly on the web portal and then talk to the SMS senders to know more about the message. The staff members then talk to the local authority to solve the problem or work on the suggestions. An analysis was made on samples of complaints received at the web portal. Majority of the complaints were about negligence to responsibility (47.1%), followed by water and sanitation (mainly unclean toilets) (11.7%); corruption and transparency (10.5%), poor maintenance (8.0%), attitude and behavior (4.0%), personal privacy (2.2%), etc. Negligence to responsibility includes unavailability of doctors or other care providers on time, not opening particular service section on time, care providers busy with personal work, doctors sleeping in duty time, ambulance driver not willing drive ambulance after midnight, etc. Complaints were received from upazila (32.9%), district (24.0%), and postgraduate teaching institute (22.5%) hospitals.

Fingerprint machines in public hospitals for remote central monitoring of staff attendance

Qualified healthcare providers, in general, feel discouraged to work in the remote or rural areas. This

is a common problem in any country. In developing countries, absenteeism of the clinical staff from the health stations is another serious problem. To improve office attendance of health staff in Bangladesh, MIS-DGHS installed remote biometric time attendance machines in all upazila and district hospitals and in some tertiary hospitals. These are low-cost fingerprint biometric machines; the recorded touch-encounters can be tracked from central office through locally-developed software. During installation, staff members' fingerprints were recorded in the database. Every day the staff members need to touch the sensor of the machine during their check-in and check-out. The machine itself can keep in memory 30,000 encounters. Connected to a local computer through USB cable, the machine becomes empty of touch-records through transferring the same to the computer when the latter is switched on. At MIS-DGHS, a webserver captures the attendance data whenever the server finds the local computers switched on. Pre-defined web-based reports can be generated on the server-side, which can be accessed through web-browser from anywhere.

mHealth for safe pregnancy and childcare

The MIS-DGHS, in coordination with Ministry of Health and Family Welfare, is carrying out a partnership program with D.Net to provide a mHealth service called MAMA (Mobile Alliance for Maternal Action). MAMA Bangladesh uses a short code '16227', provides pregnant and new mother's lifesaving information, including also advice for their newborn babies and children through SMS and IVRs. This program is supported by USAID, partnered by Smiling Sun Clinic and Save the Children and coordinated by Abt Associates and D.Net.

The MIS-DGHS is also actively working in Gaibandha district of Bangladesh with Johns Hopkins Bloomberg School of Public Health to develop and implement mCare (for pregnancy care) and mTikka (for immunization) solutions to track pregnant mothers and under-five children. The mCare project was recognized as one of the top 11 innovations of the world by a global competition operated by mHealth Alliance of the United Nations Foundation.

Bulk SMS

The innovative bulk SMS system of MIS-Health, introduced in 2009, remained an effective solution even as of now to disseminate quick and urgent messages to health staff. The use of bulk SMS was frequent and demand-driven. For bulk SMS system, mobile phone numbers of all health managers and staff members down to the grassroots level were collected and grouped. Customized text messages can be broadcast to one or multiple groups instantly.

Medical and dental admission tests being managed digitally

From 2011, the medical and dental admission tests at the public and private medical and dental colleges of Bangladesh are being digitally handled. Admission-seekers submit applications by online electronic form. The system then checks and authenticates prerequisite educational qualifications from secondary and higher secondary school examinations databases. Students then submit test-fees by mobile phone top up. On successful fee submission, a text alert informs the student for collecting the Admit Card by login to admission website. Students' admission halls and seat plans are also managed digitally. The answer-sheets of examinations use OMR (optical mark reader) technology. After the examination, all the answer-sheets are transported to Dhaka the same day from all over the country. The next day, all answer-sheets are read by OMR machine, and results are prepared with intelligent software to inform the eligible examinees for which institutions they qualified for admission. On the same day or the next day, results are sent to students' mobile phone numbers and also through website.

Leave and deputation management software

The MOHFW, with technical assistance from MIS-DGHS, developed two kinds of software—one for online leave management and another for online management of deputation of doctors for higher education.

Digital training facility

The digital training facility, inclusive of an auditorium created by MIS-Health in 2009, was efficiently used over the past years. Its attraction as one of the best meeting and seminar places continues to increase. Equipped with state-of-the-art gadgets, such as digital podium and sound, interactive board, wireless presentation, wi-fi network, video-conferencing, etc., the facility attracts the organizations to hold their workshops, meetings, and symposia. The Management Sciences for Health (MSH) presented two air conditioners for the MIS auditorium as the number of audience is on the rise requiring more efficient cooling of the auditorium. The MIS-IT Lab has also been renovated with smart sitting arrangement.

A well-connected health system

The MIS-DGHS was the hero in the entire public sector of Bangladesh that first created the Internet connectivity across all health-points down to the upazila level (~800 places) by April 2009. Now, this network is completed through expansion to all the grassroots-level health facilities (all union health centers and community clinics) and half

of the community health workers. The union health centers and community clinics have laptop computers and wireless modems. The community health workers have been given android tablets. The remaining half of the community health workers will receive android tablets within one year.

Hospital automation

In fiscal 2011-2012, automation of three public hospitals has been started. These include National Institute of Kidney Diseases and Urology (NIKDU), Government Employees' Hospital, and Azimpur Maternity Hospital. Earlier, the Bangladesh Secretariat Clinic was automated. Currently, two more hospitals, viz. National Institute of Traumatology, Orthopedics and Rehabilitation (NITOR) and National Institute of Cardiovascular Diseases (NICVD) have been included. Two more hospitals will be included in the next fiscal year. The MOHFW is jointly planning with World Bank Bangladesh to automate over 300 public hospitals under the Enabling Open Government Program (EOGP) to begin within one year. Meanwhile, an initiative has been undertaken, using DHIS 2.12 (may be replaced by Open MRS) to collect patient registration and discharge information on each patient of the hospitals to generate statistics, like patient load (outdoor, emergency, admission, bed-occupancy rate, average length of stay, morbidity and mortality profile, using ICD10 coding system).

Human resource for HIS and eHealth

Currently, there are 721 sanctioned posts of statistical staff throughout the country. These staff members are already made skilled through training and engagement in practical work since 2009. These personnel are used as dedicated HIS and eHealth staff. Other staff members are also being trained to

play their role in real-time data-entry at the source of data. The distribution of 721 statistical staff members by type of organization is shown in Table 17.1. By class category, the distribution of these sanctioned posts is as follows: Class I (103, 14%); Class II (9, 1%); Class III (602, 84%); and Class IV (7,1%).

The Government has sanctioned 32 new positions in MIS-DGHS. This will add to the existing 721 sanctioned posts of HIS and eHealth staff throughout the country. It reveals that the number of sanctioned posts is very inadequate, with many organizations lacking sanctioned posts for statistical staff. On the other hand, many posts are vacant with an overall vacancy of 26.1%. The vacancy rate among staff in different class categories is as follows: Class I: 75.7%; Class II: 22.2%; Class III: 17.9%, and Class IV: 0.0%.

The load of work done for HIS and eHealth with this limited number of technical staff is unbearable. It may be imagined: if the number of staff could be increased, the HIS and eHealth functions and outcomes could be raised to a much higher level. The available statistical staff members have graduation and/or higher secondary-level education but not in statistical discipline. To meet the current and future challenges of MIS-DGHS, it is very essential to create adequate number of positions of competent persons in all relevant areas. As an interim measure, MIS-DGHS uses an innovative solution to mobilize skilled human resource through outsourcing with technical assistance from partners.

Technical partners

In addition to the MOHFW, other technical partners, like A2I Project, World Bank, WHO, UNICEF, DFID, Rockefeller Foundation, JICA, USAID, icddr,b, Measure Evaluation, UNESCAP, JPGSPH, BRACU, JHU, MSH (SIAP), Save The

Table 17.1. Distribution of sanctioned posts of HIS and eHealth staff by type of organization and their vacancy situation (2013)

Level of facility	Upazila hospitals and health offices	District civil surgeon's offices	MIS-DGHS	Divisional health offices	Postgraduate teaching institute and hospitals	DGHS	Medical college hospitals	200-bed and 250-bed hospitals	TB clinic at Chankhar Pool of Dhaka city	Total
Sanctioned posts	483	120	59	23	13	9	10	3	1	721
(No. and%)	67.0%	17.0%	8.0%	3.0%	2.0%	1.0%	1.0%	1.0%	<1.0%	100.0%
Vacancy (%)	17.2%	48.3%	44.1%	8.7%	38.5%	33.3%	80.0%	66.7%	100.0%	26.1%

Children, D.Net, and Grameen Solution, assist MIS-DGHS to make technology-related solutions, training, and capacity-building.

Capacity-building and maintenance support

MIS-DGHS continued capacity-building through training, supply of ICT equipment, computer stationeries, payment of Internet bills, and also repair and maintenance support.

Training, workshops, and seminars

In 2012-2013, several types of training courses, workshops, and seminars of different durations were held both at MIS-DGHS office in Dhaka as well as at local hospitals/health offices. A total of 26,438 officers and staff members participated in the training courses, workshops, and seminars held under the HPNSDP 2011-2016. In the UNICEF-supported training program, another 2,061 personnel participated. With WHO's technical assistance, training was provided to 813 personnel on the use of ICD 10 and collection

of data from private and NGO hospitals. It may be mentioned here that some participants might have joined in more than one training, workshop or seminar. Detailed information on training, workshops, and seminars is provided in the annex.

Supply of ICT equipment and computer stationeries

Table 17.2 provides information on different types of hardware procured and distributed in 2011-2012 and 2012-2013.

Repair and maintenance of computers, printers, and other accessories

In fiscal 2012-2013, MIS-DGHS at its headquarter repaired 484 computers, monitors, printers, and UPSs. Table 17.3 summarizes the information.

Medical Biotechnology (MBT)

The activities relating to medical biotechnology are going on according to the plan. A number

Table 17.2. Hardware procurement from fiscal 2012-2012 to 2013-14

Hardware	Fiscal 2011-2012	Fiscal 2012-2013	Fiscal 2013-2014	Distribution
Desktop computer	600	4360	6000	Hospitals, health offices, academic and training institutions from upazila to national level across the country
Laptop computer	3465	12471	2000	- FY 2011-2012: To 3,465 community clinics - FY 2012-2013: To union health facilities and to the remaining functional community clinics - FY 2013-14: Procurement in process to go to newly-functional community clinics and other health facilities and organizations
UPS (offline - 600 VA)	500	4000	6000	Accompanies one for each desktop computer
Tablet device	3500	8400	10000	To community healthcare providers (HA, HI, and AHI)
Equipment for tertiary-level hospital automation	NIKDU	NICVD and NITOR	2 more	- A 6th hospital will be automated in fiscal 2014-2015 - From fiscal 2014-2015, Enabling Open Government Program (EOGP) with World Bank is expected to begin to automate over 300 public hospitals
Equipment for telemedicine centers and peripherals	10	10	15	- FY2009-2010: 8 telemedicine centers established in 8 hospitals and 1 coordination center at MIS-DGHS - FY 2011-2012: 10 additional telemedicine centers in 10 hospitals - FY 2012-2013: 10 additional centers in 10 hospitals - FY 2013-2014: 15 more centers in 15 hospitals
Equipment for data center	Data center start-up in Dhaka	Disaster recovery service center at Khulna	Upgradation of existing centers	- FY 2012-2012: Data center at MIS-DGHS established - FY 2012-2013: Disaster recovery service (DRS) center in Khulna established - FY 2013-2014: Data center and disaster recovery service center will be upgraded

of training programs and workshops were held in 2012-2013, where a total of 810 participants joined. A summary of the training sessions and workshops is provided in the annex. A draft of the MBT law has been prepared. Equipment have been procured to strengthen medical biotechnology laboratories in 8 medical colleges. MBT books have

been provided to a number of medical colleges, postgraduate institutions, and Bangabandhu Sheikh Mujib Medical University. Work is ongoing to set up biotechnology research laboratory at the Center for Medical Biotechnology, with collaboration of iDeshi, Bangladesh—a local science and technology philanthropy.

Table 17.3. Number of computers, monitors, printers, and UPS repaired in FY 2012-2013 at MIS-DGHS

Type of health organization	Computer	Monitor	Printer	UPS	Total
Directorate General of Health Services	72	28	27	8	135
Postgraduate teaching and/or specialized institute	7	3	3	3	16
Civil surgeon's office	30	23	38	0	91
District/general hospital	17	8	8	0	33
Upazila health complex/Upazila health office	102	33	64	10	209
Total	228	95	140	21	484

HEALTH, POPULATION AND NUTRITION UNDER HPNSDP 2011-2016

APIR 2013 summarizes progress of 32 operational plans

The Program Management and Monitoring Unit (PMMU) of the Planning Wing of MOHFW prepared the Annual Program Implementation Report in September 2013 (APIR 2013). The report summarizes performance of all 32 operational plans (OPs) for the financial year 2012-2013 and identified some critical gaps in program implementation. The Health Bulletin 2013 excerpts some of the key elements from the APIR 2013. In doing so, Alternative Medical Care, National Eye Care, and Health Education and Promotion, which were described in three separate chapters of earlier Bulletins covering progress of respective operational plans, have been merged in Chapter 18 and with its annex this year. The PMMU staff and the PMMU Technical Assistance Support Team (TAST) members (from MEASURE Evaluation and icddr,b) collected the APIR 2013 information from the line directors and their related staff of the 32 operational plans, using a structured data-reporting template customized for individual OPs.

Major findings of APIR 2013

- **Budget allocation and spending pattern:** Total fund allocation for FY 2012-2013 was Tk. 2,947.3 crore (Government: 34%, Tk. 1,012.0 crore and project aid: 66%, Tk. 1,935.3 crore). Overall spending rate was 91.4% (97% of Government and 90% of project aid fund). The overall spending rate in FY 2011-2012 was 87% (Table 18.1).
- **Programmatic achievement measured by indicators:** Of the 270 measurement indicators

(FY 2012-2013), 187 (69%) were fully achieved, 31 (12%) partially achieved, and 20 (7%) not achieved. Data on 32 (12%) indicators were “not available.” Deducting 32 indicators from 270, the rate of programmatic achievement moves up to 79% of fully-achieved indicators. In FY 2011-2012, fully-achieved indicators were 65%.

- **Training and workshops:** Out of the total expenditure during FY 2012-2013, about 10% was spent on training, workshops, seminars, and orientation, engaging a total of 1,374,293 participants [in-country training: 760,570 (55%); foreign training: 1,334 (0.1%); and workshops, seminars, orientation: 612,389 (45%)]. Among the 760,570 participants in local training, 503,062 were members of community clinic support group and community clinic management group, who received 1 to 2 day(s) training to increase their capacity in oversight of CC management and strengthening accountability for the same.
- **Key challenges:** (i) Insufficient skilled manpower to implement the OP activities; (ii) Lengthy procurement process; (iii) Inadequate fund release on a quarterly basis; (iv) Insufficient staff and logistics for monitoring and supervision activities; and (v) Vacancy in sanctioned positions.

Table 18.1 depicts OP-wise financial and physical progress of HPNSDP implementation during FY 2012-2013

Table 18.1. Financial and physical progress of OPs under HPNSDP in FY 2012-2013 (Taka in crore)

Operational plan*	Revised ADP (RADP) allocation	Fund released	Expense	% expense against release			% expense against RADP allocation	% physical progress against indicators
				GOB	PA	Total		
MNCAH	429.30	423.60	402.70	93.0	95.0	95.0	94.0	53.0
ESD	44.00	44.00	40.80	88.0	95.0	93.0	93.0	60.0
CBHC	72.50	69.70	67.20	91.0	97.0	96.0	93.0	100.0

Table 18.1 continued

Operational plan*	Revised ADP (RADP) allocation	Fund released	Expense	% expense against release			% expense against RADP allocation	% physical progress against indicators
				GOB	PA	Total		
TBLC	70.60	64.20	61.90	64.0	100.0	97.0	88.0	86.0
NASP	28.00	28.00	22.30	71.0	80.0	79.0	79.0	100.0
CDC	98.00	116.20	103.20	82.0	111.0	89.0	105.0	100.0
NCDC	122.20	115.20	112.50	94.0	99.0	98.0	92.0	54.0
NEC	4.50	4.50	4.00	97.0	82.0	90.0	90.0	100.0
HSM	434.90	434.90	423.40	93.0	99.0	97.0	97.0	67.0
AMC	13.70	13.70	12.80	98.0	67.0	93.0	93.0	50.0
IST	56.00	48.80	46.20	95.0	94.0	95.0	83.0	80.0
PSE	167.40	166.40	159.60	94.0	97.0	96.0	95.0	100.0
PMR	9.00	8.50	8.20	91.0	98.0	97.0	91.0	50.0
HIS-EH	110.30	109.70	108.40	100.0	98.0	99.0	98.0	100.0
HEP	19.50	19.50	18.00	97.0	89.0	92.0	92.0	56.0
PLSM	62.80	62.80	61.90	99.0	93.0	99.0	99.0	100.0
NNS	51.00	51.00	36.10	72.0	71.0	71.0	71.0	58.0
MCRAH	139.00	138.50	133.40	99.0	96.0	96.0	96.0	100.0
CCSD	160.80	160.60	153.70	99.0	89.0	96.0	96.0	100.0
FPFSD	212.50	211.70	84.00	97.0	26.0	40.0	40.0	63.0
PME	1.90	1.90	1.90	99.0	97.0	98.0	98.0	100.0
MIS	16.10	16.10	12.50	88.0	75.0	78.0	78.0	58.0
IEC	21.70	21.70	21.30	96.0	100.0	98.0	98.0	50.0
PSSM	13.50	13.30	12.90	100.0	75.0	97.0	95.0	80.0
TRD	20.40	20.40	15.90	97.0	63.0	78.0	78.0	71.0
NES	65.00	65.00	59.20	89.0	92.0	91.0	91.0	75.0
SDAM	3.30	1.80	0.60	90.0	28.0	34.0	19.0	100.0
PFD	482.50	482.50	481.00	100.0	100.0	100.0	100.0	25.0
HRM	5.90	5.90	4.20	80.0	68.0	70.0	70.0	25.0
SWPMM	3.20	3.20	2.80	81.0	87.0	87.0	87.0	82.0
IFM	2.50	2.50	2.40	95.0	96.0	96.0	96.0	75.0
HEF	5.50	4.90	4.60	96.0	94.0	95.0	84.0	43.0
All32Ops	2947.30	2930.70	2679.70	97.0	90.0	91.4	91.0	69.0

*See acronyms in the beginning of this Health Bulletin

History of Sector-wide Approach (SWAp) in MOHFW of Bangladesh

In 1997, during the Fourth 5-year Plan, MOHFW of Bangladesh had Health and Population Strategy of 1997—a project modality to implement development program for health and population sector. The experience helped take decision to move away to a sector-wide approach (SWAp) under the Fifth 5-year Plan. The first SWAp, thus,

began in 1998 and was known as the Health and Population Sector Program (HPSP) 1998-2003. The HPSP's main focus was to decentralize delivery of essential service package, using a 'one-stop' service model and deliver basic health and family planning services to the rural community from static community-based facility, like community clinics (CC), one for every 6,000 people. The second health SWAp titled "Health, Nutrition

Table 18.2. Three SWApS of MOHFW of Bangladesh (1998 to 2016)

Name	Duration	Funding-size (billion US\$)	GOB contribution	DP contribution
Health and Population Sector Program (HPSP)	1998-2003	2.2	62%	38%
Health, Nutrition and Population Sector Program (HNPSPP)	2003-2011	5.4	67%	33%
Health, Population and Nutrition Sector Development Program (HPNSDP)	2011-2016	7.7	76%	24%

and Population Sector Program (HNPSPP) was implemented during 2003-2011. The HNPSPP's main focus was to increase availability and utilization of user-centered, effective, efficient, equitable, affordable, accessible quality health, nutrition and population services. The third (and current) SWAp titled "Health, Population and Nutrition Sector Development Program (HPNSDP) 2011-2016" began in 2011 with the aim to strengthen health systems and improve health services. Table 18.2 shows the duration, size, and partner contributions to different SWApS.

Health, Population and Nutrition Sector Development Program (HPNSDP) 2011-2016 began in 2011 with the aim to strengthen health systems and improve health services ...

HPNSDP 2011-2016

The total estimated budget for the HPNSDP 2011-2016 is Tk. 56,993.54 crore (US\$ 7,701.83 million), inclusive of the development and non-development budget. The development budget for 32 OPs is set at Tk. 22,176.66 crore (US\$ 2,996.84 million). The DP (development partner) contribution is set at Tk. 13,573.16 crore (US\$ 1,834.21 million)—61% of the total development budget. If both development and non-development budgets of the MOHFW are combined, the GOB share stands at 76% and DP share at 24%.

There are 7 strategic priorities in HPNSDP 2011-2016:

- Revitalizing various family planning interventions to attain replacement level fertility
- Mainstreaming nutrition within the regular DGHS and DGFP services (MDG 1)

- Strengthening preventive and control programs on communicable diseases (MDG 6)
- Expanding NCD control efforts at all levels by streamlining referral systems
- Strengthening various support systems by increasing the health workforce at upazila and CC levels, including their capacity-building and enhanced focus on coordinated implementation of OPs, MIS, and M&E functions
- Increasing coverage and quality of services by strengthening coordination with other intra- and inter-sectoral and private-sector service providers
- Pursuing priority institutional and policy reforms, such as decentralization and LLP, incentives for service providers in hard-to-reach areas, PPP, and single annual work plan.

Measurement of OP's progress in FY 2012-2013

The HPNSDP 2011-2016 has two types of progress monitoring indicators, viz. results framework (RFW) for overall HPNSDP and OP indicators to measure the progress of individual OPs. Every year during the APIR, the progress on indicators are compared with the set targets to understand whether the progress is on track or not. The RFW was revised in 2011, and the updated version along with current progress is shown in the annex.

Results framework (RFW) of the HPNSDP 2011-2016 priority indicators are shown in Table 18.3.

Local Health Bulletin as a means to strengthen monitoring and supervision

The APIR 2013 mentions that, since 2012, the MIS-DGHS is helping respective organizations to produce Local Health Bulletins (LHBs), covering all public hospitals and health facilities in Bangladesh (around 550 in number), accessible online, which provide information on updated contact of facility in-charge; basic information (location, etc.); bed-capacity; number and nature of services

Table 18.3. Targets and assessment of HPNSDP priority indicators

Indicator	Baseline (as in PIP)	Progress (June 2013)	Target 2016	How much on track
Infant mortality rate (IMR)	52 (BDHS 2007)	43 (BDHS 2011)	31	Likely
Under-5 mortality rate	65 (BDHS 2007)	53 (BDHS 2011)	48	Yes
Neonatal mortality rate	37 (BDHS 2007)	32 (BDHS 2011)	21	Challenging
Maternal mortality rate	194 (BMMS 2010)	194 (BMMS 2010)	<143	Yes
Total fertility rate (TFR)	2.7 (BDHS 2007)	2.3 (BDHS 2011)	2.00	Likely
Prevalence of stunting among under-5 children	43.2% (BDHS 2007)	38.7% (UESD 2013)	38%	Yes
Prevalence of underweight among under-5 children	41.0% (BDHS 2007)	35.1% (UESD 2013)	33%	Yes
Prevalence of HIV in MARP	<1% (SS 2007)	<1.0 (SS 2011)	<1%	Yes

provided (e.g. IMCI, ANC, delivery, C-sections, and outpatient visits); HR status by category; and status of available medical equipment (e.g. functional, not installed, non-repairable, etc.) in the facility. It also provides calculated figures like bed-occupancy rate, death rates, and average length of stay in the facility. Such detailed, comparative reports from facilities are produced online as soon as the local organizations enter data into the system. If collected on a regular basis, LHBs can be effectively used for monitoring the provision of services and/or availability of equipment at the health facilities. APIR 2013 recommends, among others, to expand coverage to include DGFP facilities by collaborating with MIS-DGFP and other concerned agencies and to utilize LHBs as sources of specific OP indicators.

Monitoring Human Resources for Health (HRH), using MIS-DGHS

The APIR 2013 also recommends ensuring ownership of HR information system developed by MIS-DGHS, by MOHFW, and by the HRM-OP in particular. It also recommends maintaining data security and integrity of the system as it contains confidential information.

Local Health Bulletins can be effectively used for monitoring the provision of services and/or availability of equipment at the health facilities ...

Strengthening Routine Health Information Systems (RHIS) of MOHFW

APIR 2013 notes that, in order to improve the process of data collection and reporting at all levels and to reduce multiple redundant data collection and reporting forms for the community as well as facility-level service providers, LDs of HIS-EH and MIS-FP are provided with technical assistance (from icddr, in collaboration with MEASURE Evaluation) and have developed the Routine Health Information System (RHIS). The endeavor resulted in development of common online forms in DHIS 2.12, which are piloted at district, subdistrict, and community levels in 3 selected subdistricts and the district hospital of Tangail. Also, the piloting in all subdistricts of Habiganj, including the district hospital in collaboration with Save the Children (under USAID-supported District Health Systems Strengthening Project) is scheduled to start soon. The pilot shows that 100% reduction of manually-generated indoor patient forms in upazila and 60% reduction of monthly health assistant reports in community were possible in case of DGHS, and 50% reduction of FWA registers in union and reduction of a section of FWA register in community were possible in case of DGFP. APIR 2013 recommends harmonizing efforts to ensure inter-operability of the routine data systems from DGHS and DGFP and scale up the RHIS program based on piloting experiences.

Program Management and Monitoring Unit (PMMU)

The Program Management and Monitoring Unit (PMMU) was established by the Planning Wing in December 2011 for strengthening planning, monitoring, coordination, and evaluation of the HPNSDP. PMMU prepared the APIR for FY 2011-

2012 and FY 2012-2013, which fed into the APIR process. It prepared the revised list of indicators for all 32 OPs as per the Priority Action Plan (PAP) recommendation following APIR 2012. PMMU also produced the first HPNSDP Six-monthly Progress Report (SmPR) for July–December 2012, the first-ever such 6-monthly reporting in HNP SWAp in Bangladesh. The report identified some critical problems, which were quickly acted upon by the Planning Wing of MOHFW to hold orientation workshops for the LDs and their core staff to improve their capacity in budget planning, monitoring of program activities, and program management. This was the first time under any SWAp in MOHFW that such capacity-building activity for LDs took place.

Selected Issues on Service Delivery Activities

Nutrition

Mainstreaming of the nutrition initiative is a major step forward for expanding nutrition services all over the country. One major progress in implementation was establishing IMCI and Nutrition Corners at 130 upazila health complexes with the collaboration of MNCAH OP. NNS rolled out facility-based management of Severe and Acute Malnutrition (SAM) in 39 health facilities, for which the referral linkages need to be established with the IMCI Corners. NNS needs to focus on two to three priority activities at a time for immediate scale-up and to increase visibility of MOHFW's nutrition services at the field level.

Maternal, Newborn and Child Health

MNCH remains the priority area of HNP sector programs in Bangladesh. MNCAH OP under DGHS continues to expand the coverage of Comprehensive EmOC provision at upazila health complexes (UHCs), although it needs to gather more speed. MNCAH is funding CSBA and midwifery training activities to promote skilled birth attendance at the community level. The expansion of maternal health vouchers program into 20 new upazilas has been kept pending subject to a systematic study to evaluate the functioning and economics of the DSF program. The study has been recently completed and its findings have been shared with the stakeholders in the last week of August 2013. MNCH needs to expand DSF as planned and retain the required skill-mix in selected UHCs to provide CEmOC services.

Community clinic-based services

Till June 2013, a total of 12,527 CCs were functional; 12,063 community groups and

Early decision by MNCAH Operational Plan could potentially help improve the availability of trained CSBAs and thereby contribute to ongoing efforts to reduce the risks to maternal and neonatal life in the current situation. ...

community support groups in 13 districts are associated with the running and oversight of the CCs; 148 CCs are providing normal delivery service, even though this is not one of their scheduled functions. An extra room was built for delivery in the current FY 2012-2013 in 103 CCs with GAVI-HSS support. However, the nutrition services are not yet fully available from all CCs. The need for improved coordination between CHCPs and the other two categories of service providers at the CC, i.e., FWA and HA, is essential. Job uncertainty of the CHCPs need to be allayed to maximize their potentiality. The urgent need for mainstreaming of CCs would be facilitated by the adoption of single reporting format now under discussion with MIS and NNS, in lieu of the separate reporting done by the CHCP. The female CHCPs who constitute more than 50% of the total number offer a potential pool of qualified personnel to undertake CSBA training and thereby lead to filling in the current gap of skilled birth attendants in the rural home deliveries. Early decision by MNCAH OP could potentially help improve the availability of trained CSBAs and thereby contribute to ongoing efforts to reduce the risks to maternal and neonatal life in the current situation.

Selected Issues on Systems Strengthening

Procurement

Procurement and supply chain management system in the HNP sector improved substantially, e.g. the average time between signing of the contracts and the goods to be accepted at the hospitals got reduced from 29 months to 19.6 months; and equipment lying idle at the facilities declined from 57% to 46%. The procurement entities in MOHFW and the World Bank jointly worked on improving processes and steps so that the total procurement lead time is 58 weeks; a notification regarding this was issued in January 2013. The draft standard framework contract needs

to be finalized as per the 2012 APIR's Priority Action Plan (PAP), a biomedical engineer is to be recruited to help, among other things, to develop standardized specifications for medical equipment, and a manual of procurement procedures is to be developed.

Logistics management

Despite substantial improvement in procurement processes over the years, planning and utilization of procured items remained as major bottlenecks in Bangladesh's health sector program. The main reasons for not utilizing equipment include the lack of a comprehensive equipment planning and an absence of adequate maintenance. It is necessary to relocate, on a priority basis, unutilized equipment to facilities where these are needed to ensure effective utilization of procured items and to improve procurement planning system by creating a provision for the Directorates to adopt greater responsibility for needs assessment on the basis of field-level inputs. It is also important to create provision for adequate maintenance budget and develop effective maintenance system which would include deciding on the future role of NEMEW&TC and TEMO. Simplification of the condemnation process at district hospital and UHC levels to clear existing stockpile on regular basis and specific reporting on this activity by the managers at the district level and their supervisors are also required.

Financial management

Through the funding of HPNSDP, MOHFW contracted a firm to provide technical support to Financial Management & Audit Unit (FMAU) and to selected LDs. Connectivity with GOB's automated integrated Budgeting and Accounting System (iBAS) has been established with FMAU, which is expected to generate timely and accurate financial reports in the future. Planning and

management of OP budget and expenditure need assistance of specialists to achieve value for money and avoid irregularities.

TA coordination

The Technical Assistance Committee (TAC) needs to speed up the process of identifying and clearing TAs for smooth implementation of different OP activities.

Training in FY2012-2013

HPNSDP promotes considerable effort to improve HR capacity through training (local and foreign) and workshops, seminars, and orientation. Out of the total expenditure in FY2012-2013, Tk. 258.40 crore (10%) was spent for this purpose; out of the total training cost, Tk. 186.1 crore (72%) was spent on training and Tk. 72.2 crore (28%) on workshops, seminars, and orientation. Detailed OP-wise training cost is shown in the annex. A total of 1,374,293 participants took part in training and workshops, seminars, and orientation during FY2012-2013; among them, 760,570 (55%) participants attended in-country training programs, 1,334 (0.1%) attended training programs abroad, and the rest 6,12,389 (45%) attended workshops, seminars, and orientation. Most of the training courses (91%) were from OPs relating to health service improvement, and the rest (9%) were from strengthening health systems (SHS). However, international training was predominantly from the SHS OPs (75%). In workshops, there were a total of 612,389 participants, out of which health service improvement-related OPs conducted 60%. CBHC and MNCAH constitute 83% of the total local training participants; and PME, CDC, NCDC, and MNCAH constitute 87% of all participants in workshops, seminars, and orientation. Details are provided in the annex. Table 18.4 shows distribution of local training courses by duration.

Table 18.4. Distribution of local training courses by duration (FY 2012-2013)

Duration of training	Participants		Cost of training (Taka in lakh)	
	Number	%	Total	%
1 day	481521	63%	3162	25%
2 days	217357	29%	2712	21%
3-30 days	53529	7%	508	35%
1-6 month(s)	5885	0.8%	1998	16%
6+ months	2278	0.3%	507	4%
Total	760570	100%	12888	100%

Issues Needing Attention

- Improve visibility of NNS
- Address key human resources for health issues, like severe shortage
- Inappropriate skill-mix, deployment and retention of skilled manpower, and HR information system
- Lack of coordination among physical facility development and supply of needed HR
- Lack of coordination between five HR-related OPs, namely HRM, IST, TRD, NES, and PSE
- HSM OP to make progress in enactment of private healthcare law
- ESD OP to mobilize policy guidance for tribal health and urban health issues
- Health economics and financing OP to work out resource allocation formula, develop EGVNP strategies, and accelerate work for health insurance and healthcare financing framework
- IEC OP to strengthen activities to support the delivery of key health and family planning services
- Fine-tuning of the procurement planning of OPs and their processing, leading to timely execution
- Improve capacity of line directors and strengthen performance review of OPs
- Improve outcomes of training and workshops
- Strengthen financial management.

FINANCING HEALTHCARE

Fund utilization increased in fiscal 2012-2013

An account of the development budget and expense of the MOHFW is provided in Chapter 18 of Health Bulletin 2013. The development budget of MOHFW and its agencies comes from Health, Population and Nutrition Sector Development Program (HPNSDP) 2011-2016. In fiscal 2012-

2013, the total allocation under revised annual development program (RADP) for the Directorate General of Health Services (DGHS) was Tk 179,210.90 lakh (Figure 19.1). This allocation was distributed among 17 operational plans of DGHS as per respective work plans.

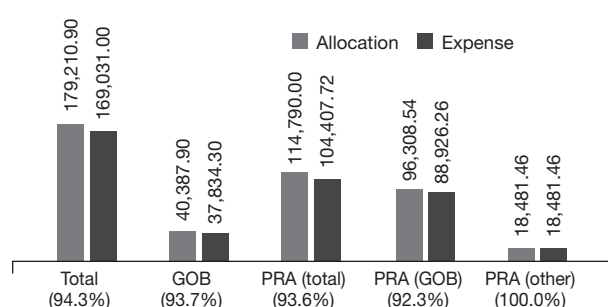


Figure 19.1. Allocation and expenditure (lakh taka) against operational plans of DGHS in fiscal 2012-2013 under HPNSDP (values in parentheses show % of fund utilization against allocation)

Figure 19.1 shows that, as of June 2013, the total expenditure was TK 169,031.00 lakh, the utilization rate being 94.3%. Of the total RADP allocation, GOB fund was Tk 40,387.90 lakh. The utilization rate of GOB fund was 93.7% (Tk 37,834.30 lakh), and that of RPA fund (RPA-GOB plus RPA-others) was 93.6% (Tk 107,407.72 lakh out of Tk 114,790.00 lakh). The RPA (GOB) fund utilization rate was 92.3% (Tk 88,926.26 lakh against allocation of Tk 96,308.54 lakh). RPA (other) fund utilization rate was 100.0% (both allocation and utilization amounted to Tk 18,481.46 lakh). In fiscal 2011-2012, the overall fund utilization rate was 84.4% (GOB: 89.4%; RPA: 82.4%; RPA-GOB: 76.0%; and RPA-other: 99.9%).

Table 19.1. Allocation in FY 2013-2014 and allocation, expenditure, and utilization in FY 2012-2013 (revised ADP) of HPNSDP 2011-2016 fund against different operational plans of DGHS (Tk in lakh)

Program	FY 2013-2014	FY 2012-2013 (revised ADP)		
	Allocation	Allocation	Expense	Utilization (%)
Essential service delivery	7,400.00	4,431.00	4,004.04	90.4
Maternal, neonatal, child and adolescent healthcare	43,700.00	42,926.00	40,273.50	93.8
Community-based healthcare	7,850.00	7,250.00	6,718.44	92.7
TB and leprosy control	5,720.00	7,062.00	6,192.35	87.7
National AIDS/STD program	3,805.00	2,800.00	2,246.00	80.2
Communicable disease control	13,200.00	9,800.00	10,366.43	105.8
Non-communicable disease control	10,147.00	12,215.00	11,254.59	92.1
National eye care (NEC)	335.00	445.90	403.22	90.4
Hospital services management	37,000.00	43,485.00	42,343.77	97.4
Alternative medical care	1,250.00	1,292.00	1,275.70	98.7
In-service training	3,964.00	5,510.00	4,625.22	83.9
Pre-service education	14,000.00	16,736.00	15,923.33	95.1
Planning, monitoring, and research	1,177.00	900.00	814.77	90.5

Table 19.1 continued

Program	FY 2013-2014	FY 2012-2013 (revised ADP)		
	Allocation	Allocation	Expense	Utilization (%)
Health information systems and eHealth	12,500.00	11,025.00	10,973.62	99.5
Health education and promotion	2,200.00	1,950.00	1,798.65	92.2
Procurement, logistics and supplies management	4,350.00	6,283.00	6,194.40	98.6
National nutrition services (NNS)	8,500.00	5,100.00	3,611.99	70.8
Total	177,098.00	179,210.90	169,031.00	94.31

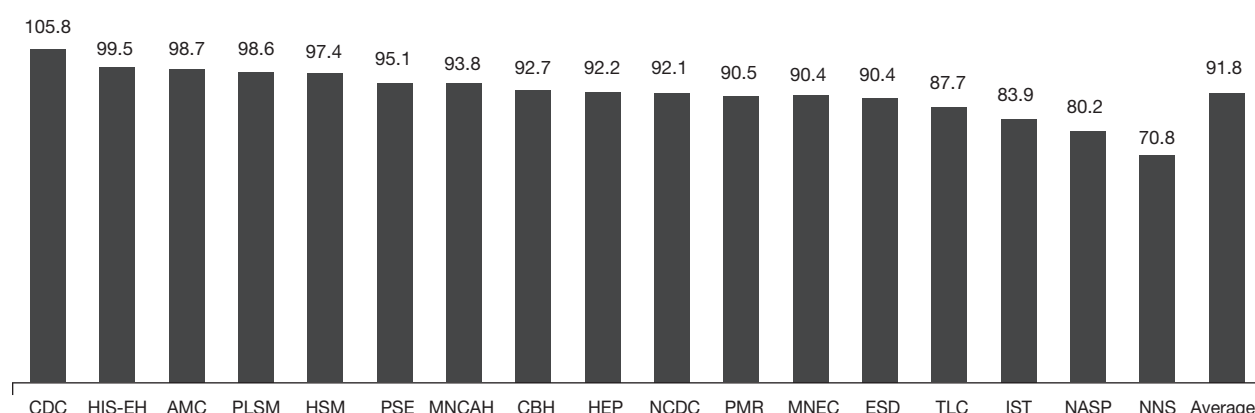


Figure 19.2. Fund utilization rate (%) of the DGHS operational plans in FY 2012-2013

Table 19.1 shows the allocation in FY 2013-2014, and allocation, expenditure, and utilization in FY 2012-2013 (revised ADP) of HPNSDP 2011-2016 fund against different operational plans of DGHS. Detailed breakdown is shown in the annex.

Figure 19.2 shows the fund utilization rate of different operational plans of DGHS in fiscal

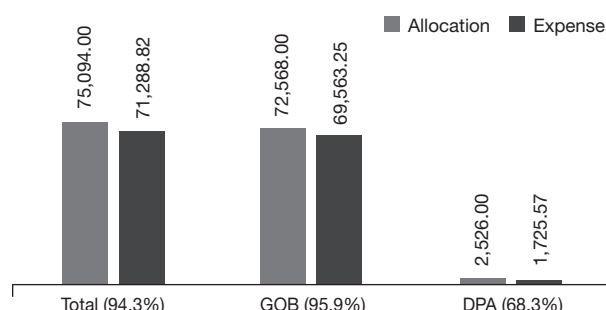


Figure 19.3. Allocation and expenses in lakh taka of 20 investment projects of MOHFW under HPNSDP in fiscal 2012-2013 (values in parentheses show fund utilization rate in %)

2012-2013 under the HPNSDP 2011-2016. The Communicable Disease Control and Health Information System and eHealth were the best performers.

Figure 19.3 shows the allocation and expense of 20 investment projects of MOHFW in fiscal 2012-2013 under HPNSDP 2011-2016. Total allocation was Tk 75,094.00 lakh, and total expense was Tk 71,288.82 lakh. The utilization rate was 94.3%. The GOB allocation was Tk 72,568.00 lakh, and expense was Tk 69,563.25 lakh. The utilization rate was 95.9%. The Direct Project Aid (DPA) allocation was Tk 2,526.00 lakh, and the expense was Tk 1,725.57 lakh. The utilization rate was 68.3%.

Table 19.2 shows the allocation in FY 2013-2014 and allocation, expenditure and utilization in FY 2012-2013 (revised ADP) under HPNSDP 2011-2016 for different investment projects of MOHFW. Detailed breakdown is shown in the annex.

Budget of the Ministry of Health and Family Welfare (MOHFW)

Figure 19.4 shows the budget of MOHFW allocated by the Ministry of Finance for fiscal 2013-2014,

Table 19.2. Allocation in FY 2013-2014 and allocation, expenditure, and utilization in FY 2012-2013 (revised ADP) under HPNSDP 2011-2016 for different investment projects of MOHFW (Tk in lakh)

Investment project	FY 2013-2014	FY 2012-2013 (revised ADP)		
	Allocation	Allocation	Expense	Utilization (%)
Establishment of 250-bed National Ophthalmology Institute and Hospital (1st Phase: 250-bed)	1,020.00	1,510.00	1,504.21	99.6
Upgrading National Institute of Cancer Research and Hospital from 50-bed to 300-bed	2,934.00	1,123.00	321.44	28.6
Establishment of National Institute of Laboratory Medicine and Referral Center	1,000.00	159.00	158.49	99.7
Extension of Dhaka Shishu (Children) Hospital Project	800.00	250.00	250.00	100.0
Establishment of Essential Drugs Company Limited, 3rd Plant, Gopalganj	300.00	292.00	290.85	99.6
Expansion and quality improvement of nursing education	1,000.00	2,500.00	2,199.00	88.0
Establishment of National Institute of Neuroscience	Transferred to revenue budget	2,896.00	2,889.43	99.8
Expansion and modernization of Dhaka Medical College Hospital	1,808.00	3,405.00	3,403.72	100.0
Establishment of National Institute of ENT (1st Phase)	600.00	3,033.00	2,530.43	83.4
ENT & Head-Neck Cancer Foundation Hospital and Institute	0	387.00	386.38	99.8
Revitalization of Community Healthcare Initiatives in Bangladesh	27,500.00	39,178.00	38,030.81	97.1
Conversion of BSMMU to a center of excellence project	5,000.00	7,500.00	7,125.32	95.0
Establishment of Sheikh Fajilatunnesa Mujib Eye Hospital and Training Institute, Gopalganj	4,800.02	5,995.00	5,664.48	94.5
Establishment of National Centre for Cervical and Breast Cancer Screening and Training at BSMMU	550.00	384.00	367.00	95.6
Establishment of Sheikh Sayera Khatun Medical College and Hospital and Nursing Institute, Gopalganj	3,000.00	900.00	899.73	100.0
Establishment of Shatkhira Medical College & Hospital	2,400.00	530.00	528.00	99.6
Establishment of Faridpur Medical College & Hospital	3,850.00	4,000.00	3,960.69	99.0
National Institute of Digestive Diseases Research & Hospital	200.00	47.00	47.00	100.0
Establishment of Kushtia Medical College	1,950.00	1,000.00	731.84	73.2
Establishment of Mother and Child Care Hospital under A K Khan Healthcare Center of Excellence	5.00	5.00	--	0.0
Total	58,717.02	75,094.00	71,288.82	94.9

fiscal 2012-2013 (revised budget), and fiscal 2011-2012 (revised budget). The revised budget is an indication of utilization pattern, which reveals that the MOHFW has utilized increased amount of fund from fiscal 2011-2012 to fiscal 2012-2013. The utilization pattern in fiscal 2013-2014 will be understood after the fiscal year ends.

Distribution of MOHFW budget

Table 19.3 shows the distribution of the MOHFW budget by economic code for fiscal 2013-2014 and 2012-2013.

Table 19.4 shows the distribution of the MOHFW budget by agency and organization for fiscal 2013-2014 and 2012-2013. Detailed breakdown is shown in the annex.

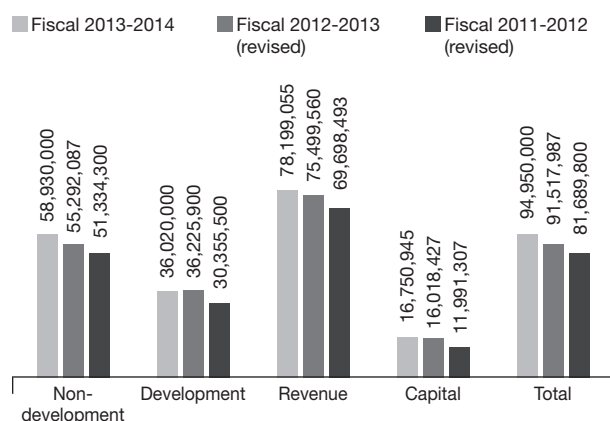


Figure 19.4. MoHFW budget allocated by Ministry of Finance of the Government of Bangladesh (in thousand taka)

Utilization of non-development budget

Table 19.5 shows the allocation and expenditure of non-development revised budget of fiscal 2012-2013 in different organizations under DGHS. Detailed breakdown is shown in the annex.

Bangladesh National Health Accounts (NHA)

The Bangladesh National Health Accounts 1997-2007 was officially published in 2010 by the Health Economics Unit (HEU) of the Ministry of Health and Family Welfare. No updated NHA is available since then. Table 19.6 shows the Total Health Expenditure (THE) and its share by different contributors in Bangladesh (2006-2007).

The excerpts from NHA 1997-2007 was published in detail in Health Bulletin 2012 which is available in our website (www.dghs.gov.bd).

Table 19.3. Distribution of MOHFW budget by economic code for fiscal 2013-2014 and 2012-2013 (Tk in thousand) (Ref: MOF 2013)

Head	Budget 2013-2014	Revised 2012-2013	Budget 2012-2013
Pay of officers (4500)	6,800,984	5,394,794	4,695,178
Pay of establishment (4600)	13,631,162	12,406,586	15,157,448
Allowances (4700)	15,530,570	13,203,772	15,036,637
Supplies and services (4800)	31,578,105	12,149,229	32,691,772
Repairs, maintenance, and rehabilitation (4900)	2,291,132	2,087,775	2,760,090
Grants in aid (5900)	1,975,220	1,898,180	1,842,180
Contributions to international organizations (6100)	9,850	22,385	5,850
Pensions and gratuities (6300)	4,200,000	6,343,500	3,870,000
Block allocations (6600)	1,281,032	78,239	1,744,048
Revenue-general (6700)	901,000	21,915,100	-
Total revenue	78,199,055	75,499,560	77,803,203
Acquisition of assets (6800)	7,450,395	1,478,370	9,134,840
Acquisition/Purchase of land and landed properties (6900)	530,000	-	80,000
Construction and works (7000)	2,607,350	4,257	5,930,057
Advances to government employees (7400)	246,000	225,000	215,000
Development import duty and VAT (7900)	40,500	12,400	387,000
Capital block allocation & misc. capital expense (7980)	5,876,700	14,298,400	-
Total capital	16,750,945	16,018,427	15,746,897
Total revenue + capital	94,950,000	91,517,987	93,550,100

Table 19.4. Distribution of the MOHFW budget by agency and organization for fiscal 2013-2014 and 2012-2013 (Tk in thousand) (Ref: MOF 2013)

Agency/organization	FY 2013-2014 budget	FY 2012-2013 (revised budget)
Secretariat (MOHFW)	20,092,812	20,896,028
Autonomous bodies	1,874,620	1,845,980
International organizations	9,850	22,385
DGHS and subordinate organizations	74,592,706	71,016,979
DGFP and subordinate organizations	16,964,485	16,587,850
Development program financed by revenue budget	-	28,608
Total	113,534,473	110,397,830

Table 19.5. Allocation and expenditure of non-development revised budget of fiscal 2012-2013 in different organizations under DGHS (Ref: MOF 2013) (in actual Taka)

Head	Revised budget	Expense	Utilization (%)
DGHS	1,565,645,000	1,830,347,777	116.9%
Divisional establishments	69,344,000	65,259,952	94.1%
Civil surgeon's offices	610,462,000	582,641,847	95.4%
Upazila health offices	5,625,080,000	5,309,497,735	94.4%
Paramedical institutes	42,602,000	34,714,391	81.5%
Medical assistant training schools	109,604,000	98,251,507	89.6%
TB control & training institutes	37,641,000	33,056,898	87.8%
District hospitals	4,072,905,000	3,942,170,458	96.8%
Other district-level hospitals	113,163,000	107,402,579	94.9%
Upazila health complexes and subcenters	8,179,716,000	7,422,875,215	90.8%
TB segregation hospitals	38,260,000	33,640,879	87.9%
Other TB hospitals	119,551,000	104,956,539	87.8%
Leprosy hospitals	42,514,000	38,286,147	90.1%
TB centers	152,413,000	144,130,715	94.6%
School health centers	34,190,000	33,529,399	98.1%
Model family planning clinics	46,352,000	45,271,341	97.7%

Table 19.6. Total health expenditure (THE) and its share by different contributors in Bangladesh (2006-2007)

Public sector		Household		Private		Insurance		NGO		Development partners		Total THE
Million Taka	% of THE	Million Taka	% of THE	Million Taka	% of THE	Million Taka	% of THE	Million Taka	% of THE	Million Taka	% of THE	Million Taka
41,318	26%	103,459	64%	1,325	1%	314	0%	2,092	1%	12,391	8%	160,899

Annex to chapter 4

Summary of private-public partnerships for community clinics

Partner organization	Working area	Key function
PLAN International	5 districts, 6 upazilas, 932 CCs	Capacity-building of Core Groups (CGs) and Community Support Groups (CSGs)
CARE	4 districts, 24 upazilas, 639 CCs	
DASCOH	1 district, 9 upazilas, 225 CCs	
SEED	Randomly-selected area	Operational research on CC performance
PHD	1 district, 7 upazilas, 148 CCs	Capacity-building of Core Groups (CGs) and Community Support Groups (CSGs)
M-CHIP	2 districts, 19 upazilas, 360 CCs	
FANTA 3600	4 districts, 27 upazilas, 602 CCs	Distribution of BCC materials and others
Health network	9 districts, 9 upazilas, 54 CCs	Capacity-building of Core Groups (CGs) and Community Support Groups (CSGs)
VSO	1 district, 8 upazilas, 285 CCs	
Spring	7 districts, 15 upazilas, 432 CCs	Capacity-building of CHCP HA, FWA, Core Groups (CGs), and Community Support Groups (CSGs) in nutrition

Data on institutional obstetric care according to UN process indicators by division-wise disaggregation (2012)

Head	Barisal	Chittagong	Dhaka	Khulna	Rajshahi	Rangpur	Sylhet	Bangladesh
Visit for ANC service (N)	73386	385946	546716	322208	205710	328898	102139	1965003
Admitted patients (N)	47319	141342	270539	613037	126556	126330	61249	1386372
Complications treated (N)	18722	49059	111988	57733	32526	28698	18549	317275
Normal delivery (N)	14036	61786	92150	54449	62858	59003	24971	369253
Forceps/Vacuum/Destructive operation (N)	33	1311	1133	518	1080	983	822	5880
Vaginal breech/Face presentation delivery (N)	136	1943	1774	737	558	1374	624	7146
Cesarean section (N)	1,035	41386	103033	35915	27872	30011	17111	271363
Total deliveries (N)	30240	106426	198090	91619	92368	91371	43528	653642
Livebirth (N)	29305	104604	196131	90317	90856	90953	41726	643892
Stillbirth (N)	1146	3968	5002	2030	2280	2176	2302	18904
Other surgeries (N)	1021	5220	10470	3060	2975	3102	1160	27008
Referred in (N)	1888	10040	12403	4619	2465	4829	2140	38384
Referred out (N)	2809	8434	12648	5343	5575	9496	4093	48398
Visit for PNC service (N)	24520	127851	245996	96505	92240	105305	54469	746886
Maternal death (N)	183	318	367	477	379	442	173	2339
Neonatal death (N)	680	401	1712	389	985	731	1010	5908
Proportion (%) of all births in EmOC facilities	17.5	10.8	20.1	28.1	24.1	27.9	21.2	21.9
Met need (%) for EmOC	72.3	33.2	75.8	118.2	56.5	58.4	60.2	70.8
Cesarean section rate as % of all births	9.3	4.2	10.5	11.0	7.3	9.2	8.3	9.1
Case-fatality rate (CFR) (%)	1.0	0.7	0.3	0.8	1.2	1.5	0.9	0.7

Data on institutional obstetric care according to UN process indicators by type of facility (2012)

UN process indicator	Med. coll. hosp.	District hosp.	Upazila health complex	MCWC	Total in govt. facility		NGO	Private clinic/ hosp.	Other	Total in non-govt. facility		Total
	No.	No.	No.	No.	No.	% of total	No.	No.	No.	No.	% of total	No.
Visit for ANC service (N)	116579	277786	691981	221423	1307769	66.6	319975	332459	4800	657234	33.4	1965003
Admitted patients (N)	136220	691711	249147	40970	1118048	80.6	51578	213127	3619	268324	19.4	1386372
Complications (N)	49583	86926	74070	5619	216198	68.1	23606	76920	551	101077	31.9	317275
Livebirth (N)	86501	105351	166832	37285	395969	61.5	57241	188467	2215	247923	38.5	643892
Stillbirth (N)	5401	4541	5327	447	15716	83.1	888	2235	65	3188	16.9	18904
Other surgeries (N)	8941	8,083	5645	530	23199	85.9	166	3589	54	3809	14.1	27008
Referred in (N)	9059	9,918	4649	1651	25277	65.9	6087	7020	-	13107	34.1	38384
Referred out (N)	786	7914	30,392	1638	40730	84.2	3815	3574	279	7668	15.8	48398
PNC service (N)	50348	132172	239,981	57997	480498	64.3	85417	178880	2091	266388	35.7	746886
Maternal death (N)	813	527	648	118	2106	90.0	177	55	1	233	10.0	2339
Neonatal death (N)	4311	375	796	22	5504	93.2	193	210	1	404	6.8	5908

Distribution of cervical and breast cancer screening centers in the country (total 300)

Name of location	No. of centers established between 2005 and 2012*	No. of centers established between 2011 and 2012**
BSMMU, medical college hospital	15	
District hospital	57	
Upazila health complex	20	35
MCHTI, MFSTC, Mother and child welfare center	61	
Union health & family welfare center	52	
URBAN Primary Health Care Project	25	
Other Non-government organizations	35	
Total	265	

*Developed by DGHS, DGFP, BSMMU, and UNFPA; **Developed by MOHFW and BSMMU

Number of colposcopy tests done in different colposcopy clinics in 2012

Name of colposcopy clinic	No. of colposcopy done N (%)
Bangabandhu Sheikh Mujib Medical University (BSMMU)	2115 (31.20)
M.A.G. Osmani Medical College Hospital (SMAGOMCH), Sylhet	752 (11.10)
Rajshahi Medical College Hospital (RajMCH)	628 (9.25)
Chittagong Medical College Hospital (CMCH)	568 (8.35)
Dhaka Medical College Hospital (DMCH)	525 (7.75)
Khulna Medical College Hospital (KMCH)	465 (6.85)
Mymensingh Medical College Hospital (MMCH)	396 (5.80)
Shaheed Suhwardy Medical College Hospital (SSMCH)	269 (3.95)
Sir Salimullah Medical College & Mitford Hospital	230 (3.40)

Table continued

Name of colposcopy clinic	No. of colposcopy done N (%)
Faridpur Medical College Hospital (FMCH)	219 (3.20)
Barisal Shere-Bangla Medical College Hospital (SBMCH)	214 (3.20)
Comilla Medical College Hospital (CoMCH)	195 (2.85)
Rangpur Medical College Hospital (RpMCH)	112 (1.65)
Dinajpur Medical College Hospital (DinajMCH)	67 (1.00)
Shaheed Ziaur Rahman Medical College Hospital (SZMCH)	30 (0.45)
Total	6785 (100)

Distribution of children aged 0-5 year(s) by division in each category of IMCI diseases, who received treatment from the IMCI facilities (2012)

Disease/health problem	Unit	Barisal	Chittagong	Dhaka	Khulna	Rajshahi	Rangpur	Sylhet	Total
Very severe disease	No.	10877	38632	58698	32448	28674	15745	25597	210671
	%	5.2	18.3	27.9	15.4	13.6	7.5	12.2	100.0
Pneumonia	No.	15775	77598	105561	36676	76487	39099	39692	390888
	%	4.0	19.9	27.0	9.4	19.6	10.0	10.2	100.0
No-pneumonia (cough and cold)	No.	86985	346406	493659	159864	337648	162559	120483	1707605
	%	5.1	20.3	28.9	9.4	19.8	9.5	7.1	100.0
Diarrhea	No.	44244	148535	153531	65935	107143	72075	78901	670364
	%	6.6	22.2	22.9	9.8	16.0	10.8	11.8	100.0
Fever-malaria	No.	770	3062	2662	623	510	2412	436	10475
	%	7.4	29.2	25.4	5.9	4.9	23.0	4.2	100.0
Fever-no malaria	No.	60763	176821	269265	93545	177432	109302	74326	961454
	%	6.3	18.4	28.0	9.7	18.5	11.4	7.7	100.0
Measles	No.	3846	15806	27305	13204	23033	13325	13292	109811
	%	3.5	14.4	24.9	12.0	21.0	12.1	12.1	100.0
Ear problem	No.	12012	34943	50298	17293	35586	21588	25634	197354
	%	6.1	17.7	25.5	8.8	18.0	10.9	13.0	100.0
Malnutrition	No.	14904	41948	45961	16838	32670	25421	31079	208821
	%	7.1	20.1	22.0	8.1	15.6	12.2	14.9	100.0
Others	No.	59415	167459	243586	77508	184275	74015	88980	895238
	%	6.6	18.7	27.2	8.7	20.6	8.3	9.9	100.0
Total	No.	309591	1051210	1450527	513934	1003458	535541	498420	5362681
	%	5.8	19.6	27.0	9.6	18.7	10.0	9.3	100.0

Distribution of children (%) aged 1-5 year(s) in each IMCI disease category by division (2012)

Disease/health problem	Barisal	Chittagong	Dhaka	Khulna	Rajshahi	Rangpur	Sylhet	Total
No. of cases	309591	1051210	1450527	513934	1003458	535541	498420	5363281
Very severe disease	3.5	3.7	4.0	6.3	2.9	2.9	5.1	3.9
Pneumonia	5.1	7.4	7.3	7.1	7.6	7.3	8.0	7.3
No-pneumonia (cough and cold)	28.1	33.0	34.0	31.1	33.6	30.4	24.2	31.8
Diarrhea	14.3	14.1	10.6	12.8	10.7	13.5	15.8	12.5

Table continued

Disease/health problem	Barisal	Chittagong	Dhaka	Khulna	Rajshahi	Rangpur	Sylhet	Total
Fever-malaria	0.2	0.3	0.2	0.1	0.1	0.5	0.1	0.2
Fever-no malaria	19.6	16.8	18.6	18.2	17.7	20.4	14.9	17.9
Measles	1.2	1.5	1.9	2.6	2.3	2.5	2.7	2.0
Ear problem	3.9	3.3	3.5	3.4	3.5	4.0	5.1	3.7
Malnutrition	4.8	4.0	3.2	3.3	3.3	4.7	6.2	3.9
Others	19.2	15.9	16.8	15.1	18.4	13.8	17.9	16.7
Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0

Distribution of IMCI diseases by age-group (summary of data received from IMCI facilities in 64 districts for 2012)

Disease/health problem	Unit	0-28 day(s)	29-59 days	2-12 months	1- 5 year(s)	Total
Very severe disease	No.	57081	81694	36878	35018	210671
	%	27	39	18	17	100
Pneumonia	No.	-	-	172433	218455	390888
	%	-	-	44	56	100
No-pneumonia (cough and cold)	No.	-	-	699262	1008343	1707605
	%	-	-	41	59	100
Diarrhea	No.	18024	46094	221731	384515	670364
	%	3	7	33	57	100
Fever-malaria	No.	-	-	3631	6844	10475
	%	-	-	35	65	100
Fever-no malaria	No.	-	-	334321	627124	961454
	%	-	-	35	65	100
Measles	No.	844	1851	3313	103803	109811
	%	1	2	3	95	100
Ear problem	No.	8386	20085	57840	111043	197354
	%	4	10	29	56	100
Malnutrition	No.	12439	25006	60813	110763	208821
	%	6	12	29	53	100
Others	No.	66992	100764	226529	501440	895238
	%	7	11	25	56	100
Total	No.	163766	275494	1816751	3107348	5362681
	%	3	5	34	58	100

Distribution of IMCI diseases (%) within each age-group of children of both sexes, who received treatment from the IMCI facilities (2012)

Disease/health problem	0-28 day(s)	29-59 days	2-12 months	1-5 year(s)	Total
Total cases (N)	163766	275494	275494	3107348	5362681
Percentage					
Very severe disease	34.9	29.7	1.1	1.1	3.9
Pneumonia	-	-	7.0	7.0	7.3
No-pneumonia (cough and cold)	-	-	32.5	32.5	31.8
Diarrhea	11.0	16.7	12.4	12.4	12.5
Fever-malaria	-	-	0.2	0.2	0.2
Fever-no malaria	-	-	20.2	20.2	17.9
Measles	0.5	0.7	3.3	3.3	2.0
Ear problem	5.1	7.3	3.6	3.6	3.7
Malnutrition	7.6	9.1	3.6	3.6	3.9
Others	40.9	36.6	16.1	16.1	16.7
Total	100.0	100.0	100.0	100.0	100.0

Annex to chapter 5

Number of sanctioned beds, free beds, departments, wards, cabins, and operation theaters in some private hospitals (arranged by bed-capacity) (2012)

Name and location of private health facility	Number					
	Sanctioned bed	Free bed	Department	Ward	Cabin	Operation theater
Jalalabad Ragib-Rabeya Hospital, Sylhet	890	9	14	18	120	10
BIRDEM Hospital, Dhaka	596	118	0	27	117	11
Ad-din Medical College Hospital, Dhaka	500	290	23	15	98	7
Shahid Mansur Ali Medical College Hospital, Uttara, Dhaka	500	200	19	13	20	6
Uttara Adhunik Medical College Hospital, Uttara, Dhaka	500	0	12	12	0	11
East West Medical College Hospital, Uttara, Dhaka	400	80	14	14	29	6
Khawja Yunus Ali Medical College and Hospital, Chowhali, Sirajganj	400	40	67	19	72	9
North Bengal M. C & Hospital, Sirajganj Sadar	400	120	12	10	30	6
Medical College for Women and Hospital, Uttara, Dhaka	350	50	11	7	23	4
Apollo Hospitals, Dhaka	304	0	31	15	49	8
Dhaka Hospital, icddr,b	300	300	2	15	29	13
Dhaka Community Hospital, Dhaka	250	75	14	3	20	2
Islami Bank Hospital, Dhaka	160	0	9	8	76	5
Christian Hospital, Chondroghona, Chittagong	125	0	10	10	10	3
BNSB Eye Hospital, Sirajganj Sadar	100	20	4	9	11	2
Z.H. Sikder Women's Medical College Hospital, Dhaka	100	0	0	2	17	2
Lion Eye Institute & Hospital, Dhaka	84	10	6	6	10	4
Monowara Hospital, Hatkhola Road, Dhaka	74	4	6	26	48	3
Islami Bank Hospital, Laxmipur, Rajshahi	63	0	8	6	30	2
Prime Hospital Ltd., Hospital Road, Majidee, Noakhali	60	2	3	4	34	2
Damien Foundation, Netrakona TB & Leprosy Hospital, Anantapur, Netrokona	52	52	1	10	0	0
Ambia Hospital, Bogra Road, Barisal	50	0	4	6	15	2
Islami Bank Hospital, Chandmary, Barisal	50	0	16	4	20	2
City Hospital Private, Majidee, Noakhali	40	2	5	2	27	2
Dr. Moklessur Clinic, Sadar Road, Barisal	40	0	4	3	12	2
Fair Health Clinic, Barisal	40	0	3	2	18	2
Good Heal Hospital, Majidee, Noakhali	40	5	3	4	22	2
Janani General Hospital, Noakhali Sadar	40	0	0	2	23	1
Luthern Health Care of Bangladesh, Dumki, Patuakhali	40	0	3	2	8	2
Modern Hospital Private, Majidee, Noakhali	40	0	5	8	32	2
Surgecare Clinic, Pirojpur Sadar	40	0	0	3	14	2
Amina Hospital, Bonpara, Natore	30	2	4	4	5	1
Avicena Hospital Ltd., Sirajganj Sadar	30	0	4	10	20	4
Dolphin Clinic, Bormalimur, Rajshahi	30	2	6	2	15	2
Dream Hospital, Begumganj, Noakhali	30	0	5	15	15	1

Table continued

Name and location of private health facility	Number					
	Sanctioned bed	Free bed	Department	Ward	Cabin	Operation theater
Medinova Hospital, Sirajganj Sadar	30	2	5	16	14	2
Mukti Clinic, Laxmipur, Rajshahi	30	0	4	3	15	2
Safeway (Pvt) Hospital, Mymensingh Sadar	28	0	0	2	6	2
Hathazari Adhunic Hospital, Chittagong	25	2	4	4	5	1
Apollo Hospital & Diagnostic Complex, Majdee Bazar, Noakhali	20	0	3	12	8	2
Bengal Community, Ullapara, Sirajganj	20	0	3	2	4	1
Eden Nursing Home, Alekanda, Barisal	20	2	3	3	8	1
Health Care Clinic, Parara Road, Barisal	20	0	4	4	8	1
Mamota Clinic, Kali Bari Road, Barisal	20	0	4	2	18	2
Modern Clinic, Munshiganj	20	0	3	3	3	1
Muslim Aid Community Hospital, Pirojpur Sadar	20	0	3	3	5	1
Nalta Hospital, Kaliganj, Satkhira	20	0	6	2	3	2
Padma Clinic, Kazihata, Rajshahi	20	1	2	2	10	1
Saiam (Pvt) Hospital, Mymensingh Sadar	20	0	0	2	9	2
Santi Private Hospital, Magura	20	0	2	14	6	1
Woodland Hospital & Diagnostic Complex, Majdee, Noakhali	20	0	3	8	12	1
Central Hospital, Sirajganj Sadar	15	3	5	12	10	2
A K Eye Hospital, Magura	10	0	1	6	4	1
Ahamedia General Hospital, Mymensingh	10	0	0	2	2	1
Ahsania Clinic, Debhata, Satkhira	10	0	2	2	2	1
Akota Clinic & Diagnostic Center, Rajshahi	10	1	3	3	6	1
Akota Clinic, Satkhira Sadar	10	0	2	2	2	1
Al Hera Private Hospital, Magura	10	0	0	3	4	1
Al-Modina Clinic, Magura	10	0	0	6	4	1
Al Modina General Hospital, Kishoreganj	10	0	2	2	3	1
Al Safa (Pvt.) Hospital, Mymensingh Sadar	10	0	0	2	4	1
Al Zannat (Pvt.) Hospital, Mymensingh Sadar	10	0	0	2	4	1
Al-Amin Nursing Home, Mymensingh Sadar	10	0	0	2	8	1
Albaraka Clinic, Laxmipur, Rajshahi	10	0	4	2	2	1
Al-Baraka Clinic, Magura	10	0	0	7	3	1
Al-Shefa Clinic, Joypurhat	10	0	2	2	4	1
Ambia Hospital, Pirojpur, Barisal	10	0	2	2	3	1
Amena Clinic, Talaimari, Rajshahi	10	0	2	4	4	1
Anowara Clinic, Satkhira Sadar	10	1	3	2	1	1
Anwara Private Hospital, Jhenaidah	10	0	2	2	3	1
Apollo Nursing Home, Sipaipara, Rajshahi	10	0	3	3	3	1
Arafat Clinic & Diagnostic, Munshiganj	10	0	2	2	3	1
Aroggo Clinic, Magura	10	0	4	6	4	1
Asha (Pvt.) Hospital, Mymensingh Sadar	10	0	0	2	1	1
Bangladesh Clinic, Upashohor, Rajshahi	10	1	4	4	1	1
Baral Clinic LTD. Pabna	10	0	1	2	3	1

Table continued

Name and location of private health facility	Number					
	Sanctioned bed	Free bed	Department	Ward	Cabin	Operation theater
Barisal Poly Clinic, Bangla Bazar, Barisal	10	0	3	2	6	1
Bashundhara (Pvt.) Hospital, Mymensingh Sadar	10	0	0	2	8	1
Bhai Bhai Private Hospital, Jhenaidah	10	0	2	2	3	1
Bonshepur Nursing Home, Shyamnagar, Satkhira	10	0	2	2	2	1
Brizee Hospital, Laxmipur, Rajshahi	10	2	3	2	2	1
Care Nursing Home, Laxmipur, Rajshahi	10	0	4	2	3	1
Chowmohani General Hospital, Begumganj, Noakhali	10	0	1	0	10	1
City Clinic, Satkhira Sadar	10	0	3	2	2	1
Cure Nursing Home, Laxmipur, Rajshahi	10	0	4	2	5	1
Dastagir Private Hospital, Narsingdi	10	0	1	2	3	1
Desh Eye Hospital, Gazipur Sadar	10	0	2	2	3	1
Dobir Uddin Hospital, Kasiadanga, Rajshahi	10	1	5	4	2	1
Doctors Care Clinic and Hospital, Barguna, Barisal	10	0	2	2	3	1
Dr. Khadem Hossain Clinic, Bangla Bazar, Barisal	10	0	3	3	2	2
Ibne Hashman (Pvt.) Hospital, Feni, Chittagong	10	0	2	2	3	1
Ehsan General Hospital, Magura	10	0	0	5	5	1
EM Center, Mohishbathan, Rajshahi	10	0	5	3	5	1
Ekushey Hospital (Pvt), Mymensingh Sadar	10	0	0	2	2	1
Faruk Al-Nasir Hospital, Kazipur, Sirajganj	10	0	-	-	-	0
Farzina Clinic, Kazipur, Sirajganj	10	0	3	2	4	1
Fatema Nursing Home, Mymensingh Sadar	10	0	0	2	7	2
Fatima Nursing Home, Laxmipur, Rajshahi	10	0	5	5	3	1
Gorib Shah Clinic, Magura	10	0	1	3	4	1
Gorib-E-Nawaz Clinic, Talaimari, Rajshahi	10	1	3	3	3	1
Hasina Clinic & Nursing Home, Magura	10	0	0	7	3	1
Impact MasudulHaque Community Health Centre, Chuadanga	10	0	2	2	1	2
Islami General Hospital, Keshorhat, Rajshahi	10	0	2	3	1	1
Islami General Hospital, Nowhata, Rajshahi	10	0	5	2	2	1
Islamia Poly Clinic, Bangla Bazar, Barisal	10	0	2	2	2	1
Jahangir Health Complex, Mymensingh Sadar	10	0	0	2	6	1
Jam-jam Islami Clinic, Laxmipur, Rajshahi	10	2	3	3	3	1
Jamuna (Pvt.) Hospital, Mymensingh Sadar	10	0	0	2	4	1
Jamuna Clinic, Kaliganj, Satkhira	10	0	2	2	2	1
Jamuna Clinic, Laxmipur, Rajshahi	10	0	3	2	4	1
Janani Clinic, Jiban Nagar, Chuadanga, Khulna	10	0	2	2	3	1
Janaseba Clinic & Nursing Home, Magura	10	2	3	3	2	1
Janaseba Clinic, Assasuni, Satkhira	10	0	3	2	2	1
Janata Clinic & Nursing Home, Magura	10	0	0	1	3	1
Janata Clinic, Shipaipara, Rajshahi	10	1	2	2	5	1
Jayeda Hospital, Bonpara, Natore	10	0	2	2	2	1
Khadiza Clinic, Kalia, Narail	10	0	2	2	3	1

Table continued

Name and location of private health facility	Number					
	Sanctioned bed	Free bed	Department	Ward	Cabin	Operation theater
Labib (Pvt.) Hospital, Mymensingh Sadar	10	0	0	2	8	1
Life Care Hospital, Mymensingh Sadar	10	0	0	3	4	1
Maya Clinic, Mymensingh Sadar	10	0	0	2	1	1
Mediplus Hospital & Diagnostic Center, Mymensingh Sadar	10	0	0	2	6	1
Micropath Diagnostic and Clinic, Laxmipur, Rajshahi	10	1	5	3	5	1
Mita Private Hospital, Narsingdi	10	0	2	2	4	1
Modern Central Hospital, Barguna, Barisal	10	0	1	3	4	1
Mohanagar Clinic, Kazihata, Rajshahi	10	1	3	3	5	1
Moin Uddin Hospital, Sirajganj Sadar	10	0	1	2	8	2
Mother Clinic, Bhurungamari, Kurigram	10	0	2	2	2	1
Motherland Clinic, Laxmipur, Rajshahi	10	2	3	2	6	1
Mother's Clinic, Uposhohor, Rajshahi	10	0	2	3	2	1
Mukta Clinic, Shibchar, Madaripur	10	0	1	2	3	1
Muktar General Hospital, Gopalpur, Lalpur, Natore	10	0	2	2	3	1
Mukti Clinic, Magura	10	0	2	2	2	1
Nabila General Hospital, Begumganj, Noakhali	10	0	4	1	7	1
Nagorick (Pvt.) Hospital, Mymensingh Sadar	10	0	0	2	5	1
Nasima Nursing Home, Mymensingh Sadar	10	0	0	2	6	3
Nazma Nursing Home, Mymensingh Sadar	10	0	0	2	5	1
New Arafat Clinic, Magura	10	0	0	4	2	1
New Surgical Clinic, Magura	10	0	0	7	3	1
Nibedita Nursing Home, Assasuni, Satkhira	10	0	2	4	1	1
Niramoya Clinic, Mymensingh Sadar	10	0	0	2	4	2
Padma Clinic, Magura	10	0	1	2	3	1
Paricharja (Pvt.) Hospital, Mymensingh Sadar	10	0	0	2	5	1
Paromita Eye Hospital, Mymensingh Sadar	10	0	0	2	4	2
PKS Clinic, Satkhira Sadar	10	0	1	1	5	1
Poly Clinic, Magura	10	0	2	6	4	1
Rafia Clinic, Ati Bazar, Keraniganj, Dhaka	10	0	1	2	3	1
Raihana Clinic, Puthia, Rajshahi	10	0	2	4	2	1
Rajdhani (Pvt.) Hospital, Mymensingh Sadar	10	0	0	2	3	1
Razzak Memorial Clinic, Alekanda, Barisal	10	0	2	2	3	1
Rokeya (Pvt) Hospital, Mymensingh Sadar	10	0	0	2	6	1
Saleha Clinic, Magura	10	0	0	2	4	1
Salma Clinic, Mymensingh Sadar	10	0	0	2	1	1
Samata Nursing Home, Laxmipur, Rajshahi	10	2	2	4	0	1
Sarmin Private Clinic & Nursing Home, Magura	10	0	2	2	4	1
Satata Clinic, Satkhira Sadar	10	1	3	2	2	1
Satata Private Hospital, Sreepur, Magura	10	0	0	8	2	1
Seba Hospital, Raipur, Laxmipur	10	0	2	2	4	1
Seba Clinic, Magura	10	0	0	2	2	1

Table continued

Name and location of private health facility	Number					
	Sanctioned bed	Free bed	Department	Ward	Cabin	Operation theater
Seba General Hospital, Ullapara, Sirajganj	10	0	2	2	1	1
Sea Side Hospital, Cox's Bazar	10	0	2	2	3	1
Shagata Clinic, Tala, Satkhira	10	0	2	2	2	1
Shahin Clinic (PVT.) Hospital, Feni, Chittagong	10	0	2	2	3	1
Shapla Nursing Home, Mymensingh Sadar	10	0	0	2	9	1
Sharmin Nursing Home, Rajshahi	10	1	2	2	7	1
Sheba Clinic, Jiban Nagar, Chuadanga, Khulna	10	0	2	2	4	1
Sheba Clinic, Kalaroa, Satkhira	10	0	2	2	2	1
Shemul Clinic, Satkhira Sadar	10	0	2	2	2	1
Shurakkha Nursing Home, Mymensingh Sadar	10	0	0	2	10	1
Soudia Hospital, Rajpara, Rajshahi	10	0	4	2	2	1
Sreepur Clinic, Sreepur, Magura	10	0	2	2	2	1
Sunmoon Clinic, Magura	10	0	3	3	4	1
Sunrise Clinic, Magura	10	0	2	2	2	1
The Akota Clinic, Ghose Para, Rajshahi	10	0	6	2	2	1
The Ibn-sina Clinic, Magura	10	0	2	2	2	1
Trauma Center and General Hospital, Mymensingh Sadar	10	0	0	2	5	1
Ullapara Hospital, Ullapara, Sirajganj	10	3	2	3	2	2
Upasam (Pvt.) Hospital, Mymensingh Sadar	10	0	0	2	3	1
Uttaran Nursing Home, Mymensingh Sadar	10	0	0	2	8	1
Uttarbanga Islami Hospital, Laxmipur, Rajshahi	10	0	3	4	4	1
Kaisar Memorial Hospital, Upashahor, Chittagong	9	0	3	2	2	1
Rabeya Banu General Hospital, Biswanath, Sylhet	6	1	11	2	0	1
Total	8401	1415	610	737	1683	305

Annex to Chapter 6

Number of beds, admissions, hospital deaths, outdoor visits, bed-occupancy rates, average length of stay, and hospital death rates in medical/dental college hospitals, postgraduate teaching hospitals, and Pabna mental hospital (2012)

Health facility	No. of beds	No. of admissions			No. of hospital deaths			No. of outdoor visits			Bed-occupancy rate (%)	Average length of stay (d)	Average daily number		Hospital death rate (%)			
		Male	Female	Child	Total	Male	Female	Child	Total	Male			Female	Child		Total	Admission	Outdoor visit
Medical college hospitals																		
Chittagong Medical College Hospital	1010	64059	38435	25118	127612	2560	2220	1974	6754	284610	203815	38705	527130	187.2	5.4	349.6	1444.2	5.3
Shahid Ziaur Rahman Medical College Hospital	500	22315	20793	7607	50715	1009	917	872	2798	127058	130012	38413	295483	142.1	5.1	138.9	809.5	5.5
Sylhet M.A.G. Osmani Medical College Hospital	900	47678	43871	20114	111663	1721	1425	1199	4345	292129	359914	94947	746990	156.4	4.6	305.9	2046.5	3.9
Shaheed Suhrawardy Medical College Hospital	850	11123	11878	6886	29887	83	63	38	184	166170	195035	24148	385353	76.7	8.0	81.9	1055.8	0.6
Faridpur Medical College Hospital	250	17381	19667	3924	40972	930	982	546	2458	46095	45538	12014	103647	159.6	3.6	112.3	284.0	6.0
Dinajpur Medical College Hospital	500	14422	13387	5053	32862	281	243	93	617	35367	37728	13567	86662	94.5	5.3	90.0	237.4	1.9
Khulna Medical College Hospital	500	18730	20580	3994	43304	1332	1164	276	2772	63264	83115	51482	197861	113.5	4.8	118.6	542.1	6.4
Dhaka Medical College Hospital	2400	49946	26296	19077	95319	5618	2790	458	8866	327089	205059	102507	634655	95.6	8.8	261.1	1738.8	9.3
Rajshahi Medical College Hospital	550	46172	53477	23768	123417	2689	1776	639	5104	181348	301599	77681	560628	228.6	3.7	338.1	1536.0	4.1
Sher-e-Bangla Medical College Hospital	1000	32935	32206	13549	78690	1101	1095	862	3058	174315	101312	78442	354069	117.6	5.5	215.6	970.1	3.9
Rangpur Medical College Hospital	1000	41292	34875	13212	89379	1955	1370	873	4198	122000	106741	8992	237733	120.0	4.9	244.9	651.3	4.7
Sir Salimullah Medical College Hospital	600	20837	32119	8442	61398	815	667	453	1935	159627	234752	82528	476907	129.5	4.6	168.2	1306.6	3.2
Mymensingh Medical College Hospital	1000	56206	52343	12225	120774	3189	2589	1897	7675	108400	134500	101051	343951	151.3	4.6	330.9	942.3	6.4
Comilla Medical College Hospital	500	17795	18099	7836	43730	511	403	702	1616	90570	87033	60703	238306	122.2	5.1	119.8	652.9	3.7
Postgraduate teaching institute hospitals																		
National Institute of Kidney Diseases & Urology	116	2563	1373	503	4439	106	72	7	185	22098	15179	3140	40417	95.8	9.2	12.2	110.7	4.2
National Center for Control of Rheumatic Fever & Heart Diseases	-	-	-	-	-	-	-	-	-	4676	8293	14510	27479	-	-	-	75.3	-
National Institute of Mental Health and Research	200	1249	679	102	2030	-	-	-	-	14959	8939	1610	25508	63.8	23.0	5.6	69.9	-
National Institute of Cardiovascular Diseases	414	30971	13588	-	44559	2156	929	-	3085	67447	34008	8291	109746	153.2	5.2	122.1	300.7	6.9
National Institute of Diseases of the Chest and Hospital	670	8235	2557	270	11062	705	192	-	897	21506	15129	6027	42662	95.2	21.1	30.3	116.9	8.1
National Institute of Ophthalmology	250	8235	7788	553	16576	-	-	-	-	66242	58304	22393	146939	81.0	-	45.4	402.6	-

Table continued

Health facility	No. of beds	No. of admissions			No. of hospital deaths			No. of outdoor visits			Bed-occupancy rate (%)	Average length of stay (d)	Average daily number		Hospital death rate (%)
		Male	Female	Child	Male	Female	Child	Male	Female	Child			Admission	Outdoor visit	
National Institute of Traumatology, Orthopedics & Rehabilitation	500	10656	6667	2023	19346	153	35	201	46109	30659	71.6	6.8	53.0	247.1	1.0
National Institute of Cancer Research & Hospital	150	1731	1289	481	3501	45	15	65	33073	26148	87.8	13.8	9.6	167.2	1.9
Institute of Child and Mother Health	200	-	7561	6115	13676	-	2	447	449	-	89.8	4.8	37.5	2939.5	3.3
Mental hospital															
Pabna Mental Hospital	500	1043	299	-	1342	1	1	-	2	13209	17199	114.1	3.7	83.3	0.1
Dental college hospital															
Dhaka Dental College Hospital	200	5540	4814	362	10716	1	1	1	3	38925	25820	23.0	29.4	208.4	0.0

Number of beds, admissions, hospital deaths, outdoor visits, bed-occupancy rates, average length of stay, and hospital death rates in district hospitals (2012)

Health facility	No. of beds	No. of admissions			No. of hospital deaths			No. of outdoor visits			Total patient days	Average length of stay (d)	Bed-occupancy rate (%)	Average daily number		Hospital death rate (%)			
		Male	Female	Child	Total	Male	Female	Child	Total	Admission				Outdoor visit					
Barisal division																			
Barguna District Hospital	100	4656	5110	2697	12463	55	53	51	159	22987	31823	34474	89284	41131.0	3.3	112.7	34.1	244.6	1.3
Bhola District Hospital	100	8186	9105	5104	22395	155	176	175	506	26317	35688	13643	75648	51897.0	2.3	142.2	61.4	207.3	2.3
Jhalokathi District Hospital	100	4294	6072	1335	11701	30	19	1	50	35896	43528	35607	115031	34974.0	3.0	95.8	32.1	315.2	0.4
Patuakhali 250-bed Sadar Hospital	250	7439	12325	5099	24863	134	136	133	403	27153	47094	47482	121729	102736.0	4.1	112.6	68.1	333.5	1.6
Perinpur District Hospital	100	3561	6583	1681	11825	71	48	42	161	30659	37951	9130	77740	39100.0	3.3	107.1	32.4	213.0	1.4
Chittagong Division																			
Bandarban District Hospital	100	2663	2755	1836	7254	23	19	34	76	13467	16350	16412	46229	20146.0	2.8	55.2	19.9	126.7	1.0
Brahmanbaria 250-bed District Sadar Hospital	250	13420	9033	4604	27057	198	174	100	472	130042	102408	53880	286330	82057.0	3.0	89.9	74.1	784.5	1.7
Chandpur 250-bed General Hospital	200	6985	8247	7624	22856	113	102	82	297	52574	57185	54471	164230	75172.0	3.3	103.0	62.6	449.9	1.3
Chittagong General Hospital	250	3351	4882	1597	9830	22	20	24	66	57745	52311	35536	145592	62096.0	6.3	68.1	26.9	398.9	0.7
Coxilla General Hospital	100	11001	14781	5336	31118	13	8	17	38	22033	32276	22420	76729	31118.0	1.0	85.3	85.3	210.2	0.1
Cox's Bazar District Hospital	100	2663	2755	1836	7254	23	19	34	76	13467	16350	16412	46229	20146.0	2.8	55.2	19.9	126.7	1.0
Feni District Hospital	250	7480	10660	6427	24567	130	145	171	446	69865	72369	68863	211097	80879.0	3.3	88.6	67.3	578.3	1.8
Khagrachhari District Hospital	100	3104	4446	2612	10162	87	74	151	312	17168	25178	22667	65013	30168.0	3.0	82.7	27.8	178.1	3.1
Lakshmipur District Hospital	100	6625	7019	1026	14670	116	41	13	170	57432	76039	15913	149384	44972.0	3.1	123.2	40.2	409.3	1.2
Noakhali 250-bed General Hospital	250	11362	10711	9606	31679	342	303	175	820	49059	64246	39984	153289	125723.0	4.0	137.8	86.8	420.0	2.6
Rangamati General Hospital	100	3529	4195	2388	10112	57	43	69	169	13353	18451	3558	35362	27531.0	2.7	75.4	27.7	96.9	1.7

Table continued

Health facility		No. of beds	No. of admissions			No. of hospital deaths			No. of outdoor visits			Total patient days	Average length of stay (d)	Bed-occupancy rate (%)	Average daily number		Hospital death rate (%)		
			Male	Female	Child	Male	Female	Child	Male	Female	Child				Admission	Outdoor visit			
Dhaka-division																			
Faridpur General Hospital	100	6072	6280	6076	18428	88	61	27	176	36321	35182	27526	99029	59021.0	3.2	161.7	50.5	271.3	1.0
	100	6463	6889	2014	15366	25	20	60	105	45123	60893	51080	157096	26689.0	1.7	73.1	42.1	430.4	0.7
	250	7021	8138	4736	19895	130	65	182	377	39848	49216	34569	123633	64831.0	3.3	71.0	54.5	338.7	1.9
	250	10574	16097	8681	35352	163	113	277	553	85868	105484	46159	237511	103053.0	2.9	112.9	96.9	650.7	1.6
	250	14063	18045	7967	40075	217	161	177	555	75952	128586	73539	278077	116826.0	2.9	128.0	109.8	761.9	1.4
	100	5070	9101	3673	17844	61	54	19	134	20131	23001	19253	62385	40621.0	2.3	111.3	48.9	170.9	0.8
	100	6736	8251	2754	17741	132	102	128	362	47126	62895	47149	157170	51484.0	2.9	141.1	48.6	430.6	2.0
	100	3808	6719	5419	15946	33	35	9	77	32864	46447	22186	101497	31244.0	2.0	85.6	43.7	278.1	0.5
	200	2161	3412	1380	6953	87	38	8	133	95740	142841	66135	304716	58576.0	8.4	80.2	19.0	834.8	1.9
	100	4092	4632	3002	11726	22	23	3	48	37997	98321	66456	202774	33746.0	2.9	92.5	32.1	555.5	0.4
Narayanganj General Hospital	100	2537	2535	1863	6935	28	30	95	153	43254	49415	22075	114744	25746.0	3.7	70.5	19.0	314.4	2.2
	100	3301	3579	1474	8354	79	51	26	156	44040	41918	31169	117127	47324.0	5.7	129.7	22.9	320.9	1.9
	100	6793	11318	2759	20870	134	69	72	275	26959	27473	10876	65308	54523.0	2.6	149.4	57.2	178.9	1.3
	100	6020	8139	2711	16870	97	70	47	214	28497	40488	11253	80238	32572.0	1.9	89.2	46.2	219.8	1.3
	100	3895	7642	1326	12863	84	50	36	170	23131	34694	26829	84654	37184.0	2.9	101.9	35.2	231.9	1.3
	100	8286	7085	5213	20584	134	79	77	290	52008	49959	17455	119422	69819.0	3.4	191.3	56.4	327.2	1.4
	250	14618	22168	9914	46700	394	260	533	1187	58590	76808	35955	171353	127503.0	2.7	139.7	127.9	469.5	2.5
	Khulna division																		
	100	4631	5251	1673	11555	77	45	64	186	32828	51038	26551	110417	37502.0	3.2	102.7	31.7	302.5	1.6
	100	7049	19773	1582	28404	313	114	95	522	46180	52249	23921	122350	67788.0	2.4	185.7	77.8	335.2	1.8
278	20289	20913	3384	44586	1033	660	285	1978	93452	128465	40706	262623	136830.0	3.1	134.8	122.2	719.5	4.4	
100	8003	11744	3227	22974	147	110	39	296	43703	67998	29375	141076	63593.0	2.8	174.2	62.9	386.5	1.3	
150	4062	2833	1830	8725	48	43	18	109	68412	60692	21600	150704	46260.0	5.3	84.5	23.9	412.9	1.2	
250	16133	16335	10253	42721	617	369	104	1090	76939	99149	59642	235730	138261.0	3.2	151.5	117.0	645.8	2.6	
100	10724	13251	9702	33677	234	129	90	453	49048	51693	49164	149905	83220.0	2.5	228.0	92.3	410.7	1.3	
100	6130	11280	4500	21910	132	70	149	351	39706	70032	38908	148646	51837.0	2.4	142.0	60.0	407.2	1.6	
100	4368	6426	3663	14457	89	66	104	259	27159	47147	24828	99134	46836.0	3.2	128.3	39.6	271.6	1.8	
100	6895	8194	5310	20399	229	142	381	752	83084	107239	41749	232072	60831.0	3.0	166.7	55.9	635.8	3.7	
Rajshahi division																			
100	7980	9235	5350	22565	69	47	133	249	34283	44984	31934	111201	22565.0	1.0	61.8	61.8	304.7	1.1	
100	7576	11583	4244	23403	175	147	62	384	45355	63880	41720	150955	68995.0	2.9	189.0	64.1	413.6	1.6	
250	5917	8022	3194	17133	47	22	8	77	87125	102705	18583	208413	85855.0	5.0	94.1	46.9	571.0	0.4	

Table continued

Health facility	No. of beds	No. of admissions			No. of hospital deaths			No. of outdoor visits			Total patient days	Average length of stay (d)	Bed-occupancy rate (%)	Average daily number		Hospital death rate (%)
		Male	Female	Child	Total	Male	Female	Child	Total	Male	Female	Child	Total	Admission	Outdoor visit	
Naogaon District Hospital	100	9031	7911	1506	18448	143	106	17	266	70529	48466	29605	148600	50.5	407.1	1.4
Natore District Hospital	100	9590	7919	4054	21563	111	69	105	285	72573	65401	21660	159634	59.1	437.4	1.3
Pabna 250-bed General Hospital	250	13939	26694	18824	59457	302	248	512	1062	58947	49812	43005	151764	162.9	415.8	1.8
Sirajganj General Hospital	100	13337	15561	11089	39987	108	120	157	385	34902	35895	39521	110318	109.6	302.2	1.0
Rangpur division																
Dinajpur District Hospital	250	4417	5549	2903	12869	42	37	60	139	57244	58545	9605	125394	35.3	343.5	1.1
Gaibandha District Hospital	100	6392	6884	4740	18016	112	76	237	425	23124	30602	17627	71353	49.4	195.5	2.4
Kurigram District Hospital	100	8012	7895	7410	23317	148	88	250	486	48083	44393	46276	138752	63.9	380.1	2.1
Lalmonirhat District Hospital	100	5453	6097	1465	13015	31	27	38	96	44290	53319	6916	104525	35.7	286.4	0.7
Nilphamari District Hospital	100	17827	19203	2130	39160	57	59	19	135	31899	42787	62800	137486	107.3	376.7	0.3
Panchagarh 100-bed District Sadar Hospital	100	4482	5092	2562	12136	120	78	27	225	25633	38213	7678	71524	33.2	196.0	1.9
Thakurgaon District Hospital	100	8168	10072	22062	40302	167	128	233	528	40094	58348	42392	140834	110.4	385.8	1.3
Sylhet division																
Habiganj District Hospital	100	10305	14396	9764	34465	156	136	451	743	43393	42934	38962	125289	94.4	343.3	2.2
Maulvibazar District Hospital	250	6832	9720	8786	25338	89	78	73	240	44422	48545	42921	135888	69.4	372.3	0.9
Shahid Shamsuddin District Hospital	100	1478	1060	111	2649	-	-	-	-	48756	59800	10721	119277	7.3	326.8	-
Sunamganj 250-bed District Sadar Hospital	250	4324	5327	6691	16342	33	39	146	218	28773	34528	33219	96520	44.8	264.4	1.3

Number of hospital beds, admissions, deaths, outdoor visits, average length of stay, bed-occupancy rates, hospital death rates, average daily admissions, and outdoor visits in upazila hospitals

Health facility	No. of beds	No. of admissions			No. of hospital deaths			No. of outdoor visits			Average length of stay (d)	Bed-occupancy rate (%)	Hospital death rate (%)	Average daily number				
		Male	Female	Child	Total	Male	Female	Child	Total	Male				Female	Child	Total	Admission	Outdoor visit
Barisal division																		
Barguna district																		
Amnali Upazila Health Complex	50	1953	2107	500	4560	6	9	7	22	5500	7004	4564	17068	2.7	67.2	0.5	12.5	46.6
Bamna Upazila Health Complex	31	695	692	104	1491	3	9	3	15	14138	12805	4320	31263	4.7	61.5	1.0	4.1	85.4
Betagi Upazila Health Complex	50	926	941	478	2345	9	8	4	21	6671	6424	3295	16390	3.6	46.2	0.9	6.4	44.8
Pathargatha Upazila Health Complex	50	1379	1721	1014	4114	18	18	10	46	5081	5806	6095	16982	3.1	70.4	1.1	11.2	46.4
Barisal district																		
Aqailhara Upazila Health Complex	50	1046	1219	1119	3384	7	8	4	19	4209	5866	5153	15228	3.2	59.2	0.6	9.2	41.6
Babuganj Upazila Health Complex	31	348	518	235	1101	1	-	-	1	4298	7923	6789	19010	5.8	56.9	0.1	3.0	51.9

Table continued

Health facility	No. of beds	No. of admissions			No. of hospital deaths			No. of outdoor visits			Average length of stay (d)	Bed- occupancy rate (%)	Hospital death rate (%)	Average daily number	
		Male	Female	Child	Total	Male	Female	Child	Total	Male				Female	Admission
Bhola district															
Bakerganj Upazila Health Complex	31	1369	1724	654	3747	3	7	1	11	13309	19188	9925	42422	10.2	115.9
Banarpara Upazila Health Complex	50	1003	2742	677	4422	2	1	1	4	9427	9479	5502	24408	12.1	66.7
Gournadi Upazila Health Complex	50	2849	2412	1969	7230	21	4	3	28	7248	15086	11987	34321	19.8	93.8
Hijla Upazila Health Complex	31	939	1027	499	2465	4	1	5	10	5568	7590	2628	15786	6.7	43.1
Mehendiganj Upazila Health Complex	31	4264	5098	704	10066	3	7	6	16	8103	9305	12605	30013	27.5	82.0
Muladi Upazila Health Complex	50	1671	1856	637	4164	7	8	5	20	3181	5726	4291	13198	11.4	36.1
Wazirpur Upazila Health Complex	50	1467	1910	1006	4383	6	1	-	7	7237	12627	6750	26614	12.0	72.7
Bhola district															
Borhanuddin Upazila Health Complex	50	3391	4271	1601	9263	17	12	11	40	13528	19052	10672	43252	25.3	118.2
Charfession Upazila Health Complex	50	3194	4472	3518	11184	43	38	57	138	13294	16430	10807	40631	30.6	110.7
Daulatkhan Upazila Health Complex	50	1600	1925	1257	4782	4	2	15	21	15576	18618	11637	45831	13.1	125.2
Laimohan Upazila Health Complex	50	2911	3121	1669	7701	14	17	7	38	16810	18818	14542	50170	21.0	137.1
Manpura Upazila Health Complex	31	588	796	554	1938	5	12	2	19	17724	22408	11792	51924	5.3	141.9
Tajumuddin Upazila Health Complex	31	1016	915	399	2330	2	3	3	8	22950	26340	14908	64198	6.4	175.4
Jhalokathi district															
Kathalia Upazila Health Complex	31	770	988	516	2274	7	10	2	19	14495	19848	11579	45922	6.2	125.5
Nalchithi Upazila Health Complex	50	1135	1730	405	3270	8	11	1	20	14960	19575	8075	42610	8.9	116.4
Rajapur Upazila Health Complex	50	1210	1559	662	3431	5	6	2	13	8380	12967	9761	31108	9.4	85.0
Patuakhali district															
Bauphal Upazila Health Complex	31	1720	1537	432	3689	6	5	4	15	34634	40162	1292	76088	10.1	207.9
Dashmina Upazila Health Complex	31	1239	1678	1265	4182	6	10	8	24	16904	28224	17888	63016	11.4	172.2
Dumki Upazila Health Complex	31	279	437	239	955	2	2	-	4	5231	7575	4777	17583	2.6	48.0
Galachipa Upazila Health Complex	50	2505	1700	699	4904	10	10	7	27	9126	13469	6666	29261	13.4	79.9
Kalapara Upazila Health Complex	50	2113	3415	1163	6691	11	11	2	24	13487	19251	6924	39662	18.3	108.4
Mirzaganj Upazila Health Complex	50	1296	1285	388	2969	13	6	8	27	8621	9008	6905	24534	8.1	67.0
Pirojpur district															
Bhandaria Upazila Health Complex	31	1566	1976	645	4187	6	6	3	15	13422	18207	4552	36181	11.4	98.9
Kawkhali Upazila Health Complex	31	830	991	220	2041	5	4	1	10	13489	17665	3892	35046	5.6	95.8
Mathbaria Upazila Health Complex	50	1680	1718	1763	5161	26	18	30	74	4022	6381	5920	16323	14.1	44.6
Nazirpur Upazila Health Complex	31	1031	3310	476	4817	2	5	-	7	19007	23007	2368	44382	13.2	121.3
Nesarabad Upazila Health Complex	50	1272	2360	1708	5340	52	39	48	139	13305	23902	17807	55014	14.6	150.3
Chittagong division															
Bandarban district															
Aikadam Upazila Health Complex	31	1662	1567	323	3552	11	6	3	20	17987	21655	7833	47475	9.7	129.7
Lama Upazila Health Complex	31	1738	2120	1440	5298	16	10	22	48	8849	21238	20087	50174	14.5	137.1
Nyongchhari Upazila Health Complex	31	1074	1100	1351	3525	1	-	8	9	6279	5030	14565	25874	9.6	70.7

Table continued

Health facility	No. of beds	No. of admissions			No. of hospital deaths			No. of outdoor visits			Average length of stay (d)	Bed-occupancy rate (%)	Hospital death rate (%)	Average daily number				
		Male		Child	Male		Female	Child	Male					Female	Child	Total	Admission	Outdoor visit
Rowangchhari Upazila Health Complex	10	285	265	108	658	-	-	2	2	4457	3880	1432	9769	2.4	43.6	0.3	1.8	26.7
Ruma Upazila Health Complex	10	434	422	148	1004	1	1	4	6	6529	5422	2173	14124	1.0	27.5	0.6	2.7	38.6
Thanchi Upazila Health Complex	31	28	18	9	55	-	-	-	-	1348	972	1364	3684	2.8	1.4	-	0.2	10.1
Brahmanbaria district																		
Akhaura Upazila Health Complex	31	768	1767	1700	4235	33	12	13	58	14766	24651	19401	58818	1.9	69.3	1.4	11.6	160.7
Bancharampur Upazila Health Complex	31	-	-	-	-	-	-	-	-	25666	33772	8821	68259	-	61.2	-	-	186.5
Kashba Upazila Health Complex	31	1501	1821	515	3837	2	2	-	4	21744	33636	24144	79524	1.6	53.2	0.1	10.5	217.3
Nabinagar Upazila Health Complex	31	1971	1767	1935	5673	9	5	8	22	16726	21767	14483	52976	1.2	62.0	0.4	15.5	144.7
Nasimagar Upazila Health Complex	50	2515	2355	897	5767	9	3	8	20	14142	18889	7074	40105	2.7	86.6	0.3	15.8	109.6
Sarail Upazila Health Complex	50	1746	2093	990	4829	4	3	-	7	22638	31126	5926	59690	1.7	44.8	0.1	13.2	163.1
Chandpur district																		
Faridganj Upazila Health Complex	31	1479	1387	1139	4005	9	5	-	14	7805	14103	10609	32517	2.1	73.9	0.3	10.9	88.8
Haimchar Upazila Health Complex	31	900	1767	523	3190	3	1	2	6	12567	11167	7281	31015	2.9	82.9	0.2	8.7	84.7
Haziganj Upazila Health Complex	50	1679	1786	865	4330	25	20	36	81	12308	22788	14900	49996	2.6	62.2	1.9	11.8	136.6
Kachua Upazila Health Complex	50	1812	2015	483	4310	8	4	6	18	6421	11817	2429	20667	2.7	64.5	0.4	11.8	56.5
Matlab (Daxin) Upazila Health Complex	50	1771	2926	1200	5897	19	7	5	31	12426	12338	8835	33599	2.7	86.6	0.5	16.1	91.8
Matlab (Uttar) Upazila Health Complex	31	512	402	303	1217	-	-	-	-	10589	9854	9179	29622	4.2	44.7	-	3.3	80.9
Saharathi Upazila Health Complex	50	1620	2765	1096	5481	10	4	6	20	9950	19353	6852	36155	2.6	77.7	0.4	15.0	98.8
Chittagong district																		
Anwara Upazila Health Complex	50	2232	3209	3395	8836	6	4	2	12	16009	31066	9286	56361	2.1	99.9	0.1	24.1	154.0
Bansikhali Upazila Health Complex	50	2704	3190	3037	8931	8	3	21	32	16117	40856	12957	69930	1.7	82.2	0.4	24.4	191.1
Boalkhali Upazila Health Complex	50	2646	3541	2295	8482	7	2	2	11	17496	34593	17691	69780	1.3	58.7	0.1	23.2	190.7
Chandanaish Upazila Health Complex	50	2145	1856	975	4976	20	1	2	23	23584	21087	21964	66635	3.6	98.3	0.5	13.6	182.1
Hathazari Upazila Health Complex	50	2202	2138	2025	6365	5	3	-	8	21515	26507	16320	64342	3.0	105.3	0.1	17.4	175.8
Lohagara Upazila Health Complex	50	3156	2654	2905	8715	11	8	7	26	18779	37654	15660	72093	2.0	94.3	0.3	23.8	197.0
Mirsarai Upazila Health Complex	50	1516	1827	1669	5012	16	5	6	27	16856	26089	19424	62369	2.2	61.0	0.5	13.7	170.4
Patiya Upazila Health Complex	50	1998	4423	3620	10041	13	7	6	26	22483	25035	13321	60839	1.8	101.2	0.3	27.4	166.2
Rangunia Upazila Health Complex	50	2705	2983	1062	6750	11	9	5	25	20323	27853	22100	70276	2.1	78.5	0.4	18.4	192.0
Roujan Upazila Health Complex	31	2037	3892	4006	9935	5	7	8	20	21985	23015	8578	53578	1.0	87.8	0.2	27.1	146.4
Sandwip Upazila Health Complex	31	905	850	1044	2799	16	11	-	27	16730	18822	17860	53412	2.3	58.0	1.0	7.6	145.9
Saitania Upazila Health Complex	31	1660	2585	2016	6261	8	6	4	18	22998	24475	21574	69047	1.8	100.0	0.3	17.1	188.7
Siakunda Upazila Health Complex	50	1922	2430	998	5350	7	11	5	23	25488	34552	16468	76508	2.3	66.9	0.4	14.6	209.0
Comilla district																		
Banura Upazila Health Complex	31	1044	1112	937	3093	4	2	7	13	9071	28806	10197	48074	2.6	70.8	0.4	8.5	131.3
Brahmanpara Upazila Health Complex	50	3111	3187	2895	9193	2	6	8	16	3543	9575	7467	20585	1.6	83.0	0.2	25.1	56.2
Burichong Upazila Health Complex	31	1460	1640	563	3663	9	3	1	13	780	9280	4163	14223	2.4	76.3	0.4	10.0	38.9

Table continued

Health facility	No. of beds	No. of admissions			No. of hospital deaths			No. of outdoor visits			Average length of stay (d)	Bed-occupancy rate (%)	Hospital death rate (%)	Average daily number				
		Male		Child	Male		Female	Child	Male					Female	Child	Total	Admission	Outdoor visit
Cox's Bazar district																		
Chowddagram Upazila Health Complex	50	4142	5347	-	9489	7	7	5	19	50484	46517	50841	147842	2.1	107.9	0.2	25.9	403.9
Chandina Upazila Health Complex	31	2836	3748	2252	8836	12	12	1	25	9389	14098	6956	30443	1.5	120.8	0.3	24.1	83.2
Daukandi Upazila Health Complex	31	1994	4012	388	6394	5	3	1	9	18334	37859	15467	71660	1.9	106.5	0.1	17.5	195.8
Debidwar Upazila Health Complex	50	3620	3181	1620	8421	10	11	14	35	15243	22135	16482	53860	2.1	96.3	0.4	23.0	147.2
Homna Upazila Health Complex	50	2100	2341	927	5368	11	14	12	37	10373	20176	17215	47764	2.1	63.2	0.7	14.7	130.5
Laksham Upazila Health Complex	50	2199	3416	1084	6699	8	4	4	16	9752	9820	5853	25425	2.2	79.6	0.2	18.3	69.5
Muradnagar Upazila Health Complex	50	1984	2374	2331	6689	24	26	20	70	13659	26277	21260	61196	2.2	81.8	1.0	18.3	167.2
Nangolkot Upazila Health Complex	50	1435	1906	1620	4961	6	1	4	11	10783	18579	16196	45558	2.4	65.0	0.2	13.6	124.5
Titas Upazila Health Complex	31	357	492	155	1004	2	7	-	9	8215	14508	5203	27926	3.7	32.9	0.9	2.7	76.3
Cox's Bazar district																		
Chakaria Upazila Health Complex	50	4179	3058	5734	12971	5	6	8	19	27823	30473	16637	74933	1.8	124.6	0.1	35.4	204.7
Kutubdia Upazila Health Complex	50	1456	2120	1635	5211	11	11	5	27	10217	12178	8238	30633	1.6	45.2	0.5	14.2	83.7
Moheshkhali Upazila Health Complex	50	4887	5317	2836	13040	6	5	7	18	8404	13255	7795	29454	1.0	71.5	0.1	35.6	80.5
Pakua Upazila Health Complex	31	879	585	84	1548	1	1	-	2	14934	11398	4530	30862	4.2	57.3	0.1	4.2	84.3
Ramu Upazila Health Complex	31	1676	3005	1098	5779	3	1	12	16	27462	32691	21824	81977	1.8	92.8	0.3	15.8	224.0
Teknaf Upazila Health Complex	50	1670	2563	3106	7339	3	3	6	12	12958	20821	17438	51217	1.4	55.2	0.2	20.1	139.9
Ukhria Upazila Health Complex	31	2315	1770	2973	7058	6	6	6	18	20143	25518	37983	83644	1.6	99.7	0.3	19.3	228.5
Feni district																		
Chhagalnaya Upazila Health Complex	50	1561	2626	2011	6198	5	-	1	6	20338	22530	24437	67305	2.6	89.7	0.1	16.9	183.9
Daganbhuia Upazila Health Complex	31	1755	2579	823	5157	10	3	-	13	11930	36485	21743	70158	2.2	99.7	0.3	14.1	191.7
Fulgazi Upazila Health Complex	31	1077	3079	207	4363	6	1	-	7	18271	27665	8142	54078	1.9	74.3	0.2	11.9	147.8
Parsuram Upazila Health Complex	50	1412	3926	1342	6680	2	4	2	8	16467	27253	13293	57013	2.8	101.5	0.1	18.3	155.8
Sonagazi Upazila Health Complex	31	1272	4547	1642	7461	7	7	5	19	25575	31201	24704	81480	2.0	128.8	0.3	20.4	222.6
Khagrachhari district																		
Dighinala Upazila Health Complex	10	849	1509	326	2684	3	1	-	4	14102	10315	4344	28761	1.8	133.5	0.1	7.3	78.6
Lakshmichhari Upazila Health Complex	31	274	315	371	960	1	-	-	1	4107	5312	5984	15403	4.8	41.0	0.1	2.6	42.1
Manikchhari Upazila Health Complex	10	582	715	945	2242	6	7	9	22	9512	23056	17108	49676	2.2	133.8	1.0	6.1	135.7
Mairanga Upazila Health Complex	31	848	771	495	2114	8	4	13	25	22395	26776	10262	59433	2.9	53.3	1.2	5.8	162.4
Mohalchhari Upazila Health Complex	31	425	293	320	1038	-	-	-	-	17310	20491	8374	46175	2.7	24.5	-	2.8	126.2
Panchhari Upazila Health Complex	10	794	888	192	1874	5	2	5	12	9154	21937	9709	40800	1.9	97.9	0.6	5.1	111.5
Ramgarh Upazila Health Complex	31	884	887	214	1985	6	3	13	22	13258	24820	10119	48197	2.5	44.5	1.1	5.4	131.7
Lakshmipur district																		
Komolnagar Upazila Health Complex	31	2957	2619	519	6095	3	1	1	5	36160	27827	8255	72242	2.7	145.6	0.1	16.7	197.4
Raipur Upazila Health Complex	50	1780	2036	716	4532	3	7	9	19	16375	31569	30880	78824	3.0	74.1	0.4	12.4	215.4
Ramganj Upazila Health Complex	31	2162	2590	652	5404	8	6	1	15	15215	22015	7595	44825	2.4	114.4	0.3	14.8	122.5
Ramgati Upazila Health Complex	31	903	872	373	2148	3	1	1	5	10956	20128	5226	36310	3.3	62.1	0.2	5.9	99.2

Table continued

Health facility	No. of beds	No. of admissions			No. of hospital deaths			No. of outdoor visits			Average length of stay (d)	Bed-occupancy rate (%)	Hospital death rate (%)	Average daily number			
		Male		Female	Male		Female	Child	Male					Female	Child	Admission	Outdoor visit
Noakhali district																	
Begumganj Upazila Health Complex	31	715	882	707	2304	2	-	-	2	17826	30349	4367	52542	73.3	0.1	6.3	143.6
Chatkhil Upazila Health Complex	50	1530	2741	2177	6448	8	7	3	18	13352	19042	12864	45258	87.0	0.3	17.6	123.7
Companiganj Upazila Health Complex	50	1711	2736	2394	6841	3	6	14	23	23079	34733	29392	87204	100.6	0.3	18.7	238.3
Hatiya Upazila Health Complex	50	1543	1426	1638	4607	29	31	75	135	11596	17532	15651	44779	76.4	2.9	12.6	122.3
Senbag Upazila Health Complex	50	1715	1366	732	3813	3	6	2	11	10830	17261	11729	39820	55.2	0.3	10.4	108.8
Sonaimuri Upazila Health Complex	50	1255	1748	1656	4659	2	3	3	8	10019	17332	4565	31916	58.7	0.2	12.7	87.2
Subarnachar Upazila Health Complex	31	819	1675	77	2571	2	-	3	5	10191	11828	4156	26175	38.2	0.2	7.0	71.5
Rangamati district																	
Baghaichhari Upazila Health Complex	31	619	630	253	1502	11	7	4	22	6580	9793	3931	20304	33.6	1.5	4.1	55.5
Barkol Upazila Health Complex	10	185	197	32	414	-	-	-	-	3505	4228	530	8263	26.0	-	1.1	22.6
Belachhari Upazila Health Complex	10	204	258	81	543	2	1	2	5	3950	5907	3264	13121	41.1	0.9	1.5	35.8
Juraichhari Upazila Health Complex	10	206	211	47	464	1	1	1	3	3921	3544	1104	8569	57.9	0.6	1.3	23.4
Kaptai Upazila Health Complex	31	552	568	446	1566	4	-	1	5	11574	12871	3979	28424	46.9	0.3	4.3	77.7
Kowkhali Upazila Health Complex	10	454	600	354	1408	2	-	1	3	7735	13101	320	21156	87.9	0.2	3.8	57.8
Lengadu Upazila Health Complex	31	605	788	223	1616	5	6	4	15	11244	14790	4723	30757	55.7	0.9	4.4	84.0
Naniarchar Upazila Health Complex	10	218	241	382	841	2	2	-	4	5226	7227	5895	18348	56.4	0.5	2.3	50.1
Rajstali Upazila Health Complex	10	237	231	188	656	2	-	2	4	5618	6502	2635	14755	49.9	0.6	1.8	40.3
Dhaka division																	
Dhaka district																	
Dhamrai Upazila Health Complex	50	1656	2582	571	4809	10	10	2	22	28335	58398	18920	105653	74.6	0.5	13.1	288.7
Dohar Upazila Health Complex	31	1415	3324	755	5494	5	4	2	11	18304	40346	24107	82757	106.4	0.2	15.0	226.1
Keraniganj Upazila Health Complex	31	718	907	446	2071	2	2	-	4	16735	35273	22539	74547	82.6	0.2	5.7	203.7
Nawabganj Upazila Health Complex	50	2171	3225	1817	7213	35	24	2	61	30130	39132	27125	96387	76.0	0.8	19.7	263.4
Savar Upazila Health Complex	50	1260	2005	419	3684	5	5	-	10	33047	44510	20148	97705	72.5	0.3	10.1	267.0
Faridpur district																	
Alfadanga Upazila Health Complex	50	2049	2458	237	4744	11	5	6	22	12898	18793	9505	41196	74.6	0.5	13.0	112.6
Bhanga Upazila Health Complex	50	2251	3275	746	6272	7	8	2	17	10528	14647	5112	30287	99.2	0.3	17.1	82.8
Boalmari Upazila Health Complex	50	3554	4014	3664	11232	19	11	33	63	19222	29323	30380	78925	113.6	0.6	30.7	215.6
Charbhadrason Upazila Health Complex	31	1018	1210	260	2488	30	21	8	59	11700	15924	7992	35616	43.9	2.4	6.8	97.3
Modukhali Upazila Health Complex	31	2530	3162	871	6563	9	3	2	14	13730	30975	21471	66176	103.0	0.2	17.9	180.8
Nagarkanda Upazila Health Complex	50	2308	2380	790	5478	8	2	1	11	6029	17588	10791	34408	94.3	0.2	15.0	94.0
Sadarpur Upazila Health Complex	50	1472	2292	178	3942	2	3	-	5	7185	16826	11019	35030	49.9	0.1	10.8	95.7
Gazipur district																	
Kaliakair Upazila Health Complex	31	1867	2904	397	5168	11	6	1	18	20758	37050	14943	72751	97.8	0.3	14.1	198.8
Kaliganj Upazila Health Complex	55	1509	2616	905	5030	11	6	-	17	16093	27036	21243	64372	72.3	0.3	13.7	175.9

Table continued

Health facility	No. of beds	No. of admissions			No. of hospital deaths			No. of outdoor visits			Average length of stay (d)	Bed-occupancy rate (%)	Hospital death rate (%)	Average daily number	
		Male	Female	Child	Male	Female	Child	Male	Female	Child				Admission	Outdoor visit
Kapasia Upazila Health Complex	50	1725	26690	236	28651	3	4	1	8		0.6	99.2	0.0	78.3	161.6
Sheepur Upazila Health Complex	31	1500	1888	495	3883	3	6	4	13		2.2	77.0	0.3	10.6	311.1
Gopalganj district															
Kassiani Upazila Health Complex	31	1207	1570	517	3294	6	4	5	15		2.0	59.4	0.5	9.0	150.0
Kotwalipara Upazila Health Complex	50	1568	2430	991	4989	21	10	10	41		2.7	74.4	0.8	13.6	126.4
Mukshedpur Upazila Health Complex	31	3851	4215	1854	9920	21	12	10	43		1.8	161.3	0.4	27.1	191.9
Tungipara Upazila Health Complex	50	1850	2110	2280	6240	8	10	3	21		2.9	99.2	0.3	17.0	232.9
Jamalpur district															
Bakshiganj Upazila Health Complex	31	1333	2129	839	4301	7	3	2	12		2.7	101.6	0.3	11.8	225.1
Dewanganj Upazila Health Complex	50	1584	2023	755	4362	9	17	9	35		4.1	97.4	0.8	11.9	148.9
Madarganj Upazila Health Complex	50	1152	1128	488	2768	8	9	3	20		4.5	68.1	0.7	7.6	138.7
Melandaha Upazila Health Complex	50	1217	1515	181	2913	1	2	-	3		3.5	56.4	0.1	8.0	205.0
Saishabari Upazila Health Complex	50	3392	4180	808	8380	26	11	7	44		2.2	101.0	0.5	22.9	102.2
Kishoreganj district															
Astagram Upazila Health Complex	50	976	1233	1114	3323	7	6	9	22		2.6	46.8	0.7	9.1	328.3
Bajipur Upazila Health Complex	31	790	1002	2376	4168	14	3	5	22		2.8	103.7	0.5	11.4	318.0
Bhairab Upazila Health Complex	50	1453	2992	1474	5919	12	17	12	41		2.7	86.7	0.7	16.2	267.0
Hossainpur Upazila Health Complex	50	2933	3200	1737	7870	2	2	-	4		1.8	78.8	0.1	21.5	171.6
Itina Upazila Health Complex	31	841	1023	509	2373	9	7	4	20		3.4	71.8	0.8	6.5	220.1
Kairiganj Upazila Health Complex	50	2270	2637	1297	6204	5	3	1	9		2.2	75.5	0.1	17.0	134.2
Kathiadi Upazila Health Complex	50	3392	5314	1487	10193	21	15	5	41		1.7	92.9	0.4	27.8	278.5
Kulliarchar Upazila Health Complex	31	1253	1163	716	3132	1	1	5	7		2.6	72.4	0.2	8.6	230.5
Mithamoin Upazila Health Complex	31	745	723	366	1834	7	3	2	12		2.8	45.7	0.7	5.0	150.0
Nikhil Upazila Health Complex	31	1361	1234	808	3403	6	3	4	13		2.5	74.8	0.4	9.3	248.8
Pakundia Upazila Health Complex	50	1214	1531	706	3451	3	2	3	8		4.4	82.4	0.2	9.4	171.3
Tarail Upazila Health Complex	50	2001	1752	1300	5053	17	10	18	45		2.7	74.2	0.9	13.8	217.0
Madaripur district															
Kalkini Upazila Health Complex	50	1524	2066	1272	4862	6	6	1	13		3.1	82.5	0.3	13.3	155.1
Rajoir Upazila Health Complex	50	2252	2685	1148	6085	70	42	17	129		2.4	80.8	2.1	16.6	295.0
Shibchar Upazila Health Complex	50	1995	2107	497	4599	-	4	-	4		2.8	70.9	0.1	12.6	58.2
Manikganj district															
Daulatpur Upazila Health Complex	31	870	928	238	2036	-	-	-	-		3.9	70.2	-	5.6	182.4
Ghor Upazila Health Complex	31	1307	1322	312	2941	5	4	1	10		3.0	79.1	0.3	8.0	140.2
Harirampur Upazila Health Complex	31	728	1485	326	2539	6	6	2	14		3.1	70.2	0.6	6.9	123.3
Saturia Upazila Health Complex	31	1603	2087	328	4018	7	1	-	8		2.9	102.4	0.2	11.0	208.9
Shibalaya Upazila Health Complex	31	2102	2724	398	5224	7	2	1	10		2.0	92.8	0.2	14.3	148.7

Table continued

Health facility	No. of beds	No. of admissions			No. of hospital deaths			No. of outdoor visits			Average length of stay (d)	Bed-occupancy rate (%)	Hospital death rate (%)	Average daily number				
		Male	Female	Child	Total	Male	Female	Child	Total	Male				Female	Child	Total	Admission	Outdoor visit
Singair Upazila Health Complex	50	1715	1352	-	3,067	5	1	-	6	14312	23728	1621	39661	2.3	37.8	0.2	8.4	108.4
Munshiganj district																		
Gazaria Upazila Health Complex	50	6380	7538	568	14486	9	4	1	14	19133	32245	17833	69211	1.0	79.4	0.1	39.6	189.1
Louhajang Upazila Health Complex	50	912	1455	410	2777	5	2	1	8	9531	22470	13807	45808	3.3	50.8	0.3	7.6	125.2
Serajdikhan Upazila Health Complex	50	1420	1826	682	3928	8	8	-	16	15782	35553	24570	75905	2.9	63.0	0.4	10.7	207.4
Sreenagar Upazila Health Complex	50	1072	1201	201	2474	6	5	4	15	15921	34102	25655	75678	6.0	81.0	0.6	6.8	206.8
Tungibari Upazila Health Complex	50	1872	2660	1759	6291	1	1	1	3	15378	27036	20098	62512	2.2	76.6	0.0	17.2	170.8
Mymensingh district																		
Bhaluka Upazila Health Complex	50	1785	3387	1292	6464	4	6	2	12	7213	7889	4810	19912	2.8	98.1	0.2	17.7	54.4
Dhobaura Upazila Health Complex	31	1984	2353	648	4985	14	18	8	40	11196	16303	6139	33638	2.3	99.6	0.8	13.6	91.9
Fulbaria Upazila Health Complex	31	1459	2066	621	4146	3	2	2	7	9137	10884	11247	31268	1.9	67.9	0.2	11.3	85.4
Fulpur Upazila Health Complex	50	2531	3775	2067	8373	12	9	5	26	9902	12124	11852	33878	2.0	92.1	0.3	22.9	92.6
Golgaon Upazila Health Complex	50	2000	2282	1994	6276	14	11	4	29	29506	23089	19934	72529	2.0	67.6	0.5	17.1	198.2
Gouripur Upazila Health Complex	31	1792	3035	1379	6206	6	7	2	15	10954	12595	10384	33933	1.9	103.4	0.2	17.0	92.7
Haluaghat Upazila Health Complex	31	2231	3547	1135	6913	7	2	3	12	9517	16416	10877	36810	1.6	100.5	0.2	18.9	100.6
Iswardganj Upazila Health Complex	50	2846	3717	1345	7908	2	5	10	17	16462	25141	5621	47224	1.9	84.2	0.2	21.6	129.0
Muktagachha Upazila Health Complex	31	2179	3649	1302	7130	5	9	2	16	18566	23021	28444	70031	1.4	87.3	0.2	19.5	191.3
Nandail Upazila Health Complex	50	2591	3512	1092	7195	13	7	15	35	11307	14207	4795	30309	2.4	95.6	0.5	19.7	82.8
Tital Upazila Health Complex	50	1885	3033	304	5222	1	2	-	3	30860	41597	13999	86456	1.9	53.5	0.1	14.3	236.2
Narayanganj district																		
Araihazar Upazila Health Complex	31	1502	1843	102	3447	4	4	-	8	38998	61889	6708	107595	2.5	75.3	0.2	9.4	294.0
Bandar Upazila Health Complex	31	312	492	289	1093	2	1	-	3	9528	45820	16325	71673	6.8	65.9	0.3	3.0	195.8
Rupganj Upazila Health Complex	50	1342	1563	-	2905	2	1	-	3	8998	17759	6624	33381	4.9	77.6	0.1	7.9	91.2
Sonargaon Upazila Health Complex	31	855	1040	35	1930	1	-	-	1	26331	40672	6378	73381	4.2	71.5	0.1	5.3	200.5
Narsingdi district																		
Belabo Upazila Health Complex	31	1491	2203	284	3978	8	-	6	14	31943	60078	11100	103121	2.6	91.3	0.4	10.9	281.8
Monohardi Upazila Health Complex	50	1685	2748	520	4953	12	4	2	18	25023	37802	16844	79669	3.1	84.6	0.4	13.5	217.7
Palash Upazila Health Complex	31	772	1528	344	2644	7	2	2	11	18623	50310	23915	92848	3.3	77.3	0.4	7.2	253.7
Raipura Upazila Health Complex	31	1431	1760	610	3801	1	3	1	5	23576	21010	20640	65226	2.1	72.0	0.1	10.4	178.2
Shibpur Upazila Health Complex	31	770	989	345	2104	1	1	2	4	15620	36868	6977	59465	3.4	63.7	0.2	5.7	162.5
Netrakona district																		
Atpara Upazila Health Complex	31	1261	1250	131	2642	6	3	1	10	31063	38927	19000	88990	4.0	92.3	0.4	7.2	243.1
Barhatta Upazila Health Complex	31	1283	1406	759	3448	4	5	9	18	26242	38882	18370	83494	2.0	60.6	0.5	9.4	228.1
Durgapur Upazila Health Complex	50	2155	2312	-	4467	12	16	16	44	18546	27419	18563	64528	2.1	50.8	1.0	12.2	176.3
Kaimakanda Upazila Health Complex	50	2337	2325	752	5414	8	4	12	24	9827	13573	6286	29686	2.9	84.9	0.4	14.8	81.1
Kendua Upazila Health Complex	50	1357	2115	1444	4916	17	8	14	39	13436	16901	7686	38023	1.6	41.8	0.8	13.4	103.9

Table continued

Health facility	No. of beds	No. of admissions			No. of hospital deaths			No. of outdoor visits			Average length of stay (d)	Bed-occupancy rate (%)	Hospital death rate (%)	Average daily number				
		Male	Female	Child	Male	Female	Child	Male	Female	Child				Admission	Outdoor visit			
Rajbari district																		
Khaliajhuri Upazila Health Complex	31	697	573	214	1484	2	1	-	3	13866	19606	10324	43796	4.0	52.7	0.2	4.1	119.7
Madan Upazila Health Complex	50	2055	1960	439	4454	11	7	5	23	6622	6318	3426	16366	2.8	68.4	0.5	12.2	44.7
Mohanganj Upazila Health Complex	50	1912	3632	214	5758	18	5	22	45	18887	17994	9579	46460	2.2	69.2	0.8	15.7	126.9
Purbadhala Upazila Health Complex	50	3246	3621	1449	8316	12	7	12	31	11334	13865	3868	29067	3.1	139.3	0.4	22.7	79.4
Rajshahi district																		
Baliakandi Upazila Health Complex	31	1314	1801	540	3655	5	4	1	10	17159	20958	5151	43268	3.0	95.9	0.3	10.0	118.2
Goalandia Upazila Health Complex	31	2316	2972	1148	6436	17	2	2	21	19203	38408	16095	73706	2.1	117.8	0.3	17.6	201.4
Pangsha Upazila Health Complex	50	2600	2760	2680	8040	18	12	10	40	9100	17204	18788	45092	2.4	107.5	0.5	22.0	123.2
Shariatpur district																		
Bhedarganj Upazila Health Complex	31	1348	1550	888	3786	5	2	10	17	16321	18587	15536	50444	2.0	67.0	0.4	10.3	137.8
Damuddya Upazila Health Complex	31	1205	1812	614	3631	13	7	3	23	16435	21017	13173	50625	2.4	78.0	0.6	9.9	138.3
Goshairhat Upazila Health Complex	31	1510	1392	1214	4116	7	10	27	44	15495	25563	22002	63060	2.7	99.7	1.1	11.2	172.3
Naria Upazila Health Complex	31	1422	1633	420	3475	7	4	2	13	9012	21310	5458	35780	2.3	69.4	0.4	9.5	97.8
Zanzira Upazila Health Complex	31	1208	1932	913	4053	5	7	4	16	23773	43766	16954	84493	2.5	89.7	0.4	11.1	230.9
Sherpur district																		
Jhenaigati Upazila Health Complex	31	1006	1089	699	2794	-	1	-	1	17821	38813	14000	70634	2.0	49.4	0.0	7.6	193.0
Nakhla Upazila Health Complex	50	1348	1548	643	3539	3	2	2	7	15776	21675	10536	47987	3.7	72.5	0.2	9.7	131.1
Nalitabari Upazila Health Complex	31	1742	1537	172	3451	15	7	11	33	28011	42569	2645	73225	3.7	113.7	1.0	9.4	200.1
Sribardi Upazila Health Complex	31	1428	1549	304	3281	2	1	1	4	42910	42351	15600	100861	3.1	89.9	0.1	9.0	275.6
Tangail district																		
Basail Upazila Health Complex	31	1008	1366	282	2656	2	2	-	4	34814	51182	27486	113482	2.5	59.6	0.2	7.3	310.1
Bhuapur Upazila Health Complex	50	2978	3583	-	6561	3	2	2	7	9213	13804	12613	35630	2.5	90.2	0.1	17.9	97.3
Daiduar Upazila Health Complex	31	980	1317	401	2698	-	1	-	1	21784	31847	11467	65098	4.9	116.7	0.0	7.4	177.9
Ghatail Upazila Health Complex	31	1977	2600	1707	6284	15	5	1	21	24994	35509	13444	73947	2.2	121.8	0.3	17.2	202.0
Gopalpur Upazila Health Complex	50	1681	1880	1401	4962	3	4	1	8	28191	32714	12582	73487	2.6	70.1	0.2	13.6	200.8
Kailhati Upazila Health Complex	50	1914	2394	805	5113	9	4	2	15	15821	15471	7775	39067	3.3	92.3	0.3	14.0	106.7
Mirzapur Upazila Health Complex	31	1002	1482	24	2508	1	2	-	3	20974	26792	17876	65642	4.3	95.7	0.1	6.9	179.3
Modhupur Upazila Health Complex	50	2989	4265	2816	10070	3	6	3	12	12230	19843	16209	48282	1.7	92.3	0.1	27.5	131.9
Nagarpur Upazila Health Complex	31	5372	8025	360	13757	13	6	7	26	15840	22032	11100	48972	1.0	121.6	0.2	37.6	133.8
Sakhipur Upazila Health Complex	50	2668	3836	1366	7870	3	5	8	16	13113	23825	17161	54099	2.3	98.0	0.2	21.5	147.8
Khulna division																		
Bagerhat district																		
Chitalmari Upazila Health Complex	31	1454	1574	636	3664	14	9	5	28	10212	14299	9272	33783	2.9	93.4	0.8	10.0	92.3
Fakirhat Upazila Health Complex	31	2512	3810	1896	8218	21	8	2	31	13351	33058	11189	57598	2.3	170.3	0.4	22.5	157.4
KachuaUpazila Health Complex	50	1317	1345	612	3274	1	2	-	3	14909	34250	22033	71192	5.5	98.6	0.1	8.9	194.5
Mollahat Upazila Health Complex	31	1727	1982	702	4411	10	9	-	19	14584	27957	9189	51730	2.5	96.9	0.4	12.1	141.3

Table continued

Health facility	No. of beds	No. of admissions			No. of hospital deaths			No. of outdoor visits			Average length of stay (d)	Bed-occupancy rate (%)	Hospital death rate (%)	Average daily number			
		Male	Female	Child	Male	Female	Child	Male	Female	Child				Admission	Outdoor visit		
Chuadanga district																	
Mondia Upazila Health Complex	50	1672	2190	263	4125	17	14	2	33	17415	29032	6715	53162	8.2	0.8	11.3	145.3
Morrelganj Upazila Health Complex	31	976	1319	1092	3387	9	15	-	24	14140	19705	14579	48424	3.5	0.7	9.3	132.3
Rampal Upazila Health Complex	50	1628	2463	451	4542	13	6	5	24	9559	16436	10110	36105	3.6	0.5	12.4	98.6
Sarakholia Upazila Health Complex	31	1130	1444	1001	3575	9	4	10	23	6799	14263	11500	32562	3.0	0.6	9.8	89.0
Chuadanga district																	
Alamdanga Upazila Health Complex	31	1678	2514	1132	5324	12	7	2	21	26880	32015	18380	77275	2.9	0.4	14.5	211.1
Damurhuda Upazila Health Complex	31	1560	2080	284	3924	4	3	1	8	17333	36156	8732	62221	2.6	0.2	10.7	170.0
Jibannagar Upazila Health Complex	31	2355	3588	867	6810	9	8	4	21	23790	35455	10504	69749	1.6	0.3	18.6	190.6
Jessore district																	
Abhoynagar Upazila Health Complex	50	2992	4524	-	7516	28	24	-	52	32219	40848	19708	92775	2.4	0.7	20.5	253.5
Bagerpara Upazila Health Complex	50	2610	1966	311	4887	3	3	1	7	12354	19666	4354	36374	3.0	0.1	13.4	99.4
Chowgachha Upazila Health Complex	50	3094	8378	2640	14112	10	8	16	34	20480	39325	21134	80939	2.4	0.2	38.6	221.1
Jhikargachha Upazila Health Complex	31	1120	2204	508	3832	7	-	-	7	17434	24707	13283	55424	2.5	0.2	10.5	151.4
Keshabpur Upazila Health Complex	50	2405	2412	1206	6023	40	21	21	82	11530	14705	16516	42751	2.6	1.4	16.5	116.8
Monirampur Upazila Health Complex	50	2159	2992	668	5819	15	11	-	26	9383	22003	16389	47775	2.5	0.4	15.9	130.5
Sarsa Upazila Health Complex	31	1750	2493	256	4499	11	6	-	17	8366	24154	14678	47198	1.6	0.4	12.3	129.0
Jhenaidah district																	
Harinakunda Upazila Health Complex	50	2239	3382	1691	7312	15	13	3	31	8477	17893	12522	38892	2.2	0.4	20.0	106.3
Kaliganj Upazila Health Complex	50	2986	3955	1893	8834	20	14	6	40	14685	34909	9507	59101	1.9	0.5	24.1	161.5
Kotchandpur Upazila Health Complex	50	2949	3740	1866	8555	28	18	11	57	15622	23415	9661	48698	1.8	0.7	23.4	133.1
Moheshpur Upazila Health Complex	50	1946	2280	1073	5299	14	14	3	31	17931	23649	15973	57553	1.8	0.6	14.5	157.2
Saikupa Upazila Health Complex	31	2592	2960	220	5772	6	7	-	13	18709	24310	890	43909	2.4	0.2	15.8	120.0
Khulna district																	
Batiaghata Upazila Health Complex	31	2935	4119	1106	8160	4	3	-	7	13348	21537	10565	45450	1.1	0.1	22.3	124.2
Dacope Upazila Health Complex	50	1910	4312	1111	7333	22	16	20	58	14080	20804	7783	42667	4.3	0.8	20.0	116.6
Dighalia Upazila Health Complex	31	851	1145	400	2396	6	5	-	11	20101	40843	20200	81144	2.8	0.5	6.5	221.7
Dumuria Upazila Health Complex	31	1740	2523	660	4923	9	19	1	29	11206	25136	12257	48599	2.2	0.6	13.5	132.8
Fulta Upazila Health Complex	50	2563	2436	609	5608	29	21	2	52	2563	2436	609	5608	3.2	0.9	15.3	15.3
Koyra Upazila Health Complex	50	1678	2344	1025	5047	21	28	4	53	6382	8642	3618	18642	3.2	1.1	13.8	50.9
Paikgacha Upazila Health Complex	50	1925	3020	1623	6568	33	20	2	55	10384	16658	9067	36109	3.6	0.8	17.9	98.7
Rupsha Upazila Health Complex	31	1165	1816	574	3555	5	1	-	6	10908	16793	5030	32721	3.4	0.2	9.7	89.4
Terakhada Upazila Health Complex	31	1032	1365	425	2822	5	4	2	11	11935	16356	4335	32626	3.2	0.4	7.7	89.1
Kushtia district																	
Bheramara Upazila Health Complex	50	1643	2181	631	4455	2	11	-	13	14770	27045	8835	50650	2.3	0.3	12.2	138.4
Daulatpur Upazila Health Complex	50	2025	4711	622	7358	7	4	4	15	12163	26305	9523	47991	2.3	0.2	20.1	131.1
Khoksha Upazila Health Complex	50	900	1650	759	3309	7	13	3	23	12896	16507	704	30107	4.1	0.7	9.0	82.3

Table continued

Health facility	No. of beds	No. of admissions			No. of hospital deaths			No. of outdoor visits			Average length of stay (d)	Bed-occupancy rate (%)	Hospital death rate (%)	Average daily number	
		Male	Female	Child	Male	Female	Child	Male	Female	Child				Admission	Outdoor visit
Kumarkhali Upazila Health Complex	50	2362	3094	1692	7148	14	5	8	27	9492	16165	8166	33823	3.0	19.5
Mirpur Upazila Health Complex	50	1547	2734	770	5051	4	2	-	6	7638	9247	1032	17917	1.7	13.8
Magura district															
Mohammadpur Upazila Health Complex	31	3508	6000	800	10308	5	4	5	14	15464	26143	10111	51718	0.9	28.2
Saika Upazila Health Complex	50	1207	1798	162	3167	6	4	2	12	9693	14784	4573	29050	2.9	8.7
Sreepur Upazila Health Complex	31	516	653	480	1649	-	1	-	1	11167	14743	5107	31017	5.5	4.5
Meherpur district															
Gangni Upazila Health Complex	50	2696	5596	397	8689	19	7	3	29	50720	64007	6169	120896	2.0	23.7
Mujibnagar Upazila Health Complex	31	248	230	41	519	1	-	-	1	12897	21666	7803	42366	2.5	1.4
Narail district															
Kalia Upazila Health Complex	50	1592	2865	1254	5711	11	6	8	25	17577	31563	10068	59208	2.7	15.6
Lohagara Upazila Health Complex	31	2000	1627	2226	5853	13	4	-	17	20997	43439	7963	72399	2.3	16.0
Satkhira district															
Assasuni Upazila Health Complex	31	1026	1358	272	2856	3	1	1	5	16812	26846	7158	50816	3.3	7.3
Debhatta Upazila Health Complex	31	1229	2608	95	3932	22	18	-	40	19804	36448	7657	63909	4.1	10.7
Kalaroa Upazila Health Complex	50	1722	2948	793	5463	17	15	2	34	14522	20750	12471	47743	3.0	14.9
Kaiganj Upazila Health Complex	50	904	1202	235	2341	3	7	1	11	12979	12419	9306	34704	5.5	6.4
Shyamnagar Upazila Health Complex	50	2534	4761	2056	9351	37	39	18	94	13536	19439	10275	43250	3.4	25.5
Tala Upazila Health Complex	50	1240	2604	698	4542	28	20	-	48	12181	13439	7688	33308	4.2	12.4
Rajshahi division															
Bogra district															
Adamdighi Upazila Health Complex	50	1609	2185	413	4207	1	2	-	3	16748	27978	16710	61436	2.3	11.5
Dhunat Upazila Health Complex	50	2642	3780	1108	7530	4	2	3	9	25170	54563	6453	86186	1.5	20.6
Dhupchachia Upazila Health Complex	50	2589	4182	1375	8146	8	3	-	11	10758	16768	6460	33986	1.6	22.3
Gabail Upazila Health Complex	50	1431	1834	797	4062	3	1	-	4	13470	32352	10817	56639	3.2	11.1
Kahaloo Upazila Health Complex	50	2079	2833	561	5473	4	3	-	7	18736	36702	10309	65747	2.9	15.0
Nandigram Upazila Health Complex	31	1325	1674	428	3427	3	1	-	4	23409	30038	13407	66854	2.7	9.4
Sariakandi Upazila Health Complex	50	1745	2066	536	4347	5	6	4	15	19838	25197	8450	53485	4.3	11.9
Shajahanpur Upazila Health Complex	31	556	824	170	1550	-	1	-	1	10751	31326	6889	48966	3.9	4.2
Sherpur Upazila Health Complex	31	2817	3463	300	6580	10	5	3	18	65700	75300	25537	166537	1.9	18.0
Shibganj Upazila Health Complex	50	2336	2744	1535	6615	2	1	1	4	19065	30699	15323	65087	2.5	18.1
Sonataia Upazila Health Complex	50	1334	1795	797	3926	2	2	2	6	19946	21222	5867	47035	3.8	10.7
Chapainowabganj district															
Bholahat Upazila Health Complex	31	922	1362	308	2592	7	5	1	13	13016	16404	12102	41522	2.7	7.1
Gomastapur Upazila Health Complex	31	1305	2485	236	4026	5	10	2	17	32886	33736	8173	74795	1.8	11.0
Nachol Upazila Health Complex	50	1351	1747	546	3644	8	1	-	9	34999	38754	12852	86605	3.6	10.0

Table continued

Health facility	No. of beds	No. of admissions			No. of hospital deaths			No. of outdoor visits			Average length of stay (d)	Bed-occupancy rate (%)	Hospital death rate (%)	Average daily number	
		Male		Female	Male		Female	Male		Female				Admission	Outdoor visit
Shibganj Upazila Health Complex															
Joypurhat district															
	50	1536	3059	2774	7369	3	3	4	10	18487	24337	18063	60887	20.1	166.4
Joypurhat district															
	50	2101	2786	525	5412	12	2	-	14	14197	24956	18541	57694	14.8	157.6
	50	1590	2542	446	4578	5	5	2	12	19273	28245	26737	74255	12.5	202.9
	50	1402	2622	286	4310	6	3	-	9	21109	31330	7495	59934	11.8	163.8
	50	1691	2160	-	3851	8	4	2	14	11860	16739	4924	33523	10.5	91.6
Naogaon district															
	50	1163	2375	713	4251	8	1	1	10	5840	8949	4928	19717	11.6	53.9
	50	784	1017	535	2336	8	2	-	10	10873	11128	8256	30257	6.4	82.7
	50	1284	1236	505	3025	6	3	-	9	9883	14748	5000	29631	8.3	81.0
	50	2011	2107	411	4529	5	7	2	14	11720	19309	11436	42465	12.4	116.0
	50	2316	2867	560	5743	20	23	6	49	14359	17941	11612	43912	15.7	120.0
	50	1436	1602	334	3372	18	8	4	30	19834	26200	7187	53221	9.2	145.4
	50	1360	2594	506	4460	22	9	4	35	8809	14552	9184	32545	12.2	88.9
	50	721	1026	489	2236	3	2	2	7	8586	15474	12205	36265	6.1	99.1
	31	1410	1561	381	3352	8	4	-	12	11569	23153	13532	48254	9.2	131.8
	50	1580	1882	148	3610	13	8	5	26	8520	10935	6638	26093	9.9	71.3
Natore district															
	31	1202	1535	247	2984	5	2	-	7	34998	45188	4851	85037	8.2	232.3
	31	1082	1550	70	2702	2	-	-	2	31676	42679	7695	82050	7.4	224.2
	50	2895	3588	1621	8104	9	14	5	28	29226	36979	11822	78027	22.1	213.2
	50	2284	3035	573	5892	19	7	3	29	11202	19390	6488	37080	16.1	101.3
	31	2610	2848	752	6210	17	5	1	23	34340	36460	7840	78640	17.0	214.9
Pabna district															
	31	1471	1759	408	3638	2	2	-	4	22740	39163	25844	87747	9.9	239.7
	31	1207	2221	539	3967	19	8	4	31	25253	31300	8808	65361	10.8	178.6
	50	4110	4322	400	8832	15	9	12	36	26853	25729	10968	63550	24.1	173.6
	50	2540	3804	1945	8289	15	22	18	55	30161	49697	35701	115559	22.6	315.7
	50	1402	1717	812	3931	7	7	8	22	11192	16046	11168	38406	10.7	104.9
	50	3442	6909	1265	11616	33	20	3	56	13602	37112	29898	80612	31.7	220.3
	50	3125	3987	470	7582	8	7	3	18	21767	35108	17557	74432	20.7	203.4
	51	2436	2713	1716	6865	5	1	1	7	7877	13224	8915	30016	18.8	82.0
Rajshahi district															
	50	1441	2768	163	4372	1	-	-	1	14777	39937	27532	82246	11.9	224.7
	31	1456	2491	991	4938	8	7	-	15	22606	44560	23559	90725	13.5	247.9
	31	2181	5133	277	7591	4	4	-	8	27040	42461	20689	90190	20.7	246.4

Table continued

Health facility	No. of beds	No. of admissions			No. of hospital deaths			No. of outdoor visits			Average length of stay (d)	Bed-occupancy rate (%)	Hospital death rate (%)	Average daily number	
		Male		Female	Male		Female	Male		Female				Admission	Outdoor visit
			Child	Total		Child	Total		Child	Total					
Sirajganj district															
Durgapur Upazila Health Complex	31	1850	2894	545	5289	4	2	2	2	8	1.7	77.7	0.2	14.5	233.5
Godagari Upazila Health Complex	31	1495	1749	471	3715	-	2	1	3	3	2.2	72.7	0.1	10.2	98.0
Mohanpur Upazila Health Complex	31	1608	1864	169	3641	2	3	1	6	6	1.6	53.1	0.2	9.9	210.3
Paba Upazila Health Complex	31	540	752	602	1894	1	-	-	1	1	3.5	57.8	0.1	5.2	241.9
Ruthia Upazila Health Complex	31	2260	3968	533	6761	6	7	-	13	13	1.8	104.8	0.2	18.5	282.7
Tanore Upazila Health Complex	50	1440	2342	748	4530	4	7	1	12	12	3.4	84.3	0.3	12.4	224.3
Sirajganj district															
Seikuchi Upazila Health Complex	31	2214	2317	1696	6227	2	2	1	5	5	1.4	79.6	0.1	17.0	166.6
Chowhali Upazila Health Complex	31	620	661	250	1531	2	-	1	3	3	4.0	54.0	0.2	4.2	106.9
Kamarkanda Upazila Health Complex	31	2735	2942	516	6193	2	-	1	3	3	1.0	54.7	0.0	16.9	214.3
Kazipur Upazila Health Complex	31	811	1120	413	2344	3	4	5	12	19149	3.0	62.3	0.5	6.4	143.1
Raiganj Upazila Health Complex	31	3649	3372	1442	8463	4	1	1	6	13309	1.0	74.8	0.1	23.1	84.8
Shahzadpur Upazila Health Complex	31	1601	4638	39	6278	3	2	-	5	23080	1.3	72.6	0.1	17.2	155.2
Tarash Upazila Health Complex	31	1196	1773	753	3722	3	2	4	9	17656	1.2	40.3	0.2	10.2	138.9
Ullapara Upazila Health Complex	31	1298	1300	82	2680	2	-	2	4	17809	2.7	63.7	0.1	7.3	160.5
Rangpur division															
Dinajpur district															
Birampur Upazila Health Complex	31	1333	2480	541	4354	4	3	6	13	24758	1.7	65.7	0.3	11.9	207.2
Birganj Upazila Health Complex	31	1478	1706	509	3693	10	8	7	25	28320	2.3	75.9	0.7	10.1	210.4
Birol Upazila Health Complex	31	1115	1208	350	2673	2	1	-	3	14752	2.6	62.1	0.1	7.3	110.7
Bochohaganj Upazila Health Complex	50	1730	2394	484	4608	10	2	1	13	16353	1.6	41.6	0.3	12.6	117.4
Chirribandar Upazila Health Complex	31	605	1006	465	2076	1	1	-	2	33287	3.6	65.6	0.1	5.7	219.8
Fulbari Upazila Health Complex	31	3155	3952	1316	8423	10	4	3	17	29070	1.5	109.9	0.2	23.0	196.2
Ghoraghat Upazila Health Complex	31	1298	1411	278	2987	-	3	3	6	29279	3.0	80.2	0.2	8.2	204.5
Hakimpur Upazila Health Complex	31	1213	1904	266	3383	6	2	3	11	18287	2.0	60.6	0.3	9.2	138.6
Kaharol Upazila Health Complex	31	1256	1537	267	3060	7	3	-	10	28270	3.0	80.4	0.3	8.4	195.8
Khansama Upazila Health Complex	31	1453	4061	705	6219	2	-	-	2	25754	1.8	101.0	0.0	17.0	208.8
Nawabganj Upazila Health Complex	31	1108	1783	605	3496	4	9	2	15	28672	2.3	69.7	0.4	9.6	225.0
Parbatipur Upazila Health Complex	50	2505	3032	948	6485	10	4	2	16	23102	2.3	82.7	0.2	17.7	156.6
Gaibandha district															
Fulchhari Upazila Health Complex	50	3461	2550	790	6801	1	-	-	1	28580	1.7	61.6	0.0	18.6	152.5
Gobindaganj Upazila Health Complex	50	3782	6926	1589	12297	17	8	7	32	8382	1.7	115.5	0.3	33.6	98.5
Sadullapur Upazila Health Complex	31	5022	4885	742	10649	5	2	8	15	18476	1.0	94.1	0.1	29.1	155.7
Shaghata Upazila Health Complex	31	940	1895	257	3092	3	3	6	12	31484	2.0	55.4	0.4	8.4	246.7
Sundarganj Upazila Health Complex	31	1139	2800	614	4553	2	1	11	14	19793	2.3	93.6	0.3	12.4	162.8

Table continued

Health facility	No. of beds	No. of admissions			No. of hospital deaths			No. of outdoor visits			Average length of stay (d)	Bed-occupancy rate (%)	Hospital death rate (%)	Average daily number				
		Male	Female	Child	Total	Male	Female	Child	Total	Male				Female	Child	Total	Admission	Outdoor visit
Kurigram district																		
Bhujungamari Upazila Health Complex	31	1641	2112	938	4691	10	11	2	23	24715	30667	12158	67540	2.1	85.5	0.5	12.8	184.5
Chilmari Upazila Health Complex	50	1850	2153	675	4678	3	4	2	9	24672	23798	18936	67406	3.1	78.3	0.2	12.8	184.2
Fulbari Upazila Health Complex	31	1206	1498	554	3258	2	-	-	2	11037	13135	8241	32413	3.7	105.7	0.1	8.9	88.6
Nageswari Upazila Health Complex	31	1369	1980	812	4161	8	2	4	14	41020	43756	26552	111328	2.7	99.2	0.3	11.4	304.2
Rajarhat Upazila Health Complex	31	892	912	160	1964	2	1	-	3	24359	22222	4360	50941	4.0	69.7	0.2	5.4	139.2
Rajibpur Upazila Health Complex	31	512	822	668	2002	7	3	4	14	23836	29939	16168	69943	3.2	57.0	0.7	5.5	191.1
Rowmari Upazila Health Complex	31	1628	2244	240	4112	20	20	3	43	24603	21598	12777	58978	2.4	88.1	1.0	11.2	161.1
Ulipur Upazila Health Complex	50	2521	3709	1104	7334	4	8	5	17	19507	24722	13059	57288	2.5	101.6	0.2	20.0	156.5
Lalmonirhat district																		
Aditmari Upazila Health Complex	50	827	1131	420	2378	-	1	-	1	11181	15279	3063	29523	3.9	51.1	0.0	6.5	80.7
Haibandha Upazila Health Complex	50	2092	3336	1428	6856	14	12	9	35	21330	26070	4740	52140	2.3	86.9	0.5	18.7	142.5
Kailiganj Upazila Health Complex	50	1226	1306	284	2816	16	7	8	31	7171	7768	2818	17757	4.0	61.6	1.1	7.7	48.5
Patgram Upazila Health Complex	31	1310	2104	653	4067	-	2	-	2	19374	18705	4396	42475	3.1	109.9	0.0	11.1	116.1
Nilphamari district																		
Dimla Upazila Health Complex	50	4845	6860	2831	14536	8	11	23	42	9590	15225	6950	31765	1.0	79.6	0.3	39.7	86.8
Domar Upazila Health Complex	50	3146	4357	1698	9201	13	15	9	37	13671	19424	15103	48198	1.7	87.4	0.4	25.1	131.7
Jaldhaka Upazila Health Complex	50	4415	10212	3842	18469	11	15	10	36	13379	17294	16090	46763	1.0	101.2	0.2	50.5	127.8
Kishoreganj Upazila Health Complex	31	1596	3842	957	6395	3	5	12	20	6231	6889	6360	19480	2.2	126.1	0.3	17.5	53.2
Panchagarh district																		
Atwari Upazila Health Complex	50	1699	2293	1268	5260	2	-	-	2	19336	21346	11870	52552	2.5	71.4	0.0	14.4	143.6
Boda Upazila Health Complex	31	1892	2149	597	4638	11	10	-	21	32165	34223	10753	77141	2.2	92.2	0.5	12.7	210.8
Debiganj Upazila Health Complex	50	1047	2930	495	4472	15	11	3	29	23978	27361	18799	70138	3.3	81.7	0.6	12.2	191.6
Teulia Upazila Health Complex	31	945	1790	244	2979	1	4	1	6	27388	30966	18173	76527	2.7	71.3	0.2	8.1	209.1
Rangpur district																		
Baderganj Upazila Health Complex	31	2810	3615	1604	8029	9	9	4	22	20625	26520	11784	58929	2.1	148.4	0.3	21.9	161.0
Gangachara Upazila Health Complex	50	1180	3748	706	5634	2	2	-	4	15521	17263	9378	42162	2.6	81.3	0.1	15.4	115.2
Kownia Upazila Health Complex	31	1236	1916	573	3725	4	5	-	9	26331	32411	20866	79608	2.0	67.3	0.2	10.2	217.5
Mithapukur Upazila Health Complex	50	2394	5252	2848	10494	4	4	3	11	19025	26577	20228	65830	1.9	106.7	0.1	28.7	179.9
Pirgachha Upazila Health Complex	31	1388	2000	300	3688	7	3	-	10	19237	24144	12505	55886	2.5	80.0	0.3	10.1	152.7
Pirganj Upazila Health Complex	31	2389	4926	2419	9734	4	5	4	13	17200	23325	11075	51600	1.4	118.8	0.1	26.6	141.0
Taraganj Upazila Health Complex	31	1236	886	1591	3713	3	2	-	5	33800	43500	12669	89969	2.0	67.1	0.1	10.1	245.8
Thakurgaon district																		
Baliadangi Upazila Health Complex	50	1954	2784	1194	5932	7	10	2	19	8070	9539	5527	23136	3.0	96.5	0.3	16.2	63.2
Haripur Upazila Health Complex	50	764	1657	280	2701	1	11	-	12	7732	13624	4746	26102	2.7	39.6	0.4	7.4	71.3
Pirganj Upazila Health Complex	50	2346	5540	1173	9059	22	24	4	50	14805	28113	14650	57568	1.9	93.6	0.6	24.8	157.3

Table continued

Health facility	No. of beds	No. of admissions			No. of hospital deaths			No. of outdoor visits			Average length of stay (d)	Bed-occupancy rate (%)	Hospital death rate (%)	Average daily number				
		Male	Female	Child	Male	Female	Child	Male	Female	Child				Admission	Outdoor visit			
Ranisankhail Upazila Health Complex	50	1661	3294	890	5845	16	8	4	28	14106	16071	11579	41756	2.1	66.4	0.5	16.0	114.1
Sylhet division																		
Habiganj district																		
Azmriganj Upazila Health Complex	31	542	1570	572	2684	11	7	21	39	25316	28957	10958	65231	2.1	50.6	1.5	7.3	178.2
Jahubai Upazila Health Complex	31	1670	2073	1447	5190	4	3	-	7	21560	25610	28905	76075	2.2	103.2	0.1	14.2	207.9
Janiachong Upazila Health Complex	31	1002	1302	1222	3526	6	1	3	10	18787	34267	22627	75681	2.5	77.3	0.3	9.6	206.8
Chunarughat Upazila Health Complex	31	2055	2265	695	5015	5	4	6	15	18784	25170	9237	53191	2.0	90.2	0.3	13.7	145.3
Lakhai Upazila Health Complex	31	945	1027	296	2268	2	2	1	5	9590	16470	4367	30427	2.7	54.0	0.2	6.2	83.1
Madhabpur Upazila Health Complex	50	2907	2949	409	6265	11	7	5	23	32720	49483	15933	98136	2.3	78.6	0.4	17.1	268.1
Nabiganj Upazila Health Complex	31	892	2506	1469	4867	1	3	-	4	10644	13020	12036	35700	1.7	72.8	0.1	13.3	97.5
Maulvibazar district																		
Barlekha Upazila Health Complex	31	2372	3906	1164	7442	10	7	22	39	25651	30878	4759	61288	1.4	93.7	0.5	20.3	167.5
Kamalganj Upazila Health Complex	31	1904	3174	1269	6347	7	4	4	15	9091	15648	6779	31518	1.5	81.8	0.2	17.3	86.1
Kulaura Upazila Health Complex	50	4456	5008	-	9464	14	15	-	29	29428	35850	7525	72803	1.3	68.9	0.3	25.9	198.9
Rajnagar Upazila Health Complex	31	1715	1851	713	4279	2	1	1	4	18901	22540	15590	57031	2.1	77.7	0.1	11.7	155.8
Sreemangal Upazila Health Complex	31	2234	2553	1597	6384	9	5	5	19	26047	32142	9357	67546	1.4	80.0	0.3	17.4	184.6
Sunamganj district																		
Biswambapur Upazila Health Complex	31	789	671	375	1835	8	3	2	13	32200	30450	1223	63873	4.4	71.9	0.7	5.0	174.5
Chhatak Upazila Health Complex	31	1064	1872	1252	4188	4	3	6	13	7797	16110	13632	37539	1.2	45.4	0.3	11.4	102.6
Deeral Upazila Health Complex	31	2734	2990	676	6400	5	6	11	22	11090	10747	983	22820	1.7	97.7	0.3	17.5	62.3
Dharmapasha Upazila Health Complex	31	2775	2356	1348	6479	18	23	18	59	32782	33309	16093	82184	1.8	102.3	0.9	17.7	224.5
Doarabazar Upazila Health Complex	31	260	359	290	909	-	-	5	5	20136	20547	6206	46889	2.4	19.1	0.6	2.5	128.1
Jagannathpur Upazila Health Complex	50	1152	1386	788	3326	5	8	3	16	6929	10221	9178	26328	1.6	29.5	0.5	9.1	71.9
Jamalganj Upazila Health Complex	31	2117	2207	883	5207	8	4	3	15	41726	49837	25283	116846	2.0	92.9	0.3	14.2	319.3
Sulia Upazila Health Complex	31	792	1006	1021	2819	8	5	13	26	9300	8869	5455	23624	3.3	82.8	0.9	7.7	64.5
Taherpur Upazila Health Complex	31	1327	1140	402	2869	10	8	8	26	12551	21276	11350	45177	2.6	66.5	0.9	7.8	123.4
Sylhet district																		
Balaganj Upazila Health Complex	31	702	1170	470	2342	4	2	5	11	10667	15367	3575	29609	2.2	45.9	0.5	6.4	80.9
Beanibazar Upazila Health Complex	50	619	4167	2138	6924	4	3	30	37	21650	35574	31170	88394	2.1	81.5	0.5	18.9	241.5
Biswanath Upazila Health Complex	31	266	351	222	839	-	-	-	-	12165	14632	19382	46179	3.4	25.4	-	2.3	126.2
Companyganj Upazila Health Complex	31	963	953	951	2867	-	1	-	1	11357	17273	14215	42845	2.0	50.1	0.0	7.8	117.1
Fenchuganj Upazila Health Complex	31	1018	2192	766	3976	1	7	7	15	8530	15991	9591	34112	2.2	76.7	0.4	10.9	93.2
Golaganj Upazila Health Complex	31	1494	1715	770	3979	2	-	3	5	16308	17134	19601	53043	1.7	61.1	0.1	10.9	144.9
Gowainghat Upazila Health Complex	50	3156	3827	2805	9788	10	8	8	26	8821	8942	8397	26160	1.9	102.4	0.3	26.7	71.5
Jointapur Upazila Health Complex	31	3905	4884	1204	9993	5	5	11	21	8298	19184	6187	33669	1.0	88.9	0.2	27.3	92.0
Kanaighat Upazila Health Complex	31	1338	1540	1640	4518	1	1	5	7	9630	15640	15646	40916	1.3	53.6	0.2	12.3	111.8
Zakiganj Upazila Health Complex	31	1276	1796	2237	5309	-	-	-	-	15313	27833	22320	65466	1.8	83.8	-	14.5	178.9

Number of beds, admissions, deaths, outdoor visits, bed-occupancy rates, average length of stay, and hospital death rates in some private hospitals and clinics (2012)

Health facility	Address	District	No. of beds	No. of admissions		No. of deaths		No. of outdoor visits			Average daily number		Bed-occupancy rate (%)	Average length of stay (d)	Hospital death rate (%)
				Male	Female	Male	Female	Male	Female	Child	Admission	Outdoor visit			
Doctors Care Clinic and Hospital	Sadar	Barguna	10	87	646	733	-	-	-	-	2.0	0	55.0	2.5	0.0
Medinova Clinic & Digital Diagnostic Center	Sadar	Barguna	10	12	200	212	-	-	-	-	0.6	0		0.0	0.0
Modern Central Hospital	Sadar	Barguna	10	81	539	620	-	-	-	-	1.7	0	42.7	2.1	0.0
Ambia Hospital	Sadar	Barisal	50	1240	992	2232	6	4	10	189	325	245	759	0.0	0.4
Islami Bank Hospital	Sadar	Barisal	50	3544	4612	8156	7	9	16	23386	29212	7392	59900	0.0	0.2
Health Care Clinic	Sadar	Barisal	20	612	301	913	1	-	1	1025	912	210	2147	0.0	0.1
Bangabondhu Hospital (Pvt.) Ltd.	Sadar	Chittagong	10	190	309	499	-	-	-	1769	4780	3645	10194	1.4	0.0
Chowdhuryhat Nursing Home	Sadar	Chittagong	20	41	1243	1284	1	7	8	-	-	-		17.6	0.6
Riad Al Khaeeb Hospital	Sadar	Chittagong	80	4310	4013	8323	72	73	145	3191	4150	988	8329	73.1	1.2
Hethazar Adhunic Hospital	Sadar	Chittagong	25	891	3053	3944	3	5	8	868	282	-	1150	43.2	0.2
Tahrim Hospital	Sadar	Chittagong	11	720	370	1090	-	-	-	3070	3725	-	6795	27.2	0.0
Abdul Hamid Memorial Hospital	Sadar	Chuadanga	10	159	237	396	-	-	-	-	-	-		28.2	0.0
Anandodham Clinic	Sadar	Chuadanga	10	144	216	360	-	-	-	-	-	-		25.5	0.0
Damurhuda Private Hospital	Sadar	Chuadanga	10	117	555	672	-	-	-	-	-	-		70.4	0.0
IMPACT Foundation NGO Hospital	Sadar	Chuadanga	10	925	850	1775	-	-	-	8153	18316	2038	28507	218.1	2.4
Islami Hospital	Sadar	Chuadanga	10	15	250	265	-	-	-	-	-	-		43.2	0.0
Ma Nursing Home	Sadar	Chuadanga	10	19	229	248	-	-	-	-	-	-		33.5	0.0
New Life Surgical Clinic	Sadar	Chuadanga	10	170	250	420	-	-	-	-	-	-		29.9	0.0
Niramoy Nursing Home	Sadar	Chuadanga	10	21	279	300	-	-	-	-	-	-		40.6	0.0
Rajdhani Clinic	Sadar	Chuadanga	10	25	281	306	-	-	-	-	-	-		49.6	0.0
Ekota Clinic	Alamdanga	Chuadanga	10	154	230	384	-	-	-	-	-	-		27.3	0.0
Kana Nursing Home	Alamdanga	Chuadanga	10	210	325	535	-	-	-	-	-	-		38.1	0.0
Lokman Hossain Clinic	Alamdanga	Chuadanga	10	180	270	450	-	-	-	-	-	-		31.8	0.0
Protigga Nursing Home	Alamdanga	Chuadanga	10	150	225	375	-	-	-	-	-	-		26.7	0.0
Shefa Clinic	Alamdanga	Chuadanga	10	185	265	450	-	-	-	-	-	-		32.1	0.0
United Clinic	Alamdanga	Chuadanga	10	167	245	412	-	-	-	-	-	-		29.2	0.0
New Digital Clinic	Chitla	Chuadanga	10	115	365	480	-	-	-	-	-	-		69.5	0.0
Upasham Nursing Home	Chuadanga	Chuadanga	10	27	305	332	-	-	-	-	-	-		53.1	0.0
Mukt Clinic, Darshana	Damurhuda	Chuadanga	10	108	432	540	-	-	-	-	-	-		76.9	0.0
Ma O Shistu Hospital	Darshana	Chuadanga	10	156	768	924	-	-	-	85	232	103	420	118.0	0.0
Modern Clinic	Darshana	Chuadanga	10	144	720	864	-	-	-	-	-	-		110.6	0.0
Shepla Clinic	Darshana	Chuadanga	10	121	719	840	-	-	-	-	-	-		108.4	0.0
Adarshaw Clinic	Jibannagar	Chuadanga	10	-	111	111	-	-	-	-	-	-		4.9	0.0
Janani Clinic	Jibannagar	Chuadanga	10	-	156	156	-	-	-	-	-	-		7.2	0.0
Janata Clinic	Jibannagar	Chuadanga	10	-	127	127	-	-	-	-	-	-		5.6	0.0
Ma Clinic	Jibannagar	Chuadanga	10	-	115	115	-	-	-	-	-	-		5.3	0.0
Mahima Clinic	Jibannagar	Chuadanga	10	-	155	155	-	-	-	-	-	-		7.0	0.0
Sheba Clinic	Jibannagar	Chuadanga	10	-	168	168	-	-	-	-	-	-		7.3	0.0
Tanjina Clinic	Jibannagar	Chuadanga	10	-	120	120	-	-	-	-	-	-		5.4	0.0

Table continued

Health facility	Address	District	No. of beds	No. of admissions		No. of deaths		No. of outdoor visits			Average daily number		Bed-occupancy rate (%)	Average length of stay (d)	Hospital death rate (%)
				Male	Female	Male	Female	Male	Female	Child	Admission	Outdoor visit			
Niramoy Clinic	Karpasdanga	Chudanga	10	97	407	504	-	-	-	-	1.4	0	66.4	4.5	0.0
Sheba Clinic	Karpasdanga	Chudanga	10	97	359	456	-	-	-	-	1.2	0	67.0	5.7	0.0
General Hospital	Sadar	Cox's Bazar	20	569	674	1243	34	31	65	-	3.4	0	19.4	0.5	5.2
Sea Side Hospital	Sadar	Cox's Bazar	10	191	1567	1758	-	-	-	-	4.8	0	90.7	4.0	0.0
Smiling Sun Clinic (FDSR) NGO	Sadar	Cox's Bazar	10	-	895	895	-	-	-	-	2.5	54	90.7	4.0	0.0
Apollo Hospitals	Sadar	Dhaka	358	7760	7005	14765	279	138	417	121820	40.5	701	100.0	8.7	2.8
Comfort Nursing Home (Pvt.) Ltd.	Sadar	Dhaka	63	2269	1382	3651	70	26	96	-	10.0	0	100.0	8.7	2.6
Dhaka Community Medical College Hospital	Sadar	Dhaka	250	2695	2864	5559	43	37	80	257325	15.2	1500	100.0	8.7	1.4
Dhaka National Medical Institute Hospital	Sadar	Dhaka	600	6175	10205	16380	58	48	106	146967	44.9	849	0.5	2.9	0.6
Rafia Clinic	AtiBazar	Dhaka	10	287	323	610	-	-	-	602	1.7	5	80.5	2.9	0.0
Sajeda Hospital	Jinjira	Dhaka	50	1124	1237	2361	1	2	3	16005	6.5	121	52.1	2.4	0.1
Hasnabad Hospital	Keraniganj	Dhaka	20	702	871	1573	-	-	-	720	4.3	6	78.0	2.1	0.0
Kalatia Central Hospital	Keraniganj	Dhaka	20	674	734	1408	-	-	-	557	3.9	4	81.3	2.4	0.0
Khalamura General Hospital	Keraniganj	Dhaka	10	195	217	412	-	-	-	340	1.1	3	43.4	2.2	0.0
Ruhitpur General Hospital	Keraniganj	Dhaka	20	612	708	1320	-	-	-	812	3.6	7	80.0	2.3	0.0
Diabetes Hospital	Sadar	Dineipur	80	1162	1172	2334	26	27	53	63485	6.4	361	19.7	1.2	2.3
Gausul Azam Eye Hospital	Sadar	Dineipur	90	1902	1551	3453	-	-	-	34001	9.5	224	26.1	1.1	0.0
Lamb Hospital	Sadar	Dineipur	115	2243	8008	10251	234	159	393	16032	28.1	164	60.8	1.9	3.8
St. Vincent Hospital	Sadar	Dineipur	100	702	3434	4136	1	3	4	2102	11.3	30	28.0	2.1	0.1
Zia Heart Foundation	Sadar	Dineipur	50	1834	1048	2882	178	90	268	4748	7.9	23	38.5	0.8	9.3
AL-Amin General Hospital	Sadar	Feni	10	205	175	380	-	-	-	220	1.0	2	84.5	0.0	0.0
AL-Kenny (Pvt.) Hospital	Sadar	Feni	30	1590	2480	4070	4	6	10	750	11.2	6	11.2	0.2	0.2
Cosmopolitan (Pvt.) Hospital	Sadar	Feni	20	200	310	510	3	2	5	550	1.4	5	1793	1.0	1.0
Dr. Haidar Clinic & Hospital	Sadar	Feni	10	280	390	670	-	-	-	299	1.8	4	1464	1.0	0.0
Ibne Hasman (Pvt.) Hospital	Sadar	Feni	10	410	760	1170	-	-	-	180	3.2	3	935	3.2	0.0
Feni Central (Pvt) Hospital	Sadar	Feni	20	850	1012	1862	-	-	-	1125	5.1	10	3545	0.0	0.0
Feni Diabetic Hospital	Sadar	Feni	25	2000	4325	6325	35	25	60	30800	17.3	237	86580	0.9	0.9
Mediscan (Pvt) Hospital	Sadar	Feni	20	390	750	1140	2	4	6	2316	3.1	17	3550	3.1	0.5
Shahin Clinic (Pvt.) Hospital	Sadar	Feni	10	110	250	360	-	-	-	260	1.0	3	1190	0.0	0.0
Upasham General Hospital	Sadar	Feni	20	2585	2024	4609	5	10	15	110	12.6	1	450	0.3	0.3
Green Hospital	Sadar	Galpur	10	264	352	616	-	-	-	1284	1.7	9	3261	1.6	0.0
Dewan General Hospital & Diagnostic Centre	Kaliakair	Galpur	10	64	67	131	-	-	-	311	0.4	3	979	1.7	0.0
Al-Baraka General Hospital	Sadar	Gazipur	10	300	360	660	-	-	-	200	1.8	2	560	2.5	0.0
Al-Noor Islami Hospital	Sadar	Gazipur	10	94	477	571	-	-	-	425	1.6	7	2553	1.8	0.0
Chandra Malika Society Hospital	Sadar	Gazipur	10	340	430	770	-	-	-	900	2.1	8	3061	2.2	0.0
Kaliakair General Hospital & Diogostic Center	Sadar	Gazipur	20	152	530	682	-	-	-	1000	1.9	9	3185	2.4	0.0
Kazi Hospital Complex	Sadar	Gazipur	10	521	729	1250	-	-	-	898	3.4	7	2721	1.5	0.0
Konabari Clinic & Diagnostic Center	Sadar	Gazipur	10	321	403	724	-	-	-	211	2.0	2	658	1.7	0.0
Meher Hospital & Diagnostic Centre	Sadar	Gazipur	10	65	230	295	-	-	-	415	0.8	3	1140	2.3	0.0
Seba General Hospital	Sadar	Gazipur	20	202	414	616	-	-	-	1102	1.7	12	414	3.4	0.0
St. Marys Catholic Mother & Child Hospital, Kaliganj	Sadar	Gazipur	10	-	108	108	-	-	-	300	0.3	19	9.1	3.1	0.0

Table continued

Health facility	Address	District	No. of beds	No. of admissions		No. of deaths		No. of outdoor visits			Average daily number		Bed-occupancy rate (%)	Average length of stay (d)	Hospital death rate (%)				
				Male		Female		Male		Female		Admission				Outdoor visit			
				Male	Female	Male	Female	Male	Female	Male	Female	Total				Total			
Sufia Hospital & Diagnostic Centre	Sadar	Gazipur	10	266	932	1198	-	-	-	-	521	727	203	1451	3.3	4	91.2	2.3	0.0
	Sadar	Gazipur	10	330	350	680	-	-	-	-	315	411	102	828	1.9	2	46.6	1.5	0.0
	Sadar	Gazipur	10	225	473	698	-	-	-	-	529	745	195	1469	1.9	4	47.2	1.8	0.0
	Sadar	Gazipur	10	234	345	579	-	-	-	-	213	289	103	605	1.6	2	20.4	0.7	0.0
	Barmi	Gazipur	10	209	687	896	-	-	-	-	413	653	105	1171	2.5	3	47.9	1.5	0.0
	Dolan Bazar	Gazipur	5	42	290	332	-	-	-	-	366	1466	-	1832	0.9	5	33.5	1.7	0.0
	Hossain Market	Gazipur	10	312	525	837	-	-	-	-	442	581	203	1226	2.3	3	60.3	1.9	0.0
	Joydebpur	Gazipur	20	279	680	959	-	-	-	-	226	501	-	727	2.6	2	71.2	4.3	0.0
	Central Hospital	Gazipur	10	250	682	932	-	-	-	-	522	892	103	1517	2.6	4	53.7	1.6	0.0
	Kaliganj	Gazipur	10	57	110	167	-	-	-	-	725	885	515	2125	0.5	6	12.7	2.1	0.0
AI-Fahim Hospital	Kapasia	Gazipur	10	270	357	627	-	-	-	-	1207	1367	1383	3957	1.7	11	45.3	1.7	0.0
	Kapasia	Gazipur	10	168	371	539	-	-	-	-	229	397	101	727	1.5	2	49.6	2.1	0.0
	Kapasia	Gazipur	10	163	381	544	-	-	-	-	1096	1826	730	3652	1.5	10	40.3	2.1	0.0
	Kapasia	Gazipur	10	180	452	632	-	-	-	-	1244	3054	1836	6134	1.7	17	39.0	1.7	0.0
	Kapasia	Gazipur	10	260	767	1027	-	-	-	-	1141	3890	1343	6374	2.8	17	53.9	1.6	0.0
	Kapasia	Gazipur	10	230	610	840	-	-	-	-	7325	6128	1907	15360	2.3	42	59.8	2.1	0.0
	Kapasia	Gazipur	10	172	205	377	-	-	-	-	215	321	106	642	1.0	2	31.0	1.6	0.0
	Khonabari	Gazipur	10	56	142	198	-	-	-	-	256	312	104	672	0.5	2	16.3	2.2	0.0
	Konabari	Gazipur	10	150	270	420	-	-	-	-	833	1167	428	2428	1.2	7	34.1	2.0	0.0
	Konabari	Gazipur	10	260	628	888	-	-	-	-	1052	1930	668	3650	2.4	10	50.4	1.4	0.0
Mawna	Gazipur	10	478	982	1460	-	-	-	-	1735	1995	-	3730	4.0	10	120.0	2.0	0.0	
	Sadar	Gazipur	10	140	133	273	-	-	-	-	650	600	50	1300	0.7	4	19.4	1.1	0.0
	Sadar	Gazipur	10	157	338	495	-	-	-	-	1146	1107	34	2287	1.4	6	27.2	1.4	0.0
	Sreepur	Gazipur	10	278	277	555	-	-	-	-	276	312	123	711	1.5	2	47.2	1.6	0.0
	Sreepur	Gazipur	10	37	148	185	-	-	-	-	840	1160	431	2431	0.5	7	15.5	2.5	0.0
	Sreepur	Gazipur	10	230	332	562	-	-	-	-	1020	1400	300	2720	1.5	7	34.5	1.4	0.0
	Sreepur	Gazipur	10	212	645	857	-	-	-	-	398	723	189	1310	2.3	4	40.4	2.3	0.0
	Tongi	Gazipur	10	-	353	353	-	-	-	-	72	128	20	220	1.0	1	28.8	3.5	0.0
	Tongi	Gazipur	20	319	527	846	-	-	-	-	521	815	218	1554	2.3	4	31.8	1.9	0.0
	Sadar	Jhenaidah	10	7	29	36	-	-	-	-	-	-	-	-	0.1	0	-	-	0.0
Anwara Private Hospital	Sadar	Jhenaidah	10	40	301	341	-	-	-	-	-	-	-	-	0.9	0	53.9	5.3	0.0
	Sadar	Jhenaidah	10	80	360	440	-	-	-	-	-	-	-	-	1.2	0	68.5	4.8	0.0
	Sadar	Jhenaidah	10	412	715	1127	-	-	-	-	-	-	-	-	3.1	0	170.9	3.8	0.0
	Sadar	Jhenaidah	10	13	164	177	-	-	-	-	-	-	-	-	0.5	0	28.7	5.6	0.0
	Sadar	Jhenaidah	10	22	249	271	-	-	-	-	-	-	-	-	0.7	0	52.0	6.4	0.0
	Sadar	Jhenaidah	10	80	630	710	-	-	-	-	-	-	-	-	1.9	0	149.0	7.1	0.0
	Kaliganj	Jhenaidah	10	135	312	447	-	-	-	-	-	-	-	-	1.2	0	80.4	4.8	0.0
	Kaliganj	Jhenaidah	10	75	230	305	-	-	-	-	-	-	-	-	0.8	0	42.8	3.9	0.0
	Kaliganj	Jhenaidah	10	37	271	308	-	-	-	-	-	-	-	-	0.8	0	58.1	6.2	0.0
	Kaliganj	Jhenaidah	10	54	460	514	-	-	-	-	-	-	-	-	1.4	0	77.6	5.0	0.0
Modern Surgical Clinic	Kaliganj	10	22	248	270	-	-	-	-	-	-	-	-	0.7	0	43.0	5.5	0.0	

Table continued

Health facility	Address	District	No. of beds	No. of admissions			No. of deaths			No. of outdoor visits			Average daily number		Bed-occupancy rate (%)	Average length of stay (d)	Hospital death rate (%)
				Male	Female	Total	Male	Female	Total	Male	Female	Child	Admission	Outdoor visit			
Najma Clinic	Kaliganj	Jhenaidah	10	36	259	295	-	-	-	-	-	-	0.8	0	55.8	6.2	0.0
National Barobazar Clinic	Kaliganj	Jhenaidah	10	43	288	331	-	-	-	-	-	-	0.9	0	44.2	4.4	0.0
New Gonib Shah Clinic	Kaliganj	Jhenaidah	10	60	444	504	-	-	-	-	-	-	1.4	0	51.8	3.3	0.0
New IBN-Sina Clinic	Kaliganj	Jhenaidah	10	69	158	227	-	-	-	-	-	-	0.6	0	39.8	4.9	0.0
Seba Clinic	Kaliganj	Jhenaidah	10	110	310	420	-	-	-	-	-	-	1.2	0	63.3	4.4	0.0
Sumon Clinic	Kaliganj	Jhenaidah	10	71	234	305	-	-	-	-	-	-	0.8	0	46.3	4.6	0.0
Grameen Clinic	Moheshpur	Jhenaidah	10	8	58	66	-	-	-	-	-	-	0.2	0			0.0
Islamic Pvt. Hospital	Moheshpur	Jhenaidah	10	4	19	23	-	-	-	-	-	-	0.1	0			0.0
Janoni Clinic	Moheshpur	Jhenaidah	10	14	62	76	-	-	-	-	-	-	0.2	0			0.0
Maa O Shishu Clinic	Moheshpur	Jhenaidah	10	8	57	65	-	-	-	-	-	-	0.2	0			0.0
Moheshpur Pvt. Hospital	Moheshpur	Jhenaidah	10	13	74	87	-	-	-	-	-	-	0.2	0			0.0
Mohiuddin Pvt. Hospital	Moheshpur	Jhenaidah	10	6	39	45	-	-	-	-	-	-	0.1	0			0.0
Moshfar Rahman Memorial Hospital	Moheshpur	Jhenaidah	10	8	37	45	-	-	-	-	-	-	0.1	0			0.0
Mukti Clinic	Moheshpur	Jhenaidah	10	12	54	66	-	-	-	-	-	-	0.2	0			0.0
Peerless Hospital	Moheshpur	Jhenaidah	10	6	51	57	-	-	-	-	-	-	0.2	0			0.0
Sajib Pvt. Hospital	Moheshpur	Jhenaidah	10	4	24	28	-	-	-	-	-	-	0.1	0			0.0
Saleha Clinic	Moheshpur	Jhenaidah	10	11	48	59	-	-	-	-	-	-	0.2	0			0.0
Sheema Clinic	Moheshpur	Jhenaidah	10	6	37	43	-	-	-	-	-	-	0.1	0			0.0
Sumi Clinic	Moheshpur	Jhenaidah	10	3	38	41	-	-	-	-	-	-	0.1	0			0.0
Sumon & Biswas Clinic	Moheshpur	Jhenaidah	10	7	41	48	-	-	-	-	-	-	0.1	0			0.0
Tamanna Clinic	Moheshpur	Jhenaidah	10	5	38	43	-	-	-	-	-	-	0.1	0			0.0
Ayesha Clinic	Sailakupa	Jhenaidah	10	55	621	676	-	-	-	-	-	-	1.9	0			0.0
Khandokar Clinic	Sailakupa	Jhenaidah	10	76	515	591	-	-	-	-	-	-	1.6	0			0.0
Ma O Shishu Clinic	Sailakupa	Jhenaidah	10	75	234	309	-	-	-	-	-	-	0.8	0			0.0
Tanisa Clinic	Sailakupa	Jhenaidah	10	60	250	310	-	-	-	-	-	-	0.8	0			0.0
Al-Shefa Clinic	Sadar	Joypurhat	10	94	170	264	-	-	-	-	-	-	0.7	0	28.9	2.6	0.0
Anowara Clinic & Diagnostic Centre	Sadar	Joypurhat	10	150	200	350	-	-	-	100	120	50	1.0	1	0.3		0.0
Bandhan Clinic & Diagnostic Centre	Sadar	Joypurhat	10	180	300	480	-	-	-	-	-	-	1.3	0	52.6	2.5	0.0
Doctors Clinic	Sadar	Joypurhat	10	250	300	550	-	-	-	600	500	50	1.5	3	15.1	0.7	0.0
Ernu General Hospital	Sadar	Joypurhat	10	90	695	785	-	-	-	201	703	103	2.2	3	69.5	2.7	0.0
Islami Hospital & Diagnostic Centre	Sadar	Joypurhat	8	19	170	189	-	-	-	452	1062	69	0.5	4	19.4	2.6	0.0
Kamal Memorial Hospital	Sadar	Joypurhat	10	80	396	476	-	-	-	-	-	-	1.3	0	39.1	2.5	0.0
Khanjahanpur Mission Hospital	Sadar	Joypurhat	10	1722	1899	3621	-	-	-	9402	11222	3417	9.9	66	99.2	0.5	0.0
Modern General Hospital & Diagnostic Centre	Sadar	Joypurhat	10	160	200	360	-	-	-	-	-	-	1.0	0	29.6	1.7	0.0
New Doctors Clinic	Sadar	Joypurhat	10	300	350	650	-	-	-	-	-	-	1.8	0	71.2	2.2	0.0
Rawshan Clinic	Sadar	Joypurhat	10	65	122	187	-	-	-	104	384	-	0.5	1	15.4	2.0	0.0
Rawshan Clinic & Diagnostic Centre	Sadar	Joypurhat	10	112	212	324	-	-	-	520	820	110	0.9	4	7.3	0.6	0.0
Sadman Clinic & Nursing Home	Sadar	Joypurhat	10	85	305	390	-	-	-	730	1095	365	1.1	6	53.4	3.9	0.0
Square Clinic & Hospital	Sadar	Joypurhat	10	60	240	300	-	-	-	-	-	-	0.8	0	24.7	2.4	0.0
City Nursing Home	Sadar	Khulna	20	57	327	384	-	-	-	1741	1305	-	1.1	8	66.0	10.0	0.0
Cure Home General Hospital	Sadar	Khulna	20	457	333	790	1	2	3	1934	1458	982	2.2	12	47.9	2.2	0.4

Table continued

Health facility	Address	District	No. of beds	No. of admissions		No. of deaths		No. of outdoor visits			Average daily number		Bed-occupancy rate (%)	Average length of stay (d)	Hospital death rate (%)
				Male	Female	Male	Female	Male	Female	Child	Admission	Outdoor visit			
Doctors Clinic	Sadar	Khulna	10	177	586	763	-	-	-	-	2.1	2	74.9	1.0	0.0
Dr. Badsha Mia Memorial Clinic	Sadar	Khulna	10	365	313	678	3	2	5	5144	1082	28	81.6	2.3	0.7
Gazi Medical College & Hospital	Sadar	Khulna	250	1603	1095	2698	54	39	93	1870	1248	9	9.7	1.3	3.4
Gorib Newaj Clinic	Sadar	Khulna	20	392	394	786	9	2	11	760	722	4	27.3	1.2	1.4
Islami Bank Hospital	Sadar	Khulna	60	2891	4007	6898	31	33	64	20156	31251	165	54.5	0.8	0.9
Matrimangol & Health Center	Sadar	Khulna	10	343	243	586	21	14	35	1785	1415	9	86.4	2.6	6.0
Shandhami Clinic	Sadar	Khulna	20	561	687	1248	6	2	8	121	253	1	42.4	1.3	0.6
Shapla Clinic	Sadar	Khulna	10	10	436	446	-	-	-	110	1212	4	40.2	3.2	0.0
Abedin Hospital (Pvt. Ltd.)	Sadar	Kishoreganj	20	710	1255	1965	2	3	5	1752	2190	12	80.8	1.9	0.3
Al Modina General Hospital	Sadar	Kishoreganj	10	99	254	353	1	1	2	1265	2875	12	29.2	2.3	0.6
Al-Shefa General Hospital (Pvt.)	Sadar	Kishoreganj	30	654	945	1599	1	1	2	3695	4981	26	30.5	0.9	0.1
Anowara General Hospital (Pvt.)	Sadar	Kishoreganj	20	427	1258	1685	1	-	1	4331	10333	40	63.6	2.0	0.1
Diabetic Hospital	Sadar	Kishoreganj	10	2	555	557	-	-	-	1781	1226	14	47.3	3.1	0.0
Ishakh Nursing Home	Sadar	Kishoreganj	10	149	1460	1609	-	-	-	590	950	5	132.3	2.8	0.0
Kishoreganj Clinic	Sadar	Kishoreganj	27	627	1400	2027	-	-	-	3650	4204	25	60.7	2.1	0.0
Ma-Shishu General Hospital	Sadar	Kishoreganj	20	435	715	1150	2	1	3	1678	3125	14	43.2	1.8	0.3
Matirika Adhunik General Hospital	Sadar	Kishoreganj	20	213	389	602	1	3	4	1389	2158	10	24.6	1.9	0.7
Mediab Health Centre	Sadar	Kishoreganj	10	520	920	1440	2	1	3	-	-	0	96.1	1.6	0.2
NGO Hospital	Sadar	Kishoreganj	6	142	215	357	1	2	3	425	1054	4	43.2	1.8	0.8
Padma General Hospital	Sadar	Kishoreganj	10	12	76	88	1	1	2	954	1546	7	11.0	2.9	2.3
Popular General Hospital	Sadar	Kishoreganj	10	60	125	185	1	1	2	812	1524	7	19.5	2.9	1.1
Syed Yusuf General Hospital	Sadar	Kishoreganj	10	253	490	743	3	2	5	2145	2934	14	60.9	1.9	0.7
The Khidma Hospital	Sadar	Kishoreganj	20	396	624	1020	-	-	-	1536	2784	12	43.3	1.9	0.0
Central Hospital	Bhairab	Kishoreganj	40	216	326	542	1	2	3	1245	2658	15	11.6	1.9	0.6
Mahbub Clinic	Bhurungaman	Kurigram	10	177	353	530	-	-	-	-	-	0	57.3	2.7	0.0
Mother Clinic	Bhurungaman	Kurigram	10	90	363	453	-	-	-	-	-	0	49.6	3.3	0.0
Alekjan Community Clinic	Ulipur	Kurigram	14	1039	1164	2203	10	16	26	2180	2305	13	79.3	0.9	1.2
Doctor's Community Clinic	Ulipur	Kurigram	10	375	405	780	-	-	-	150	200	1	42.6	1.0	0.0
Morium Eye Hospital	Ulipur	Kurigram	50	1504	1571	3075	-	-	-	2230	2050	13	33.5	1.0	0.0
Matree Chaya Hospital	Raipur	Laxmipur	10	672	952	1624	5	5	10	11494	13993	89	174.3	2.3	0.6
Meghna Hospital	Raipur	Laxmipur	10	30	54	84	-	-	-	200	345	2	9.2	2.9	0.0
Meherunnesa Hospital	Raipur	Laxmipur	10	95	956	1051	-	-	-	250	2247	7	138.6	4.6	0.0
Modern Hospital	Raipur	Laxmipur	10	148	524	672	-	-	-	720	1240	6	69.6	3.2	0.0
Seba Hospital	Raipur	Laxmipur	10	70	230	300	-	-	-	400	1095	4	31.9	3.1	0.0
Noor Clinic & Diagnostic Centre	Kalkini	Madaripu	10	17	266	283	-	-	-	-	-	0	0.2	0.0	0.0
Chowdhury Clinic	Madaripur	Madaripu	20	-	2008	2008	-	-	-	6000	12000	99	0.7	0.0	0.0
Dhaka Progressive Lions Eye Hospital	Rajoir	Madaripu	10	254	348	602	-	-	-	3818	3913	22	49.5	1.7	0.0
Medinova Clinic	Shibchar	Madaripu	8	15	213	228	-	-	-	117	1214	5	19.8	2.3	0.0
Medipath Clinic	Shibchar	Madaripu	8	13	381	394	-	-	-	607	710	5	33.9	2.4	0.0
Mukta Clinic	Shibchar	Madaripu	10	32	140	172	-	-	-	560	940	6	12.2	2.0	0.0
Popular Clinic	Shibchar	Madaripu	8	10	245	255	-	-	-	150	3220	12	26.0	2.9	0.0

Table continued

Health facility	Address	District	No. of beds	No. of admissions			No. of deaths			No. of outdoor visits			Average daily number		Bed-occupancy rate (%)	Average length of stay (d)	Hospital death rate (%)
				Male	Female	Total	Male	Female	Total	Male	Female	Child	Admission	Outdoor visit			
Adhunik Hospital	Sadar	Munshiganj	10	245	654	899	-	-	-	201	324	215	740	2.5	128.7	3.7	0.0
Anan Clinic & Diagnostic	Sadar	Munshiganj	10	247	657	904	-	-	-	214	365	258	837	2.5	137.3	4.1	0.0
Arafat Clinic & Diagnostic	Sadar	Munshiganj	10	230	578	808	-	-	-	265	451	210	926	2.2	141.9	4.7	0.0
Gazaria General Hospital	Sadar	Munshiganj	15	245	714	959	-	-	-	457	857	542	1856	2.6	93.3	4.0	0.0
Halimanur General Hospital	Sadar	Munshiganj	15	236	815	1051	-	-	-	4125	5462	1452	11039	2.9	100.6	3.3	0.0
Hossneara Hospital & Clinic	Sadar	Munshiganj	10	265	654	919	-	-	-	248	451	256	955	2.5	130.3	3.5	0.0
Jansheba Clinic	Sadar	Munshiganj	10	298	784	1082	-	-	-	254	369	236	859	3.0	137.8	3.3	0.0
Modem Clinic	Sadar	Munshiganj	20	321	365	686	2	1	3	1965	4521	1324	7810	1.9	106.6	8.1	0.4
Mollah Clinic & Diagnostic	Sadar	Munshiganj	20	214	698	912	2	1	3	2356	5698	1562	9616	2.5	117.4	6.0	0.3
Munshiganj Adhunik Hospital	Sadar	Munshiganj	20	322	459	781	2	2	4	3694	5421	1254	10369	2.1	100.1	6.6	0.5
Padma Hospital and Lab	Sadar	Munshiganj	20	216	562	778	-	-	-	2356	5698	1562	9616	2.1	101.2	7.1	0.0
Ranin Clinic	Sadar	Munshiganj	10	296	457	753	-	-	-	258	452	211	921	2.1	130.0	4.4	0.0
Baligaon Clinic	Baligaon	Munshiganj	10	214	354	568	-	-	-	195	314	192	701	1.6	136.1	5.4	0.0
Takerhat General Hospital	Rajoir	Munshiganj	10	54	530	584	-	-	-	131	764	52	947	1.6	74.3	4.4	0.0
Wuk Mahani Hospital	Rajoir	Munshiganj	10	-	136	136	-	-	-	100	300	-	400	0.4	14.9	4.0	0.0
Popular Hospital	Serajdikhan	Munshiganj	10	216	458	674	-	-	-	201	546	214	961	1.8	101.9	3.6	0.0
Tangbari Clinic	Tangbari	Munshiganj	20	235	568	803	-	-	-	345	489	311	1145	2.2	109.3	7.3	0.0
United Clinic	Tangbari	Munshiganj	35	224	789	1013	-	-	-	475	724	425	1624	2.8	100.7	9.9	0.0
Reg. Private Clinic	Manda	Naogaon	18	412	503	915	-	1	1	-	-	-	-	2.5	16.8	0.7	0.1
Kalia Surgical Clinic & Diagnostic Center	Sadar	Narail	10	3	18	21	-	-	-	-	-	-	-	0.1	4.1	6.2	0.0
Khadiza Clinic	Kalia	Narail	10	8	40	48	-	-	-	-	-	-	-	0.1	9.2	5.8	0.0
BRAC Clinic	Sadar	Narsingdi	10	-	346	346	-	-	-	337	1791	184	2312	0.9	47.4	5.0	0.0
Dasagiri Pvt. Hospital	Sadar	Narsingdi	10	95	220	315	-	3	3	1476	2918	498	4892	0.9	31.3	2.4	1.0
Efa Hospital	Sadar	Narsingdi	10	70	463	533	-	-	-	1640	1191	287	3118	1.5	60.3	3.5	0.0
Faith Hospital	Sadar	Narsingdi	10	256	303	559	-	-	-	4852	3663	7723	16238	1.5	27.6	1.1	0.0
Holy Crescent Hospital	Sadar	Narsingdi	10	632	1120	1752	-	-	-	10608	7980	13560	32148	4.8	174.7	2.6	0.0
Jomjom Pvt. Hospital	Sadar	Narsingdi	10	180	270	450	-	-	-	870	1330	830	3030	1.2	61.6	3.0	0.0
Mita Pvt. Hospital	Sadar	Narsingdi	10	77	445	522	-	-	-	2902	6715	280	9897	1.4	66.4	4.3	0.0
Modem Pvt. Hospital	Sadar	Narsingdi	10	140	276	416	-	-	-	1610	1302	406	3318	1.1	57.0	3.3	0.0
Mukti Pvt. Hospital	Sadar	Narsingdi	10	520	739	1259	-	-	-	1575	1753	458	3786	3.4	150.5	2.3	0.0
Narsingdi Dental Hospital	Sadar	Narsingdi	10	286	610	896	-	-	-	2572	1165	345	4082	2.5	98.2	2.7	0.0
National Hospital	Sadar	Narsingdi	10	835	1024	1859	-	-	-	497	423	630	1550	5.1	254.7	2.8	0.0
New Sulia Hospital	Sadar	Narsingdi	7	142	342	484	-	-	-	1620	1080	220	2920	1.3	70.2	2.8	0.0
Niramon Hospital	Sadar	Narsingdi	10	145	300	445	-	-	-	300	370	100	770	1.2	44.8	2.7	0.0
Nurjahan Pvt. Hospital	Sadar	Narsingdi	10	169	425	594	-	-	-	2322	2284	522	5128	1.6	69.1	2.9	0.0
Prime Madhabdi	Sadar	Narsingdi	10	339	960	1299	-	-	-	5540	7122	2834	15496	3.6	132.6	3.0	0.0
Prime Tarua	Sadar	Narsingdi	30	232	1140	1372	1	1	2	2043	3916	555	6514	3.8	47.8	3.3	0.1
Square Pvt. Hospital	Sadar	Narsingdi	10	250	480	730	-	-	-	420	570	230	1220	2.0	73.2	2.7	0.0
United Hospital	Sadar	Narsingdi	10	183	1614	1797	-	-	-	514	1667	279	2460	4.9	246.2	4.5	0.0
Alpona Clinic	Sadar	Natore	10	251	398	649	-	-	-	107	218	86	411	1.8	20.0	1.2	0.0
Care Medical Center	Sadar	Natore	10	288	482	770	-	-	-	2312	5336	770	8418	2.1			0.0

Table continued

Health facility		Address	District	No. of beds	No. of admissions		No. of deaths		No. of outdoor visits			Average daily number		Bed-occupancy rate (%)	Average length of stay (d)	Hospital death rate (%)	
					Male	Female	Male	Female	Total	Male	Female	Child	Total				Admission
Chalan Bli Clinic	Sadar	Natore	10	249	395	644	-	-	-	510	714	240	1464	1.8	4	19.7	0.0
	Sadar	Natore	9	31	529	560	-	-	-	201	300	100	601	1.5	2	17.1	0.0
	Sadar	Natore	10	81	259	340	-	-	-	103	162	-	265	0.9	1	39.7	0.0
Satata Clinic	Sadar	Natore	10	75	242	317	-	-	-	1152	1640	335	3127	0.9	9		0.0
	Sadar	Natore	10	80	250	330	-	-	-	1050	1560	340	2950	0.9	8	14.9	0.0
	Bonpara	Natore	30	230	1525	1755	3	6	9	1520	1630	525	3675	4.8	10		0.5
Jayeta Hospital	Bonpara	Natore	10	98	340	438	1	1	2	580	620	120	1320	1.2	4		0.5
	Bonpara	Natore	10	75	250	325	1	-	1	430	550	110	1090	0.9	3		0.3
	Bonpara	Natore	30	198	1415	1613	2	5	7	1374	1392	487	3253	4.4	9		0.4
Shadesh Diagnosis & Clinic	Bonpara	Natore	10	85	370	455	1	1	2	560	730	170	1460	1.2	4		0.4
	Gopalpur	Natore	20	123	937	1060	-	-	-	734	1348	680	2762	2.9	8	29.8	1.8
	Gopalpur	Natore	10	1520	1920	3440	-	-	-	532	1180	102	1814	9.4	5	276.6	2.1
Cosmos General Hospital	Gopalpur	Natore	10	65	243	308	-	-	-	-	-	-	-	0.8	0	15.6	1.4
	Gopalpur	Natore	10	130	406	536	-	-	-	40	213	35	288	1.5	1	2.3	0.1
	Gurudaspur	Natore	10	56	324	380	-	-	-	95	131	28	254	1.0	1	19.6	1.1
Muktar General Hospital	Gurudaspur	Natore	10	95	295	390	2	-	2	490	690	120	1300	1.1	4		0.5
	Jonail	Natore	10	65	243	308	-	-	-	-	-	-	-	0.8	0	15.6	1.4
	Lalpur	Natore	10	36	228	264	-	-	-	3187	5543	423	9153	0.7	25		0.0
DSK Maternity & Laboratory	Durgapur	Netrakona	16	36	228	264	-	-	-	3187	5543	423	9153	0.7	25		0.0
	Durgapur	Netrakona	10	108	305	413	1	2	3	1743	5351	2435	9529	1.1	26	217.0	14.4
	Sadar	Pabna	10	130	170	300	-	-	-	650	750	420	1820	0.8	5	27.1	1.8
Baral Clinic Ltd.	Sadar	Pabna	20	1057	1125	2182	-	1	1	4257	5418	1125	10800	6.0	30	68.7	1.2
	Chatmohor	Pabna	10	293	360	653	-	-	-	55	315	14	384	1.8	1	72.6	2.4
	Chatmohor	Pabna	10	72	240	312	-	-	-	60	480	96	636	0.9	2	70.0	6.0
Bhargura Health Care Ltd.	Sujanagar	Pabna	8	5	144	149	-	-	-	-	-	-	-	0.4	0		0.0
	Sadar	Patuakhali	40	178	602	780	1	-	1	674	2803	1701	5178	2.1	14	0.1	0.1
	Sadar	Pirojpur	10	6	7	13	-	-	-	123	472	551	1146	0.0	3	0.3	0.4
Muslim Aid Hospital	Sadar	Pirojpur	20	103	196	299	-	-	-	1984	2330	2056	6370	0.8	17	0.1	0.0
	Sadar	Pirojpur	10	70	67	137	-	-	-	215	320	105	640	0.4	2	0.1	0.0
	Sadar	Rajshahi	30	1664	1836	3500	-	-	-	4863	5187	1342	11392	9.6	31	57.4	0.9
Dolphin Clinic	Sadar	Rajshahi	20	1226	1318	2544	1	-	1	2578	2755	562	5895	7.0	16	44.6	0.7
	Sadar	Rajshahi	31	1598	2076	3674	2	1	3	3658	3782	1291	8731	10.1	24	41.5	0.7
	Sadar	Rajshahi	20	825	764	1589	-	-	-	1854	1689	588	4131	4.4	11	31.7	0.8
Padma Clinic	Sadar	Rajshahi	100	2114	3900	6014	35	32	67	7114	19845	5107	32066	16.5	88	76.4	3.0
	Chandraghona	Rangamati		428	574	1002	13	10	23	2747	1878	755	5380	2.7	15	106.7	2.2
	Langadu	Rangamati		599	381	980	9	6	15	2971	1951	478	5400	2.7	15	49.3	0.8
Central Hospital	Sadar	Sirajganj	10	1084	561	1645	6	6	12	4110	3014	589	7713	4.5	21	77.7	0.6
	Sadar	Sirajganj	8	30	35	65	-	-	-	15810	3981	1289	21080	0.2	58		0.0
	Sadar	Sirajganj	30	1620	1380	3000	15	13	28	8782	6623	1113	16518	8.2	45	41.4	0.8
Medinova Hospital Ltd.	Sadar	Sirajganj	400	10120	7595	17715	55	36	91	30510	19914	-	50424	48.5	138		0.5
	Sadar	Sirajganj	10	325	865	1190	-	-	-	2070	3615	700	6385	3.3	17		0.0
	Sadar	Sirajganj	10	45	49	94	-	-	-	14091	7810	1210	23111	0.3	63		0.0

Table continued

Health facility	Address	District	No. of beds	No. of admissions		No. of deaths		No. of outdoor visits			Average daily number		Bed-occupancy rate (%)	Average length of stay (d)	Hospital death rate (%)		
				Male	Female	Male	Female	Total	Male	Female	Child	Admission				Outdoor visit	
				16420	17426	33846	390	406	796	74384	86445	40207	201036	92.7	551	0.1	2.4
	Jalalabad Ragib-Rabeya Medical College Hospital	Sylhet	925	8760	13505	22265	103	78	181	59205	93742	11513	164460	61.0	451	60.4	0.8
	Sylhet Women's Medical College Hospital	Sylhet	500	14421	22928	37349	263	423	686	82134	110242	37168	229544	102.3	629	73.9	1.8
	Mirzapur	Tangail	750														

Annex to Chapter 8

Mortality profiles 2012 from various medical college hospitals, with names of fatal diseases and conditions

Sher-e-Bangla Medical College Hospital (total deaths: 3,058)

ICD-10 code with name of disease/condition	No. of cases	%
J441 Chronic obstructive pulmonary disease with acute exacerbation, unspecified	217	7.10
I612 Intracerebral hemorrhage in hemisphere, unspecified	206	6.74
P219 Birth asphyxia, unspecified	174	5.69
I110 Hypertensive heart disease with (congestive) heart failure	166	5.43
J159 Bacterial pneumonia, unspecified	154	5.04
T600 Organophosphate and carbamate insecticides	122	3.99
I210 Acute transmural myocardial infarction of anterior wall	110	3.60
N189 Chronic renal failure, unspecified	99	3.24
V011 Pedestrian injured in collision with pedal cycle, traffic accident	86	2.81
G049 Encephalitis, myelitis, and encephalomyelitis, unspecified	65	2.13

Chittagong Medical College Hospital (total deaths: 3,241)

ICD-10 code with name of disease/condition	No. of cases	%
V860 Driver of all-terrain or other off-road motor vehicle injured in traffic accident	782	24.13
T670 Heatstroke and sunstroke	617	19.04
P139 Birth injury to skeleton, unspecified	510	15.74
S420 Fracture of clavicle	379	11.69
X85 Assault by drugs, medicaments, and biological substances	327	10.09
K761 Chronic passive congestion of liver	195	6.02
A267 Erysipelothrix septicemia	190	5.86
G04 Encephalitis, myelitis, and encephalomyelitis	128	3.95
J10 Influenza due to identified influenza virus	79	2.44
G00 Bacterial meningitis, NEC	34	1.05

Comilla Medical College Hospital (total deaths: 1,623)

ICD-10 code with name of disease/conditions	No. of cases	%
P522 Intraventricular (non-traumatic) hemorrhage, grade 3, of fetus and newborn	473	29.14
I219 Acute myocardial infarction, unspecified	229	14.11
J150 Pneumonia due to <i>Klebsiella pneumoniae</i>	138	8.50
G039 Meningitis, unspecified	120	7.39
T571 Phosphorus and its compounds	97	5.98
V011 Pedestrian injured in collision with pedal cycle, traffic accident	95	5.85
A419 Septicemia, unspecified	86	5.30
O159 Eclampsia as to unspecified time period	32	1.97
R101 Pain localized to upper abdomen	23	1.42
K561 Intussusception	21	1.29

Shaheed Suhrawardy Medical College Hospital (total deaths: 184)

ICD-10 code with name of disease/condition	No. of cases	%
P522 Intraventricular (non-traumatic) hemorrhage, grade 3, of fetus and newborn	473	29.14
G46 Vascular syndromes of brain in cerebrovascular diseases	35	19.02
A419 Septicemia, unspecified	28	15.22
E117 Non-insulin-dependent diabetes mellitus with multiple complications	17	9.24
C76 Malignant neoplasm of other and ill-defined sites	15	8.15
K729 Hepatic failure, unspecified	12	6.52
G03 Meningitis due to other and unspecified causes	11	5.98
E649 Sequelae of unspecified nutritional deficiency	10	5.43
A199 Miliary tuberculosis, unspecified	9	4.89
T81 Complications of procedures, NEC	8	4.35
V029 Pedestrian injured in collision with two- or three-wheeled motor vehicle, unspecified whether traffic or non-traffic accident	7	3.80

Faridpur Medical College Hospital (total deaths: 1,912)

ICD-10 code with name of disease/condition	No. of cases	%
G46 Vascular syndromes of brain in cerebrovascular diseases	446	23.33
P210 Severe birth asphyxia	315	16.47
P07 Disorders related to short gestation and low birthweight, NEC	97	5.07
I219 Acute myocardial infarction, unspecified	94	4.92
T44 Poisoning by drugs primarily affecting the autonomic nervous system	55	2.88
J441 Chronic obstructive pulmonary disease with acute exacerbation, unspecified	47	2.46
V73 Bus occupant injured in collision with car, pick-up truck or van	35	1.83
R100 Acute abdomen	33	1.73
J46 Status asthmaticus	25	1.31
N189 Chronic renal failure, unspecified	17	0.89

Dhaka Medical College Hospital (total deaths: 8,866)

ICD-10 code with name of disease/condition	No. of cases	%
V99 Unspecified transport accident	1502	16.94
I64 Stroke, not specified as hemorrhage or infarction	1255	14.16
S069 Intracranial injury, unspecified	920	10.38
T142 Fracture of unspecified body region	850	9.59
I11 Hypertensive heart disease	520	5.87
A419 Septicemia, unspecified	478	5.39
X85 Assault by drugs, medicaments, and biological substances	402	4.53
J189 Pneumonia, unspecified	352	3.97
N19 Unspecified renal failure	290	3.27
J449 Chronic obstructive pulmonary disease, unspecified	270	3.05

Sir Salimullah Medical College Hospital (total deaths: 1,953)

ICD-10 code with name of disease/condition	No. of cases	%
P071 Other low birthweight	150	7.68
G048 Other encephalitis, myelitis, and encephalomyelitis	110	5.63
I461 Sudden cardiac death, so described	95	4.86
P219 Birth asphyxia, unspecified	93	4.76
J441 Chronic obstructive pulmonary disease with acute exacerbation, unspecified	93	4.76
A419 Septicemia, unspecified	90	4.61
I516 Cardiovascular disease, unspecified	90	4.61
I608 Other subarachnoid hemorrhage	76	3.89
I238 Other current complications following acute myocardial infarction	71	3.64
J188 Other pneumonia, organism unspecified	63	3.23

Mymensingh Medical College Hospital (total deaths: 7,675)

ICD-10 code with name of disease/condition	No. of cases	%
P20 Intrauterine hypoxia	960	12.51
I62 Other non-traumatic intracranial hemorrhage	960	12.51
I21 Acute myocardial infarction	444	5.79
P05 Slow fetal growth and fetal malnutrition	324	4.22
N18 Chronic renal failure	300	3.91
A41 Other septicemia	252	3.28
V70 Bus occupant injured in collision with pedestrian or animal	240	3.13
G04 Encephalitis, myelitis, and encephalomyelitis	204	2.66
J44 Other chronic obstructive pulmonary disease	204	2.66
T44 Poisoning by drugs primarily affecting the autonomic nervous system	132	1.72

Khulna Medical College Hospital (total deaths: 2,772)

ICD-10 code with name of disease/condition	No. of cases	%
I619 Intracerebral hemorrhage, unspecified	472	17.03
I21 Acute myocardial infarction	105	3.79
X689 Intentional self-poisoning by and exposure to pesticides, unspecified place	65	2.34
I501 Left ventricular failure	42	1.52
J441 Chronic obstructive pulmonary disease with acute exacerbation, unspecified	38	1.37
I500 Congestive heart failure	35	1.26
N185 Chronic kidney disease, stage 5	28	1.01
G042 Bacterial meningoencephalitis and meningomyelitis, NEC	23	0.83
K721 Chronic hepatic failure	21	0.76
O15 Eclampsia	14	0.51

Shahid Ziaur Rahman Medical College Hospital (total deaths: 2,798)

ICD-10 code with name of disease/condition	No. of cases	%
P210 Severe birth asphyxia	288	10.29
I679 Cerebrovascular disease, unspecified	203	7.26
I219 Acute myocardial infarction, unspecified	164	5.86
I509 Heart failure, unspecified	125	4.47
V099 Pedestrian injured in unspecified transport accident	104	3.72
J449 Chronic obstructive pulmonary disease, unspecified	88	3.15
X44 Accidental poisoning by and exposure to other and unspecified drugs, medicaments and biological substances	71	2.54
P220 Respiratory distress syndrome of newborn	65	2.32
J159 Bacterial pneumonia, unspecified	51	1.82
O15 Eclampsia	23	0.82

Dinajpur Medical College Hospital (total deaths: 617)

ICD-10 code with name of disease/condition	No. of cases	%
I64 Stroke, not specified as hemorrhage or infarction	107	17.34
X87 Assault by pesticides	88	14.26
J159 Bacterial pneumonia, unspecified	62	10.05
P070 Extremely low birthweight	55	8.91
I21 Acute myocardial infarction	52	8.43
P210 Severe birth asphyxia	38	6.16
A41 Other septicemia	35	5.67
J449 Chronic obstructive pulmonary disease, unspecified	30	4.86
N189 Chronic renal failure, unspecified	30	4.86
G039 Meningitis, unspecified	13	2.11

Rangpur Medical College Hospital (total deaths: 3,325)

ICD-10 code with name of disease/condition	No. of cases	%
I679 Cerebrovascular disease, unspecified	449	13.50%
R090 Asphyxia	421	12.66%
B949 Sequelae of unspecified infectious or parasitic disease	397	11.94%
P071 Other low birthweight	208	6.26%
I679 Cerebrovascular disease, unspecified	196	5.89%
G049 Encephalitis, myelitis, and encephalomyelitis, unspecified	179	5.38%
I219 Acute myocardial infarction, unspecified	133	4.00%
S099 Unspecified injury of head	127	3.82%
E148 Unspecified diabetes mellitus with unspecified complications	73	2.20%
G039 Meningitis, unspecified	52	1.56%

M.A.G. Osmani Medical College Hospital, Sylhet (total deaths: 4,345)

ICD-10 code with name of disease/condition	No. of cases	%
I64 Stroke, not specified as hemorrhage or infarction	495	11.39
A419 Septicemia, unspecified	248	5.71
I21 Acute myocardial infarction	182	4.19
P219 Birth asphyxia, unspecified	145	3.34
S099 Unspecified injury of head	138	3.18
A85 Other viral encephalitis, NEC	101	2.32
T159 Foreign body on external eye, part unspecified	94	2.16
P369 Bacterial sepsis of newborn, unspecified	83	1.91
T609 Pesticide, unspecified	45	1.04
G039 Meningitis, unspecified	40	0.92

Annex to Chapter 9

Year-wise malaria cases by type and death rate

Year	Total cases	<i>P. falciparum</i>		<i>P. vivax</i>		Death	
	No.	No.	%	No.	%	No.	%
2000	54223	39272	72.4	14951	27.6	478	0.88
2001	54216	39274	72.4	14942	27.6	490	0.90
2002	62269	46418	74.5	15851	25.5	588	0.94
2003	54654	41356	75.7	13298	24.3	577	1.06
2004	58894	46402	78.8	12492	21.2	535	0.91
2005	48121	37679	78.3	10442	21.7	501	1.04
2006	32857	24828	75.6	8029	24.4	307	0.93
2007	59857	46791	78.2	13066	21.8	228	0.38
2008	84690	70281	83.0	14409	17.0	1544	1.82
2009	63873	57020	89.3	6853	10.7	47	0.07
2010	55873	52049	93.2	3824	6.8	37	0.07
2011	51773	49194	95.0	2579	5.0	36	0.07
2012	29518	27819	94.2	1699	5.8	11	0.04
Average per year	54678	44491	81.4	10187	18.6	414	0.76

Distribution of malaria cases and deaths in endemic districts of Bangladesh (2012)

District	Uncomplicated malaria	Severe malaria	Vivax malaria	Total cases	Death
Sherpur	46	-	27	73	-
Mymensingh	165	2	1	168	-
Netrakona	266	-	19	285	-
Kurigram	38	-	63	101	-
Sylhet	364	-	72	436	-
Habiganj	60	-	12	72	-
Sunamganj	467	32	41	540	-
Maulvibazar	235	49	137	421	-
Chittagong	991	5	99	095	-
Khagrachhari	5443	329	225	5997	6
Rangamati	7650	72	259	7981	1
Bandarban	7659	457	345	8461	3
Cox's Bazar	2978	511	399	3888	1
Total	26362	1457	1699	29518	11

List of 34 districts endemic for filariasis according to ICT card test survey (May-June 2004) and microfilaria survey till 2008

Division	District	Division	District	Division	District	Division	District
Rangpur	Panchagarh Thakurgaon Nilphamari Lalmonirhat Rangpur Kurigram Dinajpur	Rajshahi	Rajshahi Chapainowabganj Sirajganj Pabna Bogra	Khulna	Meherpur Narail Bagerhat Chuadanga Jhenaidah Kushtia Feni Laxmipur Bandarban	Dhaka	Dhaka Gopalganj Munshiganj Narsingdi Gazipur Jamalpur Narayanganj
			Pirojpur Jhalokathi Barguna Barisal Patuakhali				
				Sylhet	Habiganj		

Year-wise mass drug administration rounds (2001-2012) and coverage

Year	No. of districts	Total population (million)	Reported coverage (%)	Observed coverage by survey (%)	Actual coverage among eligible (%)
2001	2001	0.81	95.5	93.0	ND*
2002	2002	5.18	93.6	83.2	87.3
2003	2003	8.73	93.3	77.9	81.9
2004	2004	11.75	98.6	ND*	ND*
2005	2005	20.16	90.3	78.0	82.2
2006	2006	23.92	92.2	78.2	82.2
2007	2007	31.0	91.5	82.4	84.3
2008	2008	42.0	90.53	79.38	83.06
2009	2009	35.0	96.87	83.33	85.76
2010	2010	35.0	92.47	60.23	62.98
2011	2011	29.70	97.14	92.35	94.90
2012	2012	16.67	98.11	89.76	92.78

Reported year-wise coverage of school deworming program through albendazole

Year	Round	No. of districts covered	No. of children		Reported coverage (%)
			Targeted	Treated	
2008	November	64	15743159	15482778	94.0
2009	May	64	19303404	19101496	98.0
	November	64	19303404	18782212	97.0
2010	May	64	19837612	19440860	98.0
	November	64	21971611	21745757	99.0
2011	May	64	22070512	21735040	98.5
	November	64	22082923	21992383	99.6
2012	May	64	22263213	22040581	99.0
	November	64	22263192	22038334	99.0

List of 37 districts under 4 divisions where mass dog vaccination was done in 2012

District	No. of wards	Dogs counted	Dogs vaccinated	Coverage in %
Rangpur division				
Rangpur	15	2203	1907	86.6
Gaibandha	9	705	538	76.3
Kurigram	9	1105	845	76.5
Lalmonirhat	9	886	733	82.7
Nilphamari	9	1668	1353	81.1
Dinajpur	12	2036	1601	78.6
Panchagarh	9	2036	815	40.0
Thakurgaon	12	1150	950	82.6
Sub-total in 8 districts	84	10784	9001	83.5
Rajshahi division				
Naogaon	9	1627	1485	91.3
Chapainowabganj	15	1298	1054	81.2
Bogra	21	3012	2591	86.0
Joypurhat	9	821	621	75.6
Natore	9	1257	954	75.9
Sirajganj	15	1662	1446	87.0
Pabna	15	1125	1084	96.4

Table continued

District	No. of wards	Dogs counted	Dogs vaccinated	Coverage in %
Rajshahi	12	5172	4397	85.0
Sub-total in 8 districts	105	15974	13632	85.3
Sylhet division				
Habiganj	9	575	524	91.1
Maulvibazar	9	454	429	94.5
Sunamganj	9	629	560	89
Sub-total in 3 districts	27	1658	1513	91.3
Dhaka division				
Narayanganj	27	4243	3323	78.3
Gazipur	9	1826	1570	86.0
Gazipur (Tongi only)	21	3642	3036	83.4
Narsingdi	9	1320	1250	94.7
Shariatpur	9	1055	976	92.5
Rajbari	9	346	313	90.5
Madaripur	9	770	625	81.2
Gopalganj	9	611	527	86.3
Faridpur	9	1253	1059	84.5
Tangail	21	1183	956	80.8
Kishoreganj	9	1490	1341	90.0
Mymensingh	21	1761	1510	85.7
Netrakona	9	1350	1173	86.9
Sherpur	9	755	635	84.1
Jamalpur	9	1125	1024	91.0
Manikganj	9	1249	1097	87.8
Munshiganj	9	1278	1139	89.1
Sub-total in 17 districts	207	25257	21554	85.3
Total in 36 districts	423	53673	45700	85.1

Year-wise number of rabies-related deaths at the Infectious Disease Hospital, Mohakhali, Dhaka

Year	No. of rabies-related deaths
2006	167
2007	166
2008	165
2009	164
2010*	58
2011	109
2012	88

*(Jul-Dec)

Nipah or Nipah-like viralencephalitis in Bangladesh (2001 to 2012)

Duration	District	No. of cases	No. of deaths	Case fatality rate
April-May 2001	Meherpur	13	9	69%
January 2003	Naogaon	12	8	67%
January 2004; April 2004	Rajbari; Faridpur	31; 36	23; 27	74%; 75%
Jan-March 2005	Tangail	12	11	92%
Jan-Feb 2007; March 2007; April 2007	Thakurgaon; Kushtia, Pabna, Natore; Naogaon	7; 8; 3	3; 5; 1	43%; 63%; 33%
February 2008; April 2008	Manikganj; Rajbari and Faridpur	4; 7	4; 5	100%; 71%
Jan 2009	Gaibandha, Rangpur, and Nilphamari, Rajbari	3; 1	0; 1	0%; 100%
Feb-Mar 2010	Faridpur, Rajbari, Gopalganj, and Madaripur	16	14	87.5%
Jan-Feb 2011	Lalmohirhat, Dinajpur, Comilla, Nilphamari, and Rangpur	44	40	91%
Jan-Mar 2011	Joypurhat, Rajshahi, Natore, Rajbari, and Gopalganj	12	10	83%
Jan-Feb 2012	Joypurhat, Rajshahi	18	13	72.2%
	Total	215	164	76.3%

Year-wise tuberculosis case notification by type (2006-2012)

Year	Residence of population	No. of tuberculosis cases				Total
		Smear-positive		Smear-negative	Extra pulmonary	
		New	Replase	New	New	
2006	Rural/upazila	89704	2645	16717	9707	11773
	Urban/metropolitan	9255	1279	5409	3499	19442
	CDC	2806	287	2375	1155	6623
	Total	101765	4211	24501	14361	144838
2007	Rural/upazila	91606	2517	15852	10861	120836
	Urban/metropolitan	10264	1049	5449	4164	20926
	CDC	2437	222	1934	1093	5686
	Total	104307	3788	23235	16118	147448
2008	Rural/upazila	93659	2753	15069	12825	124306
	Urban/metropolitan	10289	1165	5660	4486	21600
	CDC	2425	220	1463	1048	5156
	Total	106373	4138	22192	18359	151062
2009	Rural/upazila	96333	2692	17759	15768	132552
	Urban/metropolitan	10390	1136	5829	4872	22227
	CDC	2171	150	1548	1225	5094
	Total	108894	3978	25136	21865	159873
2010	Rural/upazila	93937	2101	15539	17255	128832
	Urban/metropolitan	9977	770	4788	4943	20478
	CDC	1858	129	1298	1308	4593
	Total	105772	3000	21625	23506	153903
2011	Rural/upazila	87743	1889	16433	20340	126405
	Urban/metropolitan	9391	698	4442	5648	20179
	CDC	1814	114	1046	1341	4315
	Total	98948	2701	21921	27329	150899
2012	Rural/upazila	95132	2135	18856	22506	138629
	Urban/metropolitan	10068	820	4640	6849	22377
	CDC	1640	112	955	1194	3901
	Total	106840	3067	24451	30549	164907

Distribution of newly-detected leprosy cases by division of Bangladesh (2009-2012)

Division	Year	No. of cases				Registered prevalence per 10,000 population
		Population	MB	PB	Total	
Barisal	2009	9170109	10	1	11	0.01
	2010	9254080	4	-	4	0.0
	2011	9338999	1	1	2	0.002
	2012	9424877	2	2	4	0.004
Chittagong	2009	28901453	400	358	758	0.19
	2010	29387800	226	202	428	0.16
	2011	29883564	229	141	370	0.122
	2012	30388960	241	227	468	0.111
Dhaka	2009	47845021	652	930	1582	0.26
	2010	48795514	542	897	1439	0.24
	2011	49771330	563	893	1546	0.227
	2012	50776361	454	7785	1239	2.47
Khulna	2009	17163458	50	15	65	0.34
	2010	17410525	52	16	68	0.03
	2011	17661430	49	12	610	0.037
	2012	17916236	49	9	58	0.03
Rajshahi	2009	35208052	916	1537	2453	0.53
	2010	35702832	697	1269	1966	0.44
	2011	36204971	697	1007	1704	0.378
	2012	36714583	729	919	1648	0.41
Sylhet	2009	9350784	218	151	369	0.45
	2010	9496717	199	79	278	0.41
	2011	9644939	169	118	287	0.384
	2012	9795488	168	91	259	0.344
Total	2009	147638877	2246	2992	5238	0.28
	2010	150047468	1720	2463	4183	0.24
	2011	152505233	1798	2172	3970	0.216
	2012	155016505	1643	2033	3676	0.223

Division-wise leprosy cases who completed MDT (multidrug treatment) in Bangladesh (2010-2012)

Division	Mycobacterium (MB method) (>5 lesions)			Phosphate buffer (PB method) (1 to 5 lesions)			Division		
	2010	2011	2012	2010	2011	2012	2010	2011	2012
Dhaka	644	539	454	583	816	785	1227	1355	1239
Barisal	2	2	2	0	1	2	2	3	4
Chittagong	281	346	241	220	185	227	501	531	468
Sylhet	172	196	168	121	79	91	293	275	259
Khulna	42	44	49	16	20	9	58	64	57
Rajshahi	826	798	729	1425	1010	919	2259	1808	1648
Total	1967	1925	1643	2365	2111	2033	4332	4036	3676
(%)	(45.41)	(47.7)	(44.5)	(54.59)	(52.3)	(54.5)	(100)	(100)	(100)

New cases of HIV infection by occupation in Bangladesh (2012)

Occupation	New HIV	%
Barber and Bar dancer	3	0.9
Drug seller	1	0.3
Guard	1	0.3
Hijra	1	0.3
Painter	1	0.3
Child	2	0.6
Cook	2	0.6
Hawker	2	0.6
Tailor	2	0.6
Transport worker	3	0.9
Garments worker	4	1.2
Paper collector	4	1.2
Farmer	5	1.5
Rickshaw-puller	5	1.5
Driver	6	1.8
Tokai	7	2.1
Student	11	3.3
Sex worker	12	3.6
Ex. migrant worker	17	5.0
Laborer	17	5.0
Businessmen	25	7.4
Service-holder	25	7.4
Housewife	72	21.3
Unemployed	72	21.3
Other	26	7.7

Annex to Chapter 11

Number of admissions, outdoor visits, average length of stay, and bed-occupancy rates at the National Institute of Cardiovascular Diseases (NICVD) (2002-2012)

Year	Admission (N)		Outdoor visit (N)					Average length of stay (d)	Bed-occupancy rate (%)
	Total	Daily average	Male	Female	Child	Total	Daily average		
2002	17081	47	52740	29532	4674	86944	238	6.9	129.6
2003	20083	55	54550	31939	5150	91639	251	7.1	157.8
2004	21522	59	56482	31250	4857	92589	253	6.9	164.0
2005	22419	62	59950	34608	5497	100055	274	6.5	160.4
2006	24376	67	61565	34861	6060	102486	281	6.5	175.8
2007	29147	80	76732	41792	7417	125941	345	5.5	174.8
2008	33946	93	91147	47889	8534	147570	403	5.2	147.7
2009	41554	114	99102	51539	9367	160008	438	5.2	141.8
2010	42779	117	100868	51364	9726	161958	444	5.4	152.8
2011	43275	119	103930	50081	9802	163813	449	5.4	146.6
2012	44559	122	113157	51488	9721	174366	476	5.2	153.2

Number of exercise tolerance tests (ETTs) done in the National Institute of Cardiovascular Diseases (NICVD) from 2001 to 2012

Year	Male		Female		Total	
		%		%		%
2001	210	89.7	24	10.3	234	100.0
2002	254	49.9	55	10.8	509	100.0
2003	731	87.8	102	12.2	833	100.0
2004	828	83.2	167	16.8	995	100.0
2005	823	82.1	180	17.9	1003	100.0
2006	1233	79.3	321	20.7	1554	100.0
2007	1437	82.7	301	17.3	1738	100.0
2008	1798	84.1	339	15.9	2137	100.0
2009	1610	85.2	288	15.2	1889	100.0
2010	1549	88.0	212	12.0	1761	100.0
2011	1353	80.6	323	19.2	1678	100.0
2012	1945	79.6	497	20.4	2442	100.0

Number of myocardial perfusion imaging done in the National Institute of Cardiovascular Diseases (NICVD) from 2003 to 2012

Year	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012
No.	8	21	16	0	21	69	30	104	152	147

Number of different cath lab procedures performed in the NICVD from 2003 to 2012

Procedure		Year									
		2003	2004	2005	2006	2007	2008	2009	2010	2011	2012
Coronary angiography		2827	3210	2780	3105	3266	3980	4437	4711	4426	4881
Cardiac cath		308	225	227	229	295	380	340	334	251	256
Angiography	Renal	13	69	6	-	-	-	1	6	12	1
	Peripheral	42	93	85	106	87	112	112	124	124	120
	Total	55	162	91	106	87	112	113	130	136	121
Angioplasty	Renal	-	-	-	-	-	-	9	7	17	6
	Peripheral	-	-	4	7	43	23	3	18	12	22
	Total	-	-	4	7	43	23	12	25	29	28
Other interventions	PCI	371	599	488	584	574	889	1149	1312	1254	1681
	PTMC	189	273	295	280	20	130	154	187	117	137
	TPM	646	715	708	675	850	741	950	647	905	1090
	PPM	320	333	368	321	359	414	487	402	418	461
	EPS&RFA	-	-	-	161	204	113	177	66	72	56
	Deviceclosure	-	-	-	1	-	-	-	SS0	-	-
	Others	12	13	11	4	-	18	40	56	34	97
	Total	1538	1933	1870	2026	2007	2305	2957	2670	2800	3522

Number of heart and vascular surgeries performed in the NICVD from 2000 to 2012

Year	Open-heart surgery					Closed- heart surgery	Vascular surgery		
	CABG	Valve	Congenital	Other	Total		Routine	Emergency	Total
2000	44	133	88	26	291	186	74	213	287
2001	60	134	133	3	330	157	100	193	293
2002	112	89	210	4	415	151	114	232	346
2003	170	142	162	22	496	140	69	153	222
2004	180	159	205	17	561	95	92	208	300
2005	267	102	237	20	626	93	90	206	296
2006	226	113	255	28	622	70	95	405	500
2007	188	165	256	46	655	58	121	447	568
2008	233	182	327	21	763	63	152	840	992
2009	218	264	364	11	857	71	219	1001	1220
2010	152	304	365	37	859	88	254	1036	1290
2011	101	207	342	67	717	98	183	1640	1823
2012	175	249	468	57	949	82	254	1274	1528
Total	2126	2243	3412	359	8141	1352	1817	7848	9665

Number of patients attending the outdoor of National Center for Control of Rheumatic Fever and Heart Diseases (NCCRFHD) in different months of 2012 (showing age and sex distribution)

Month	Age and sex																	
	1-4 year(s)			5-14 years			15-24 years			25-49 years			50+ years			Grand total		
	M	F	Total	M	F	Total	M	F	Total	M	F	Total	M	F	Total	M	F	Total
Jan	14	27	41	349	427	776	316	693	1009	151	330	481	30	32	62	860	1509	2369
Feb	23	11	34	350	346	696	372	592	964	207	431	638	42	48	90	994	1428	2422
Mar	14	17	31	421	508	929	365	746	1111	156	371	527	27	19	46	983	1661	2644
Apr	20	6	26	374	327	701	360	575	935	165	429	594	45	25	70	964	1362	2326
May	9	27	36	413	550	963	346	789	1135	155	343	498	23	31	54	946	1740	2686
June	19	49	68	278	439	717	265	571	836	236	458	694	56	61	117	854	1578	2432

Table continued

Month	Age and sex																	
	1-4 year(s)			5-14 years			15-24 years			25-49 years			50+ years			Grand total		
	M	F	Total	M	F	Total	M	F	Total	M	F	Total	M	F	Total	M	F	Total
July	17	5	22	362	321	683	429	607	1036	188	388	576	62	52	114	1058	1373	2431
Aug	-	-	-	189	337	526	399	535	934	55	32	87	25	34	59	668	938	1606
Sep	3	10	13	343	464	807	431	656	1087	242	347	589	62	58	120	1081	1535	2616
Oct	13	1	14	296	407	703	294	568	862	218	386	604	65	79	144	886	1441	2327
Nov	11	9	20	244	352	596	298	541	839	194	314	508	41	37	78	788	1253	2041
Dec	1	16	17	234	315	549	220	513	733	53	207	260	13	7	20	521	1058	1579
Total	144	178	322	3853	4793	8646	4095	7386	11481	2020	4036	6056	491	483	974	10603	16876	27479

M=Male, F=Female

Number of outdoor and indoor patients from 2009 to 2012 at the National Institute of Kidney Diseases & Urology (NIKDU), with male, female and child disaggregation

OPD/ Indoor	2009				2010				2011				2012			
	M	F	C	Total	M	F	C	Total	M	F	C	Total	M	F	C	Total
OPD (old & new pts)	32522	17081	3595	53198	33246	17406	3056	53708	33988	16948	3176	54112	32098	15179	3140	50417
OPD (new pts)	32160	12240	3548	38584	25579	13510	3036	42125	25018	13193	3176	41387	27663	12913	3130	43706
OPD (old pts)	7777	3961	495	12233	7667	3866	20	11553	8970	3855	0	12825	4435	2266	10	6711
Indoor	2397	1462	447	4306	2309	1357	418	4084	2370	1378	476	4224	2563	1373	503	4439

M=Male, F=Female, C=Child

Number of new outdoor patients and the number of emergency visits and admissions in the NIMHR from 2007 to 2012

Year	Sex and age-group	OPD (new patients)	Indoor patients	Emergency patients
2007	Male	8959	671	-
	Female	5175	349	-
	Total	14134	1020	-
2008	Male	12692	749	-
	Female	9209	427	-
	Total	21901	1176	-
2009	Male	12427	876	-
	Female	9478	527	-
	Total	21905	1403	-
2010	Male	6506	489	382
	Female	4710	300	242
	Total	11216	789	624
2011	Male	13420	1136	1017
	Female	9968	636	594
	Total	23388	1772	1611
2012	Male	14959	1249	1159
	Female	8939	679	667
	Child	1610	102	127
	Total	25508	2030	1953

Annex to Chapter 15

Bangladesh Medical Research Council (BMRC)

Seminars, workshops, and symposia

1. Dissemination Seminar on the Pilot for Testing Delivery Mechanisms for Newborn Vitamin A Supplementation (NBVAS) in Bangladesh, held on 31 January 2012 at BMRC Bhaban, BMRC, Mohakhali, Dhaka. In total, one hundred participants attended the seminar.
2. Meeting of South Asian Forum for Health Research (SAFHR), held on 5-7 February 2012 at India International Centre, New Delhi, India. Prof. Dr. Mohammad Moazzam Hossain, Senior Vice Chairman; Professor Dr. Md. Habibe Millat, Director (Honorary); and Dr. Mahmood-uz-Jahan, Deputy Director from BMRC, Mohakhali, Dhaka, participated in the meeting. The program was inaugurated by Shri Ghulam Nabi Azad, Hon'ble Minister of Health and Family Welfare, Government of India, as the Chief Guest. Shri S. Gandhiselvan, Hon'ble Minister of State for Health and Family Welfare, Government of India and Sreemoti Preneet Kaur, Hon'ble Minister of State for External Affairs, Government of India, were the the Guests of Honour. Dr. V.M. Katoch, Secretary, Department of Health Research, Ministry of Health and Family Welfare and Director General, Indian Council of Medical Research, delivered welcome address in the meeting.
3. Video Conference on Autism, held on 15 February 2012 at BMRC Bhaban, Mohakhali, Dhaka, with Mrs. Saima Wazed, Chairman, National Committee on Autism as the Chief Guest, and Prof. (Dr.) Syed Modasser Ali, Adviser to the Honorable Prime Minister on Health and Family Welfare and Social Welfare, Government of the People's Republic of Bangladesh, Dr. Captain (Rtd.) Mujibur Rahman Fakir, MP, Honorable State Minister for Health and Family Welfare, Government of the People's Republic of Bangladesh as Special Guests. Professor Dr. Pran Gopal Datta, Vice Chancellor, BSMMU, Shahbagh, Dhaka, was also present. The program was presided over by Professor Dr. Khondaker Md. Shefyetullah, Director General of Health Services, Mohakhali, Dhaka.
4. Workshop on Research Management, held on 1 April to 15 May 2012 at BMRC Bhaban, BMRC, Mohakhali, Dhaka. In total, twenty participants from medical university, different medical colleges, dental colleges, postgraduate institutions and upazila health complexes attended the workshop. Professor Dr. Mahmud Hasan, Chairman, BMRC, distributed certificates among the participants.
5. Training of Trainers' (TOT) Program on Health Inter-network Access to Research Initiative (HINARI), held on 16-19 April 2012 at BMRC, Mohakhali, Dhaka. Chief Guest was Prof. (Dr.) Syed Modasser Ali, Adviser to the Honorable Prime Minister on Health and Family Welfare and Social Welfare, Government of the People's Republic of Bangladesh. Special Guest was Professor Dr. AFM Saiful Islam, Additional Director General (Administration), Directorate General of Health Services, Mohakhali, Dhaka. In total, 10 participants attended from medical university, icddr,b, BMRC, different medical colleges, and postgraduate institutions. Dr. Mahmood-uz-jahan, Deputy Director, BMRC, distributed certificates among the participants.
6. Seminar on Dissemination of Manuals "Reduction in Maternal Mortality for Community Health Workers", held on 20 May 2012 at BMRC, Mohakhali, Dhaka. Chief Guest was Professor Dr. Gowher Rizvi, Adviser to the Honorable Prime Minister on International Affairs, Government of the People's Republic of Bangladesh, and Special Guest was H.E. Svend Olling, Ambassador of Denmark, Royal Danish Embassy, Dhaka, Bangladesh. Mr. Nawazes Ali Khan, Adviser (Programme), ATN Bangla television, Dhaka, also attended the seminar. The program was presided over by Professor Dr. Mahmud Hasan, Chairman, BMRC, Mohakali, Dhaka. Participants were principals of different medical colleges; directors of medical Institutions; medical doctors of various disciplines, community healthcare providers, and BMRC personnel.
7. Workshop on Health Inter-network Access to Research Initiative (HINARI), held on 28-31 May 2012; 4-7 June 2012; 11-14 June 2012; 18-21 June 2012; 25-28 June 2012 at BMRC, Mohakali, Dhaka. In each batch, there were

twenty participants. One hundred participants attended from medical university, different medical colleges, postgraduate institutes, dental colleges, civil surgeons' offices, Sadar hospitals, and upazila health complexes. Chief Guest was Professor Dr. AFM Saiful Islam, Additional Director General (Admin.), Directorate General of Health Services, Mohakhali, Dhaka, and Special Guest was Dr. Md. Shah Nawaz, Director (Planning & Research), Directorate General of Health Services, Mohakhali, Dhaka. The workshop was presided over by Professor Dr. Mahmud Hasan, Chairman, BMRC, Mohakhali, Dhaka.

8. Workshop on Project Development, Data Collection, Data Analysis, and Report Writing, held on 8 July to 20 September 2012; 18 November 2012 to 17 January 2013 at BMRC, Mohakhali, Dhaka. Forty participants attended from medical university, different medical colleges, postgraduate institutions, general hospitals, and upazilas health complexes. Chief Guest was Professor Dr. AFM Saiful Islam, Additional Director General (Admin.), Directorate General of Health Services, Mohakhali, Dhaka, and Special Guest was Dr. Md. Shah Nawaz, Director (Planning & Research), Directorate General of Health Services, Mohakhali, Dhaka. In the Certificate distribution ceremony in Batch-1, Dr. Khaled Hassan was Chief Guest. In Batch-2, Acting WHO Representative to Bangladesh was Chief Guest; and Professor Dr. Abul Kalam Azad, Additional Director General (Planning & Development), Directorate General of Health Services, Mohakhali, Dhaka, was Special Guest. Professor Dr. Mahmud Hasan, Chairman, BMRC, Mohakhali, Dhaka, who presided over the workshop, distributed certificates among the participants.
9. Sub-regional Workshop on Using Research Evidence for Policy-making, held on 16-21 September 2012 at Kathmandu, Nepal. Professor Dr. Md. Habibe Millat, Honorary Director from BMRC, Mohakhali, Dhaka, participated in that meeting. The workshop's beginning session focused on the unique attributes of health system policy-making in relation to using research evidence. The final session focused on what is currently being done (and what could be done) to promote the use of research evidence in health system policy-making in the South-East Asia region and in each of the selected countries. Welcome address, introductions, and overview of the

workshop were made by WHO Representative of Nepal, representatives from MOHFW, and of other participating countries.

10. Workshop on How to Review a Scientific Paper, held on 11 August 2012 at BMRC, Mohakhali, Dhaka. Nineteen participants, including professors, associate professors, and assistant professors from medical university, different medical colleges, postgraduate institutions, and BMRC personnel attended the workshop.
11. Workshop on Primary Healthcare, held on 23 September, 25 September, 27 September, and 30 September 2012. One hundred nineteen participants (community healthcare providers) from different community clinics attended the workshop.
12. Meeting of the Editors of Research Journals of the Islamic Countries in Citation and Indexing, held in Shiraz, Iran, during 23-24 October 2012. Dr. Mahmood-uz-Jahan, Deputy Director and Member of Editorial Board, BMRC, Mohakhali, Dhaka, participated in that meeting.
13. Workshop on Capacity-building on Health Systems Research, held on 17-19 December 2012.
14. Another workshop, in 2 batches each comprising 20 participants, were held during 22-24 December, 2012 at BMRC Bhaban, Mohakhali, Dhaka. Forty participants from different medical colleges, dental colleges, postgraduate institutes, upazila health complexes, and private health facilities attended the workshop.

Research Grants (48 research protocols)

1. Occular Injury: Incidence and prevalence in Rural Bangladesh
2. Comparison of high-dose and low-dose atorvastatin in the reduction of carotid intimal-medial thickness in patients with coronary artery diseases
3. Prevalence of metabolic syndrome and its correlation with cardiovascular disease and diabetes mellitus in an urban community of Bangladesh
4. Biochemical profile of the drug abusers undergoing detoxification (TC, DC, ESR, RBS, and CRP)
5. Influence of perceived service quality and socioeconomic predictors on utilization of community clinic

6. National asthma prevalence study in Bangladesh, 2011
7. Pattern of contact dermatitis among bakery workers in Dhaka city
8. Moving beyond cancer: immediate impact of mastectomy on the quality of life of breast cancer patients
9. Association between periodontal disease and elevated blood glucose level: a case-control study
10. Impact of educational booklet on cardiac rehabilitation and secondary prevention of cardiovascular diseases
11. Man's role to make pregnancy safer to reduce maternal morbidity and mortality in rural Bangladesh
12. Evaluation of iron store by serum ferritin in voluntary blood donors of Bangladesh
13. Hepatitis B virus and sexually transmitted infections (STIs) among injection drug abusers undergoing detoxification
14. Factors affecting postnatal care coverage in rural Bangladesh
15. Factors contributing to poor compliance with anti-TB treatment among tuberculosis patients in rural Bangladesh
16. Incidence and prevalence of spinal cord injury (SCI) and its associated risk factors in Bangladesh
17. Knowledge, attitudes and practices (KAP) of health service providers towards cerebral palsy in Bangladesh
18. Effects of statins, alone or in combination, on lowering serum lipoprotein-A and LDL cholesterol in patient with hyperlipidemia
19. Assess the knowledge and awareness of tribal community regarding existing spread of HIV/AIDS in Bangladesh
20. Policies, practices, and barriers regarding evidence-based delivery and newborn practice in Bangladesh: A qualitative research
21. Appearance of microalbuminuria among type 2 diabetes according to duration of diabetes mellitus
22. Prevalence of dyspepsia and peptic ulcer disease in general population of Bangladesh
23. Prasugrel and clopidogrel-induced platelet inhibition in diabetic and non-diabetic high-risk NSTEMI patients undergoing coronary intervention (PCI: DHAKA)
24. Impact of educational intervention in reducing NCD risk indicators
25. Impact of climate change over priority-based infectious diseases in northeastern part of Bangladesh
26. Serum biomarkers in non-invasive diagnosis of endometriosis
27. Practice of inhaler-use among the chronic asthma patients in Bangladesh
28. Influencing factors of failure of glycemic control among diabetic patients attending the outpatient department of BIRDEM Hospital
29. Determinants of patient-provider interfacing in the community clinics of Bangladesh
30. Risk factors for urinary bladder carcinoma
31. Circadian variation in the onset of acute myocardial infarction and clinical presentations of the patients attending Cardiology Department of Rajshahi Medical College
32. Effect of Metformin and Atorvastatin combination vs Metformin alone on inflammatory biomarkers and insulin resistance in patients with type 2 diabetes
33. Sociodemographic and glycaemic status of type II diabetic patients with their compliance to management protocol in Sirajganj municipality area: a community-based study
34. Comparison of risk factors between pulmonary and extra-pulmonary tuberculosis in rural Bangladesh
35. Fast food-related health awareness of the secondary-level students in Chittagong City Corporation
36. Current status of service centers serving autism spectrum disorder in Dhaka city
37. Oral health status among the patients of major non-communicable diseases attending different hospitals in Dhaka city
38. Emergency dental services in public and private hospitals in Dhaka city
39. Oral health status of pregnant women in some selected Mother and Child Welfare Centers (MCWC) in Bangladesh
40. Role of social leaders in current health service management at primary healthcare level of Bangladesh

41. A study on follicular variant of papillary thyroid carcinoma on the basis of histopathological and immunohistochemical diagnosis
42. Association between clinically-diagnosed lumbar intervertebral disc prolapse and magnetic resonance image findings
43. Seroprevalence of hepatitis B, hepatitis C, and human immunodeficiency virus among multi-transfused thalassaemic children
44. Dental health facilities in upazilla health complex in Bangladesh
45. Assessing the quality of OPD service management of NIDCH
46. Oral and dental health status among the diabetic patients attending a selected hospital in Dhaka city (BIRDEM)
47. Outpatient service management in Laxmipur District Hospital
48. Inpatient service management in Narsingdi District Hospital

Publications in BMRC Bulletin during 2012

Issue 38 (1), April 2012: Original articles

1. CYP3A genotypes in Bangladeshi tuberculosis patients
2. A postmortem study on the volume of the human thyroid gland
3. Elevated serum homocysteine level has a positive correlation with serum cardiac troponin I in patients with acute myocardial infarction
4. Comparison between prostate volume and intravesical prostatic protrusion in detecting bladder outlet obstruction due to benign prostatic hyperplasia
5. Chronic kidney disease-specific cardiovascular risk factors among non-dialytic patients with chronic kidney disease (Stage V) : An experience of a specialized hospital
6. A simple biological marker to differentiate the types of herpes simplex viruses in resource-limited settings
7. What cannot be measured cannot be done; risk factors for childhood tuberculosis: a case-control study
8. Symptomatic overlap in patients with diarrhea-predominant irritable bowel syndrome and microscopic colitis in a subgroup of Bangladeshi population

Letter to the editor

1. Congenital absence of one tonsil without any other congenital abnormality: a rare presentation
2. Back pain: an unusual presentation of Castleman's Diseases
3. Constipation-presenting compliant and clinical marker of Parkinson's disease

Issue 38 (2), August 2012: Original articles

1. Topical tazarotene cream (0.1%) in the treatment of facial acne: an open clinical trial
2. Impaired fasting glucose and impaired glucose tolerance in rural population of Bangladesh
3. Recurrence of cancer cervix in patients treated by radical hysterectomy, followed by adjuvant external beam radiotherapy
4. Study on association of cutaneous tuberculosis with pulmonary tuberculosis
5. Oral health status of disabled children
6. Serum triglyceride level in type 2 diabetes mellitus patients with or without frozen shoulder
7. Preoperative detection of ovarian cancer by color doppler ultrasonography and CA 125

Issue 38 (3), December 2012: Original articles

1. Surgery for primary intracerebral haemorrhage: is it safe and effective?
2. Intraoperative serum parathyroid hormone level is an indicator of hypocalcaemia in total thyroidectomy patients
3. Efficacy of lower cutoff value of serum prostate-specific antigen in diagnosis of prostate cancer
4. Pattern of cutaneous tuberculosis among children and adolescents
5. Community-acquired pneumonia in diabetic and non-diabetic hospitalized patients: presentation, causative pathogens, and outcome
6. Hepatitis B surface antigen-negative but HBV DNA-positive patients in Bangladesh
7. Prevalence and risk factors for gastro-esophageal reflux disease in the northeastern part of Bangladesh
8. Serotype and serovar distribution of *Neisseria gonorrhoeae* isolated from high-risk populations in Bangladesh
9. Effect of short-term recombinant human erythropoietin therapy in the prevention of anemia of prematurity in very low-birthweight neonates

Ethical clearance

1. Evaluation of the effect of bicarbonate buffer on the immune response to oral poliovirus vaccine in Bangladesh: a randomized controlled trial
2. Assessment of patients' costs and patients' perspectives of TB treatment
3. Designing and evaluation of parenting education program targeting fathers regarding holistic childhood development of 0-3 year(s) old children in rural Bangladesh
4. Maternal nutrition and its role in child development in Bangladesh
5. A qualitative study of acceptability of the safety turtle device to prevent drowning
6. Improving effectiveness of the primary healthcare through operationalising community clinics in rural Bangladesh
7. Integration of post-abortion care (PAC), menstrual regulation (MR), and FP services in Bangladesh
8. Laser osteoperforation: a new and minimal invasive modality for the treatment of avascular necrosis of femoral head
9. Policies, practices, and barriers regarding evidence-based delivery and newborn practice in Bangladesh: a qualitative research
10. Obstetric fistula patients in Dhaka, Bangladesh: understanding healthcare-seeking behaviour: a mixed method approach
11. Non-communicable diseases as risk factors for adult pulmonary tuberculosis in rural Bangladesh community
12. Quality assessment in smear sputum microscopy at district level under the National TB Control Programme (NTP) of Bangladesh
13. Risk factors for treatment delay in pulmonary tuberculosis
14. Assessing risk of multidrug-resistant tuberculosis in pulmonary tuberculosis patients, using clinical prediction rule
15. Surveillance of multidrug-resistant *Mycobacterium tuberculosis* infection among HIV/AIDS patients in Bangladesh
16. Stigmas, healthcare-seeking behavior and access to diagnosis and treatment initiation among leprosy patients in an endemic rural area of Bangladesh
17. Early leprosy case detection through extended contact survey
18. Risk factors for extra-pulmonary tuberculosis: a case-control study
19. Epidemiological and clinical attributes of multidrug-resistant tuberculosis in Bangladesh
20. Healthcare-seeking behavior and knowledge about tuberculosis among the patients in a selected rural community
21. Awareness regarding pulmonary tuberculosis and DOTS Program among slum population in city corporations of Bangladesh
22. Efficacy and safety of artemether-lumefantrine for the treatment of uncomplicated *Plasmodium falciparum* malaria in Teknaf upazila of Cox's Bazar district, Ima Upazila of Bandarban district, and Kalmakanda upazila of Netrakona district in Bangladesh
23. Pilot study for dendritic cell-based therapy in patients with hepatocellular carcinoma
24. Retraining the community health workers in rural areas of Bangladesh for providing selected NCD-related care and services
25. Pilot study evaluation of the effect of probiotics supplements on the immune response to rotavirus vaccine (RotaTeq) in Bangladesh: a randomized controlled trial
26. Survey of surgical and anesthesia capacity in Bangladesh
27. Clinical analysis role of Neostigmine and short-term outcome of patients with potka fish (Tetrodotoxin) poisoning
28. A pilot study to test the acceptability of ultra-rice fortified with micronutrients among the consumers in Bangladesh
29. Work stress among readymade garment workers in Gulshan, Dhaka
30. Analysis of the Indications for caesarean section in Thakurgaon district of Bangladesh
31. A randomised, double-blind, double dummy-active comparator-controlled study investigating the efficacy and safety of linagliptin co-administered with metformin QD at evening time versus metformin BID over 14 weeks in treatment-naïve patients with type 2 diabetes mellitus and insufficient glycaemic control
32. Impact assessment of the credit program for the tenant farmers

33. Improving effectiveness of diagnosis and treatment of diabetes, cardiovascular diseases (CVD), and related conditions through primary healthcare in Bangladesh
34. Screening for depression among patients with diabetes melitus in Bangladesh
35. Customer Satisfaction Survey at the National Malaria Control Program
36. Community access to HIV treatment care and support services study (CAT-S) in six Asian countries
37. An open-label phase I placebo-controlled, dose escalation study assessing the ocular and systemic safety and tolerability of OC-10X
38. Comparative bioavailability study of generic Atorvastatin 40 mg tablet, a cholesterol-lowering drug, with the reference product Lipitor 40 mg in healthy human volunteers
39. Bioavailability study of tablet Omeprazole 20 mg and tablet Ketorolac 10 mg in Bangladeshi health human volunteers
40. Reducing exposure of children to second-hand smoking: a pilot study in selected areas in Bangladesh
41. Estimation of nail selenium and arsenic among arsenic-exposed populations in arsenic-affected areas of Bangladesh
42. A phase III study with a novel vaccine containing both hepatitis B surface antigen and hepatitis B core antigen in patients with chronic hepatitis B
43. Bangladesh Prospective Urban and Rural Epidemiological Study
44. The combined effect of chemoprophylaxis with rifampicin and immunoprophylaxis with BCG in the prevention of leprosy in contacts: a randomized controlled trial
45. Comparison of nutritional status and morbidity of children aged 6-12 years between slums of tannery area and other slums and their determinants in Dhaka city
46. Impact of nutrition education on reducing overweight in 6-12 years old children of affluent society in Dhaka city
47. Impact of health and nutrition education on healthy lifestyle, nutritionally-sound eating practice of elderly to improve their food intake, nutritional status and reduce morbidity in urban Dhaka

48. Evaluation of nutritional status of pulmonary tuberculosis patients by giving nutrition education and hygiene behavior
49. A study on impact of nutrition education on pre-school children's nutritional status (1-4 yrs.) of middle- and lower-income families of Dhaka city
50. Impact of nutrition education on exclusively breastfed infants compared to non-exclusively breastfed infants (1-5 month(s)) on growth and illness

International Centre for Diarrhoeal Disease Research, Bangladesh (icddr,b)

Publications 2012

Internal Publication Series

Annual report

1. Annual Report 2011. Cravioto A, Judd G, Ahmed N, Christian J, editors, 2012. 66 p.
2. Financial Statements for the Year Ended December 31, 2011, 2012. 27 p.

Special publications

Anwar ATMI, Rahman A, Ferdous J, Ahmed S, Akhter S, Hossain SAS, Huda FA, Mamun MA, Haque NA, Islam N, Islam SS. Manual for executive training on how to apply research evidence in policy-making for reproductive health in Bangladesh. Dhaka: International Centre for Diarrhoeal Disease Research, Bangladesh. 84 p. (icddr,b special publication no. 137)

Scientific reports

1. Rahman MM, Alam MN, Razzaque A, Streatfield PK. Health and demographic surveillance system: Matlab. V. 44. Registration of health and demographic events 2010, 2012. 75 p. (Scientific report no. 117)
2. Scobie H, Ahmed S, Ram M, Hossain MS, Prue CS, Khyang J, Sullivan D, Streatfield PK, Khan WA, Sack D. Initial demographic surveillance system report for the icddr,b Bandarban field site, 2012. 25 p. (Scientific report no. 119)
3. Khan SI, Reza MM, Gourab G, Hussain MI, Deb PK, Ahmed T, Shahriar A, Alam MN, Alam MS, Rana AKMM, Azim T. Counting the numbers of males who have sex with males, male sex workers, and *Hijra* in Bangladesh to provide HIV preventing services, 2012. 23 p. (Scientific report no. 120)

4. Health and demographic surveillance system: Matlab. V. 45. Registration of health and demographic events 2011, 2012. 71 p. (Scientific report no. 121)

Working paper

Naved RT, Amin S, editors. Growing up safe and healthy (SAFE): baseline report on sexual and reproductive health and rights and violence against women and girls in Dhaka slums, 2012. (icddr,b working paper no. 176)

Manoshi working papers

1. Alam K, Khan J, Walker DG. Impact of dropout of female volunteer community health workers in Dhaka urban slums, 2012. 20 p. (Manoshi working paper no.15)
2. Alam B, Sarker BK, Shafi S, Banu M, Ahmed A, Mridha MK. Assessment of performance of community health workers of Manoshi, 2012. 31 p. (Manoshi working paper no.16)
3. Sarker BK, Mridha MK, Dasgupta SK, Islam N, Reichenbach L. The effect of behavior change communication (BCC) interventions on maternal, neonatal and child health (MNCH) knowledge in urban slums of Bangladesh, 2012. 49 p. (Manoshi working paper no.17)
4. Sarker BK, Higgins J, Malay Mridha MK, Ferdous J, Dasgupta SK, Ahmed S, Khan JAM, Reichenbach L. Caesarean delivery in urban slums of Dhaka city: indications and consequences, 2012. 74 p. (Manoshi working paper no.18)

Monograph

Wahed T, Rasheed S, Bhuiya A, editors. Doctoring the village doctors: giving attention where it is due. Dhaka: International Centre for Diarrhoeal Disease Research, Bangladesh, 2012. 73 p. (icddr,b monograph no. 10)

Journal and newsletters

1. News 'n' Views V. 6, no. 1-26, 2012
2. Glimpse V. 34, no. 1-4, 2012
3. Chronic Disease News V. 4, no. 1-2, 2012
4. Shasthya Sanglap V. 20, no. 3, Chaitra 1418, and V. 21, no. 1-2, Srabon 1419
5. Health and Science Bulletin V. 10, no. 1-4, 2012
6. Journal of Health, Population and Nutrition V. 30, No. 1-4, 2012
7. Shahjadpur Integrated MNH Project Newsletter Issue 7, 2012

8. CRH Knowledge Translation Brief no. 10, 2012

Original papers, including review articles and short reports in journals

1. Ahmad SAI, Bari SMN, Mohiuddin M. Biofilm: multicellular living of the unicellular bacteria. *Int J Biosci* 2012 Jun;2(6):59-71
2. Ahmed A, Reichenbach LJ, Alam N. Symptoms of sexually transmitted infections and care-seeking behaviors of male clients of female sex workers in Bangladesh. *Sex Transm Dis* 2012 Dec;39(12):979-84
3. Ahmed AMS, Ahmed T, Roy SK, Alam N, Hossain MI. Determinants of undernutrition in children under 2 years of age from rural Bangladesh. *Indian Pediatr* 2012 Oct;49(10):821-4
4. Ahmed D, Hoque A, Elahi MS, Endtz HP, Hossain MA. Bacterial aetiology of diarrhoeal diseases and antimicrobial resistance in Dhaka, Bangladesh, 2005-2008. *Epidemiol Infect* 2012 Sep;140(9):1678-84
5. Ahmed S, Ahsan KB, Kippler M, Mily A, Wagatsuma Y, Hoque AMW, Ngom PT, Arifeen SE, Raqib R, Vahter M. *In utero* arsenic exposure is associated with impaired thymic function in newborns possibly via oxidative stress and apoptosis. *Toxicol Sci* 2012 Oct;129(2):305-14
6. Ahmed T, Mahfuz M, Ireen S, Ahmed AMS, Rahman S, Islam MM, Alam N, Hossain MI, Rahman SMM, Ali MM, Choudhury FP, Cravioto A. Nutrition of children and women in Bangladesh: trends and directions for the future. *J Health Popul Nutr* 2012 Mar;30(1):1-11
7. Ahmed T, Michaelsen KF, Frem JC, Tumvine J. Malnutrition report of the FISPUGHAN Working Group. *J Pediatr Gastroenterol Nutr* 2012 Nov;55(5):626-31
8. Akond MA, Shirin M, Alam S, Hassan SMR, Rahman MM, Hoq M. Frequency of drug-resistant *Salmonella* spp. isolated from poultry samples in Bangladesh. *Stamford J Microbiol* 2012 Dec;2(1):15-9
9. Akter J, Thriemer K, Khan WA, Sullivan DJ, Jr., Noedl H, Haque R. Genotyping of *Plasmodium falciparum*, using antigenic polymorphic markers and to study anti-malarial drug resistance markers in malaria-endemic areas of Bangladesh. *Malar J* 2012 Nov 22;11(1):386
10. Akter SM, Roy SK, Thakur SK, Sultana M, Khatun W, Rahman R, Saliheen SS, Alam N. Effects of third trimester counseling on pregnancy weight

- gain, birthweight, and breastfeeding among urban poor women in Bangladesh. *Food Nutr Bull* 2012 Sep;33(3):194-201
11. Alam DS, Chowdhury MAH, Siddiquee AT, Ahmed S, Hossain MD, Pervin S, Streatfield K, Cravioto A, Niessen LW. Adult cardiopulmonary mortality and indoor air pollution: a 10-year retrospective cohort study in a low-income rural setting. *Global Heart* 2012 Sep;7(3):215-21
 12. Alam K, Khan J, Walker DG. Impact of dropout of female volunteer community health workers: an exploration in Dhaka urban slums. *BMC Health Serv Res* 2012 Aug 17;12(1):260
 13. Alam K, Tasneem S, Oliveras E. Performance of female volunteer community health workers in Dhaka urban slums. *Soc Sci Med* 2012 Aug;75(3):511-5
 14. Alam K, Tasneem S, Oliveras E. Retention of female volunteer community health workers in Dhaka urban slums: a case-control study. *Health Policy Plan* 2012 Sep;27(6):477-86
 15. Alam M, Islam MT, Rashed SM, Fatema-tuz Johura, Bhuiyan NA, Delgado G, Morales R, Mendez JL, Navarro A, Watanabe H, Nur-A Hasan, Colwell RR, Cravioto A. *Vibrio cholerae* classical biotype strains reveal distinct signatures in Mexico. *J Clin Microbiol* 2012 Jul;50(7):2212-6
 16. Alam MA, Chowdhury MZ, Ahmed F, Alam A, Hossain MA. Serotype and serovar distribution of *Neisseria gonorrhoeae* isolated from high-risk populations in Bangladesh. *Bangladesh Med Res Counc Bull* 2012 Dec;38(3):114-8
 17. Alam MS, Chakma S, Khan WA, Glass GE, Mohon AN, Elahi R, Norris LC, Podder MP, Ahmed S, Haque R, Sack DA, Sullivan DJ, Norris DE. Diversity of Anopheline species and their *Plasmodium* infection status in rural Bandarban, Bangladesh. *Parasit Vectors* 2012 Jul 27;5:150
 18. Alam MS, Kato H, Fukushige M, Wagatsuma Y, Itoh M. Application of RFLP-PCR-based identification for sandfly surveillance in an area endemic for kala-azar in Mymensingh, Bangladesh. *J Parasitol Res* 2012;2012:467821
 19. Alam N, Lindeboom W, Begum D, Streatfield PK. The association of weather and mortality in Bangladesh from 1983-2009. *Glob Health Action* 2012;5:19121
 20. Ali IK, Haque R, Alam F, Kabir M, Siddique A, Petri WA, Jr. Evidence for a link between locus R-R sequence type and outcome of infection with *Entamoeba histolytica*. *Clin Microbiol Infect* 2012 Jul;18(7):E235-7
 21. Ali IKM, Haque R, Siddique A, Kabir M, Sherman NE, Gray SA, Cangelosi GA, Petri WA, Jr. Proteomic analysis of the cyst stage of *Entamoeba histolytica*. *PLoS Negl Trop Dis* 2012 May;6(5):e1643
 22. Arifeen SE, Mullany LC, Shah R, Mannan I, Rahman SM, Talukder MRR, Begum N, Al-Kabir A, Darmstadt GL, Santosham M, Black RE, Baqui AH. The effect of cord-cleansing with chlorhexidine on neonatal mortality in rural Bangladesh: a community-based, cluster-randomised trial. *The Lancet* 2012 Mar 17;379(9820):1022-8
 23. Arifuzzaman M, Rashu R, Leung DT, Hosen MI, Bhuiyan TR, Bhuiyan MS, Rahman MA, Khanam F, Saha A, Charles RC, LaRocque RC, Weil AA, Clements JD, Holmes RK, Calderwood SB, Harris JB, Ryan ET, Qadri F. Antigen-specific memory T cell responses after vaccination with an oral killed cholera vaccine in Bangladeshi children and comparison with natural cholera. *Clin Vaccine Immunol* 2012 Aug;19(8):1304-11
 24. Ashraf H, Alam NH, Chisti MJ, Mahmud SR, Hossain MI, Ahmed T, Salam MA, Gyr N. A follow-up experience of 6 months after treatment of children with severe acute malnutrition in Dhaka, Bangladesh. *J Trop Pediatr* 2012 Aug;58(4):253-7
 25. Ashraf H, Alam NH, Chisti MJ, Salam MA, Ahmed T, Gyr N. Observational follow-up study following two cohorts of children with severe pneumonia after discharge from day care clinic/hospital in Dhaka, Bangladesh. *BMJ Open* 2012 Jul 27;2(4). pii: e000961. doi: 10.1136/bmjopen-2012-000961
 26. Azziz-Baumgartner E, Alamgir A, Rahman M, Homaira N, Sohel BM, Sharker MAY, Rashid Uz Zaman, Dee J, Gurley ES, Al Mamun A, Mah-E-Muneer S, Fry AM, Widdowson MA, Bresee J, Lindstrom S, Azim T, Brooks A, Podder G, Hossain MJ, Rahman M, Luby SP. Incidence of influenza-like illness and severe acute respiratory infection during three influenza seasons in Bangladesh, 2008-2010. *Bull World Health Organ* 2012 Jan 1;90(1):12-9
 27. Azziz-Baumgartner E, Dao C, Nasreen S, Bhuiyan MU, Mah-E-Muneer S, Mamun AA, Sharker MAY, Zaman RU, Cheng P-Y, Klimov AI, Widdowson M-A, Uyeki T, Luby SP, Mounts A, Bresee J. Seasonality, timing, and climate drivers of influenza activity worldwide. *J Infect Dis* 2012 Sep 15;206(6):838-46

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James P Grant School of Public Health (JPGSPH) (2006-2012)

Research/tool/monograph

1. Building and Strengthening the Technical and Institutional Capacities of Government, NGOs, and Implementing Partners (GF 913 Package). Dr. Timothy Evans, Ismat Bhuiya, Asiful Haider Choudhury, Maliha Bassam, Gayatri Choudhury, and J.N. Zahiduzzaman. Partners:, JPGSPH, BRAC University (As lead agency), Population Council, NIPSOM PIAC Bangladesh, and CWFD
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7. Review of Maternal, Neonatal and Child Health Projects led by Dr. Malabika Sarker; this project brings together a team of researchers and experts from JPGSPH, AMDD (Columbia University), and icddr,b. UNICEF. December 2011-August 2012
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10. Updating Frequently Asked Questions Databank on Adolescent Sexual and Reproductive Health in Bangladesh. Dr. Quamrun Nahar (icddr,b), Ismat Bhuiya, Asiful Haider Chowdhury, Fariha Haseen (icddr,b). Icddr,b. USAID. April 2012 – March 2013
11. Governance for Urban Health Equity: Mobilizing Demand for Primary Healthcare among the Urban Poor in Bangladesh. Dr. Sabina F. Rashid, Dr. Alayne Adams, and Anuradha Hashemi. The International Development Research Centre. May 01, 2012- 2015

12. Exploring the Role of Community in Preventing Early Marriage and Early Childbearing: An Exploratory Study. Dr. Quamrun Nahar (icddr,b), Ismat Bhuiya, Asiful Haider Chowdhury, Maliha Bassam. USAID. June 2012 - May 2013

Publications

1. Feeding Practices of Under-5 Children in a Semi-Urban Village in Bangladesh. A.Z. Moe, F. Sharmin, and S. Zaman
2. Perceptions of Mental Illness among Adolescents in Bagnibari Village. E. Noykhovich, P. Bulage and Md. D. Hossain
3. Perceptions of Occupational Health Risks, Illnesses and Coping Strategies among Workers in a Factory in Savar, Bangladesh. M. Iacobelli, M. Chowdhury, and T.K. Nagbe
4. Perceptions and Practices of Food Consumption of Adult Women in Bagnibari Village, Bangladesh. S. Chaudhury and S. Yeasmin
5. Pharmaceutical Use for Common Cough and Cold in Under-Five Children. J.A. Purser and K.I. Sanin
6. Utilization of a Local Government Health Center: Community and Provider Perceptions from a Bangladeshi Village. S.A. Zahan, M.R. Gyawali, and C. Vernon
7. Tobacco and Substance Use: Perceptions and Practices among Men in Bagnibari, Bangladesh, K.A. Morris and A. Aftab
8. Traditional Healing Beliefs and Practices in Bagnibari and Samair Villages: Implications for Public Health in Rural Bangladesh. L.A. Camp, J. Pervin, and F. Raza
9. Use of Contraceptive Methods among Married Women of Bagnibari Village: Perceptions, Preferences and Behaviors. K. Parvin, K. Thaemert, and S. Turkmani

Institute of Epidemiology, Disease Control and Research (IEDCR) (2012)

Research

1. Assessment of the impact of Hepatitis B vaccination in Bangladesh: a sero-prevalence study; Principal Investigator: Professor Mahmudur Rahman; Co-Investigators: Dr. M. Mushtuq Husain *et al.*
2. Safety and efficacy of Liposomal Amphotericin B (Ambisome) in Bangladeshi patients with visceral

leishmaniasis—A phase III clinical trial; Principal Investigator: Professor Mahmudur Rahman

3. HIV, syphilis, and hepatitis among pregnant women in selected health facilities of Greater Sylhet area of Bangladesh; Principal Investigator: Professor Mahmudur Rahman; Co-Investigators: Dr. M. Mushtuq Husain, Dr. A.S.M. Alamgir, Dr. Selina Khatun, Dr. Farhana Haque *et al.*
4. Assessment of the vulnerability of population and the health system in Bangladesh to the impact of climate change; Principle Investigator: Professor Mahmudur Rahman; Co-Investigators: Dr. M. Mushtuq Husain, Dr. Farhana Haque *et al.*
5. Effectiveness of Ghagra shak poisoning prevention campaign in Sylhet: A post-test only intervention control study; Principle Investigator: Dr. Farhana Haque; Co-Principal Investigator(s): Dr. Stephen P. Luby; Co-Investigators: Emily S. Gurley, Saiful Islam, Md. Kamal Hossain, Dr. M. Mushtuq Husain, Professor Mahmudur Rahman, Dr. Selina Khatun
6. Mitigating the impact of climate change to reduce the burden of climate-sensitive illnesses through strengthening health systems, collaborative networking and enhanced disease surveillance in Bangladesh (2012-2014); Principal Investigator: Professor Mahmudur Rahman; Co-Principal Investigator: Dr. Farhana Haque; Co-Investigators: Dr. M. Mushtuq Husain, Dr. Sabbir Haider, Dr. A.S.M. Alamgir, Dr. A.K.M. Samsuzzaman, Dr. Khaleda Islam, Dr. Selina Khatun, Dr. Mujadded Ahmed, Dr. Imtiaz Ashraf Chowdhury, Dr. Shamim Jubayer, Dr. Raihan Sharif, Nuzhat Nasreen Banu, Dr. Md. Waliur Rahman
7. Assessing prevalence and risk factors of mild/asymptomatic H5N1 infections among persons exposed to H5N1 infected poultry; Principal Investigator: Professor Mahmudur Rahman; Co-Investigators: Dr. A.S.M. Alamgir *et al.*
8. Estimate the risk of mild human infection among persons exposed to H5N1-infected poultry; Principal Investigator: Professor Mahmudur Rahman; Co-Investigators: Dr. A.S.M. Alamgir *et al.*
9. Preparedness of primary healthcare to respond to climate-sensitive infectious disease outbreaks in flood-prone areas of Bangladesh (2012); Principal Investigator: Dr. Farhana Haque; Co-investigators: Professor Mahmudur Rahman, Dr. M. Mushtuq Husain, Dr. M. Sabbir Haider, Dr. A.S.M. Alamgir, Dr. A.K.M. Samsuzzaman, Dr.

Khaleda Islam, Dr. Selina Khatun, Dr. Mujadded Ahmed, Dr. Imtiaz Ashraf Chowdhury, Dr. Shamim Jubayer, Dr. Raihan Sharif, Nuzhat Nasreen Banu, Dr. Md. Waliur Rahman.

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National Institute of Cardiovascular Diseases (NICVD)

Research

1. Reduction of peri-procedural myocardial injury by bolus dose of atorvastatin before PCI
2. Impact of blood glucose status at admission on clinical outcome of thrombolysed patients with ST-segment elevation in AMI.
3. Study of risk factors and in hospital outcome AMI in young adults
4. Angiographic correlation of quantitative ST segment depression in 12 lead ECG in non-ST segment elevation ACS
5. Association of hypertension with development of coronary collaterals in server CAD
6. Angiographic severity of acute STEMI patients with or without metabolic syndrome
7. Clinical and echo evaluation of inferior MI with or without RV infarction
8. Correlation of magnitude of ST segment elevation in acute inferior MI with the proximity of RCA lesion
9. Echo evaluation of LV diastolic dysfunction after PCI
10. Angiographic status of prolonged QRS duration in patients with acute MI

11. Assessment of ventricular dyssynchrony in heart failure with normal QRS complex
12. Effect of statin therapy in ventricular arrhythmia in patients with acute anterior MI
13. Effect of PTMC on pulmonary function
14. Coronary angiographic correlation with troponin-I level in acute STEMI
15. Impact of blood glucose level on contrast induced nephropathy after PCI in patients not known to be diabetic with ACS
16. Increased arterial stiffness correlates with the severity of coronary artery disease
17. In-hospital outcome of comptonised small side branch after percutaneous coronary intervention
18. Risk factor profile of pre-menopausal woman with acute coronary syndrome
19. Correlation between elevated B type natriuretic peptide levels with extent of coronary artery disease in patient with unstable angina
20. Comparison of in hospital outcome between transradial and transfemoral PCI
21. Correlation of cardiac troponin I with left ventricular systolic function in a patient with acute ST segment elevated myocardial infarction
22. Aortic pulse wave velocity correlates with the severity of coronary artery disease in patient with STEMI
23. Association of atherosclerotic renal artery stenosis with coronary artery disease
24. Association of severity of CAD with mitral annular calcification under 65 years of age
25. Association of left atrial volume index and in hospital outcome in patients with anterior myocardial infarction
26. Association of hyperglycemia with angiographic severity of coronary artery disease in patients with acute coronary syndrome
27. Effect of tranexamic acid on post-operative bleeding following myocardial revascularization (CABG) using cardiopulmonary bypass
28. Early outcome of OPCAB in patients with multivessel diseases
29. Effect of peri-operative oral sildenafil on pulmonary hypertension after congenital heart surgery
30. In-hospital outcome after primary total correction of tetralogy of fallot in different age-groups
31. Early outcome of prolonged cardiopulmonary bypass time on renal function in patients of open heart surgery at NICVD, Dhaka
32. Early outcome of CABG surgery in patients with pre-operative elevated level of HbA1c with undiagnosed DM
33. Early outcome of OPCAB in patients with multivessel disease with or without prior myocardial infarction
34. Pre-operative high-sensitive C reactive protein level predicts early outcome after coronary artery bypass graft surgery
35. Evaluation of pulmonary function between off-pump and on-pump CABG surgery
36. Evaluation of myocardial preservation following ante-grade and retrograde cardioplegia in AVR
37. Early outcome of mitral valve replacement in patients having mitral stenosis with severe pulmonary hypertension
38. Early outcome of peri-operative hyperglycemia in patients undergoing CABG with or without DM
39. Immediate post-operative outcome in double valve replacement patients with or without preservation of mitral subvalular structures
40. Atorvastatin pre-treatment diminishes the myocardial ischemic markers early after CABG operation
41. Early outcome of VSD with pulmonary hypertension
42. Association of HbA1C with severity of CAD in non-diabetic patients with NSTEMI
43. In-hospital outcome and angiographic finding in acute inferior MI with STEMI in posterior chest leads (V7 ,V8, V9) following thrombolysis
44. Effect of PCI on QT dispersion in patient with angina
45. Association of aortic pulse-wave velocity with the severity of CAD in patient with NSTEMI
46. Association of left atrial spontaneous echo-contrast to inflammatory markers in patients with MS
47. Pre-procedural high-sensitivity CRP is a predictor of in-hospital outcome of patients undergoing PCI
48. Association of renal impairment and severity of CAD

49. Comparison of CAD severity between pre- and post-menopausal women in ACS
50. Association of GRACE risk score with angiographic severity of coronary artery diseases in patient with AMI
51. Association of p-wave dispersion with LV-diastolic dysfunction in patients with IHD
52. Early outcome of conventional CABG with pre-operative hyperglycemia in diabetic and non-diabetic patients in NICVD
53. Early outcome of mitral valve replacement in patients having mitral stenosis, with moderate to severe pulmonary hypertension
54. Evaluation of myocardial preservative following AVR with antigrade verves both anigrade and retrograde cardioplegia
55. Comparison of early surgical outcome of VSD with mild versus moderate to severe pulmonary hypertension
56. Effect of preoperative high-dose Atorvastation on the level of myocardial ischemia marker (Troponin-I) early after CABG

Institute of Child & Mother Health (ICMH)

Studies

1. Allyestrenol versus 17 Alpha hydroxyprogesterone caproate for prevention of preterm birth. Dr. Dilruba Akhter
2. Survey on selected health parameters in the two villages at Jalkuri area of the Institute of Child & Mother Health (ICMH). Prof. Md. Abidul Haque
3. Risk factors of autism: a cross-sectional study. Prof. SM Shahnawaz Bin Tabib
4. Analysis of predisposing factors and outcome of preterm birth. Prof. Saria Tasnim
5. Montelukast versus inhaled corticosteroids in the management of mild persistent asthma in children. Dr. Md. Delwar Hossian
6. Improvement of complementary feeding (CF) practices by strengthening the peer mothers group counselling in the community. Dr. Wahida Khanam
7. Management of severe acute malnutrition (SAM) among under-five children according to national guideline and its outcome, ICMH. Dr. Tariq Hassan
8. Maternal stress in adolescent pregnancy and fetal outcome. Dr. Shahin Akter
9. Factors contributing to caesarean section at the Institute of Child & Mother Health (ICMH). Prof. Md. Abidul Haque
10. Medico-social dimensions of menopause in women attending a periurban hospital. Prof. Saria Tasnim
11. Evaluation of indications of caesarean section in primigravida. Dr. Shireen Akter
12. Perinatal outcome in fetal mal presentation at term. Dr. Farhana Sultana
13. Prevalence of iron-deficiency anemia in normal pregnancy. Dr. Jeasmin Sultana
14. Outcome of induction of labour in postdated pregnancy. Dr. Bushrah Yasmin
15. Comparative study between sonographic estimation of foetal weight in relation to femoral length at the term and actual birthweight in ICMH in 100 patients. Dr. Nurun Nahar Zannat
16. Maternal and perinatal outcome of adolescent pregnancy from viable age to term. Dr. Tahmina Khan
17. Serum albumin, creatinin and uric acid levels in pregnancy and its association with pre-eclampsia. Dr. Joyanta Kumar Das
18. Colposcopic evaluation of cervix with persistent inflammatory PAP smear. Dr. Kamrun Nahar
19. Abdominal hysterectomy vs vaginal hysterectomy in patients with benign non-descent uterus. Dr. Tanzia Akhter
20. Use of rectal misoprostol in the prevention of postpartum haemorrhage following caesarian section. Dr. Samina Akter
21. Routine partographic assessment of progress of labour and pregnancy outcome. Dr. Mariam Akter

Annex to Chapter 16

Division-wise distribution of sanctioned, filled-up and vacant posts under the DGHS (December 2012)

Division	Class		Sanctioned	Filled-up				Vacant	
				Male	Female	Total	Filled-up as % of sanctioned posts	No.	Vacant as % of sanctioned posts
Barisal	Class I	Doctors	1579	812	150	962	61%	617	39%
		Non-doctors	25	3	0	3	12%	22	88%
	Class II		1098	53	902	955	87%	143	13%
	Class III		4105	2324	1124	3448	84%	657	16%
	Class IV		1830	1146	331	1477	81%	353	19%
	Total		8637	4338	2507	6845	79%	1792	21%
Chittagong	Class I	Doctors	3745	1980	530	2510	67%	1235	33%
		Non-doctors	56	13	0	13	23%	43	77%
	Class II		2358	142	1630	1772	75%	586	25%
	Class III		9948	6179	1923	8102	81%	1846	19%
	Class IV		4213	2477	746	3223	77%	990	23%
	Total		20320	10791	4829	15620	77%	4700	23%
Dhaka	Class I	Doctors	8203	5066	1792	6858	84%	1345	16%
		Non-doctors	253	101	41	142	56%	111	44%
	Class II		5852	479	4556	5035	86%	817	14%
	Class III		15946	10160	3659	13819	87%	2127	13%
	Class IV		9674	5709	2155	7864	81%	1810	19%
	Total		39928	21515	12203	33718	84%	6210	16%
Khulna	Class I	Doctors	2250	1128	240	1368	61%	882	39%
		Non-doctors	44	18	5	23	52%	21	48%
	Class II		1550	35	1313	1348	87%	202	13%
	Class III		5904	3605	1331	4936	84%	968	16%
	Class IV		2446	1342	558	1900	78%	546	22%
	Total		12194	6128	3447	9575	79%	2619	21%
Rajshahi	Class I	Doctors	2695	1421	419	1840	68%	855	32%
		Non-doctors	53	19	5	24	45%	29	55%
	Class II		2123	140	1569	1709	80%	414	20%
	Class III		6793	4782	1265	6047	89%	746	11%
	Class IV		3692	2174	799	2973	81%	719	19%
	Total		15356	8536	4057	12593	82%	2763	18%
Rangpur	Class I	Doctors	2188	1126	275	1401	64%	787	36%
		Non-doctors	36	7	2	9	25%	27	75%
	Class II		1446	74	952	1026	71%	420	29%
	Class III		5551	3825	846	4671	84%	880	16%
	Class IV		2394	1430	475	1905	80%	489	20%
	Total		11615	6462	2550	9012	78%	2603	22%

Table continued

Division	Class		Sanctioned	Filled-up				Vacant	
				Male	Female	Total	Filled-up as % of sanctioned posts	No.	Vacant as % of sanctioned posts
Sylhet	Class I	Doctors	1460	823	160	983	67%	477	33%
		Non-doctors	23	2	0	2	9%	21	91%
	Class II		994	45	521	566	57%	428	43%
	Class III		3631	2277	660	2937	81%	694	19%
	Class IV		1777	1080	386	1466	82%	311	18%
	Total		7885	4227	1727	5954	76%	1931	24%
	Grand total		115935	61997	31320	93317	80%	22618	20%
Seven divisions	Class I	Doctors	22120	12356	3566	15922	72%	6198	28%
		Non-doctors	490	163	53	216	44%	274	56%
	Class II		15421	968	11443	12411	80%	3010	20%
	Class III		51878	33152	10808	43960	85%	7918	15%
	Class IV		26026	15358	5450	20808	80%	5218	20%
	Total		115935	61997	31320	93317	80%	22618	20%
	Grand total		115935	61997	31320	93317	80%	22618	20%

Institutions offering postgraduate medical courses and titles of courses, with the number of seats in each course (December 2012)

Name of institution	MS	MD	M. Phil	Diploma	MPH	MTM	MMED	Total
Government (autonomous) (No. of institutions: 1)								
Bangabandhu Sheikh Mujib Medical University, Shahbag, Dhaka	140	150	70	106	0	10	0	476
Total	140	150	70	106	0	10	0	476
Government (No. of institutions: 22)								
Centre for Medical Education (CME), Mohakhali, Dhaka	0	0	0	0	0	0	15	15
Chittagong Medical College, Chittagong	37	48	29	48	03	0	0	165
Dhaka Dental College, Mirpur 14, Dhaka	22	0	0	0	0	0	0	22
Dhaka Medical College, Dhaka	70	110	86	82	06	0	0	354
Institute of Child & Mother Health (ICMH), Matuail, Dhaka	10	10	0	30	0	0	0	50
Institute of Nuclear Medicine and Ultrasound, Block D, BSMMU Campus, Shahbag, Dhaka	0	0	0	10	0	0	0	10
Mymensingh Medical College, Mymensingh	22	40	33	59	0	0	0	154
National Institute of Cancer Research and Hospital, Mohakhali, Dhaka	06	12	0	0	0	0	0	18
National Institute of Cardiovascular Diseases (NICVD), Sher-e-Bangla Nagar, Dhaka	20	20	0	14	0	0	0	54
National Institute of Chest Diseases and Hospital (NIDCH), Mohakhali, Dhaka	06	15	0	20	0	0	0	41
National Institute of Child Health, Sher-e-Bangla Nagar, Dhaka	10	15	0	15	0	0	0	40
National Institute of Kidney Diseases and Urology (NIKDU), Sher-e-Bangla Nagar, Dhaka	06	09	0	0	0	0	0	15
National Institute of Mental Health, Sher-e-Bangla Nagar, Dhaka	0	06	0	0	0	0	0	06
National Institute of Ophthalmology, Sher-e-Bangla Nagar, Dhaka	10	0	0	10	0	0	0	20
National Institute of Preventive and Social Medicine (NIPSOM), Mohakhali, Dhaka	0	0	07	0	166	0	0	173
National Institute of Traumatology and Orthopedic Rehabilitation (NITOR), Sher-e-Bangla Nagar, Dhaka	30	0	0	15	0	0	0	45
Rajshahi Medical College, Rajshahi	10	19	25	41	05	0	0	100

Table continued

Name of institution	MS	MD	M. Phil	Diploma	MPH	MTM	MMED	Total
Rangpur Medical College, Rangpur	08	08	08	22	0	0	0	46
SZR Medical College, Bogra	0	0	0	10	0	0	0	10
Sher-e-Bangla Medical College, Barisal	04	0	08	22	0	0	0	34
Sir Salimullah Medical College, Dhaka	21	36	18	40	05	0	0	120
M.A.G Osmani Medical College, Sylhet	20	12	28	40	0	0	0	100
Total	312	360	242	478	185	0	15	1592
Private (No. of institutions: 10)								
Bangladesh College of Physicians and Surgeons, Mohakhali, Dhaka*	-	-	-	-	-	-	-	-
Bangladesh Institute of Research and Rehabilitation in Diabetes, Endocrine and Metabolic Disorders (BIRDEM), Shahbag, Dhaka	10	22	15	14	0	0	0	61
Chittagong Maa O Shishu & General Hospital. Agrabad, Chittagong	0	0	0	06	0	0	0	06
Institute of Child Health and Shishu (Children) Hospital, Shishu Shasthya Foundation, Bangladesh, Mirpur 2, Dhaka	0	0	0	06	0	0	0	06
Institute of Community Ophthalmology, Chittagong	0	0	0	08	0	0	0	08
Institute of Health Sciences (under USTC), Foy's Lake, Chittagong	0	05	0	45	0	0	0	50
Lions Eye Institute and Hospital, Lions Bhaban, Rokeya Sarani, Agargaon, Dhaka	0	0	0	06	0	0	0	06
Mirza Ahmed Ispahani Institute of Ophthalmology and Islamia Hospital, Sher-e-Bangla Nagar, Dhaka	0	0	0	10	0	0	0	10
National Heart Foundation, Mirpur 2, Dhaka	05	05	0	0	0	0	0	10
United Hospital Ltd. Gulshan 2, Dhaka	06	06	0	0	0	0	0	12
Total	21	38	15	95	0	0	0	169
Grand total (No. of institutions: 33)								
Grand total (no. of seats)	473	473	327	327	327	25		2237

*Offers FCPS and MCPS courses. Number of seats are not fixed and not included in this count

Number of fellowships and memberships offered by Bangladesh College of Physicians and Surgeons in various disciplines from 2007 to 2012

Discipline	FCPS						MCPS						
	2007	2008	2009	2010	2011	Total	2007	2008	2009	2010	2011	2012	2012
Anesthesiology	9	6	7	11	12	45	7	10	10	7	8	10	52
Biochemistry	1	0	0	0	1	2	0	0	0	0	0	0	0
Cardiology	2	1	0	3	3	9	0	0	0	0	0	0	0
Clinical Pathology	0	0	0	0	0	0	5	3	4	2	4	2	20
Conservative Dentistry	0	0	0	3	3	6	0	0	0	0	0	0	0
Dental Surgery	0	0	0	0	0	0	4	5	2	6	2	3	22
Dermatology & Venereology	3	6	3	5	7	24	2	4	10	8	6	1	31
Family Medicine	0	0	1	0	0	1	2	1	1	1	2	0	7
Forensic Medicine	0	0	0	0	0	0	0	1	1	0	6	3	11
Gastroenterology	0	1	0	0	0	1	0	0	0	3	0	0	3
Hematology	1	3	3	4	2	13	0	0	0	0	0	0	0
Histopathology	1	1	1	42	3	48	0	0	0	0	0	0	0
Medicine	13	57	57	34	66	227	3	12	19	13	11	12	70
Microbiology	1	2	1	2	1	7	0	0	0	0	0	0	0
Neonatology	1	0	1	0	0	2	0	0	0	0	0	0	0
Obstetrics & Gynecology	52	46	67	99	85	349	40	32	40	40	45	39	236

Table continued

Discipline	FCPS						MCPS					
	2007	2008	2009	2010	2011	Total	2007	2008	2009	2010	2011	2012
Ophthalmology	23	11	15	12	20	81	4	7	14	5	11	11
Oral & Maxilla-Facial Surgery	3	0	3	5	3	14	0	0	0	0	0	0
Orthodontics & Dentofacial Orthopedic	1	2	3	5	9	20	0	0	0	0	0	0
Otolaryngology	9	14	11	24	20	78	2	1	3	5	9	6
Pediatric Surgery	0	1	0	0	1	2				7	0	0
Pediatrics	16	33	26	25	33	130	1	5	6	0	2	7
Physical Medicine & Rehabilitation	1	2	4	10	10	27	0	0	0	0	0	0
Prosthodontics	0	1	0	1	0	2	0	0	0	0	0	0
Psychiatry	3	1	3	5	4	16	4	2	1	2	4	4
Pulmonology	0	0	0	0	0	0	0	0	0	0	0	0
Radiology & Imaging	4	4	9	2	2	21	1	6	2	2	1	3
Radiotherapy	3	0	0	0	4	7	0	0	0	0	1	2
Surgery	25	23	22	35	29	134	4	4	12	7	6	5
Thoracic Surgery	0	0	1	0	0	1	0	0	0	0	0	0
Urology	0	1	1	1	1	4	0	0	0	0	0	0
Plastic and Reconstructive Surgery	0	0	0	0	1	1	0	0	0	0	0	0
Total	172	216	239	288	320	1235	79	93	125	108	118	108

Government institutions offering MBBS degrees, with number of seats (December 2012)

Serial No	Name of college	Year of establishment	No. of seats as of 2012 (+10% of 2012)
1	Dhaka Medical College, Dhaka	1948	197
2	Mymensing Medical College, Mymensing	1962	197
3	Chittagong Medical College, Chittagong	1962	197
4	Rajshahi Medical College, Rajshahi	1962	197
5	M.A.G. Osmani Medical College, Sylhet	1966	197
6	Sher-e-Bangla Medical College, Barisal	1968	197
7	Rangpur Medical College, Rangpur	1972	197
8	Sir Salimullah Medical College, Mitford, Dhaka	1972	197
9	Comilla Medical College, Comilla	1992	114
10	Khulna Medical College, Khulna	1992	142
11	Shaheed Ziaur Rahman Medical College, Bogra	1992	142
12	Faridpur Medical College, Faridpur	1992	114
13	Dinajpur Medical College, Dinajpur.	1992	142
14	Shaheed Suhrawardy Medical College, Sher-e-Bagla Nagar, Dhaka	2005	142
15	Pabna Medical College, Pabna.	2008	58
16	Noakhali Medical College, Noakhali.	2008	58
17	Cox's Bazar Medical College, Cox's Bazar	2008	58
18	Jessore Medical College, Jessore	2010	58
19.	Satkhira Medical College, Satkhira	2011	52
20	Shohid Syed Nazrul Islam Medical College Kishoreganj	2011	52
21	Kushtia Medical College. Kushtia	2011	52
22	Shaikh Shahera Khatun Medical College. Gopalganj	2011	52
	Total		2812

Private institutions offering MBBS degree, with the number of seats (December 2012)

Serial No	Code no.	Name of college	No. of seats	Year of establishment
01	41	Bangladesh Medical College, Road # 14/A, Dhanmondi, Dhaka	110	1985
02	42	SamajVittic Medical College, Mirza Nagar, Via Savar Cant., Dhaka	100	1989
03	43	Institute of Applied Health Sciences, Foy's lake, Chittagong	200	1990
04	44	Jahurul Islam Medical College, Bajitpur, Kishoreganj	100	1992
05	45	Medical College for Women & Hop, Rd # 8-9 Set-1, Uttara Model Town, Dhaka	80	1992
06	46	Z.H. Sikder, Women Medical College, Monica Estate, Western Dhanmondi, Dhaka	90	1992
07	47	Dhaka National Medical College, 53/1 Jonson Road, Dhaka	125	1995
08	48	Community Based Medical College, 161 K.B.Ismail Road, Mymensingh	120	1995
09	49	Jalalabad Ragib-Rabeya Medical College, Pathantola, Sylhet	160	1996
10	50	Shaheed Monsur Ali Medical College, Plot # 26, Rd# 10, St-11, Uttara, Dhaka	95	1998
11	51	North East Medical College, South Surma, Sylhet	115	1998
12	52	Holy Family Red Cressent Medical College, 1 Eskaton Garden Road, Dhaka	120	2000
13	53	International Medical College, Sataish Bazar, Gushuli, Tongi, Gazipur	110	2000
14	54	North Bengal Medical College, JC Road, Dhanbandi, Sirajganj	60	2000
15	55	East West Medical College, Aichi Nagar, JBCS Sarani, Horirampur Turag, Dhaka	100	2000
16	56	Kumudini Medical College, Mirzapur, Tangail	110	2001
17	57	Tairunnessa Medical College, Targas, Kunia, Board Bazar, Gazipur	85	2001
18	58	Ibrahim Medical College, Ibrahim Sarani, Segun Bagicha, Dhaka	110	2002
19	59	BGC Trust Medical College, Kanchannagar, Chandanaish, Chittagong	100	2002
20	60	Shahabuddin Medical College, Rd. # 113/A, Plot # 12, Gulshan Model Town, Dhaka	85	2003
21	61	Enam Medical College, Parbatinagar, Thana Road, Savar, Dhaka	120	2003
22	62	Islami Bank Medical College, Nowdapara, Safura, Airport Road, Rajshahi	75	2004
23	63	IBN Sina Medical College, H # 48, Rd # 9/A, Satmoshjid Rd., Dhanmondi, Dhaka	65	2005
24	64	Central Medical College, Comilla Tower, Laksham Road, Comilla	65	2005
25	65	Eastern Medical College, Race Course, Comilla	85	2005
26	66	Khawja Eunus Medical College, Enayetpur, Sirajganj	95	2005
27	67	Chottogram Ma O Shishu Medical College, Agrabad, Chottogram 4100	90	2006
28	68	Sylhet Women Medical College, Mirbox Tolla, Sylhet	90	2006
29	69	Nightingale Medical College, Ashulia, Sarker Market, Dhaka	50	2006
30	70	Southern Medical College, Mozaffor Ahmed Chy Rd., East Nasirabad, Chittagong	70	2006
31	71	Northern International Medical College, House # 81, Rd. # 7, Dhanmondi, Dhaka	60	2006
32	72	Uttara Adhunik Medical College, Uttara, Dhaka	80	2007
33	73	Delta Medical College, Mirpur, Dhaka	75	2008
34	74	Addin Women Medical College, 2- Boro Mogbazar, Dhaka	75	2008
35	75	Dhaka Community Medical College, 190 Boro Mogbazar, Dhaka	90	2008
36	76	TMSS Medical College. Bogra	100	2008
37	77	Anwar Khan Modern Medical College, Dhanmandi, Dhaka	90	2008
38	78	Prime Medical College, Pirjabad, Rangpur	120	2008
39	79	Rangpur Community Hospital Medical College, Medical East Gate, Rangpur	120	2008
40	80	Northern Private Medical College, Dhap, Chiklibata Burirhat Road, Rangpur	70	2006
41	81	Faridpur Diabetic Association Medical College, Ziltuli, Faridpur	60	2010
42	82	Green Life Medical College, Dhanmondi, Dhaka	90	2010
43	83	Popular Medical College, Road # 02, House # 25, Dhanmondi, Dhaka	75	2010
44	84	MH Shamaria Medical College, 13/A & 89/1 PanthaPath, Dhaka 1215	75	2011

Table continued

Serial No	Code no.	Name of college	No. of seats	Year of establishment
45	85	Moonno Medical College, Manikganj	70	2011
46	86	Central International Medical College, 2/1 Ring Road, Shyamoli, Dhaka	70	2011
47	87	Dr. Sirajul Islam Medical College, Mogbazar, Dhaka	70	2011
48	88	Marks Medical College, Mirpur, Dhaka	50	2011
49	89	Moinamoti Medical College, Baropara, Comilla	70	2012
50	90	Ad-din-Sakina Medical College, 15, Rail Road, Jessore	50	2012
51	91	Gazi Medical College, Sonadanga, Khulna	75	2012
52	92	Barind Medical College, Shershah Road, Laksumpur, Rajshahi	65	2012
53	93	City Medical College, Eta Hata, Block-B, Tangail Road, Gazipur	70	2012

Government institutions offering BDS degree, with the number of seats (December 2012)

Serial No	Name of college	Established	Seats
01	Dhaka Dental College, Mirpur14, Dhaka	1960	97
02	Chittagong Medical College Dental Unit, Chittagong	1990	60
03	Rajshahi Medical College Dental Unit, Rajshahi	1989	59
05	Shahid Suhrawardhy Medical College Dental Unit, Dhaka	2012	56
04	Sir Salimullah Medical College Dental Unit, Dhaka	2012	52
06	Mymensingh Medical College Dental Unit, Mymensingh	2012	52
07	M.A.G. Osmani Medical College Dental Unit, Sylhet	2012	52
08	Sher-e-Bangla Medical College Dental Unit, Barishal	2012	52
09	Rangpur Medical College Dental Unit, Rangpur	2012	52
	Total		532

Private institutions offering BDS degree, with the number of seats (December 2012)

Serial No	Code no.	Name of college	No. of seats	Year of establishment
01	21	Pioneer Dental College, 111, Malibag, DIT Road, Dhaka	90	1995
02	22	City Dental College, 1085/1 Malibag Chowdhury Para, Dhaka	75	1998
03	23	University Dental College, 120 Siddeswari Outer Circular Road, Century Arcade, Mogbazar, Dhaka	85	1996
04	24	Bangladesh Dental College, Road # 14/A, Dhanmondi, Dhaka	70	1997
05	25	Sapporo Dental College, Plot-12, Road-1/B, Sector-9, Uttara Model Town, Dhaka	80	2000
06	26	Rangpur Dental College, Medical East gate, Rangpur	100	2008
07	27	Chittagong International Dental College, 206/1, Hazi Chandmia Road, Samsherpara, Chandgaon, Chittagong	50	2005
08	28	SamajVittik Dental College, Miza Nagar, Via Savar Cant, Dhaka	50	1997
09	29	Marks Dental College, A/3 Main Road, Section 14, Mirpur, Dhaka	50	2008
10	30	Update Dental College, 162, Atish Dipankar Road, West Mugdha, Dhaka	70	2008
11	31	Udayan Dental College, Rajshahi	50	2008
12	32	Shaphena Dental College, Boro Mogbazar, Dhaka	75	2010
13	33	Mandi Dental College, 295/jha/14 Sikdar Real Estate, Dhanmondi (West), PS-Hazaribag, Dhaka 1209	55	2010

Table continued

Serial No	Code no.	Name of college	No. of seats	Year of establishment
14	34	MH Shamarita Medical College Dental Unit. 13/A & 89/1 PanthaPath, Dhaka1215	50	2010
15	35	Kumudini Medical College Dental Unit, Mirzapur, Tangail	25	2011
16	36	Holy Family Red Crescent Medical College, 1 Eskaton Garden Road, Dhaka	30	2012
17	36	TMSS-Bogra Medical College Dental Unit, Bogra	30	2011
18	37	Community Medical College Dental Unit, 190 Boro Mogbazar, Dhaka	30	2012
		Total	1065	

Government nursing colleges offering four-year Basic BSc nursing course (December 2012)

Division	Name of nursing college	Degree	No. of seats
Under the Ministry of Health and Family Welfare			
Chittagong	College of Nursing, Chittagong Medical College, Chittagong	BSc Nursing	100
Dhaka	College of Nursing, Dhaka Medical College Hospital, Dhaka	BSc	100
	College of Nursing, Mymensingh Medical College, Mymensingh	BSc	100
Rajshahi	College of Nursing, Rajshahi Medical College, Rajshahi	BSc	100
Rangpur	College of Nursing, Rangpur Medical College, Rangpur	BSc	100
Sylhet	College of Nursing, M.A.G. Osmani Medical College, Sylhet	BSc	100
Barisal	College of Nursing, Sher-e-Bangla Medical College, Barisal	BSc	100
Total no. of nursing colleges under MOHFW=7		Total seats	700
Dhaka	Armed Forces Medical Institute, Dhaka Cantonment, Dhaka	BSc	60
	Faculty of Nursing, BSMMU, Dhaka		25
Total no. of nursing colleges under the Ministry of Defense=1		Total seats	60
Total no. of nursing colleges in the government sector=8		Grand total	785

Government Post-basic B.Sc nursing colleges (December 2012)

Division	Name of institution	Degree	No. of seats
Under the Ministry of Health and Family Welfare			
Dhaka	Nursing University, Mohakhali, Dhaka	Post-basic BSc	125
Chittagong	Foujderhat Nursing College, Chittagong	Post-basic BSc	125
Rajshahi	Bogra Nursing College, Bogra	Post-basic BSc	125
Khulna	Khulna Nursing College, Khulna	Post-basic BSc	125
Total no. of Post-basic BSc nursing colleges=4		Total seats	500

Private nursing colleges offering Post-basic B.Sc Nursing degree (December 2012)

Division	Name of nursing college	No. of seats
Dhaka	East West Nursing College, Turag, Dhaka	35
	Kumudini Nursing College, Kumudini Hospital, Tangail	25
	Faculty of Nursing, City University, Banani, Dhaka (no admission)	-
	Square Nursing College, Square Hospital, Dhaka (Closed now)	-
	State College of Health Sciences, Dhanmondi, Dhaka	30
	United College of Nursing, Gulshan, Dhaka	20
	BIRDEM Nursing College, 122 Kazi Nazrul Islam Avenue, Shahbag, Dhaka	50
	TMMC Nursing College, Targas, Board Bazar, Gazipur	25
Rajshahi	TMSS Nursing College, Thengamara, Bogra	30
Sylhet	North East Nursing College, Telihaor, Sylhet	25
	Begum Rabeya Khatun Chowdhury Nursing, College, Sylhet	30
Rangpur	Prime Nursing College, Rangpur	50
Total no. of colleges=10		Total seats 345

Private nursing colleges offering Basic B.Sc Nursing degree (December 2012)

Division	Name of nursing college	No. of seats
Dhaka	State College of Health Science, Dhaka	30
	United College of Nursing, Gulshan, Dhaka	40
	East West Nursing College, Turag, Dhaka	35
	International Medical College, Gazipur	20
	Kumudini Nursing College, Mirzapur, Tangail	25
	TMMC Nursing College, Targas, Board Bazar, Gazipur	25
	BIRDEM Nursing College, 122 Kazi Nazrul Islam Avenue, Shahbag, Dhaka	50
	CRP Nursing College, Chapain, Savar, Dhaka	40
	Square Nursing College, Square Hospital, Dhaka (Closed now)	50
Rajshahi	TMSS Nursing College, Thengamara, Bogra	25
Sylhet	North East Nursing College, Telihaor, Sylhet	25
	Begum Rabeya Khatun Chowdhury Nursing College, Sylhet	25
Rangpur	Prime Nursing College, Rangpur	50
Total=13		Total seats 440

Private institutions offering specialized nursing courses (December 2012)

Division	Name of nursing college	No. of seats
Dhaka	Cardiac Nursing, National Heart Foundation, Mirpur, Dhaka	20
	Cardiac Nursing, Ibrahim Cardiac Hospital, Shahbag, Dhaka	20
	Pediatric Nursing, Dhaka Shishu Hospital, Dhaka	20
Rangpur	Cardiac Nursing, Institute of Nursing Science, Dinajpur, (Zia Heart Foundation)	20
Total no. of institutions in the private sector=4		Total seats 80

Government nursing institutions, with number of seats (December 2012)

Division	Name of institution	No. of seats
Nursing institutions attached with medical college hospitals		
Chittagong	1. Nursing institutions attached with Comilla Medical College Hospital, Comilla	80
Dhaka	2. Nursing institutions attached with Faridpur Medical College Hospital, Faridpur	80
	3. Nursing institutions attached with SSMC Hospital, Mitford, Dhaka	80
	Total	240
Nursing institutions attached with general hospitals		
Barisal	1. Nursing institutions attached with Patuakhali General Hospital	80
Chittagong	2. Nursing institutions attached with Noakhali General Hospital	80
	3. Nursing institutions attached with Rangamati General Hospital	80
Dhaka	4. Nursing institutions attached with Tangail General Hospital	80
Khulna	5. Nursing institutions attached with Jessore General Hospital	80
	6. Nursing institutions attached with Khulna General Hospital	80
	7. Nursing institutions attached with Kushtia General Hospital	80
Rajshahi	8. Nursing institutions attached with Mohammad Ali Hospital, Bogra	80
	9. Nursing institutions attached with Dinajpur General Hospital	80
	10. Nursing Institutions attached with Pabna General Hospital	80
	11. Nursing institutions attached with Sirajganj General Hospital	50
Total		850
Nursing institutions attached with district hospitals		50
Barisal	1. Nursing institutions attached with Bhola District Hospital	50
	2. Nursing institutions attached with Pirojpur District Hospital	50
	3. Nursing institutions attached with Barguna District Hospital	70
Chittagong	4. Nursing institutions attached with Brahmanbaria District Hospital	50
	5. Nursing institutions attached with Cox's Bazar District Hospital	50
	6. Nursing institutions attached with Feni District Hospital	50
	7. Nursing institutions attached with Chandpur District Hospital	50
Dhaka	8. Nursing institutions attached with Munshiganj District Hospital	50
	9. Nursing institutions attached with Netrakona District Hospital	50
	10. Nursing institutions attached with Rajbari District Hospital	50
	11. Nursing institutions attached with Gopalganj District Hospital	50
	12. Nursing institutions attached with Madaripu District Hospital	50
	13. Nursing institutions attached with Jamalpur District Hospital	50
	14. Nursing institutions attached with Kishoreganj District Hospital	50
	15. Nursing institutions attached with Sherpur District Hospital	50
Khulna	16. Nursing institutions attached with Bagerhat District Hospital	50
	17. Nursing institutions attached with Chuadanga District Hospital	50
	18. Nursing institutions attached with Magura District Hospital	70
	19. Nursing institutions attached with Satkhira District Hospital	50
	20. Nursing institutions attached with Jhenaidah District Hospital	50
Rajshahi	21. Nursing institutions attached with Chapainowabganj District Hospital	50
	22. Nursing institutions attached with Joypurhat District Hospital	50
	23. Nursing institutions attached with Naogaon District Hospital	
Sylhet	24. Nursing institutions attached with Maulvibazar District Hospital	50
	25. Nursing institutions attached with Habiganj District Hospital	50

Table continued

Division	Name of institution	No. of seats
Rangpur	26. Nursing institutions attached with Kurigram District Hospital	50
	27. Nursing institutions attached with Thakurgaon District Hospital	50
	28. Nursing institutions attached with Nilphamari District Hospital	50
	29. Nursing institutions attached with Panchagarh District Hospital	50
Total seats		1490
Grand total		2580

Private nursing institutions, with the number of seats (December 2012)

Division	Name of nursing institution	No. of seats
Chittagong	1. Nursing Institute, Christian Hospital, Chondroghona	30
	2. Jemison Red Crescent Nursing Institute, 395 Andorkillah, Chittagong	50
	3. Nursing Institute, Chottogram Ma O Shishu Hospital, Agrabad, Chittagong	25
	4. Comilla Diabetic Hospital Nursing Institute, Comilla	40
	5. Begum Osman Ara College of Nursing, (BGC Trust), Chandanaish, Chittagong	50
Dhaka	6. Fatima Nursing Institute, Mogbazar, Dhaka	50
	7. Kumudini Nursing School, Mirzapur, Tangail	50
	8. Christian Health Project Nursing Institute, Joyramkura, Haluaghat, Mymensingh	20
	9. CRP Nursing Institute, Savar, Dhaka	50
	10. Diabetic Association Nursing Institute, Jhiltuli, Faridpur	40
	11. B.A. Siddiqui Nursing Institute, Holy Family Red Crescent Medical College Hospital, Mogbazar, Dhaka	50
	12. Jahurul Islam Nursing Institute, Bajitpur, Kishoreganj	50
	13. Munnu Nursing Institute, Manikganj	40
	14. Prime Bank Nursing Institute, Kuril Bishwaroad, Dhaka	40
	15. Japan-Bangladesh Friendship Nursing Institute, Mirpur, Dhaka	40
	16. Central Hospital Nursing Institute, Green Road, Dhanmondi, Dhaka	50
	17. Nursing Institute, Shishu Shasthya Foundation Hospital, Mirpur, Dhaka	20
	18. Nursing Institute, Medical College for Women and Hospital, Uttara, Dhaka	25
	19. TMMC Nursing Institute, Targas, Board Bazar, Gazipur	50
	20. Green Life Hospital Nursing Institute, Dhanmondi, Dhaka	40
	21. Shaheed Monsur Ali Nursing Institute, Uttara, Dhaka	20
	22. Dhaka Community Nursing Institute, Wireless Gate, Mogbazar, Dhaka	30
	23. Community Based Nursing Institute, Mymensingh	50
	24. East West Nursing Institute, Turag, Dhaka	40
	25. Gramin Calidunian College of Nursing, Mirpur, Dhaka	50
	26. IBN Sina Nursing Institute, Kalyanpur, Dhaka	50
Khulna	27. Ad-Din Nursing Institute, Jessore	30
	28. CSS Nursing Institute, Khulna (Not admitted)- Seat no: 30	-
	29. GMR Nursing Institute, Sonadanga, Khulna	50
	30. Safina Nursing Institute, Kushtia	30
Rajshahi	31. Nursing Institute attached with Khaja Yunus Ali Medical College Hospital, Enayetpur, Sirajganj	50
	32. Nursing Institute attached with Rajshahi Missionary Hospital	20
	33. Pabna Community Nursing Institute, Sathia, Pabna	40
	34. Islami Bank Medical College Hospital Nursing Institute, Rajshahi	80
	35. Shah Maghdum Nursing Institute, Boalia, Rajshahi	30

Table continued

Division	Name of nursing institution	No. of seats
	36. TMSS Nursing College, Thengamara, Bogra	50
	37. Mojibur Rahman Foundation Nursing Institute, Joypurhat	40
	38. Diabetic Association Nursing Institute, Rajshahi	40
	39. Ideal Nursing Institute, Sherpur Road, Bogra	30
	40. Safa-Macca Nursing Institute, Sirajganj	40
Sylhet	41. North East Nursing Institute, Sylhet	70
Rangpur	42. Begum Rabeya Khatun Chowdhury Nursing Institute, Sylhet	30
	43. Rangpur Community Nursing Institute, Rangpur	50
	44. Prime Nursing College, Rangpur	40
	45. Lamb Nursing Institute, Parbotipur, Dinajpur	50
	46. Institute of Nursing Science, Dinajpur	50
	47. Saint Vincent Nursing Institute, Dinajpur	40
Total seats		1910

Private-sector institutions to produce midwives
Junior midwifery institutions, with the number of seats in each (December 2012)

Division	Name of junior midwifery institution	No. of seats
Chittagong	1. Junior Midwifery Institute, Red Crescent Matrisadan Hospital, Chandpur	20
	2. Jemison Red Crescent Midwifery Institute, Agrabad, Chittagong	50
	3. Christian Hospital, Chandroghona, Rangamati	20
	4. Junior Midwifery Institute, Memon Hospital, City Corporation, Chittagong	30
Dhaka	5. Junior Midwifery Institute, Holy Family Red Crescent Hospital, Dhaka	60
	6. Junior Midwifery Institute, Shaheed Moyez Uddin Memorial Red Crescent Matrisadan Hospital, Bangla Bazar, Dhaka	20
	7. Junior Midwifery Institute, Kumudini Hospital, Mirzapur, Tangail	20
	8. Central Hospital Nursing Institute, Green Road, Dhanmondi, Dhaka	20
Khulna	9. Junior Midwifery Institute Ad-Din Matrisadan Hospital, Jessore	20
	10. Junior Midwifery Institute, Fatema Hospital, Jessore	20
Rajshahi	11. Junior Midwifery Institute, Christian Hospital, Bogra	20
Rangpur	12. Prime Nursing College, Rangpur	20
Total seats		320

Government Medical Assistant Training Schools (MATs), with the number of seats (December 2012)

Division	Name of MATs	No. of seats
Chittagong	Medical Assistant Training School, Comilla	52
	Medical Assistant Training School, Noakhali	102
Dhaka	Medical Assistant Training School, Faridpur	52
	Medical Assistant Training School, Tangail	102
Khulna	Medical Assistant Training School, Bagerhat	152
	Medical Assistant Training School, Kushtia	102
	Medical Assistant Training School, Jhenaidah	52
Rajshahi	Medical Assistant Training School, Sirajganj	102
Total seats		716

Private Medical Assistant Training Schools (MATS), with the number of seats (December 2012)

Division	Name of institution	Year of establishment	No. of seats
Chittagong	Comilla Institute of Technology and MATS, Thakurpara, Comilla	2008	50
	Chittagong Institute of Medical Technology, Chittagong	2011	50
	Moynamoti Medical Assistant Training School, Comilla	2011	50
	Noakhali Paramedical Centre (NPCMATS)	2011	50
	Brahmanbaria Medical Assistant Training School, Brahmanbaria	2011	50
	Chandpur Medical Assistant Training School, Chandpur	2011	50
	Advance Medical Assistant Training School, Green Road, Dhaka	2010	50
Dhaka	AR Medical Assistant Training School, Mohammadpur, Dhaka	2008	100
	Bangladesh Medical Assistant Training School, Uttara, Dhaka	2009	75
	Dhaka Medical Assistant Training School, Mirpur, Dhaka	2009	50
	New Pilot Medical Assistant Training School, Tangail Sadar	2008	100
	Rabeya Medical Assistant Training School, Savar, Dhaka	2009	50
	Rampura Medical Assistant Training School, Rampura, Dhaka	2008	75
	Rumdo Medical Assistant Training School, Mymensingh	2008	80
	SAIC Institute of Medical Assistant, Mirpur, Dhaka	2008	75
	SIMT Medical Assistant Training School, Kalabagan, Dhaka	2008	40
	Spark SIMT Medical Assistant Training Academy, Mirpur, Dhaka	2008	100
	SPKS Medical Assistant Training School, Mirpur, Dhaka	2008	60
	Sumona Medical Assistant Training School, Sadarghat, Dhaka	2008	75
	The Medical Assistant Training School, Mirpur, Dhaka	2007	60
	Trauma Medical Assistant Training School, Mohammadpur, Dhaka	2008	125
	Institute of Medical Assistants, Faridpur	2010	100
	Eden Medical Assistant Training School, Mirpur, Dhaka	2010	50
	Tangail Medical Assistant Training School, Kumudini College Road, Tangail	2010	50
	Shyamoly Medical Assistant Training School, Mohammadpur, Dhaka	2010	125
	Taleb Ali Medical Assistant Training School, Natun Bazar, Mymensingh	2010	100
	Rajbari Community Medical Assistant Training School, Rajbari	2010	50
	National IM&DT MATS, Mohammadpur, Dhaka	2010	50
	Prince Medical Assistant Training School, Saver, Dhaka	2010	50
	Unilab Medical Assistant Training School, Magura	2010	50
	Khondoker Abdul Mannan Medical Institute (MATS), Kishoreganj	2010	50
	Reliable Medical Assistant Training School, Mirpur, Dhaka	2011	50
	Dr. Halima Khatun Medical Assistant Training School, Mymensingh	2011	50
	Jashimuddin Medical Assistant Training School, New College Road, Jamalpur	2011	50
	Shahid SA Memorial Medical Institute, Uttara, Dhaka	2011	50
	Paramedical Institute, Chandona, Gazipur	2011	50
	Institute of Medical Technology & MATS, Narayanganj	2011	50
	Fortune Institute of Medical Technology, Dour Rajbari, Kamarpara Road, Turag Thana, Dhaka	2011	50
	New Turag General Hospital Private Limited, Station Road, Tongi, Gazipur	2011	50
	Rajdhani Medical Assistant Training School (Rajdhani MATS), Mirpur, Dhaka	2011	50
	Dhaka Microlab Institute of Medical Technology (IST, MATS)	2011	50
	Nidasa Medical Assistant Training School, 20/24 North South Road, Siddik Bazar, Dhaka	2011	50
	Bibartan Medical Assistant Training School, Mirpur, Dhaka	2011	50
	Matri Sheba Medical Training School (MATS), Kona Bari, Gazipur	2012	30
	Ideal Medical Training Institute & Health Technology, Mymensingh Road, Sabalia, Tangail	2012	25
	Dr. Rubi Medical Assistant Training School, Shyamoly, Dhaka	2011	50

Table continued

Division	Name of institution	Year of establishment	No. of seats
	Scholar Medical Assistant Training School (MATS), Maskanda, Mymensingh	2011	50
	Eitam Welfare Organization, Mohammadpur, Dhaka	2012	50
	Firoza Medical Assistant Training School, Sholakia, Kishoreganj	2012	50
	Jamuna Medical Assistant Training School, Tangail Sadar	2012	50
	Health Ways Medical Assistant Training School, Bogra	2013	30
Rajshahi	Rajshahi Medical Assistant Training School, Rajshahi	2008	50
	SIMT Medical Assistant Training School, Nishindho, Bogra	2008/2011	100
	TMSS Medical Assistant Training School, Bogra	2008	30
	Udayan Medical Assistant Training School, Rajshahi	2008	100
	Medical Assistant Training School, Joypurhat	2008	30
	Ideal Medical Technology, Sherpur Road Koloni, Bogra	2008	100
	People's International Medical Assistant Training School, Airport Sarak, Sapura, Rajshahi	2008	180
	Galaxy Medical Assistant Training School, Sapura, Rajshahi	2010	50
	Pabna Community Medical Assistant Training School, Bisnopur, Sathia, Pabna	2011	50
	Bangladesh Institute of Medical Technology, HetemKha, Bowalia, Rajshahi	2011	50
	PIMT Medical Assistant Training School, (MATS), Bogra	2011	50
	Natore Medical Assistant Training School, Natore	2011	50
	ASI Medical Assistant Training School, (MATS), Sirajganj	2011	50
	NDC Medical Assistant Training School, (NDC MATS), Paharpur Road, Khanjonpur, Joypurhat	2011	50
	Pabna Medical Assistant Training School, Mujib Palace, PP Road, Singa, Pabna	2011	50
	Sirajganj Modern Medical Training School, Coddamore, Sirajganj	2011	50
	Anwara Medical Assistant Training School, Dinajpur	2011	75
	TS Medical Assistant Training School, New Bagura Road, Sirajganj	2011	75
	Morning Glory Medical Assistant Training School, Shibola, Chapainowabganj	2012	50
	SDDL Medical Assistant Training School (MATS), Bogra	2012	50
	Prime Institute of Medical Technology, 213/A Talaimari, Rajshahi	2011	50
	State Medical Assistant Training Academy, Mill Gate Sarak, Kaliganj, Jhenaidah	2011	50
	Ideal Medical Assistant Training School (MATS), Poura College Para, Chuadanga	2012	25
	Uttar Banga Medical Assistant Training School, Uttar Banga MATS	2011	50
	Khulna Medical Assistant Training School (MATS), Khulna	2011	50
Khulna	Dr. Liza Raton Medical Assistant Training School, 42/1 NS Road, Kushtia	2011	50
	Alo Medical Assistant Training School, Alobhaban, NS Road, Kushtia	2011	50
	Ideal Medical Assistant Training School (MATS), Alamdanga Road, Pouro College Para, Chuadanga	2012	25
	Central Medical Assistant Training School (MATS), Rangpur	-	-
	Ranine Medical Assistant Training School, Lalmonirhat	2012	25
	Green International Medical Assistant Training School, Rangpur	-	-
	Disable Welfare Foundation Medical Assistant Training School, Sabujbagh, Patuakhali	2012	50
Barisal	Morning Sun Assistant Training School (MATS), Nobogram Road, Barisal	2012	25
	Prime Medical Assistant Training School, Rangpur	2012	25
	Rangpur CT MATS, Kelabond C O Bazar, Rangpur	2012	25
Rangpur	Rangpur Medical Assistant Training School, Rangpur	2010	50
	Jalalabad Medical Assistant Training School, Sylhet	2012	75
	Janata Medical Assistant Training School, Nageshori, Kurigram	2011	50
Sylhet	Jalalabad Medical Assistant Training School, Sylhet	2011	50
	Maulvibazar Medical Assistant Training School, Kushumbag, Maulvibazar	2011	50
	Sylhet Medical Assistant Training School, Sylhet	2008/2012	40

**List of Government Institutes of Health Technology, with number of seats by discipline
(December 2012)**

Division	Name of institute with location	Estd.	Discipline								Total
			LAB	RDL	PTY	SI	DENT	PHAR	RTY	FF&TR	
Dhaka	Institute of Health Technology, Mohakhali, Dhaka	1963	50	50	50	50	50	50	20	5+2	327
Rajshahi	Institute of Health Technology, Rajshahi.	1976	50	50	50	50	50	50	20	5+1	327
	Institute of Health Technology, Bogra	2006	65	55	50	50	55	55	20	5+2	357
Chittagong	Institute of Health Technology, Chittagong	2011	50	50	50	50	50	50	20	5+2	327
Barisal	Institute of Health Technology, Barisal	2011	50	50	50	50	50	50	20	5+2	327
Rangpur	Institute of Health Technology, Rangpur	2011	50	50	50	50	50	50	20	5+2	327
Khulna	Institute of Health Technology, Jhenaidah,	2012	50	50	50	50	50	50	50	50	327
Sylhet	Institute of Health Technology, Sylhet	2012	50	50	0	0	0	0	0	-	100
Total institutes=8		Total seats	415	405	350	350	355	355	170	-	2419

LAB= Laboratory; RDL=Radiology; PTY=Physiotherapy; SI=Sanitary Inspection; DENT=Dentistry; PHAR=Pharmacy; RTY=Radiotherapy; FF&TR=Children of freedom fighters and tribal students

**Updated list of private Institutes of Health Technology, with the number of seats by discipline
(December 2012)**

Division	Name of institute with location	Estd.	Discipline							Total
			LAB	RDL/ RDT	PTY	DENT	PHAR	Other 1	Other 2	
Chittagong	C.S.C.R. Institute of Medical Technology, Golpahar, Chittagong	2008	35	0	25	25	30	0	0	115
	Chittagong Institute of Medical Technology, Halishahar, Chittagong	2005	50	0	0	50	50	0	0	150
	Comilla Institute of Medical Technology, Thakurpara, Comilla.	2007	25	50	0	25	50	0	0	150
	Ilah College of Medical Technology, NaharKutir, East Bank of RanirDighi, Comilla	2005	25	0	0	0	25	0	0	50
	Institute of Medical Technology, 180 Firingibazar, City Corporation, Chittagong	2003	50	50	0	50	0	0	0	150
	Cox's Bazar Institute of Medical Technology, Cox's Bazar	2011	30	30	0	30	0	0	0	90
	United Care Institute of Medical Technology, Madhyapara, Brahmanbaria	2010	25	0	0	15	0	0	0	40
	Compact Medical Institute, Hazari Road, Feni	2011	25	0	0	50	25	0	0	50
Dhaka	A.R. Institute of Medical Technology, Nabodoy Housing Society, Mohammadpur, Dhaka	2008	50	25	50	50	50	0	0	225
	Ahsania Mission Institute of Health Technology, Mirpur, Dhaka	2008	25	25	25	0	0	0	0	75
	Armed Forces Institute of Medical Technology, Dhaka Cantonment, Dhaka	2010	25	25+25	25	25	25	10 (OTA)	15 (ICA)	175
	Bangladesh Health Profession Institute, Mirpur, Dhaka	1996	50	50	50	0	0	50 (Occupational)		200

Table continued

Division	Name of institute with location	Estd.	Discipline							Total
			LAB	RDL/ RDT	PTY	DENT	PHAR	Other 1	Other 2	
	Bangladesh Institute of Medical & Dental Technology, Iqbal Road, Mohammadpur, Dhaka	1997	85	20	20	25	0	25 (B.Sc. in Lab)	25 (B.Sc. in Dentistry)	200
	Bangladesh Medical College, Dhaka	-	0	0	25	0	0	0	0	25
	Center for Rehabilitation of the Paralyzed, Savar, Dhaka	1999	50	50	50	0	0	0	Occu: 50	200
	Dhaka Institute of Health Technology, Humayun Road, College Gate, Mohammadpur, Dhaka	2008	50	0	25	30	40	0	0	145
	Fortune Institute of Medical Technology, Jasimuddin Road, Uttara, Dhaka	2007	50	25	0	50	50	0	0	175
	Gonoshasthya Institute of Health Sciences, Tengra, Sreepur, Gazipur	2006	50	0	25	50	0	0	0	150
	Green View IHT, Green Road, Dhanmondi, Dhaka	2002	50	0	0	40	25	50 (B.Sc in Lab)	0	165
	Institute of Medical Technology, Rajbari	2010	50	0	0	0	50	0	0	100
	Institute of British Colombia Medical Technology, Uttara, Dhaka	2008	40	0	25	25	35	0	0	125
	Institute of Community Health Bangladesh, Mogbazar, Dhaka	2005	50	0	25	25	25	0	0	125
	Institute of Medical Technology, Mirpur, Dhaka	2000	100	0	0	50	50	0	0	200
	Institute of Medical & Dental Technology, Tangail	2007	50	25	0	25	50	0	0	150
	International Institute of Health Sciences, Shankar Bus-stand, Dhaka	2006	70	25	25	50	40	0	0	210
	Institute of Medical Technology, Tamizuddin Road, Jhiltuli, Faridpur	2005	50	0	0	50	25	0	0	125
	Jefri Institute of Health Sciences & Technology, Dhanmondi, Dhaka	2009	50	50	50	50	50	0	0	250
	Marks Institute of Medical Technology, Mirpur, Dhaka	2002	50	25	0	50	50	0	0	175
	Millennium Institute of Medical Technology, Bacharam Dewry, Dhaka	2007	25	25	0	25	0	0	0	75
	National Institute of Medical & Dental Technology, Mohammadpur, Dhaka	2005	34	0	0	0	31	0	0	65
	National Institute of Medical Technology, Uttara Model Town, Dhaka	2003	50	0	0	50	50	0	0	150
	New Lab Institute of Medical Tech, Asad Gate, Mohammadpur, Dhaka	2005	70	0	0	30	30	0	0	130
	Prince Institute of Medical Technology, Savar, Dhaka	2008	45	0	0	30	40	0	0	115
	Prof. Suhrabuddin Institute of Medical Technology, Sabalia, Tangail	2007	75	0	25+25 (B.Sc in PTY)	45	55	50 (B.Sc in Lab)	25 (B.Sc in Dent)	300
	Radiant Institute of Medical Technology Green Road, Dhaka	2003	40	0	0	40	0	0	0	80
	Rumdo Institute of Medical Technology, Boundary Road, Mymensingh	2007	60	0	0	0	25	0	0	85
	SAIC Institute of Medical Technology Mirpur, Dhaka	2005& 2008	50	0	0	25	50	0	0	125
	Shahid S.A. Memorial Institute of Medical Technology, Uttara, Dhaka	2007	40	0	0	25	25	0	0	90

Table continued

Division	Name of institute with location	Estd.	Discipline							Total
			LAB	RDL/ RDT	PTY	DENT	PHAR	Other 1	Other 2	
Dhaka	Shyamoly Ideal Institute of Medical Technology, Dhaka	2010	50	0	50	50	50	0	0	200
	State University of Bangladesh, iqbal Road, Mohammadpur, Dhaka	2008	0	0	0	0	0	0	50 (Op-tometry)	50
	Sumona Institute of Medical Technology, Sadarghat, Dhaka	2007	50	0	0	30	50	0	0	130
	Trauma Institute of Medical Technology, Shyamoly, Dhaka	2008	75	50	0	25	50	75	50	325
	Christian Institute of Medical Technology, West Tejuri Bazar, Tejgaon, Dhaka	2010	30	0	0	0	30	0	0	60
	Rampura Institute of Medical Technology, Rampura, Dhaka	2010	60	0	0	0	50	0	0	110
	Central Institute of Health Science (Diploma Course), Mirpur, Dhaka	2012	25	25	0	0	0	25 (B.Sc. in Lab)	25 (B.Sc. in PTY)	100
	Dhaka Microlab Institute of Medical Technology, Shahjadpur, Gulshan, Dhaka	2010	30	0	0	15	30	40 (B.Sc. in Lab)	30 (B.Sc. in Lab)	145
	Institute of Medical Technology, Jalkuri, Narayanganj	2010	25	0	0	15	25	0	0	65
	Bhairab Institute of Medical Technology, Kishoreganj	2010	25	0	0	15	25	0	0	65
	Uttara Crescent Institute of Medical Technology, Dhaka	2011	25	25	25	25	25	0	0	125
	Dialab Institute of Medical Technology, Lalbag, Dhaka	2011	25	25	25	0	25	0	0	100
	Mymensingh BNSB Institute of Community Ophthalmology, Mymensingh	2011	0	0	0	0	0	MLOP Asstt.20	0	20
	SPKS Medical Assistant Training School (IHT also), Mirpur, Dhaka	2011	25	0	0	0	25	0	0	50
	Florence Institute of Medical Technology, Maskanda, Mymensingh	2012	40	0	0	40	0	0	0	80
	Genemi Institute of Health Technology, Paik Kandi, Gopalganj	2012	30	30+30	0	0	0	0	0	90
Khulna	Ad-Din Women's Institute of Health Technology, Jessore	2007	25	25	0	25	25	0	0	100
	SAIC Institute of Medical Technology, Khulna	2010	50	0	30	50	50	0	0	180
Barisal	Advance Institute of Medical & Dental Technology, Barisal	2010	50	25	25	25	50	0	0	175
	Disable Welfare Foundation Science and MT Institute, Sabujbag, Patuakhali	2012	25	0	0	25	0	0	0	50
	Jamjam Institute of Health Technology, Kajipara, C&B Road, Barisal	2012	25	25	0	25	0	0	0	75
Rajshahi	Bangladesh Institute of Medical Technology Haji Mohsin Road, Dilalpur, Pabna	2007	30	0	30	30	30	25 (B.Sc. in Lab)	0	145
	DOD Institute of Medical Technology, Dinajpur	2010	50	50	0	0	0	0	0	100
	Health Ways Institute of Medical Technology, Bogra	2002	100	0	0	31	50	0	0	181
	Islami Bank Institute of Medical Technology, Rajshahi	2007	50	25	0	25	50	0	0	150
	Janata Institute of Medical Technology, Bogra	2002	50	40	0	40	25	0	0	155

Table continued

Division	Name of institute with location	Estd.	Discipline							Total
			LAB	RDL/ RDT	PTY	DENT	PHAR	Other 1	Other 2	
	Joypurhat Institute of Medical Technology, Joypurhat	2010	50	0	0	0	50	0	0	100
	Prime Institute of Medical Technology, 213/A Talaimari, Rajshahi	2006	100	0	0	50	50	0	0	200
	Brahmanbaria Institute of Medical Technology, Baharampur, Rajshahi	2010	25	0	0	0	25	0	0	50
	Naogaon Institute of Medical Science & Technology, Kazirmoor, Naogaon	2011	25	0	0	25	50	0	0	100
	Rajshahi Institute of Medical Technology, Laxmipur, Rajshahi	2002	50	0	0	50	70	0	0	170
	Shah Maghdum Institute of Medical Technology, Bowalia, Rajshahi	2011	30	-	-	20	20	-	-	70
	SAIC Institute of Medical Technology, Bogra	2008	50	0	0	25	50	0	0	125
	TMSS Institute of Medical Technology, Thengamara, Bogra	2007	100	40	0	30	50	0	0	220
	City Institute of Medical Technology, Rajshahi	2010	25	0	0	0	25	0	0	50
	Bangladesh Institute of Medical Technology, Bowalia, Rajshahi	2010	25	0	0	0	25	0	0	50
	NDC Institute of Medical Technology, Joypurhat	2010	50	0	0	0	50	0	0	100
	Sirajganj Institute of Medical Technology, Sirajganj	2010	25	0	0	15	25	0	0	65
	Morning Glory Medical Assistant Training School, Shibtola, Chapainowabganj	2011	25	0	0	0	25	0	0	50
Rangpur	Birampur Institute of Medical Technology, (IHT), Birampur, Rajshahi	2012	50	0	0	0	50	0	0	100
	Prime Institute of Science & Medical Technology, Rangpur	2007	100	0	0	25	50	0	0	175
	Rangpur Medical Assistant Training School	2011	40	25	25	30	25	0	0	145
	Rangpur CT IMT, Kelabond CO Bazar, Rangpur	2012	0	0	0	30	25	40 (Diploma in Lab)	0	95
Total no. of institutions=82		Total seats								10231

LAB= Laboratory; RDL=Radiology; PTY=Physiotherapy; SI=Sanitary Inspection; DENT=Dentistry; PHAR=Pharmacy; RTY=Radiotherapy; FF&TR=Children of freedom fighters and tribal students

List of govt. and private institutions offering certificate courses in medical technology, with number of seats by discipline (December 2012)

Division	Name of institution with location	Estd.	Optometrist	Refraction	Ophthalmic assistant	Ophthalmic nursing assistant	Cathlab tech	Total
Chittagong	Bangladesh Jatiyo Andho Kallyan Samity, Comilla	2008	25	0	25	0	0	50
Dhaka	Bangladesh Islamia Eye Hospital, Dhaka	2008	25	25	25	25	0	100
	Fashion Eye Hospital Limited, Fashion Tower, 98/60A Boro Mogbazar, Dhaka	2008	0	10	10	0	0	20
	NICVD&H	2010	0	0	0	0	10	10
Total no. of institutions=4		Total seats	50	35	60	25	10	180

Government institutions offering BSc courses in Medical Technology, with name of discipline and the number of seats (December 2012)

Division	Name of institution with location	Estd.	Physiotherapy	Laboratory Medicine	Dental	Total
Dhaka	NITOR, Sher-e-Bangla Nagar, Dhaka	1993	25	0	0	25
	Institute of Health Technology, Mohakhali, Dhaka	2007	30	30	60	120
Rajshahi	Institute of Health Technology, Rajshahi	2007	30	30	60	120
No. of institutions=3		Total seats	85	60	120	265

Private institutions offering BSc and MSc courses in Medical Technology (December 2012)

Division	Name of institution with location	Estd.	Physiotherapy	Laboratory Medicine	Dentistry	Occupational therapy	Others	Total
Dhaka	Bangladesh Health Professionals Institute, Savar, Dhaka (BSc)	2007	20	0	0	10	15 (Speech therapy)	45
	Bangladesh Medical College, Dhanmondi, Dhaka (BSc)	2008	0	0	0	0	0	0
	Bangladesh Shishu Shasthya Institute, Sher-e-Bangla Nagar, Dhaka (BSc)	2008	0	25	0	0	0	25
	GonoShasthya University, Savar, Dhaka (BSc)	2005	60	0	0	0	0	60
	Institute of Medical Technology, Mirpur, Dhaka(BSc)	2007	0	30	30	0	0	60
	International Institute of Health Science, Shewrapara, Dhaka (BSc)	2010	30	30	30	0	0	90
	Mark's Institute of Medical Technology, Mirpur, Dhaka(BSc)	2008	0	50	50	0	0	100
	New Lab Institute of Medical Technology, Iqbal Road, Mohammadpur, Dhaka (BSc)	2010	0	30	30	0	0	60
	SAIC Institute of Medical Technology, Dhaka (BSc)	2007	50	50	50	0	0	150
	State University, Mohammadpur, Dhaka (BSc)	2006	50	30	30	0	50 (Optometry)	160
	The People's University, Dhanmondi, Dhaka (BSc)	2007	25	0	0	0	0	25
	GonoSyasthaUniversity, Savar, Dhaka (MSc)	2005	20	0	0	0	0	20
Chittagong	Bangladesh Health Professionals Institute, Savar, Dhaka (MSc)	2007	15	0	0	0	0	15
	Chittagong Institute of Medical Technology, Halishahar, Chittagong	2008	0	50	50	0	50 (Ph)	150
Rajshahi	Institute of Health Technology, TuniBhaban, Rajshahi	2007	0	25	0	0	0	25
	Prime Institute of Health Technology, Talaimari, Rajshahi	2007	50	0	50	0	0	100
	Prime Institute of Science & Technology, Rangpur	2008	0	75	0	0	0	75
Total no. of institutions=15		Total seats	320	395	320	0	115	1160

Training/workshop/seminar (FY 2012-2013)

Topic/subject of the training/workshop/seminar	Duration	No. of batches	No. of participants
Local Training			
Short-term			
Essential Service Delivery (ESD)			
ESD Refresher Training for Field Service Providers	6 days	59	1298
Training on Nutrition for Field Service Providers	6 days	53	1166
Training on Emergency Medical and Surgical Care for Doctors	7 days	30	660
Training on Medical and Surgical Emergency Management for Support Staff	7 days	24	600
Training on Primary Management & Prevention of kidney & Urological Diseases for Physicians	6 days	13	312
Training on Kidney & Urological Diseases for Nurses	6 days	9	225
Orientation Training on Kidney & Urological Diseases for Health Workers	3 days	41	1025
Training Program for Doctors on Mental Health	6 days	26	486
Training Program for Health Workers on Mental Health	14 days	26	640
Orientation on Early Detection of Breast and Cervical Cancer for Doctors and Nurses	2 days	24	625
Orientation on Cervical and Breast Cancer Awareness for Opinion Leaders	1 days	25	625
Training on Primary Eye Care for Doctors	6 days	16	304
Training on Primary Eye Care for Nurses and Paramedics	6 days	24	604
Training for Doctors on Violence against Women and Girls	6 days	16	368
Orientation for Awareness-building on Violence against Women for Health Workers (HA, AHI, HI, SI, etc.)	1 day	25	625
Training Asthma Prevention and Management for Doctors	5 days	13	234
Training for Health Care Providers (Doctors and Nurses) on Youth-friendly Health Services	3 days	22	484
Training on Basic Dental Healthcare for Primary Healthcare Providers	5 days	13	260
Training on Recent Advances in Dentistry for Dental Surgeons	6 days	8	160
Orientation on Autism Awareness for Health Personnel and Opinion Leaders at Upazila Level	1 day	103	2575
Breastfeeding Counseling Training for Healthcare Providers (HAs/Field Service Providers)	3 days	25	625
Basic Training (Management & Clinical) for Medical Assistants	6 days	12	291
Training for Doctors on Rational Use of Antimicrobials	3 days	20	395
Orientation for Awareness building on Fistulae Prevention and Care for Field Service Providers and Social Representatives	1 day	25	625
Total for Essential Service Delivery (ESD) Training		652	15212
Management Training			
Refresher Training on Basic Service Management for Doctors	3 days	40	960
Management Training on Cardiac Emergency for Health Personnel at Division, District and UZ level	7 days	12	276
Training on Improved Financial Management for Personnel Working at Division, District, Upazila, and Specialized Institutions, TTU and Others	6 days	16	445
Training on Office Management for Office Staff	5 days	30	753
English Language Course for Nurses	28 days	27	373
Arabic Language Course (Training) for Nurses	28 days	27	402
Basic Computer Training for health personnel	28 days	19	266
Refresher Computer Training on Operating System, Installation, Internet, etc. for the Personnel of MOHFW, DGHS, and Autonomous Institutions	14 days	28	353
Training for SSN, SNs, ASN, MTs, and others on Proper Use and Preventive Maintenance of Medical Equipment	3 days	25	607
Training on Standard Operating Procedures (SOP) regarding IPD, OPD, OT, Emergency, House-keeping, Record-keeping, Nursing Services, Diagnostic Services, etc. for Service Providers of Primary, Secondary and Tertiary Hospitals	5 days	22	550
Basic Training on Hospital Waste Management for Support Staff	3 days	21	546
Training on Store Management for Store-keepers	5 days	13	325

Table continued

Topic/subject of the training/workshop/seminar	Duration	No. of batches	No. of participants
Training on Gender Issue and Poverty Alleviation for HEO/HE/Other Related Officers	2 days	11	275
Training on Gender Issue for Field Staff (HA, AHI, HI, SI, etc.)	3 days	56	1400
Organization on Joint Simulation Exercise with BDRCS at Most Cyclone-prone Districts (Multi-sectoral Approach) on Emergency Preparedness Response	2 days	20	524
Training for MOs and Field Staff on Disaster Mitigation/Post-disaster Hazards	2 days	22	550
Training Course on Mass casualty Management for Hospital-level Staff	2 days	20	500
Basic Training on Patient-care and Hospital Management for Nurses, MTs	15 days	14	350
Training on Health Statistics for Statistical Personnel Working at Different Levels of Health services	7 days	5	130
Total of Management Training		428	9585
Orientation for the Members of DTCC and DUTT	1 day	40	919
Upgradation of Training Management Information System (TMIS)	Gross	0	0
Training Implemented by ICMH	6 days	25	437
Training Implemented by IPH	5 days	24	428
Training Implemented by NIPSOM	6 days	8	211
Training Implemented by IEDCR	3 days	5	97
Training Implemented by BCPS	1-5 days	31	658
Training Implemented by CME	3-7 days	6	118
Sub-total cost center & others		139	2868
(a) Sub-total local training		1219	27665
b. Overseas Training			
Different Clinical Specialties			
Short-term (4 weeks or less) Clinical Training for Health Service Providers	1-4 weeks	12	100
Short-term (4 weeks or less) Training for Basic Science and Para-clinical Medical Teachers	1-4 weeks	2	21
Medium-term Training (3-6 months) in Clinical Services, like Eye, Orthopedics, Cancer, Critical Care, Pain Management, Burn, and bone Marrow, Other Transplantation among Service Providers, including Nurses	3-6 months	12	95
Long-term Fellowship (More than 6 months)/Diploma for Health Personnel, including Basic & Para-clinical Medical Teachers	More than 6 months	0	0
Different Management & Public Health Specialties			
Short-term (4 weeks or less) Training on Training and Teaching Technology, Hospital Management, Personnel Management, Waste Management, Exposure Visit of Teachers for Curriculum Development	1-4 weeks	12	120
Masters and PG Diploma in Public Health	More than 6 months	0	0
Specialized Overseas Training			
A. Short-term (4 weeks or less) Hands-on Clinical Training for Health Service Providers in Local Institutions (Resource persons from abroad)	1-4 weeks	-	225
B. Exchange Visits among the Institutions of Home and Abroad	1-4 weeks	0	0
(b) Sub-total of Overseas Training		38	561
C. Orientation/Workshop/Seminar			
Workshop	1-7 days	0	0
Seminar	1-7 days	0	0
Orientation	1-7 days	0	0
Advocacy	1-7 days	0	0
(c) Sub-total		0	0
Grand total (a+b+c)		1257	28226

Percent distribution of new male and female medical and dental doctors produced from various medical and dental colleges during the 5-year period from 2008 to 2012

Name of medical or dental college	2008		2009		2010		2011		2012	
	M	F	M	F	M	F	M	F	M	F
Dhaka Medical College	69.60	30.40	59.20	40.80	53.80	46.20	47.19	52.81	59.90	40.09
Sir Salimullah Medical College	61.40	38.60	49.70	50.30	61.60	38.40	57.06	42.94	61.07	38.92
Rajshahi Medical College	61.20	38.80	49.60	50.40	32.90	67.10	56.10	43.90	63.90	36.09
Rangpur Medical College	38.60	61.40	46.10	53.90	58.40	41.60	52.17	47.83	55.60	44.39
Mymensingh Medical College	50.80	49.20	53.10	46.90	60.30	39.70	51.61	48.39	61.35	38.64
Chittagong Medical College	52.00	48.00	50.70	49.30	55.80	44.20	54.42	45.58	60.00	40.00
M.A.G. Osmani Medical College, Sylhet	48.80	51.20	55.90	44.10	56.90	43.10	54.19	45.81	57.14	42.85
Sher-e-Bangla Medical College, Barisal	54.40	45.60	42.00	58.00	56.00	44.00	48.17	51.83	63.15	36.84
Faridpur Medical College	50.00	50.00	62.70	37.30	42.40	57.60	53.33	46.67	49.57	50.42
SZR Medical College, Bogra	50.00	50.00	50.90	49.10	32.10	67.90	43.01	56.99	52.30	47.69
Dinajpur Medical College	51.20	48.80	52.00	48.00	34.60	65.40	48.00	52.00	52.94	47.06
Khulna Medical College	51.10	48.90	47.50	52.50	54.40	45.70	52.50	47.50	69.82	30.17
Comilla Medical College	27.50	72.50	44.20	55.80	43.30	56.70	42.86	57.14	36.05	63.94
Dhaka Dental College	49.30	50.70	55.70	44.30	53.60	46.40	46.51	53.49	41.67	58.34
Chittagong Dental College	0.00	0.00	22.20	77.80	44.40	55.60	28.81	71.19	44.18	55.82
Rajshahi Dental College	0.00	0.00	46.00	54.10	33.30	66.70	30.77	69.23	57.47	42.56
Overall	54.70	45.30	50.80	49.20	48.60	51.60	50.18	49.82	55.68	44.32

M= Male; F=Female

Annex to Chapter 17

11 COIA Indicators for Maternal and Child Health

1. Maternal mortality ratio (deaths per 100,000 livebirths)
2. Under-five child mortality, with the proportion of newborn deaths (deaths per 1,000 livebirths)
3. Under-five children who are stunted (percentage of children below five years of age whose height-for-age is below minus two standard deviations from the median of the WHO Child Growth Standards)
4. Met need for contraception (proportion of women aged 15–49 years, who are married or in union and who have met their need for family planning, i.e. who do not want any more children or want to wait at least two years before having a baby, and are using contraception)
5. Antenatal care coverage (percentage of women aged 15–49 years, with a livebirth, who received antenatal care from a skilled healthcare provider at least four times during pregnancy)
6. Antiretroviral prophylaxis among HIV-positive pregnant women to prevent vertical transmission of HIV, and antiretroviral therapy for women who are treatment-eligible
7. Skilled attendance at birth (percentage of livebirths attended by skilled health personnel)
8. Postnatal care for mothers and babies (percentage of mothers and babies who received postnatal care visit within two days after childbirth)
9. Exclusive breastfeeding for six months [percentage of infants aged 0–5 month(s) who are exclusively breastfed]
10. Three doses of the combined diphtheria/pertussis/tetanus vaccine (percentage of infants aged 12–23 months, who received three doses of diphtheria/pertussis/tetanus vaccine)
11. Antibiotic treatment for pneumonia [percentage of children aged 0–59 month(s) with suspected pneumonia receiving antibiotics]

Health Workforce Information System in Bangladesh—Problems and Solutions

Why we need health workforce information system?

The human resource management departments at ministries, agencies, and organizations frequently need to know information on human resource distribution, vacancies, leave status, etc. The health planners and development agencies emphasize the health workforce information for the whole country, both in public and private sectors. Bangladesh's MOHFW has much over 100,000 regular staff members across different agencies. Without an integrated electronic system, it is difficult to make available all information on the ministry-wide human resources. Detailed human resource information is important for making long-term human resource planning, production, placement, and management. It is important to know health workforce information not only for those who are working under MOHFW but also for those health workforce who work under other ministries and those working in the private sector, including those who constitute the self-employed health workforce. Even the informal healers are a matter of concern. A recent study

done in Gaibandha district of Bangladesh by JiVita Project of Johns Hopkins University, USA, among women suffering from threatened abortion reveals that about 70% of them first went to informal healers. When some of them had complications, they consulted qualified healers. During treatment under qualified healers, if they felt a bit better, they again returned to the informal healers. This kind of to-and-fro movement was noticed four times, on average, which may be attributed to financial implication and/or travel for consulting a qualified healer. However, the study shows that informal healers are factors to influence health situation in Bangladesh.

What are the data elements and complexities?

The most frequent information profile that all agencies under MOHFW require include: sanctioned posts, filled-up posts, vacancies, retirement, leave, deputation, lien, death, resignation, disciplinary action, migration, untraced, salary, training, etc. with regard to distribution by staff category, such as skill mix, demographic information, and workplace (health

organization, urban, rural, administrative boundaries, hard-to-reach, poverty stricken, etc.). There are multiple agencies (MOHFW, DGHS, DGFP, NIPORT, DNS, DGDA, and HED), and there are many types of posts in each agency. The naming of sanctioned posts suffer from lack of uniformity even within the same organization. For example, under DGHS, there are over 1,200 categories of posts. Some posts are the same but have different names. The same posts have different payscales. The human resource structure is also not standardized, having variation between same kinds of organizations. The HR management is done separately in each agency by many vertical authorities. For example, the DGHS staff members are managed or administered by over 600 authorities (MOHFW, DGHS, divisional offices, district offices, upazila offices, medical colleges, secondary/tertiary hospitals and institutions) across countries by paper-based office orders. The office copies of the orders are maintained in paper folders in silos and are often difficult to trace. In most cases, these folders become untraceable as time passes. Some information, viz. on death, retirement, untraced employees, etc. as these events happen in local organizations, takes a long time to reach to central offices. The MOHFW and all agencies face a common problem of duplicate posting in the same post due to lack of exact information on who is where.

Evolutions in ministry's HR-MIS

The ministries deal with management of Class I category of staff and so want to make a solution to update information on Class I staff in a common place of each agency. So, DGHS and DGFP were asked to start a program in 1990s on personal data-sheet (a personal resume of each doctor in DGHS and of each Class I officer in DGFP). Each of the officers was required to submit a paper (PDS form) every quarter to MIS headquarter. Staff members in MIS-headquarter used to put the information in stand-alone Microsoft Access database. The compliance was not satisfactory, and the process also appeared as burden over the doctors. Thus, in 2008, DGHS converted the stand-alone database to an online system where the doctors could update their own information as and when required. For submission of PDS for official purpose, the officer requires to take a print-out and signature of local authority on the PDS for verification and authentication of data. However, ensuring the compliance remains as a problem. Doctors feel an urge to update PDS only when they are required to submit the PDS for official reason. As the PDSs are individual resumes, there is no way to track relationship with sanctioned posts to estimate vacancy by type of post, geographic location, or by

organization. DGFP still maintains the PDS system and makes manual approach to making the system updated at intervals. The Directorate of Nursing recently developed an HR Management Information System, which is also resume-based. PDSs are only for officers and do not provide information on other staff and on sanctioned posts.

New HR-MIS in DGHS—triggered from a debate in National Parliament

On 7 February 2012, there was a debate in the Parliament, accusing MOHFW for not distributing medical doctors equitably throughout the country. Honorable Minister, in response, decided to redistribute doctors to create a reasonably equitable availability throughout the country, particularly in upazila levels. This demanded collection of data quickly from all over the country. A new online HR-MIS system was developed quickly with a view to collecting a core set of information on all categories of staff. The respective organizations have been given responsibility for entering data and keeping the system updated. The system guaranteed reliability of data as it is updated by the respective organizations. The system was initially designed for health workforce of facilities (70,000 staff members) under DGHS and could successfully capture data of about 60,000 staff members. The data on community health workers and community clinics staff are maintained in separate databases. The system has been designed in such a way that all kinds of human resource function, such as posting, transfer, release, joining, leave, vacation, disciplinary action, retirement, resignation, termination, abstinence, death, etc. can be managed online across all organizations backed by alert system so that accurate and up-to-date HR information can be made available. Health workforce distribution with age, sex, skill-mix, qualification, and other disaggregation is also possible. However, the HRM departments did not use the system and continued to use the paper-based transfer and other human resource management functions. As a result, this new HR-MIS also could not be made successful in providing updated real-time HR information. However, this new system also lacks capability to track information in relation to sanctioned posts.

Weaknesses in all existing HR information systems

The DGHS was required to develop not only the above-mentioned HR-MISs but also other short-term databases to quickly meet the ad-hoc requirements of the policy-makers. The electronic HR-MIS in each agency has been developed using non-standard coding systems and without considering interoperability issues between agencies. Therefore, it is not possible to merge data from them electronically

to produce relational analysis. All databases were built for tracking individual staff under their own agency—without considering need for tracking relationship with sanctioned posts, which required to know vacancy status and distribution of sanctioned posts by category. Also, no consideration has been given to include HWF in non-state sectors. None of the systems can provide real-time updated information on existing staff as compliance has not been ensured.

DGHS developed an integrated HR-MIS as building block of an integrated National HIS

The MIS-DGHS has done large-scale nationwide deployment of IT projects for national health information systems and eHealth. One of its core activities under rapid progress is the development of a National eHealth Enterprise Architecture (eHEA), incorporating several electronic registries (viz. geo-location, health organization, health professional, every citizen's life-time EHR, asset, etc.) connected through virtual electronic Health Information Exchange (HIE) with ultimate aim to work in unison with the Civil Registration and Vital Statistics Systems and National Population Register to be jointly developed by the MOHFW, Ministry of Local Government and National Statistics Office (BBS). We already completed the development of geo-location and organization registry. Health professional registry for DGHS is almost completed. Other organizations and private sector will be added incrementally. Individual records of 120 million citizens have been collected using machine-readable forms, the soft version of which will be available by November 2013. An integrated HR-MIS is part of this large-scale long-term project. We have heavily experimented several human resource software available in the world, including iHRIS, OrangeHRM, etc. However, the complexities and diversities of HWF and its management processes in MOHFW and Bangladesh did not allow us to make a satisfactory choice from any of the globally-available well-reputed software systems. In this context, our developers decided to build our own customized system as building-block of the National eHealth Enterprise Architecture. The newest integrated HR-MIS is now populating data from the existing HR-MIS and already updated all sanctioned posts country-wide for DGHS. This system meets international standards and interoperability framework. It is ready for capturing HWF data for all agencies under MOHFW and for all other ministries and private sectors, including self-employed care providers and informal healers incrementally. It is capable of reporting agency- or organization-wise disaggregated data in all human resource-related parameters. It is capable of showing relationship between sanctioned and

available posts. The design may start with simple profile to scale up to any level of complexity, i.e. automation for all kinds of HWF functions and processes, including automated alert systems, synchronized with biometric time attendance machine and payroll. Given due consideration that adequate policy support may not be available for using the system as a ministry-wide application, it has been planned to use it, in such cases as a system for DGHS, MOHFW, other ministries, and non-state sector.

Other considerations for a national integrated HR-MIS

A national integrated HR-MIS requires a data warehouse having adequate data security, back-up, privacy-protection, and technical personnel to support, maintain, recover from disaster, and upgrade. The MIS-DGHS has already established state-of-the-art data center and a remote (400 km away) disaster recovery system, in addition to various online and offline back-up systems at the national level. An outsourced IT company, responsible for managing data center of a mobile phone company, looks after our national data center. We have three Internet service providers; each transmits Internet connectivity through two fiber-optic lines through two separate routes to ensure 99.9% connectivity and server uptime. We have created an experienced and skilled developer and support team. Our existing resources can easily be shared among multiple agencies and can further be built to cater to requirements of multiple agencies and multiple functionalities. Such use by multiple stakeholders will increase cost-effectiveness, efficiency, and data quality.

Design and process agreement for shared HR-MIS

For a shared resource, following requirements are essential to follow:

- Common exhaustive online form that captures all required data elements of every agency and stakeholder; Required data elements may vary between agencies or stakeholders; Intelligent design may enable view of only required fields relevant for organization or agency
- Common process for each agency
- Common coding system for common data elements
- Common updating mechanism by all users; Responsibility (how and who to update)
- Accountability and feedback mechanism to ensure compliance.

The health sector of Bangladesh is a huge enterprise

with health workforce managed through many agencies, stakeholders, organizations, and locations. A shared centralized database accessible online to all end-users from any geographic location and conducting all human resource functions and processes through automated system could be the best option for managing its HR information. Integrating the information system with payroll may be a good incentive and conditional option to ensure that the HR-MIS will remain updated.

How to capture data from non-state service providers?

The MOHFW may be able to exert influence on the registered private health institutions, hospitals, clinics, diagnostic centers, etc. for providing data on their HWF. However, it is difficult to get data from the unregistered private health institutions, hospitals, clinics, diagnostic centers, self-employed caré providers, etc. The regulatory boards, viz. Bangladesh Medical and Dental Council, Bangladesh Nursing Council, Bangladesh Pharmacy Council, State Medical Faculty, Homeopathic and Unani Board, etc. will have to play roles to impose binding on all registered health professionals to provide updated information annually, with a tracking system to identify, communicate, apply accountability framework and update information of the defaulters. The tracking mechanism may also be started from registration of health professionals from their graduation and/or certification level directly through

academic/training institutions. The registration and updating of information of informal healers may be done through the community clinics and government CHWs.

Guaranteeing the use and updating of integrated HR-MIS

Without strong policy decision and strict enforcement, it would not be possible to implement and maintain a successful integrated health HR-MIS in Bangladesh. The role of HRM departments is the most crucial in this respect. Until and unless the HRM departments under the MOHFW and agencies will own and be serious about implementation of effective HR-MIS, any measure may not be fully successful. Currently, the HRM departments conduct most human resource functions, using manually-produced paper-based office orders without using databases. Building a fully-functional integrated HR-MIS is a gradual process and should be developed incrementally as part of a long-term integrated solution. Any isolated ad-hoc measure will simply waste money and time and will end up in failure. Rather than duplicating the same effort, it is the best idea to pay attention to already-existing resources, to use these as shared resource and to build upon these. Provision of mandatory use of online system for all human resource management function will ensure real-time updates of HR information and analysis.

The recognition by the United Nations of MIS-DGHS's eHealth success is still a pleasant memory

The personnel of MIS-DGHS across the country still remember the pleasant memory of the recognition by the United Nations. The remarkable success in eHealth of MIS-DGHS made Bangladesh a recipient of the United Nations "Digital Health for Digital Development Award" in 2011. Honorable Prime Minister Sheikh Hasina formally received the award during the 66th United Nations General Assembly held in September 2011 from Dr. Hamadoun, Secretary-General of the International Telecommunication. The ceremony was held on 19 September 2011 in New York, USA



Local training/workshops/seminars held in fiscal 2012-2013

Training/workshops/seminars supported with HPNSDP fund

Type of training	Venue	Duration (day)	Batch (No.)	Participants (No.)
eHealth Advocacy Seminar	MIS-DGHS	1	3	140
Consultative Workshop for Developing Training Manual & Module for Training of CHCP	MIS-DGHS	4	1	10
TOT for Training of Health Workers on HIS and eHealth, Inclusive of the Use of Mobile Devices	MIS-DGHS	2	1	141
Training of Health Workers on HIS and eHealth, Inclusive of the Use of Mobile Phone Devices	Upazilas	2	206	7238
Computer Training for Doctors and Health Staff	Dhaka	14	58	2004
Consultative Workshop on HIS and eHealth	MIS-DGHS	1	3	95
Training for Head-Office Staff	MIS-DGHS	5	4	120
Annual MIS Conference (National Level) for All Tertiary Hospitals and MCH	MIS-DGHS	2	1	79
Annual MIS Conference (Divisional Level) for All Tertiary Hospitals and MCH	Divisions	2	7	1967
Consultative Workshop with District-level ICT Focal Points	Dhaka	1	1	91
Consultative Workshop with Upazila-level ICT Focal Points	Districts	2	64	547
Young Professionals' Orientation and Innovation Seminar of Young Doctors, Medical Students, and other groups	Dhaka	1	1	60
Training of CHCPs on HIS and eHealth	Upazilas	2	482	12500
PDA TOT in district level	Districts	2	64	1446
			Total	26438

Training/workshops supported by UNICEF

Type of training	Venue	Duration (day)	Batch (No.)	Participants (No.)
MIS Training for Statisticians, Statistical Assistants on IMCI and EmOC Web-based Software	MIS-DGHS	2 days	16	316
Divisional Review Meeting on IMCI and EmOC Performance and MIS Reporting	Divisions	1 day	7	145
Upazila MIS Review Meeting for IMCI and EmOC	Upazilas	1 day	46	1416
Consultative Meeting on Special Care Newborn Unit (SCNU) Register/Record and Newborn Indicators	MIS-DGHS	1 day	3	44
Consultative Meeting on Automation of 29 Special Care Newborn Unit	MIS-DGHS	1 day	3	60
Consultative Meeting on GIS Representation Platform and Integration with DGHS Website	MIS-DGHS	2 days	4	80
			Total	2061

Training supported by WHO

Type of training	Venue	Duration (day)	Batch (No.)	Participants (No.)
ICD 10 TOT	Dhaka	2 days	25	753
Data Collection of Private NGO Hospital	Dhaka	2 days	2	60
			Total	813

Training/workshops on medical biotechnology

Type of training	Venue	Duration (day)	Batch (No.)	Participants (No.)
Core Group Consultative Workshop	Dhaka	1	1	20
Training Workshop for Journalists	Dhaka	2	1	25
Consultative Workshop for Development of Updated MBT Plan	Dhaka	1	2	30
Consultative Workshop of Selection Board for Identifying a Group of Potential Young Health Professionals for Building Career in MBT	Dhaka	1	1	20
Sensitization Workshop	Dhaka	2	10	300
Consultative Workshop for MBT	Dhaka	1	3	60
Consultative Workshop on Legislation	Dhaka	1	4	60
Consultative Workshop for Updating Medical Curriculum	Dhaka	1	7	105
Hands-on Training for Medical Teachers and Scientists	Dhaka	14	4	40
Training Workshop for Medical Teachers	Dhaka	2	5	150
			Total	810

Annex to Chapter 18

HPNSDP 2011-2016 results framework (RFW) indicators (updated 2011 version)

Sl. No.	Performance Indicator	Means of verification and timing	Baseline and source	Update 2013	Target 2016
Goal: Ensure quality and equitable healthcare for all citizens of Bangladesh					
GI 1	Infant mortality rate (IMR)	BDHS, every 3 years	52, BDHS 2007	43, BDHS 2011	31
GI 2	Under-5 mortality rate (U5MR)	BDHS, every 3 years	65, BDHS 2007	53, BDHS 2011	48
GI 3	Neonatal mortality rate (NMR)	BDHS, every 3 years	37, BDHS 2007	32, BDHS 2011	21
GI 4	Maternal mortality ratio (MMR)	BMMS, every 5 years	194, BMMS 2010	194, BMMS 2010	<143
GI 5	Total fertility rate (TFR)	BDHS, every 3 years	2.7, BDHS 2007	2.3, BDHS 2011	2
GI 6	Prevalence of stunting among under-5 children	BDHS, every 3 years	43.2%, BDHS 2007	38.7%, UESD 2013	38%
GI 7	Prevalence of underweight among under-5 children	BDHS, every 3 years	41.0%, BDHS 2007	35.1%, UESD 2013	33%
GI 8	Prevalence of HIV in MARP	Sero-Surveillance (SS), every 2 years	<1%, SS 2007	0.7%, SS 2011	<1%
Program development objective: Increase availability and utilization of user-centered, effective, efficient, equitable, affordable and accessible quality HPN services					
Strategic objective: Improve access to and utilization of essential health, population and nutrition services, particularly by the poor					
Component 1: Service delivery improved					
Result 1.1: Increased utilization of essential HPN services					
1.1.1.	% of delivery by skilled birth attendants	BDHS, every 3 years	26.%, UESD 2010 18%, BDHS 2007	34.4%, UESD 2013	50%
1.1.2.	Antenatal care coverage (at least 4 visits)	BDHS, every 3 years	19.9%, UESD 2010 20.6%, BDHS 2007	25.0%, UESD 2013	50%
1.1.3.	Postnatal care within 48 hours (at least 1 visit)	BDHS, every 3 years	20.9%, UESD 2010 18.5%, BDHS 2007	27.6%, UESD 2013	50%
1.1.4.	Contraceptive prevalence rate (CPR)	BDHS, every 3 years	61.7%, UESD 2010 55.8%, BDHS 2007	62.0%, UESD 2013	72%
1.1.5.	Unmet need for FP	BDHS, every 3 years	17.1%, BDHS 2007	13.5%, BDHS 2011	9.00%
1.1.6.	Measles immunization coverage by 12 months	CES, annual	82.4%, CES 2009	81.9%, UESD 2013	90%
1.1.7.	% of under-5 children with pneumonia receiving antibiotics	BDHS, every 3 years	38.0%, UESD 2010 37.1%, BDHS 2007	87.0%, UESD 2013	50%
1.1.8.	% of children (6-59 months) receiving vitamin A supplementation in the last 6 months	BDHS, every 3 years	82.6%, UESD 2010 88.3%, BDHS 2007	74.8%, UESD 2013	90%
1.1.9.	TB case detection rate	NT Program, annual	74%, NTP 2009	99 per 100,000 people, Health Bulletin 2012	75%
Result 1.2: Improved equity in essential HPN service utilization (MDG 1, 4, 5, and 6)					
1.2.1.	Proportion of births in health facilities by wealth quintiles	BDHS, every 3 years	Q1:Q5=8.0:59.5, UESD 2010 Q1:Q5=4.4:43.4, BDHS 2007	14.3:57.2=(1:4), UESD 2013	Q1:Q5=<1:4
1.2.2.	Use of modern contraceptives in low-performing areas	BDHS, every 3 years	Sylhet: 35.7%, Chittagong: 46.8%, UESD 2010 Sylhet: 24.7%, Ctg: 38.2%, BDHS 2007	Sylhet: 39.4%, Chittagong: 43.9%, UESD 2013	Sylhet and Chittagong: 50.0%
1.2.3.	No. of upazilas with women targeted by improved voucher scheme for having institutional deliveries	DSF Monitoring Reports, annual	31 DSF upazilas (+9 universal), DSF Monitoring Report 2010	73, APIR 2013	50 DSF upazilas

Table continued

Sl. No.	Performance Indicator	Means of verification and timing	Baseline and source	Update 2013	Target 2016
Result 1.3: Improved awareness of healthy behavior (MDG 1, 4, 5)					
1.3.1.	Rate of exclusive breastfeeding in infants up to 6 months	BDHS, every 3 years	43.0%, BDHS 2007	59.7%, UESD 2013	50%
1.3.2.	% of children of 6-23 months fed following appropriate IYCF practices	BDHS, every 3 years	41.5%, BDHS 2007	31.6%, UESD 2013	52%
Result 1.4: Improved PHC-CC systems					
1.4.1.	No. of community clinics (CCs) with increasing number of service contacts over time	CC Project/MIS/MOHFW, annual	Registers and forms supplied to CCs in June 2012	12,527, APIR 2013	13,500
1.4.2.	% of upgraded union-level facilities able to provide basic EmOC services	Health Facility Survey (BHFS), every 2 years	15.5%, BHFS 2009	4.3%, BHFS 2011	50%
Component 2: Strengthened health systems					
Result 2.1: Strengthened planning and budgeting procedures					
2.1.1.	% of MOHFW budget allocated to upazila level or below	Public expenditure review, annual	52%, PER 2006/2007	47%, PER 2008/2009	60%
2.1.2.	% of annual work plans with budgets submitted by LDs by defined time period (July/Aug)	Planning Wing, annual	NA	100%, Planning Wing 2013	100% (target by 2013)
Result 2.2: Strengthened monitoring and evaluation systems					
2.2.1.	MIS reports on service delivery published and disseminated annually	HIS, eHealth, and MIS-FP, annual	NA	Both MISs have published but not disseminated; APIR 2013	100%
2.2.2.	Performance report of OPs reviewed with policy-makers, MOHFW, Directorates, and DPs, six months and annually	Planning Wing, six-monthly	NA	100%, Planning Wing 2013	100% (achieved by 2013)
Result 2.3: Improved human resources—planning, development, and management					
2.3.1.	Proportion of service provider positions functionally vacant at district level and below, by category	DGHS/DGFP MIS, annual; BHFS, every 2 years	Physicians: 45.7%; Nurses: 24.7%; FWV/SACMO/MA: 16.9%, BHFS 2009	Physicians: 46.1%; Nurses: 19.59%; FWV/SACMO/MA: 21.2%, BHFS 2011	Physicians: 22.8%; Nurses: 15.0%; FWV/SACMO/MA: 8.50%
2.3.2.	No. of additional providers trained in midwifery at upazila health facilities	HRD/MOHFW, annual	NA	596, APIR 2013	3,000
2.3.3.	Number of comprehensive EmOC facilities with functional round-the clock services covering all districts	MIS/EOC BHFS, every 2 years	132, MIS/DGHS 2009	75, APIR 2013	20
Result 2.4: Strengthened quality assurance and supervision systems					
2.4.1.	Case-fatality rate among admitted children with pneumonia in upazila health complexes	DGHS MIS, annual	8%, Health Bulletin 2009	NA	6.20%
Result 2.5: Sustainable and responsive procurement and logistic system					
2.5.1.	% of health facilities, by type, without stock-outs of essential medicines at a given point in time	BHFS, every 2 years	66.1%, BHFS 2009	74%, BHFS 2011	75%
2.5.2.	% of facilities without stock-outs of contraceptives at a given point in time	BHFS, every 2 years	58.1%, BHFS 2009	55.1%, BHFS 2011	70%
Result 2.6: Improved infrastructure and maintenance					
2.6.1.	% of facilities (excluding CCs) having separate, improved toilets for female clients	BHFS, every 2 years	51.0%, BHFS 2009	44.5%, BHFS 2011	75%

Table continued

Sl. No.	Performance Indicator	Means of verification and timing	Baseline and source	Update 2013	Target 2016
Result 2.7: Sector management and legal framework					
2.7.1.	Regulatory framework for accreditation of health facilities, including hospitals (both in the public and private sectors) reviewed and updated	MOHFW	1982 Regulatory Act	Accreditation document finalized; MOU with 2 private hospitals signed, APIR 2013	Reviewed (by 2012)
Result 2.8: Decentralization through LLP procedures					
2.8.1.	No. of districts/upazilas having functional LLP procedures	Respective agencies, annual	NA	14 upazilas (PME OP) and 42 upazilas (PMR OP), APIR 2013	Piloting completed and reviewed for scale-up
Result 2.9: SWAp and improved DP coordination (deliver on the Paris Declaration)					
2.9.1.	No. of non-pool DPs submitting quarterly expenditure reports	Planning Wing FMAU	Irregular	3 out of 16 DPs, Planning Wing 2013	100%
Result 2.10: Strengthened Financial Management Systems (funding and reporting)					
2.10.1.	% of project aid fund (e.g. development budget) disbursed annually and quarterly	FMAU	79.4%, FMAU 2009/2010	99%, APIR 2013	100% (by 2013)
2.10.2.	% of OPs with spending >80% of ADP allocation (annually)	FMAU/Planning Wing	44.7%, FMAU 2003-2011	78.1%, APIR 2013	100% (by 2013)
2.10.3.	% of serious audit objections settled within the last 12 months	FMAU	7%, FMAU 2007/2008	61%, World Bank 2013	>80%

Physical Progress of HPNSDP In FY 2012-2013

APIR 2013 describes the physical progress of the OPs in bundle. For example, improving health services (IHS)-related major activities or policy, guideline, strategy, etc. for health system strengthening. Physical progress of individual OPs was merged under common heading. For writing this section, an effort was made to relate physical progress to individual OPs. However, we apprehend that there may be misrepresentation in some cases. Based on our exercise, as mentioned here, the physical progress of HPNSDP for fiscal 2012-2013 excerpted from APIR 2013 is presented in this annex.

Health Service Improvement

Maternal, neonatal, child and adolescent health (MNCAH), nutrition and primary healthcare (PHC)

MNCAH

- Number of UHCs providing round-the clock comprehensive EmOC services has been reduced to 75 due to human resource issues, like transfer, posting, and promotion of skilled manpower. However, vacancies will be filled up shortly with the recently-trained obs-gyne, anesthetist pairs, which will result in the rise of CEmOC to 110. All 12

obstetric drugs are available in 20 upazilas of 10 low-performing districts.

- Use of Misoprostol was scaled up in 8 districts, and Active Management of Third Stage of Labor (AMTSL) is available in all upazilas of 52 districts; 869,032 Misoprostol tablets were distributed for AMTSL.
- A total of 471 CSBAs (440 from public sector and 31 from private sector) completed training; training of further 493 CSBAs started in December 2012; 363 post-basic midwifery training sessions held; process initiated for training of private CSBAs; 144 private CSBAs completed training and accredited by competent authority; present number of CSBAs in the public sector is 7,710 and in private sector is 234.
- The Helping Babies Breathe initiative expanded to 45 districts and two city corporations for the management of birth asphyxia by skilled birth attendants. A total of 5,459 doctors, nurses, CSBAs, and FWVs trained for the purpose. Operations research on community-based management of neonatal sepsis implemented in 4 upazilas.

- Development of “National Neonatal Strategy and Guidelines” and “Action plan” completed and being implemented. Home-based essential newborn care interventions by trained CHWs scaled up to 32 districts in accordance with Neonatal Action Plan. Country-specific strategy for the prevention and management of pneumonia and diarrhea developed.
- Expansion of facility-based IMCI completed in 460 upazilas, 59 new district hospitals, and 8 medical college hospitals, with the provision of training to 468 doctors and 1,314 paramedics (cumulative total of 3,729 doctors and 10,175 paramedics till June 2013) on IMCI and Emergency Triage Assessment and Treatment (ETAT). Community-based IMCI expanded to 28 new upazilas (total 157 upazilas), with the provision of training to 2,175 staff members (total 4,625 staff members trained till June 2013).

Nutrition

- Draft National Nutrition Policy, National Food Safety Policy, and dietary guideline developed.
- Completed 3 days’ basic nutrition training for medical officers and field workers of DGFP for establishment of Nutrition Units/Corners in upazila health complexes and community clinics of 50 upazilas; instructions sent to respective upazilas.
- Weighing scales (11,900 in total), weighing trousers, MUAC tape, height-length boards, and GMP cards were supplied to all district hospitals, MCWCs, and UHCs of 105 upazilas in 15 districts.
- In UHCs, 132 medical officers trained in nutrition service delivery; weighing scales provided to 8,000 CCs for starting GMP activities; community-based nutrition (CBN) started in 9 hard-to-reach upazilas of Barisal and Khulna districts.
- An action plan with guideline developed for provision of PHC services to districts of Chittagong Hill Tracts (CHT). Training of field workers in 22 batches completed in three hill districts. With participation of local traditional leaders in tribal health, 3 workshops on micro-planning for satellite and mobile medical team in hard-to-reach area completed in three hill districts.

Community clinic

- Reportedly 12,527 community clinics made functional with trained manpower and over 92% availability of medicines; 148 CCs conducting safe delivery.

- A total of 193,912 members of community support groups of 12,071 CCs trained and 28,798 CC management committee meetings held.

Alternative medicine

- Alternative medical care (i.e., unani, ayurvedic, and homeopathic) services expanded to 45 district hospitals, Government Unani and Ayurvedic Medical College Hospital, Government Homeopathic College Hospital, and Government Tibbia College.

Essential Service Delivery

- Service providers of 25 UHCs, 82 subcenters/ rural dispensaries, 152 UHFWCs trained on adolescent-friendly health service concepts are providing services to adolescent groups in their respective health facilities; 998 health service providers and 1,651 secondary/higher secondary school teachers trained in ensuring availability of adolescent-friendly sexual and reproductive health (SRH), and school health services, including counseling and health education at union and upazila levels ongoing.
- A module developed for school nutrition education programs. Steps have been taken to initiate school nutrition education programs in collaboration with the school health program of ESD, CDC, and the Ministry of Education.

Family planning service

- A total of 640,000 family planning satellite clinics organized; refreshers’ training to 2,957 field-level service providers given; 10,264 personnel trained in clinical contraception.
- A total of 323,000 couples referred by FWAs for LAPMs; 900 NSV and 3,570 IUD acceptors trained as referral agents; 249,767 VSCs performed; 281,121 IUDs and 263,285 implants inserted.
- Postpartum family planning (PPFP) started in 21 districts.

Communicable and non-communicable disease control

Communicable disease control

- Due to reducing case loads, 15 filariasis-endemic districts stopped Mass Drug Administration (MDA).

Tuberculosis and leprosy control

- Only 1,536 multidrug-resistant (MDR) tuberculosis patients identified; they were managed; treatment success rate among detected new smear-positive (NSP) TB cases is 92%.

STD/AIDS control

- Number of health facilities with capacity to deliver care to HIV-positive people increased to 4, with a capacity to treat 200 for opportunistic infections. Number of eligible HIV-positive cases covered with ART is 820.

Non-communicable disease control (NCDC)

- NCD prevention, control and management strategy developed and being implemented; health facilities of upazila level and below in 140 upazilas doing hypertension and diabetes screening.
- A total of 21,000 service providers trained on major NCDs in district level and below; 10,500 persons of 9 districts in 3 divisions, and of 95 industries and factories provided occupational health safety and awareness training; training of health personnel on disaster preparedness in 28 disaster-prone upazilas completed.

National eye care (NEC)

- A total of 1,355 adult cataract patients per million population underwent surgery, and 700 patients with diabetic retinopathy screened and managed; 4,700 child cataract surgeries performed annually, and 170 hospitals follow standard protocols.
- A total of 6 district committees formed and are functioning on the Vision 2020 (Right to Sight); 26 vision centers established.
- Training of eye care providers given to 12 ophthalmologists on microsurgery, to 18 nurses on OT and ward management, to 4 batches each consisting of 30 participants of primary healthcare workers on primary eye care in Gopalganj district.
- Orientation given to community leaders (1 batch with 100 per batch in Kotalipara upazila under Gopalganj district.
- Procurement, distribution, installation, and maintenance of eye care equipment (in Bhola, Kishoreganj, Rajbari, and Narail district Sadar Hospital) done.
- Uninterrupted supply of MSR in 14 district hospitals (Dhaka Division: Gazipur, Tangail, Kishoreganj, Jamalpur, Gopalganj, Faridpur; Chittagong division: Brahmanbaria, Cox's Bazar; Khulna division: Narail, Kushtia, Jessore, Satkhira; and Barisal division: Bhola and Jhalokathi) made.
- Conduction of cataract screening completed and surgical camps held at Sreepur upazila under Gazipur district; Hatiya upazila under Noakhali district; Gouripur upazila under Mymensingh

district; Kotalipara upazila under Gopalganj district; and Sadar under Sirajganj district. Cataract surgical rate among adults increased from 1,164 in 2009 to 1,300 in 2012 per million population, and that among children increased from 4,000 in 2009 to 5,000 in 2012 annually).

- Demand-side financing (DSF) ensured for the poor cataract patients in selected district for ocular surgery (600 cataract patients operated).
- National survey on prevalence of blindness activity started (Protocol signed between WHO and Director, NIOH).

Behavior change communication (BCC)/ Information, education, and communication (IEC): health education and promotion

BCC/IEC

- Broadcasting of docudrama on MNCH, named Sonali Alo, started in Bangladesh TV.
- Health workers conducting courtyard meetings on essential newborn care on a regular basis to develop awareness and generate demand.
- Production, distribution, and display of 400,000 IEC materials, 5 TV spots, 5 documentary films, and establishment of 128 model health education and promotion villages completed.
- A total of 640 advertisements published in the national dailies with different messages on FP activities; 10,080 programs on FP-MCH, RH, and GI broadcast through the Population Cell of Bangladesh Betar, and 336 programs on FP-MCH and RH telecast through the Population Cell of Bangladesh TV; 3,500 film shows organized through 23 audiovisual vans, with emphasis on hard-to-reach/low-performing areas.

Health education and promotion (IEC) for nutrition, communicable disease, mental health, injury, etc. (OPs other than HEP may have role)

- BCC materials (e.g., promotion of IYCF) on nutrition developed and started broadcasting. Collaborative activities with the Ministry of Industries for awareness-creation among relevant stakeholders on food fortification are going on.
- Three batches of advocacy workshops/meetings with professional healthcare providers (GO/NGO) organized to create awareness about appropriate attitude and behavior towards mental patients.
- Awareness on dengue prevention and control measures made through workshops in 30 batches with 1,200 participants.

- Awareness campaigns on injury (traffic, child and other injuries) completed in 270 upazilas.
- 10 health education and promotion (HEP) service packages procured.

National eye care

- Behavior change communication (BCC) made to create awareness on blindness and its prevention (in Kotalipara and Tongipara), and BCC activities conducted in 6 districts to create awareness on blindness and its prevention.
- Country Wide World Sight Day 2012 observed with collaboration of international NGOs and private eye health organizations (with rally, talk show at Channel I, scientific seminar, discussion, and free cataract screening camp).
- TV spots and radio spool developed.
- Billboards installed on Shahid Samsuddin Sadar Hospital, Sylhet; Tungipara Upazila Health Complex, Gopalganj; and NIOH, Dhaka.

Strengthening Health Systems

Physical facility development

- A total of 132 existing FWCs upgraded to UHFWCs for providing improved MCH service.
- A total of 17 mother and child welfare centers/ general hospitals (each 10-bed) constructed.
- Shishu Bikash Kendra established in 11 medical college hospitals.
- Safe blood transfusion services introduced in 206 healthcare facilities (14 medical college hospitals, 7 specialized hospitals, 64 district hospitals, 14 combined military hospitals, and 107 upazila hospitals).
- Standard in-house medical waste management introduced in 6 medical college hospitals, 7 specialized hospitals, and 8 district hospitals. 11 DHs and 88 UHCs introduced quality assurance. 7 MCHs and 26 district hospitals introduced structured referral system.
- Modernization of Drug Testing Laboratory was completed and made functional; 3,795 drug samples tested in FY-2012-2013. An adverse drug reaction (ADR) cell was established.
- Draft accreditation guideline and regulations of Bangladesh Nursing Council (BNC) is completed and waiting for stakeholder consultation.

Procurement, supplies and logistics management

- CMSD awarded 90% of contracts within initial bid validity period and materialized 30% of procurement within the timeframe after receiving requests from the LDs.
- Sixty percent of training on procurement management conducted; 80% on-line storage and distribution system established to track all goods procured in CMSD; 90% of items monitored through electronic inventory system.
- Procurement management training given to 75 persons in CMSD.

Procurement for family planning services

- Commodity audit conducted in 21 warehouses, 6 MCWCs, 42 UHFWCs, 42 NGOs, and 126 private pharmacies.
- A total of 15,000 DDS kits; adequate quantity of drugs for treatment of RTI/STI and adolescents' health problems; other drugs; and 8,000 MR (menstrual regulation) kits procured.
- Training and workshops conducted through Management Science for Health on improved access to pharmaceuticals and services on framework contract, supply manual, user acceptance testing; training on supply chain management portal (SCMP), and a 5-day training on procurement at ESCB, upazila inventory management system (UIMS) completed; Training for UFPO and UFPAs and on procurement procedure manual completed for DGFP.
- Procurement and supply of multimedia for 40 UHCs under Barisal division, civil works for data center at Khulna divisional health office and restructuring physical design for appropriate ICT set-up, etc. completed.
- A total of 630 mother and baby kits were distributed, and 188 tool kits sent to the LLP upazilas.

Medical education and in-service training

Training relating to medical education

- Medical Education Units established in 22 medical and 3 dental colleges, and HR capacity developed. Laboratory and library facilities improved in 12 medical and 2 dental colleges. Strengthening facilities of 8 medical assistant training schools and 8 institutes of health technology completed.
- Quality assurance scheme expanded to 22 medical colleges, 3 dental colleges, 8 IHTs, and 8

MATS. Medical Education Units and Medical Skill Centers of 22 medical colleges, 3 dental colleges, 8 IHTs, and 8 MATS improved.

- An infection control manual and infection training module developed for intervention. Training completed in 2 government hospitals.
- Residential Field Site Training (RFST) given to 2,225 fourth-year medical and dental students in 8 batches.
- A total of 450 teachers from medical and dental colleges trained in quality medical education.

Training related to nursing and midwifery

- A diploma in midwifery curriculum developed. Modification and review of BSc (Nursing) curriculum completed and is awaiting approval. The preparation of draft MSc Nursing curriculum is underway.
- 1,929 BSc nurses produced, and 525 students admitted in midwifery diploma course (3 years diploma); 710 nurses trained as certified midwives under a 6-month midwifery course; 54 nurses posted in different places after completion of 6-month training.
- A proposal sent to the MOHFW to create 3,000 posts for midwives.
- Short-term foreign training (on midwifery and TOT) given to 18 batches; 26 nurses sent to Mahidol University, Thailand, for obtaining Masters in Nursing degree; 4 batches of local training sessions completed.

Training related to reproductive, maternal child health

- Basic training given to 3,166 FWVs, FWAs, and FPIs; refreshers' training given to 2,171 FWVs, SACMOs, HAs, FWAs, and FPIs; training on FWA register and reporting formats provided to 30,500 FWAs.
- A total of 141 workshops (9,948 participants) conducted, and refreshers' training given to 316 paramedics for capacity development in clinical contraception service delivery.
- Reproductive and child health training (IUD, IP, essential newborn care, and ELCD) given to 93 batches; on education, information and communication technology to 8 batches; on management development to 34 batches; and also 6 workshops completed.

Training relating to essential service delivery

- Training completed for 292 batches (15,212

participants) on essential service delivery; 428 batches given management training; specialized hands-on clinical training provided to 181 personnel bringing resource persons from abroad.

- A total of 998 service providers trained on adolescent-friendly SRH services.

Training related to eye care

- Training of eye care providers was completed in 4 upazilas of Gopalganj district. Training of 12 ophthalmologists on microsurgery, 18 nurses on OT and ward management, and primary healthcare workers on primary eye care completed in four batches (30 persons per batch) at Gopalganj district.

Training relating to non-communicable disease control (NCDC)

- Capacity-building training sessions were organized for 13 batches of nurses, SACMOs, and pharmacists on mental health and related activities. One workshop was held in Bandarban district.

Training relating to health education and promotion (HEP)

- Foreign training to 17 health education and promotion personnel and local training to 2,400 personnel given.

Training relating to research and development

- Training conducted in 6 batches on planning, monitoring for health personnel at district level and below and in another 10 batches on research methodology.

Information systems

MIS-DGHS

- A set of core health indicators developed with definition of data and process of data collection.
- All LDs provided financial data in the web-based template of the online ADP Monitoring System.
- Health Bulletin 2012, Annual Report 2012 (a requirement by National Information Commission), Voice of MIS Newsletter, IMCI Newsletter, Our eHealth/Amader Shaystha printed and distributed. All past and new MIS reports made available on website.
- Collection of health service data, using DHIS 2.12 (District Health Information System, version 2.12) strengthened—individual tracking module for registering and tracking pregnant women and under-5 children added to this system.

- 18 telemedicine centers established; 10 more will start operation soon.
- An SMS-based citizens' complaint-suggestion box, carrying citizens' feedback from all public hospitals real-time on a web-based platform with the provision of immediate response by dedicated staff, has been established.

IEDCR

- IEDCR established a web-based disease surveillance system which is receiving data from all district and upazila hospitals.

MIS-DGFP

- Monthly FP-MCRAH reports, inclusive of logistics information published.
- About 96,000 FWA registers (7th and 8th edition) and reporting forms printed and started distribution.

Sector-wide project management

- A draft M&E plan and reporting format on nutrition was developed.

Planning, monitoring and financial management

- Annual Program Review (APR) 2012 completed successfully and Prioritized Action Plan (PAP) of 2012 APR being followed up.
- MOHFW's Planning Wing continuously working with DPs for mobilizing Project Aid (PA) allocations.
- Six-monthly Progress Report (Jan-Dec 2012) of HPNSDP published. Preparation work for the APR 2013 commenced in May 2013.
- Two LCG meetings, 20 meetings of different Task Groups, including M&E Task Group, 3 meetings on JFA/JCA, 5 meetings of the TA Committee, 11 of APR Steering Committee, 7 of OPIC held.
- PMMU made functional with GOB and TA officials in position.

Research by BMRC

- A total of 60 research studies conducted by BMRC.

Local-level planning

- A total of 146 upazila local-level plans (LLP) prepared, inclusive of hard-to-reach and low-performing areas. A total of 1,366 CCs incorporated with LLP activities; 415 motivational workshops, 7 district-level meetings, 270 CC-level meetings, and 3,255 ward-level meetings conducted.

Monitoring and evaluation

- 52 NGOs providing FP services were supervised, monitored, and renewed; 3 new NGOs were affiliated for implementing FP activities.

Financial management

- A healthcare financing strategy developed and approved by the MOHFW. A draft implementation plan developed and shared with DPs.
- A total of 155 persons trained in 9 batches; and 5 workshops conducted on healthcare financing, with 290 participants.
- Pre-piloting of a legal study completed in connection with health insurance piloting. Preparation for piloting is at the final stage. Documents and operation manuals developed, such as SSK benefit package with pricing, IEC strategy, manual for drug management, hospital manual for SSK, grievance mechanism, plan of operation of SSK.
- IFM adopted the interim unaudited financial report (IUFR), and annual financial reports prepared regularly.
- Refreshers' courses for nine batches conducted, and 3 workshops on audit and financial management system held.

Distribution of training costs (in crore Tk) and training participants by operational plan

Operational plan (OP)	Total OP expense	Total training expense	Training expense as % of OP expense	No. of training participants (Local)	No. of training participants (Foreign)	No. of participants in workshops/seminars
All OPs	2677.9	258.3	10	760570	1334	612389
MNCAH	402.7	34.0	8	165011	0	60670
ESD	40.8	3.3	8	2186	50	851
CBHC	67.2	47.3	70	467413	89	0
TBLC	61.9	4.9	8	22876	0	230
NASP	22.3	7.5	34	225	0	30438
CDC	103.2	7.4	7	14734	0	165202
NCDC	112.5	42.6	38	0	0	72000
NEC	4.0	0.4	9	156	0	275
HSM	423.4	6.3	1	0	64	8705
AMC	12.8	0.8	6	0	5	1006
IST	46.2	45.1	98	27665	561	0
PSE	159.6	2.1	1	150	64	0
PMR	8.2	1.4	17	1040	0	1610
HIS-EH	108.4	2.2	2	813	0	3243
HEP	18.0	1.7	9	2460	18	0
PLSM	61.9	0.3	1	176	0	340
NNS	36.1	4.4	12	12029	0	1633
MCRAH	133.4	2.9	2	2297	37	1600
CCSD	153.7	3.8	2	316	70	9948
FPFSD	84.0	0.6	1	0	0	7353
PME	1.9	1.2	65	0	0	234855
MIS-FP	12.5	2.0	16	30950	0	359
IEC	21.3	2.1	10	-	-	-
PSSM	12.9	0.2	1	2	2	70
TRD	15.9	9.8	61	8045	34	1180
NES	59.2	17.8	30	1025	240	250
SDAM	0.6	0.04	6	44	0	0
PFD	481.0	1.6	0	226	24	0
HRM	2.4	2.1	90	367	45	239
SWPMM	2.8	1.0	34	0	13	1141
IFM	2.4	1.0	40	144	8	565
HEF	4.6	0.7	16	220	10	306

Annex to Chapter 19

Allocation in FY 2013-2014 and allocation, expenditure, and utilization in FY 2012-13 (revised ADP) under HNPSDP 2011-2016 fund against different operational plans of DGHS (Ref: DGHS 2013) (Tk in lakh)

Program	FY 2013-2014 (Tk in lakh)				FY 2012-13 (Tk in lakh)				FY 2012-13 (Tk in lakh)				Utilization (%)	
	Allocation				Allocation				Expense					
	Total	GOB	RPA (GOB)	DPA	Total	GOB	RPA	Other	Total	GOB	RPA	Other		
Essential service delivery	7,400.00	1,400.00	6,000.00	-	4,431.00	1,400.00	3,000.00	-	31.00	4,004.04	1,178.54	2,825.49	-	90.4
Maternal, neonatal, child and adolescent healthcare	43,700.00	4,500.00	23,400.00	15,800.00	42,926.00	2,041.00	9,977.54	18,481.46	12,426.00	40,273.50	1,895.82	8,024.27	18,481.46	93.8
Community-based healthcare	7,850.00	500.00	6,850.00	500.00	7,250.00	500.00	6,250.00	-	500.00	6,718.44	453.39	6,041.73	-	92.7
TB and leprosy control	5,720.00	500.00	804.00	4,416.00	7,062.00	586.00	700.00	-	5,776.00	6,192.35	377.62	684.38	-	87.7
National AIDS/STD program	3,805.00	300.00	3,305.00	200.00	2,800.00	200.00	2,500.00	-	100.00	2,246.00	185.00	1,961.00	-	80.2
Communicable disease control	13,200.00	2,500.00	5,000.00	5,700.00	9,800.00	1,800.00	5,000.00	-	3,000.00	10,366.43	1,468.21	4,024.03	-	105.8
Non-communicable disease control	10,147.00	3,200.00	6,947.00	-	12,215.00	4,015.00	8,000.00	-	200.00	11,254.59	3,756.37	7,426.22	-	92.1
National eye care (NEC)	335.00	200.00	135.00	-	445.90	252.90	193.00	-	-	403.22	246.88	156.34	-	90.4
Hospital services management	37,000.00	8,000.00	28,500.00	500.00	43,485.00	9,000.00	33,985.00	-	500.00	42,343.77	8,328.62	33,515.15	-	97.4
Alternative medical care	1,250.00	1,000.00	250.00	-	1,292.00	1,092.00	200.00	-	-	1,275.70	1,142.60	133.10	-	98.7
In-service training	3,964.00	600.00	3,364.00	-	5,510.00	510.00	5,000.00	-	-	4,625.22	484.38	4,140.84	-	83.9
Pre-service education	14,000.00	4,000.00	10,000.00	-	16,736.00	6,736.00	9,900.00	-	100.00	15,923.33	6,560.33	9,363.00	-	95.1
Planning monitoring and research	1,177.00	200.00	811.00	166.00	900.00	150.00	650.00	-	100.00	814.77	136.63	636.14	-	90.5
Health information systems and eHealth	12,500.00	3,500.00	8,500.00	500.00	11,025.00	3,825.00	7,000.00	-	200.00	10,973.62	3,824.92	6,999.13	-	99.5
Health education and promotion	2,200.00	500.00	1,500.00	200.00	1,950.00	850.00	1,000.00	-	100.00	1,798.65	822.85	875.80	-	92.2
Procurement, logistics and supplies management	4,350.00	4,000.00	350.00	-	6,283.00	6,030.00	253.00	-	-	6,194.40	5,958.50	235.90	-	98.6
National nutrition services (NNS)	8,500.00	500.00	7,500.00	500.00	5,100.00	1,400.00	2,700.00	-	1,000.00	3,611.99	1,002.77	1,883.73	-	70.8
Total	177,098.0	35,400.00	113,216.00	28,482.00	179,210.90	40,387.90	96,308.54	18,481.46	24,033.00	169,031.00	37,834.30	88,926.26	18,481.46	91.8

Allocation in FY 2013-2014 and allocation, expenditure, and utilization in FY 2012-2013 (revised ADP) under HPNSDP 2011-2016 in different investment projects of MOHFW (Ref: DGHS 2013) (Tk in lakh)

Investment project	FY 2013-2014 (Tk in lakh)			FY 2012-2013 (revised ADP) (Tk in lakh)					Utilization (%)	
	Allocation			Allocation						
	GOB	DPA	Total	GOB	DPA	Total	GOB	DPA		Total
Establishment of 250-bed National Ophthalmology Institute and Hospital (1st Phase: 250 bed) Upgrading National Institute of Cancer Research and Hospital from 50-bed to 300-bed Establishment of National Institute of Laboratory Medicine and Referral Center Extension of Dhaka Shishu (Children) Hospital Project Establishment of Essential Drugs Company Limited, 3rd Plant, Gopalganj Expansion and quality improvement of nursing education Establishment of National Institute of Neuroscience Expansion and modernization of Dhaka Medical College Hospital Establishment of National Institute of ENT (1st Phase) ENT & Head-Neck Cancer Foundation Hospital and Institute Revitalization of community healthcare initiatives in Bangladesh Conversion of BSMMU to a center of excellence project Establishment of Sheikh Fajilatunnesa Mujib Eye Hospital and Training Institute, Gopalganj Establishment of National Centre for Cervical and Breast Cancer Screening and Training at BSMMU Establishment of Sheikh Sayera Khatun Medical College and Hospital and Nursing Institute, Gopalganj Establishment of Shaikhira Medical College & Hospital Establishment of Faridpur Medical College & Hospital National Institute of Digestive Diseases Research & Hospital Establishment of Kushtia Medical College Establishment of Mother and Childcare Hospital under A K Khan HealthCare Center of Excellence	20.00	1,000.00	1,020.00	10.00	1,500.00	1,510.00	4.21	1,500.00	1,504.21	99.6
	147.00	2,787.00	2,934.00	97.00	1,026.00	1,123.00	96.86	224.58	321.44	28.6
	1,000.00	-	1,000.00	159.00	-	159.00	158.49	-	158.49	99.7
	800.00	-	800.00	250.00	-	250.00	250.00	-	250.00	100.0
	300.00	-	300.00	292.00	-	292.00	290.85	-	290.85	99.6
	1,000.00	-	1,000.00	2,500.00	-	2,500.00	2,199.00	-	2,199.00	88.0
	-	-	-	2,896.00	-	2,896.00	2,889.43	-	2,889.43	99.8
	1,808.00	-	1,808.00	3,405.00	-	3,405.00	3,403.72	-	3,403.72	100.0
	600.00	-	600.00	3,033.00	-	3,033.00	2,530.43	-	2,530.43	83.4
	-	-	-	387.00	-	387.00	386.38	-	386.38	99.8
	27,500.00	-	27,500.00	39,178.00	-	39,178.00	38,030.81	-	38,030.81	97.1
	5,000.00	-	5,000.00	7,500.00	-	7,500.00	7,125.32	-	7,125.32	95.0
	4,800.02	-	4,800.02	5,995.00	-	5,995.00	5,664.48	-	5,664.48	94.5
	550.00	-	550.00	384.00	-	384.00	367.00	-	367.00	95.6
	3,000.00	-	3,000.00	900.00	-	900.00	899.73	-	899.73	100.0
	2,400.00	-	2,400.00	530.00	-	530.00	528.00	-	528.00	99.6
	3,850.00	-	3,850.00	4,000.00	-	4,000.00	3,960.69	-	3,960.69	99.0
	200.00	-	200.00	47.00	-	47.00	47.00	-	47.00	100.0
	1,950.00	-	1,950.00	1,000.00	-	1,000.00	730.85	0.99	731.84	73.2
	5.00	-	5.00	5.00	-	5.00	-	-	-	0.0
Total	54,930.02	3,787.00	58,717.02	72,568.00	2,526.00	75,094.00	69,563.25	1,725.57	71,288.82	94.9

Distribution of the MOHFW budget (total of non-development and development budget) by organization in fiscal 2013-2014 and fiscal 2012-2013 (revised budget) (Ref: MOF 2013) (Tk in thousand)

Agency	Organization with organization code	FY 2013-2014	FY 2012-2013 (revised)
MOHFW	Secretariat (MOHFW) (2701)	20,092,812	20,896,028
	MOHFW Total	20,092,812	20,896,028
Autonomous bodies	Autonomous bodies & other institutions (2705)	1,874,620	1,845,980
	Autonomous bodies Total	1,874,620	1,845,980
International organizations	International organizations (2706)	9,850	22,385
	International organizations Total	9,850	22,385
DGHS and subordinate organizations	DGHS (2711)	21,444,169	20,963,245
	Divisional establishments (2712)	74,665	69,344
	Civil surgeons' offices (2713)	652,647	610,462
	Upazila health offices (2714)	5,959,055	5,625,080
	DG Drug Administration (2715)	140,090	79,472
	Directorate of Nursing (2716)	4,631,940	1,319,264
	Health Engineering Department (2717)	1,347,660	1,335,570
	Medical colleges (2721)	1,901,101	1,702,275
	Paramedical institutes (2722)	45,164	42,602
	Medical Assistants Training Schools (2723)	116,499	109,604
	TB Control & Training institute (2724)	40,176	37,641
	Dental colleges (2725)	71,881	67,440
	Colleges of Nursing (2726)	20,087	19,082
	Govt. Tibbia College, Sylhet (2728)	8,132	7,456
	Govt. Unani & Ayurvedic Degree College & Hospital (2729)	48,615	46,632
	Govt. Homeopathic Degree College & Hospital (2730)	48,508	43,864
	Center for medical education (2733)	2,317,711	2,093,329
	Medical college hospitals (2741)	4,157,923	4,437,463
	District hospitals (2742)	3,935,816	4,072,905
	Other district-level hospitals (2743)	102,714	113,163
	Upazila health complexes and subcenters (2744)	8,087,857	8,179,716
	Dental college hospitals (2747)	16,387,242	16,906,715
	Specialized hospitals and institutes (2751)	2,617,113	2,721,310
	Epidemic disease control center (IEDCR) (2761)	30,081	28,526
	TB centers (2771)	162,169	152,413
	School health clinics (2772)	36,051	34,190
	Other facilities (2775)	207,640	198,216
	DGHS and subordinate organizations Total	74,592,706	71,016,979
DGFP and subordinate organizations	DGFP (2781)	5,900,297	6,453,458
	Divisional family planning offices (2783)	14,524	14,534
	District family planning offices (2785)	259,274	248,590
	Upazila family planning offices (2787)	10,123,041	9,216,352
	Family planning hospitals & dispensaries (2789)	241,961	229,303
	Other family welfare facilities under DGFP (2791)	425,388	425,613
	DGFP and subordinate organizations Total	16,964,485	16,587,850
Development program financed by revenue budget	Development program financed by revenue budget (2796)	-	28,608
	Development program financed by revenue budget Total	-	28,608
	Grand Total	113,534,473	110,397,830

Allocation and expenditure of non-development revised budget of fiscal 2012-2013 by organization under DGHS (Ref: MOF 2013) (Tk in thousand)

Head		Revised budget	Expense	Utilization (%)
DGHS		1,565,645,000	1,830,347,777	116.9
Divisional establishments		69,344,000	65,259,952	94.1
Civil surgeon's offices		610,462,000	582,641,847	95.4
Upazila health offices		5,625,080,000	5,309,497,735	94.4
Paramedical institutes	Medical and surgical supplies	1,000,000	-	0.0
	Others	41,602,000	34,714,391	83.4
	Total	42,602,000	34,714,391	81.5
Medical assistant training schools	Medical and surgical supplies	100,000	-	0.0
	Others	109,504,000	98,251,507	89.7
	Total	109,604,000	98,251,507	89.6
TB control & training Institutes	Medical and surgical supplies	2,000,000	1,983,986	99.2
	Others	35,641,000	31,072,912	87.2
	Total	37,641,000	33,056,898	87.8
District hospitals	Medical and surgical supplies	1,200,000,000	1,210,704,141	100.9
	Others	2,872,905,000	2,731,466,317	95.1
	Total	4,072,905,000	3,942,170,458	96.8
Other district-level hospitals	Total	113,163,000	107,402,579	94.9
Upazila health complexes and subcenters	Medical and surgical supplies	1,500,000,000	1,088,396,957	72.6
	Others	6,679,716,000	6,334,478,258	94.8
	Total	8,179,716,000	7,422,875,215	90.8
TB segregation hospitals	Medical and surgical supplies	8,000,000	6,347,241	79.3
	Others	30,260,000	27,293,638	90.2
	Total	38,260,000	33,640,879	87.9
Other TB Hospitals	Medical and surgical supplies	15,000,000	6,788,965	45.3
	Others	104,551,000	98,167,574	93.9
	Total	119,551,000	104,956,539	87.8
Leprosy hospitals	Medical and surgical supplies	6,000,000	4,885,322	81.4
	Others	36,514,000	33,400,825	91.5
	Total	42,514,000	38,286,147	90.1
TB centers	Medical and surgical supplies	21,000,000	16,609,478	79.1
	Others	131,413,000	127,521,237	97.0
	Total	152,413,000	144,130,715	94.6
School health centers	Medical and surgical supplies	4,400,000	4,064,479	92.4
	Others	29,790,000	29,464,920	98.9
	Total	34,190,000	33,529,399	98.1
Model family planning clinics	Total	46,352,000	45,271,341	97.7



Government of the People's Republic of Bangladesh
Ministry of Health and Family Welfare